Holomorphic Feynman diagrams in four dimensions and the quantum BPS ring

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Abstract

We compute, at arbitrary loop order, the kernels of the holomorphic Feynman graphs that carry the loop corrections to supersymmetry transformations and to regularized products of local operators in four-dimensional $\mathcal{N} = 1$ supersymmetric gauge theories. Building on the holomorphic graph integrals of Wang and the operatope of Budzik, Gaiotto, Kulp, Wu and Yu, we read the operatope current equations as those of an electrical network, with Schwinger times as resistances and holomorphic momenta as injected currents. Series, parallel and star–triangle reductions, used as changes of variables on Schwinger space, reduce each graph step by step to one loop; iterating them gives the fan, spider and cubic product-forest kernels at arbitrary loop order, and extends the reduction scheme of earlier work while giving it a physical interpretation. These kernels determine the loop-corrected, anomalous supersymmetry transformation of composite operators and the renormalized quantum product that defines the BPS ring. As an explicit application, we give every perturbative correction to the supersymmetry transformation in $\mathcal{N} = 4$ Yang–Mills theory, including its action on operators with arbitrary holomorphic derivatives. Using the marked higher operations of Gaiotto, Kulp and Wu, we also construct and compute the quantum product of operators and prove that, on $\mathbf{Q}$-cohomology, it is associative, graded commutative and unital, supplying the multiplication of the quantum BPS ring. As a consequence of this ring structure, every multi-graviton operator lifts to a cocycle of the full quantum supercharge, with an explicit and finite all-order completion. We also report a computer census of fortuitous classes and of their lifting: new fortuitous classes and one-loop lifts in $\mathcal{N} = 4$ Yang–Mills theory with several gauge groups, where we have found no two-loop lift so far, two-loop lifts in the Leigh–Strassler and β-deformations of $\mathcal{N} = 4$ Yang–Mills theory, two-, three- and four-loop lifts in abelian gauge theories, and two- and three-loop lifts in $\mathcal{N} = 2$ $SU(2)$ and $SU(3)$ gauge theories and their singlet deformations.

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1 Introduction

In a unitary superconformal theory, the cohomology of a nilpotent supercharge Q is isomorphic to the corresponding BPS sector. Each class has a unique harmonic representative, annihilated by Q and its radial adjoint $Q^\dagger = S$. For $Q = Q^4_-$ in $\mathcal{N} = 4$ Yang–Mills theory, this is the 1/16-BPS sector [1, Section 2]. The scaling dimensions of these states are fixed by their angular momenta and R-charges through the BPS bound. Quantum corrections to the supercharge, the anomalous supersymmetry transformation, determine which classical classes persist in the quantum BPS spectrum and constrain the microscopic states of dual AdS black holes [2].

In the physical theory, a complete treatment requires a careful definition of the renormalized products of local operators and of their counterterms, which is technically demanding. The holomorphic twist of the theory is a much more efficient framework. The holomorphic twist organizes supersymmetry cohomology in the Batalin–Vilkovisky (BV) formalism [3, 4]. Homotopy transfer takes the physical BV complex to the complex of holomorphic local operators, where the action and the BV differential[1] simplify considerably. The graph integrals of the twisted theory are ultraviolet finite [5, Theorem 5 and Corollary 4]. The anomalous supersymmetry transformation of composite operators is then a quantum correction to the BV differential, organized into the higher brackets of the twisted theory [4, 6]. Its perturbative cohomology determines the perturbative quantum BPS states, and computing it requires the higher brackets at every loop order.[2] Quantum corrections to operator relations already appear in the Konishi anomaly and the four-dimensional chiral ring [7, 8]; the complete one-loop correction Q_1 was computed in [4].

We further develop the holomorphic graph-integral and higher-operation constructions of [4–6, 9] to evaluate these corrections at higher loops. The graph integrals that evaluate the higher brackets are integrals over a region of Schwinger space called the operatope [9].[3] Intuitively, these are cut Feynman diagrams, a familiar structure in the computation of anomalies [4, 5].

We observe that the current equations and the loop constraints of the operatope [9, Equations (2.26), (2.31), (2.33)] are Ohm's and Kirchhoff's laws of an electrical network: the Schwinger time of each edge becomes a resistance, the holomorphic momenta of the output letters become

[1] Here the BV differential is the sum of the Koszul–Tate differential, which imposes the equations of motion, the gauge BRST transformation, and the nilpotent supercharge under study.

[2] The exact spectrum may also receive nonperturbative corrections: instanton effects can lift perturbative cohomology classes [3, Section 1] or, equivalently, correct the supercharges [4, Section 2.2.1].

[3] Wang defined the heat-kernel regularized graph integrals on $\mathbb{C}^d$ and established their ultraviolet finiteness, the holomorphic locality of their boundary integrals and the Kirchhoff structure of their Schwinger integrands [5, Sections 2–4]; the operatope supplied the combined Schwinger-space propagator, the edge-current variables and the Laman constraints [9, Section 2]; the higher operations are those of Gaiotto, Kulp and Wu [6], and the holomorphic supercharge construction is that of [3, 4].

injected currents, and Wang's weighted Laplacian [5, Definition 12, Remark 6] becomes the conductance matrix.[4] Building on this analogy and on the locality of Ohm's law, we use the circuit reductions iteratively, transferring the sources at each step, as changes of variables on Schwinger space. They cancel the spanning-tree denominators of the earlier formulations and reduce a higher-loop graph step by step to lower loops, and finally to one loop or to a tree.[5] Our circuit iteration interprets and extends the cube-integral recursion of Gui, Wang and Williams [13, Theorem 4.13]: it gives the recursion an explicit physical meaning as series and parallel reductions of resistors, and adds the star–triangle transformations.

The operatope is defined in arbitrary dimension d, for all graphs that satisfy the Laman condition together with the corresponding bound on every subgraph. Two families of graphs are of particular interest in the holomorphic twist of four-dimensional $\mathcal{N} = 1$ gauge theories, the fans and the spiders: they give all perturbative corrections to the supersymmetry transformation of composite operators, at arbitrary loop order, in every such theory with a Lagrangian description whose superpotential is treated perturbatively, as a polynomial or a formal power series. They are defined in Sections 5 and 7. In a theory whose superpotential is at most cubic, with an arbitrary Kähler potential, such as $\mathcal{N} = 4$ Yang–Mills theory, every loop correction to the supersymmetry transformation is a fan. We derive the fan and spider kernels in closed form at every loop order. For these graphs, the successive series and parallel reductions, read in reverse, construct the positive edge resistances from dimensionless ratios, the fractions in which a current divides between parallel resistors, and the network reduction gives the kernels in these simplex coordinates.[6]

The anomalous supersymmetry transformation[7] violates the Leibniz rule: at one loop it acts on two letters at once, like a second-order differential operator, and at n loops on $n + 1$ letters. The holomorphic twist also provides a perturbative quantum product, built from the higher brackets with a pair of auxiliary fields [6] and analogous to normal ordering in an interacting conformal field theory. When the quantum master equation holds, that is, when the gauge anomaly cancels, the perturbative BV differential obeys the Leibniz rule for this product. We prove that the analogy is exact: the quantum product is cohomologous to the integral over an arbitrary three-sphere against the Bochner–Martinelli form, the propagator. It is associative and graded commutative on $\mathbf{Q}$-cohomology, but neither on the full BV complex. The quantum product is not needed to compute the cohomology as a vector space; it supplies the ring structure. For example, the test of quantum lifting [3, Section 1] then needs to be carried out only on the generators of the BPS ring. The explicit form of the product also shows that the multi-gravitons of the 1/16-BPS sector of $SU(N)$ $\mathcal{N} = 4$ Yang–Mills theory remain quantum cocycles at arbitrary loop order. The quantum product is evaluated by the same kind of graph integrals; it involves more types of kernels than the supersymmetry transformation. From three loops on, the quantum product also contains interaction forests that are neither fans nor spiders; for renormalizable theories, series, parallel, and zero-source star–triangle moves keep their currents polynomial.[8]

[4] The reading of Schwinger parameters as resistances of a network is classical for Feynman integrals [10, 11].

[5] A single star–triangle move is a classical change of Schwinger variables [12, Lemma 2.4.23]; here, series, parallel, and star–triangle moves are iterated.

[6] In these coordinates, the currents are polynomial and the spanning-tree denominator of the Kirchhoff solution cancels, so every coefficient in the kernel's displacement expansion is a polynomial integral over explicit products of intervals and simplices, hence a finite factorial sum.

[7] We use the terms anomalous supersymmetry transformation, loop correction to the supercharge and quantum BV differential interchangeably; strictly they are different notions, but here they refer to the same operator.

[8] In the original Schwinger times the same integrands are rational, with powers of the spanning-tree polynomial in the denominator [5, Corollaries 1–2 and proof of Theorem 6], while the operatope coordinates simplify the integrand at the price of an intricate integration region [9, Sections 2.3 and 2.6]. Fans and spiders are Laman graphs of type I′; Gui, Wang and Williams wrote their weights recursively from the L_∞ relations as integrals over unit cubes with polynomial integrands at each order in the displacements [13, Theorem 4.13], leaving the relation of the cube parameters to Schwinger parameters open [13, Remark 4.14], and evaluated the fans explicitly through three loops, the last with computer algebra [13, Sections 4.4–4.6]. We supply this relation and its physical

As an explicit application, we give the all-order supercharge of $\mathcal{N}=4$ Yang–Mills theory in (8.3). We also improve the enumeration algorithm for fortuitous classes and find new classes in $\mathcal{N}=4$ Yang–Mills theory with gauge groups $SU(N)$, $SO(2N+1)$, $Sp(N)$ and G_2. The results and a description of the algorithm are collected in the repository [14]. With the explicit $Q_2, Q_3, \ldots$ derived here, we test the lifting of these classes beyond one loop. We find no two-loop lift in $\mathcal{N}=4$ Yang–Mills theory. We do find many one-loop lifts in the $SO(2N+1)$ theories; through the levels computed, they restore the agreement of the $SO(2N+1)$ and $Sp(N)$ spectra required by S-duality. We also find two-loop lifts in the Leigh–Strassler deformation [15] of $SU(3)$ $\mathcal{N}=4$ Yang–Mills theory, with superpotential

$$W = \operatorname{tr}(\gamma^1\gamma^2\gamma^3) - q\operatorname{tr}(\gamma^1\gamma^3\gamma^2) + \frac{h}{3}\left[\operatorname{tr}((\gamma^1)^3) + \operatorname{tr}((\gamma^2)^3) + \operatorname{tr}((\gamma^3)^3)\right]$$

two-, three- and four-loop lifts in abelian theories of SQED type, and two- and three-loop lifts in $\mathcal{N}=2$ $SU(2)$ and $SU(3)$ gauge theories and their singlet deformations.

Section 2 reviews the holomorphic twist of the physical theory, for readers who have not met it. Section 3 reviews the higher brackets and their reduction to a current integral, the starting point of our computations, and proves constraints on the contributing graphs in four dimensions and their classification. Section 4 summarizes our results; the proofs are in the later sections. Section 5 computes the fan kernel, the kernel most used in practice; further details are in Appendix A. Section 6 treats the quantum product and its Leibniz rule, with explicit products through two loops. Section 7 gives the further kernels, those of spiders and cubic forests, which enter the quantum product and theories with superpotentials of higher degree. Section 8 writes the n-loop supercharge Q_n of $\mathcal{N}=4$ Yang–Mills theory as a differential operator of order $n+1$. Section 9 proves that the multi-gravitons remain closed under the full supercharge and sets up the successive homotopy transfer on which the practical computation of the cohomology ring rests. Section 10 presents the tests of lifting at one loop and beyond. As a by-product, we find the first fortuitous class of $SU(4)$, at $E + J_L = 14$. Appendix A gives the iterated circuit reductions in detail, and Appendix B the charge sectors and gradings used in the lifting tests.

2 The BV twist and transfer to holomorphic fields

This section reviews the holomorphic twist for readers who have not met it [3, 4]. The starting point is the BV formulation of the physical theory, in which the fields, the ghosts and their antifields carry the differential $(S_{\mathrm{BV}}, \cdot\,)$; its cohomology consists of the gauge-invariant operators modulo the equations of motion. Adding one supercharge Q_- to this differential gives a complex whose cohomology consists of the operators annihilated by Q_- modulo Q_--exact ones, that is, the BPS sector.

The twist simplifies this complex in three steps. First, a canonical change of BV variables removes the Q_--exact part of the action and exhibits most fields as doublets (u, v) with $(S_{\mathrm{BV}}, u) \propto v$; they do not contribute to the cohomology and are contracted (Section 2.2). Second, the remaining fields and antifields assemble into four Dolbeault forms $\mathsf{c}, \mathsf{b}, \gamma, \beta$ on $\mathbb{C}^2$ with a first-order action (Section 2.3); their zero-form components are the familiar BPS letters (Section 2.4). Third, the free differential $\bar\partial$ is contracted, so that local operators become polynomials in the zero-form fields and their holomorphic derivatives (Section 2.5). The interaction then acts on these operators through the graph operations of Section 3.

interpretation.

2.1 Coordinates and fields

The index A labels the Lie algebra and i the matter representation. With the continuation $x^0 = ix^0_E$, we use Euclidean spinor coordinates and the correspondingly continued sigma matrices

$$x^{a\dot{a}} = x^\mu_E \bar{\sigma}^{\dot{a}a}_\mu. \tag{2.1}$$

The coordinates $x^{+\dot{a}}$ are holomorphic and $x^{-\dot{a}}$ antiholomorphic. In these coordinates,

$$\begin{aligned} &\frac{\partial}{\partial x^{a\dot{a}}} = -\frac{1}{2}\partial_{a\dot{a}}, \qquad \epsilon^{\dot{+}\dot{-}} = 1 = -\epsilon_{\dot{+}\dot{-}}, \qquad dx^-_{\dot{a}} = \epsilon_{\dot{a}\dot{b}} dx^{-\dot{b}}, \\ &\bar{\partial} = dx^{-\dot{a}} \frac{\partial}{\partial x^{-\dot{a}}} = -\frac{1}{2} dx^{-\dot{a}} \partial_{-\dot{a}}, \qquad \Omega = 2\, dx^{+\dot{+}} \wedge dx^{+\dot{-}}. \end{aligned} \tag{2.2}$$

Let c^A be the gauge ghost, V^A the gauge field, $\lambda^A, \bar{\lambda}^A$ the gauginos, and D^A the gauge auxiliary field. A chiral matter multiplet has components $\phi^i, \psi^i, \mathcal{F}^i$, with $\mathcal{F}^i$ the auxiliary field.

$$\begin{aligned} [t_B, t_C] &= iC^A{}_{BC} t_A, \qquad C_{ABC} = \delta_{AD} C^D{}_{BC} = C_{[ABC]}, \\ f^A_{\mu\nu} &= \partial_\mu V^A_\nu - \partial_\nu V^A_\mu + g C^A{}_{BC} V^B_\mu V^C_\nu, \\ f^A_{++} &= \partial_{+\dot{a}} V^{A\dot{a}}_+ + \frac{g}{2} C^A{}_{BC} V^B_{+\dot{a}} V^{C\dot{a}}_+. \end{aligned} \tag{2.3}$$

Here $\partial_\mu = \partial/\partial x^\mu_E$ and $\partial_{a\dot{a}} = \sigma^\mu_{a\dot{a}} \partial_\mu$; undotted indices use $\epsilon^{+-} = 1 = -\epsilon_{+-}$. Write $\bar{W}^A_{\dot{a}}(x, \theta, \bar{\theta})$ for the antichiral field-strength superfield, with $\theta^a, \bar{\theta}^{\dot{a}}$ the odd superspace coordinates [16, Eqs. (27.2.12), (27.3.16)]. Formula (3.1) follows from the identification $dx^{-\dot{a}} = 4\bar{\theta}^{\dot{a}}$ in (2.12).

2.2 Canonical changes of variables

In this subsection, we start from the original physical action and perform a canonical transformation ("adding Q-exact terms to the action").

We use the $N = 1$ off-shell superfield formulation in the gauge vector representation [16, Chapters 26–27].

Write $\Phi^{\mathcal{A}}$ for fields and gauge ghosts, $\Phi^\ddagger_{\mathcal{A}}$ for antifields, and s for the gauge BRST differential. The BV bracket is

$$(F, G) = \int d^4x \left[\frac{\delta_R F}{\delta \Phi^\ddagger_{\mathcal{A}}} \frac{\delta_L G}{\delta \Phi^{\mathcal{A}}} - \frac{\delta_R F}{\delta \Phi^{\mathcal{A}}} \frac{\delta_L G}{\delta \Phi^\ddagger_{\mathcal{A}}} \right]. \tag{2.4}$$

Let S_{free} be the free quadratic kinetic action. For the chosen nilpotent supercharge Q_- and with the gauge BRST transformation denoted by s, the physical and BV actions have the decomposition

$$\begin{aligned} S_{\text{phys}} &= \int d^4x \Big[\frac{1}{8} \nabla_- \nabla_+ \bar{\nabla}^2 K(\bar{\Phi}, \Phi) + \frac{1}{2} \nabla_- \nabla_+ W(\Phi, W_a) \\ &\qquad + \frac{1}{4} \bar{\nabla}^2 \bar{W}(\bar{\Phi}, \bar{W}_{\dot{a}}) \Big] \Big|, \\ Q_- \Psi &= \int d^4x\, \nabla_- \left[\frac{1}{8} \nabla_+ \bar{\nabla}^2 K(\bar{\Phi}, \Phi) + \frac{1}{2} \nabla_+ W(\Phi, W_a) \right] \Big| \\ &= \int d^4x \left[\frac{1}{8} \nabla_- \nabla_+ \bar{\nabla}^2 K(\bar{\Phi}, \Phi) + \frac{1}{2} \nabla_- \nabla_+ W(\Phi, W_a) \right] \Big| \\ &= S_{\text{phys}} - \frac{1}{4} \int d^4x\, \bar{\nabla}^2 \bar{W}(\bar{\Phi}, \bar{W}_{\dot{a}}) \Big|. \end{aligned}$$

$$
\begin{aligned}
S_{\text{phys}} &= Q_-\Psi + \int d^4x\,[\bar{W}(\bar{\Phi},\bar{W}_{\dot{a}})]_{\bar{\mathcal{F}}},\\
\Psi &= \int d^4x\left[\frac{1}{8}\nabla_+\bar{\nabla}^2 K(\bar{\Phi},\Phi)+\frac{1}{2}\nabla_+ W(\Phi,W_a)\right]\Big|,\\
S_{\text{BV}} &= S_{\text{phys}} + \int d^4x\,((s+Q_-)\Phi^{\mathcal{A}})\Phi^{\ddagger}_{\mathcal{A}},\\
Q_{\text{BV}} &= (S_{\text{free}},\,\cdot), \qquad s\Psi = 0, \quad (s+Q_-)^2 = 0.
\end{aligned}
\tag{2.5}
$$

Here K is the Kähler potential, and $W(\Phi, W_a)$ is the chiral superpotential including its gauge kinetic term, with W_a the chiral field-strength superfield. The derivatives $\nabla_a, \bar{\nabla}_{\dot{a}}$ are gauge-covariant superspace derivatives, and $[\bar{W}]_{\bar{\mathcal{F}}} = \frac{1}{4}\bar{\nabla}^2\bar{W}|$ is the antichiral top component. The vertical bar evaluates at $\theta = \bar{\theta} = 0$, taking the lowest component.

We now perform the BV canonical transformation generated by Ψ (Ψ can be regarded as a gauge-fixing fermion).

$$
\begin{aligned}
\Phi^{\ddagger}_{\mathcal{A},\text{new}} &= \Phi^{\ddagger}_{\mathcal{A},\text{old}} + \frac{\delta_L\Psi}{\delta\Phi^{\mathcal{A}}}, \qquad \Phi^{\mathcal{A}}_{\text{new}} = \Phi^{\mathcal{A}}_{\text{old}}\\
S_{\text{BV,new}} &= S_{\text{BV,old}}\left(\Phi, \Phi^{\ddagger}_{\text{new}} - \frac{\delta_L\Psi}{\delta\Phi}\right)\\
&= \int d^4x\left([\bar{W}(\bar{\Phi},\bar{W}_{\dot{a}})]_{\bar{\mathcal{F}}} + ((s+Q_-)\Phi^{\mathcal{A}})\Phi^{\ddagger}_{\mathcal{A},\text{new}}\right).
\end{aligned}
\tag{2.6}
$$

This removes the Q_--exact term from the action, which cancels almost all terms in the physical action, leaving only the antichiral superpotential.

Many of the fields can be arranged into the contractible pairs listed in the table below, which we will contract.

Table 2.1: Contractible pairs of the BV complex.

u	v	$(S_{\text{BV}}, u) = av$
$V^A_{+\dot{a}}$	$\bar{\lambda}'^A_{\dot{a}} = \bar{\lambda}^A_{\dot{a}} - \frac{1}{2}\nabla_{+\dot{a}}c^A$	$a = -2$
λ^A_+	$D'^A = D^A + gC^A{}_{BC}c^B\lambda^C_+ - if^A_{+-}$	$a = -1$
ϕ^i	$\psi'^i_- = \psi^i_- - \frac{g}{\sqrt{2}}c^A(t_A)^i{}_j\phi^j$	$a = -i\sqrt{2}$
ψ^i_+	$\mathcal{F}'^i = \mathcal{F}^i + \frac{g}{\sqrt{2}}c^A(t_A)^i{}_j\psi^j_+$	$a = i\sqrt{2}$

Nilpotency gives $(S_{\text{BV}}, v) = 0$.

We perform an upper-triangular coordinate change, with $\bar{\lambda} \to \bar{\lambda}', D \to D', \psi_- \to \psi'_-, \mathcal{F} \to \mathcal{F}'$.

Let $\mathcal{T}$ contain these pairs and their spacetime derivatives. The remaining fields $\mathcal{S} = \{c, V_-, \lambda_-; \bar{\phi}, \bar{\psi}, \bar{\mathcal{F}}\}$ are unchanged.

The key point is that the $\mathcal{S}$ letters transform only among themselves (without $\mathcal{T}$ dependence), and the $\mathcal{T}$ letters form contractible pairs.

We remind the reader that we will contract $\mathcal{T}$ and keep the $\mathcal{S}$ letters and their antifields; however, the $\mathcal{S}$ letters are not the familiar "BPS letters".

On field functionals the contraction is

$$
\begin{aligned}
h_{\mathcal{T}}\mathcal{O} &= \int_0^1 \frac{dt}{t}\left(\sum_{(u,v)}\frac{u}{a}\frac{\partial_L\mathcal{O}}{\partial v}\right)\Bigg|_{(u,v)\mapsto(tu,tv)},\\
\Psi' &= h_{\mathcal{T}}\int d^4x\,[\bar{W}(\bar{\Phi},\bar{W}_{\dot{a}})]_{\bar{\mathcal{F}}},\\
(s+Q_-)\Psi' &= \int d^4x\,[\bar{W}]_{\bar{\mathcal{F}}} - \int d^4x\,[\bar{W}]_{\bar{\mathcal{F}}}\Big|_{\mathcal{T}=0}.
\end{aligned}
\tag{2.7}
$$

The sum includes all spacetime derivatives of each pair. The second antifield translation, generated by Ψ' with the sign in (2.6), removes the $\mathcal{T}$ dependence of the antichiral action. It gives

$$\begin{gathered} S_{\mathrm{BV,new}} = S_{\mathcal{S}} + S_{\mathcal{T}}, \qquad S_{\mathcal{T}} = \int d^4x \sum_{(u,v)} av\, u^{\ddagger}, \\ (S_{\mathcal{S}}, S_{\mathcal{T}}) = 0, \qquad (S_{\mathcal{S}}, S_{\mathcal{S}}) = (S_{\mathcal{T}}, S_{\mathcal{T}}) = 0. \end{gathered} \tag{2.8}$$

Here $S_{\mathcal{S}}$ depends only on the retained fields and their antifields.

Including the antifields and all spacetime jets, the contraction satisfies

$$\begin{gathered} (S_{\mathrm{BV}}, v^{\ddagger}) = (-1)^{|u|} a u^{\ddagger}, \qquad (S_{\mathrm{BV}}, u^{\ddagger}) = 0, \\ h_{\mathcal{T}}\mathcal{O} = \int_0^1 \frac{dt}{t} \sum_{(u,v)} \left(\frac{u}{a}\frac{\partial_L}{\partial v} + \frac{(-1)^{|u|} v^{\ddagger}}{a} \frac{\partial_L}{\partial u^{\ddagger}} \right) \mathcal{O}\Big|_{\mathcal{T}\mapsto t\mathcal{T}}, \\ (S_{\mathrm{BV}}, h_{\mathcal{T}}\mathcal{O}) + h_{\mathcal{T}}(S_{\mathrm{BV}}, \mathcal{O}) = \mathcal{O} - \mathcal{O}|_{\mathcal{T}=0}, \qquad [h_{\mathcal{T}}, \partial_\mu] = 0. \end{gathered} \tag{2.9}$$

For $(S_{\mathrm{BV}}, \mathcal{O}) = 0$, this gives

$$\mathcal{O} = \mathcal{O}|_{\mathcal{T}=0} + (S_{\mathrm{BV}}, h_{\mathcal{T}}\mathcal{O}). \tag{2.10}$$

The representative on the right contains only retained fields and antifields in the complex $\mathcal{S}$.

In particular,

$$\nabla_{+\dot{a}} \longmapsto \partial_{+\dot{a}}, \qquad \bar{\lambda}^A_{\dot{a}} \longmapsto \frac{1}{2}\partial_{+\dot{a}} c^A = -\frac{\partial c^A}{\partial x^{+\dot{a}}}. \tag{2.11}$$

2.3 Assemble the retained fields into Dolbeault forms

In the remaining complex $\mathcal{S}$, the fields and antifields assemble as

$$\begin{aligned} \mathsf{c}^A &= c^A - 2\bar{\theta}_{\dot{a}} V_-^{A\dot{a}} + 2i\bar{\theta}_{\dot{a}}\bar{\theta}^{\dot{a}} \lambda_-^A, \\ \gamma_i &= \bar{\phi}_i - \sqrt{2}\bar{\theta}_{\dot{a}}\bar{\psi}_i^{\dot{a}} + \bar{\theta}_{\dot{a}}\bar{\theta}^{\dot{a}}\bar{\mathcal{F}}_i, \\ \beta^i &= \bar{\mathcal{F}}^{\ddagger i} + \sqrt{2}\bar{\theta}_{\dot{a}}\bar{\psi}^{\ddagger \dot{a} i} + \bar{\theta}_{\dot{a}}\bar{\theta}^{\dot{a}}\bar{\phi}^{\ddagger i}, \\ \mathsf{b}_A &= -\frac{i}{2}\lambda_A^{\ddagger -} + \bar{\theta}_{\dot{a}} V_A^{\ddagger -\dot{a}} + \bar{\theta}_{\dot{a}}\bar{\theta}^{\dot{a}} c_A^{\ddagger}, \\ \bar{\partial} &= 2\bar{\theta}_{\dot{a}}\partial_-{}^{\dot{a}}, \qquad dx^{-\dot{a}} = 4\bar{\theta}^{\dot{a}}. \end{aligned} \tag{2.12}$$

Here $\bar{\psi}^{\ddagger i}_{\dot{a}}$ is BV-dual to $\bar{\psi}^{\dot{a}}_i$, and $\bar{\psi}^{\ddagger \dot{a} i}$ in β^i has its dotted index raised as in (2.2).

$$\begin{gathered} \bar{\psi}^{\ddagger i}_{\dot{a}}((s+Q_-)\bar{\psi}^{\dot{a}}_i) = -((s+Q_-)\bar{\psi}_{i\dot{a}})\bar{\psi}^{\ddagger \dot{a} i}, \\ \sqrt{2}\bar{\theta}_{\dot{a}}\bar{\psi}^{\ddagger \dot{a} i} = -\sqrt{2}\bar{\psi}^{\ddagger i}_{\dot{a}}\bar{\theta}^{\dot{a}}. \end{gathered}$$

The identification $dx^{-\dot{a}} = 4\bar{\theta}^{\dot{a}}$ and the volume relation

$$\Omega \wedge dx^{-\dot{+}} \wedge dx^{-\dot{-}} = -8\, d^4x_E, \qquad d^4x_E = dx^0_E \wedge dx^1_E \wedge dx^2_E \wedge dx^3_E \tag{2.13}$$

give the action S_{tw} in (3.3).

Its free kinetic part is

$$\begin{aligned} S_{\mathrm{free}} &= \int_{\mathbb{C}^2} \Omega \wedge [\mathsf{b}_A \bar{\partial}\mathsf{c}^A + (\bar{\partial}\gamma_i)\beta^i]_{(0,2)}, \\ Q_{\mathrm{BV}} &= (S_{\mathrm{free}}, \cdot) = \bar{\partial}, \qquad (S_{\mathrm{tw}}, \cdot) = Q_{\mathrm{BV}} + (I, \cdot). \end{aligned} \tag{2.14}$$

The original-coordinate embedding keeps the antifield translations:

$$\begin{aligned}\beta^{(0)i} &= \bar{\mathcal{F}}^{\ddagger i}_{\rm old} + \frac{\delta_L(\Psi+\Psi')}{\delta\bar{\mathcal{F}}_i},\\ \mathsf{b}^{(0)}_A &= -\frac{i}{2}\left(\lambda^{\ddagger -}_{A,\rm old} + \frac{\delta_L(\Psi+\Psi')}{\delta\lambda^A_-}\right).\end{aligned} \tag{2.15}$$

For canonical matter and quadratic gauge kinetic terms, the combined translation has generator

$$\begin{aligned}\Psi+\Psi' = \int d^4x \Big[&-\frac{1}{\sqrt{2}}\phi^i\nabla_{+\dot a}\bar\psi^{\dot a}_i + g\bar\phi_j(t_A)^j{}_i\phi^i\lambda^A_+\\ &-\frac{i}{\sqrt{2}}\bar{\mathcal{F}}_i\psi^i_+ - \frac{i}{\sqrt{2}}\psi^i_+\frac{\partial W(\phi,0)}{\partial\phi^i}\\ &+\frac{i}{2}\delta_{AB}f^A_{++}\lambda^B_- - \frac{1}{2}\delta_{AB}(D^A+if^A_{+-})\lambda^B_+\Big].\end{aligned} \tag{2.16}$$

They give

$$\begin{aligned}\beta^{(0)i} &= \bar{\mathcal{F}}^{\ddagger i}_{\rm old} - \frac{i}{\sqrt{2}}\psi^i_+,\\ \mathsf{b}^{(0)}_A &= -\frac{i}{2}\lambda^{\ddagger -}_{A,\rm old} + \frac{1}{4}\delta_{AB}f^B_{++}.\end{aligned} \tag{2.17}$$

For the quadratic gauge kinetic term used below,

$$-\frac{1}{4}\delta_{AB}\mathsf{w}^A_{\dot a}\mathsf{w}^{B\dot a} = -\frac{1}{16}\delta_{AB}\partial_{+\dot a}(\mathsf{c}^A\partial_+{}^{\dot a}\mathsf{c}^B), \tag{2.18}$$

so it is a total derivative.

2.4 BV completion of the Chang–Yin letters

For the canonical kinetic terms of (2.16), the Chang–Yin letters are the scalar $\bar\phi(z)$, chiral fermion $\psi_+(z)$, self-dual curvature $f_{++}(z)$ and antichiral gaugino $\bar\lambda(z)$. Their component complex uses the auxiliary $\bar{\mathcal{F}}$ and gaugino λ^- equations of motion (which are exactly the antifield parts of β and b). In the present conventions these equations are

$$\mathcal{F}^i + \frac{\partial\bar W(\bar\phi,0)}{\partial\bar\phi_i} = 0, \qquad \nabla_{+\dot a}\bar\lambda^{A\dot a} - \sqrt{2}g\delta^{AB}\bar\phi_j(t_B)^j{}_i\psi^i_+ = 0. \tag{2.19}$$

The original antifields implement these equations within the BV complex:

$$\begin{aligned}(S_{\rm BV},\bar{\mathcal{F}}^{\ddagger i}_{\rm old}) &= igc^A(t_A)^i{}_j\bar{\mathcal{F}}^{\ddagger j}_{\rm old} - \mathcal{F}^i - \frac{\partial\bar W(\bar\phi,0)}{\partial\bar\phi_i},\\ (S_{\rm BV},\lambda^{\ddagger -}_{A,\rm old}) &= -gC^B{}_{AC}c^C\lambda^{\ddagger -}_{B,\rm old} + i\delta_{AB}\nabla_{+\dot a}\bar\lambda^{B\dot a}\\ &\quad - i\sqrt{2}g\bar\phi_j(t_A)^j{}_i\left(\psi^i_+ + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger i}_{\rm old}\right).\end{aligned} \tag{2.20}$$

The completed fermion and curvature are precisely the retained letters of (2.17), with their fixed component factors:

$$\psi^i_+ + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger i}_{\rm old} = i\sqrt{2}\beta^{(0)i}, \qquad f^A_{++} - 2i\delta^{AB}\lambda^{\ddagger -}_{B,\rm old} = 4\delta^{AB}\mathsf{b}^{(0)}_B. \tag{2.21}$$

Using $Q_-\psi_+^i = i\sqrt{2}\mathcal{F}^i$ and $Q_- f_{++}^A = -2\nabla_{+\dot a}\bar\lambda^{A\dot a}$ gives

$$
\begin{aligned}
&(S_{\rm BV}, \psi_+^i + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger i}_{\rm old}) \\
&\quad = igc^A(t_A)^i{}_j\left(\psi_+^j + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger j}_{\rm old}\right) - i\sqrt{2}\frac{\partial\bar W(\bar\phi,0)}{\partial\bar\phi_i}, \\
&(S_{\rm BV}, f_{++}^A - 2i\delta^{AB}\lambda^{\ddagger-}_{B,\rm old}) \\
&\quad = -gC^A{}_{BC}c^B\left(f_{++}^C - 2i\delta^{CD}\lambda^{\ddagger-}_{D,\rm old}\right) - 2\nabla_{+\dot a}\bar\lambda^{A\dot a} \\
&\qquad - 2i\left[i\nabla_{+\dot a}\bar\lambda^{A\dot a} - i\sqrt{2}g\delta^{AB}\bar\phi_j(t_B)^j{}_i\left(\psi_+^i + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger i}_{\rm old}\right)\right] \\
&\quad = -4g\delta^{AB}C^C{}_{BD}c^D\mathsf{b}_C^{(0)} - 4ig\delta^{AB}\bar\phi_j(t_B)^j{}_i\beta^{(0)i}.
\end{aligned}
\tag{2.22}
$$

The auxiliary field and the gaugino divergence cancel, leaving the familiar classical terms and keeping the $\mathcal{S}$ complex closed under $(S_{\rm BV},\cdot)$.

At the insertion $x=0$, physical operators are gauge-invariant words independent of the undifferentiated ghost $c^A(0)$.

We can now build the full dictionary between twisted letters and Chang–Yin letters (jets).

The symbol $\longleftrightarrow$ denotes the image after the doublet contraction and projection to holomorphic zero-form jets. Parentheses symmetrize the dotted indices with weight one, and $n \geq 0$.

$$
\begin{aligned}
\nabla_{+(\dot a_1}\cdots\nabla_{+\dot a_n)}\bar\phi_i &\longleftrightarrow \partial_{+\dot a_1}\cdots\partial_{+\dot a_n}\gamma_i^{(0)}, \\
\nabla_{+(\dot a_1}\cdots\nabla_{+\dot a_n)}\left(\psi_+^i + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger i}_{\rm old}\right) &\longleftrightarrow i\sqrt{2}\partial_{+\dot a_1}\cdots\partial_{+\dot a_n}\beta^{(0)i}, \\
\nabla_{+(\dot a_1}\cdots\nabla_{+\dot a_n)}\left(f_{++}^A - 2i\delta^{AB}\lambda^{\ddagger-}_{B,\rm old}\right) &\longleftrightarrow 4\delta^{AB}\partial_{+\dot a_1}\cdots\partial_{+\dot a_n}\mathsf{b}_B^{(0)}, \\
2\nabla_{+(\dot a_1}\cdots\nabla_{+\dot a_n}\bar\lambda^A_{\dot a_0)} &\longleftrightarrow \partial_{+\dot a_0}\cdots\partial_{+\dot a_n}\mathsf{c}^{A(0)}.
\end{aligned}
\tag{2.23}
$$

Ordered covariant derivatives reduce to symmetrized derivatives and curvature terms. Completing each f_{++}^A to $f_{++}^A - 2i\delta^{AB}\lambda^{\ddagger-}_{B,\rm old}$ gives representatives in the retained complex.

For ordered derivatives, the curvature commutator determines the additional b term. Since $\nabla_{+\dot a}\bar\phi_i = \partial_{+\dot a}\bar\phi_i + igV^A_{+\dot a}(t_A)^j{}_i\bar\phi_j$, the full BV variation is

$$
\begin{aligned}
(S_{\rm BV}, \nabla_{+\dot a}\nabla_+{}^{\dot b}\bar\phi_i) = &-igc^A(t_A)^j{}_i\nabla_{+\dot a}\nabla_+{}^{\dot b}\bar\phi_j \\
&- 2ig\Big[\frac{1}{2}\left(\nabla_{+\dot a}\bar\lambda^{A\dot b} + \nabla_+{}^{\dot b}\bar\lambda^A_{\dot a}\right)\bar\phi_j + \bar\lambda^A_{\dot a}\nabla_+{}^{\dot b}\bar\phi_j + \bar\lambda^{A\dot b}\nabla_{+\dot a}\bar\phi_j\Big](t_A)^j{}_i \\
&- ig\delta_{\dot a}{}^{\dot b}(\nabla_{+\dot c}\bar\lambda^{A\dot c})(t_A)^j{}_i\bar\phi_j.
\end{aligned}
\tag{2.24}
$$

The last term has the free gaugino equation as its linear field factor. The complementary antifield term has variation

$$
\begin{aligned}
&\left(S_{\rm BV}, g\delta_{\dot a}{}^{\dot b}\delta^{AB}\lambda^{\ddagger-}_{B,\rm old}(t_A)^j{}_i\bar\phi_j\right) \\
&\quad = -ig^2\delta_{\dot a}{}^{\dot b}\delta^{AB}c^C\lambda^{\ddagger-}_{B,\rm old}(t_C)^j{}_i(t_A)^k{}_j\bar\phi_k \\
&\qquad + ig\delta_{\dot a}{}^{\dot b}(\nabla_{+\dot c}\bar\lambda^{A\dot c})(t_A)^j{}_i\bar\phi_j \\
&\qquad - i\sqrt{2}g^2\delta_{\dot a}{}^{\dot b}\delta^{AB}\bar\phi_k(t_B)^k{}_\ell\left(\psi_+^\ell + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger\ell}_{\rm old}\right)(t_A)^j{}_i\bar\phi_j.
\end{aligned}
\tag{2.25}
$$

The divergence cancels between these two equations.

$$\begin{aligned}
&\nabla_{+\dot{a}}\nabla_+{}^{\dot{b}}\bar{\phi}_i + g\delta_{\dot{a}}{}^{\dot{b}}\delta^{AB}\lambda^{\ddagger-}_{B,\mathrm{old}}(t_A)^j{}_i\bar{\phi}_j \\
&\quad= \frac{1}{2}\left(\nabla_{+\dot{a}}\nabla_+{}^{\dot{b}} + \nabla_+{}^{\dot{b}}\nabla_{+\dot{a}}\right)\bar{\phi}_i + \frac{ig}{2}\delta_{\dot{a}}{}^{\dot{b}}\left(f^A_{++} - 2i\delta^{AB}\lambda^{\ddagger-}_{B,\mathrm{old}}\right)(t_A)^j{}_i\bar{\phi}_j \\
&\quad\longleftrightarrow \partial_{+\dot{a}}\partial_+{}^{\dot{b}}\gamma^{(0)}_i + 2ig\delta_{\dot{a}}{}^{\dot{b}}\delta^{AB}\mathsf{b}^{(0)}_B(t_A)^j{}_i\gamma^{(0)}_j.
\end{aligned} \tag{2.26}$$

The antisymmetric gaugino jet gives the other relation used in [1, Eq. (2.4)]. Equation (2.20) gives the explicit BV boundary

$$\begin{aligned}
&\nabla_{+\dot{a}}\bar{\lambda}^{A\dot{b}} + \frac{ig}{2}\delta_{\dot{a}}{}^{\dot{b}}\delta^{AB}C^C{}_{BD}c^D\lambda^{\ddagger-}_{C,\mathrm{old}} \\
&\quad- \left[\frac{1}{2}\left(\nabla_{+\dot{a}}\bar{\lambda}^{A\dot{b}} + \nabla_+{}^{\dot{b}}\bar{\lambda}^A_{\dot{a}}\right) + \frac{g}{\sqrt{2}}\delta_{\dot{a}}{}^{\dot{b}}\delta^{AB}\bar{\phi}_j(t_B)^j{}_i\left(\psi^i_+ + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger i}_{\mathrm{old}}\right)\right] \\
&\quad= \left(S_{\mathrm{BV}}, -\frac{i}{2}\delta_{\dot{a}}{}^{\dot{b}}\delta^{AB}\lambda^{\ddagger-}_{B,\mathrm{old}}\right).
\end{aligned} \tag{2.27}$$

The representative in square brackets transfers as

$$\begin{aligned}
&\frac{1}{2}\left(\nabla_{+\dot{a}}\bar{\lambda}^{A\dot{b}} + \nabla_+{}^{\dot{b}}\bar{\lambda}^A_{\dot{a}}\right) + \frac{g}{\sqrt{2}}\delta_{\dot{a}}{}^{\dot{b}}\delta^{AB}\bar{\phi}_j(t_B)^j{}_i\left(\psi^i_+ + i\sqrt{2}\bar{\mathcal{F}}^{\ddagger i}_{\mathrm{old}}\right) \\
&\qquad\longleftrightarrow \frac{1}{2}\partial_{+\dot{a}}\partial_+{}^{\dot{b}}\mathsf{c}^{A(0)} + ig\delta_{\dot{a}}{}^{\dot{b}}\delta^{AB}(t_B)^j{}_i\gamma^{(0)}_j\beta^{(0)i}.
\end{aligned} \tag{2.28}$$

The induced classical differential acts on the four zero-form fields by

$$\begin{aligned}
&Q_0\mathsf{c}^A = -\frac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C, && Q_0\gamma_i = -ig\mathsf{c}^A(t_A)^j{}_i\gamma_j, \\
&Q_0\beta^i = ig\mathsf{c}^A(t_A)^i{}_j\beta^j - \frac{\partial\bar{W}(\gamma,0)}{\partial\gamma_i}, && Q_0\mathsf{b}_A = -gC^B{}_{AC}\mathsf{c}^C\mathsf{b}_B - ig(t_A)^j{}_i\gamma_j\beta^i.
\end{aligned} \tag{2.29}$$

Here all fields denote their zero-form parts, and $[Q_0, \partial_{+\dot{a}}] = 0$ supplies the transformations of every jet.

2.5 Dolbeault contraction and Wick fibre contraction

We now prove that, with the twisted action S_{tw}, the antiholomorphic parts of the letters are indeed contractible.

We use the perturbative homotopy transfer of Section 9.3, one step before its page E_0: we first take the $\bar{\partial}$-cohomology of the letters.

For polynomial Dolbeault jets, let ι include holomorphic representatives and π project onto them. Here $\iota_{-\dot{a}}$ contracts a differential form with $\partial/\partial x^{-\dot{a}}$; N counts antiholomorphic coordinate and form degree:

$$\begin{aligned}
&\{\bar{\partial}, \iota_{-\dot{a}}\} = \frac{\partial}{\partial x^{-\dot{a}}}, \qquad N = \{\bar{\partial}, \iota_{x^{-\dot{a}}\partial/\partial x^{-\dot{a}}}\} = x^{-\dot{a}}\frac{\partial}{\partial x^{-\dot{a}}} + dx^{-\dot{a}}\iota_{-\dot{a}}, \\
&\pi X = X\big|_{x^{-\dot{a}}=dx^{-\dot{a}}=0}, \qquad h_{\bar{\partial}} = \begin{cases} \iota_{x^{-\dot{a}}\partial/\partial x^{-\dot{a}}}N^{-1}, & N > 0, \\ 0, & N = 0, \end{cases} \\
&\bar{\partial}h_{\bar{\partial}} + h_{\bar{\partial}}\bar{\partial} = 1 - \iota\pi, \qquad h^2_{\bar{\partial}} = h_{\bar{\partial}}\iota = \pi h_{\bar{\partial}} = 0.
\end{aligned} \tag{2.30}$$

The contraction removes antiholomorphic coordinates and forms, leaving the holomorphic jets

of the zero-form fields. For a Dolbeault-closed output field, translation therefore acts as

$$\begin{aligned}\pi X(x_v+x) &= \pi\exp\left(x_v^{+\dot a}\frac{\partial}{\partial x^{+\dot a}}+x_v^{-\dot a}\frac{\partial}{\partial x^{-\dot a}}\right)X(x)\\ &= \exp\left(x_v^{+\dot a}\frac{\partial}{\partial x^{+\dot a}}\right)\pi X(x).\end{aligned}\tag{2.31}$$

This is the source of the momentum substitution in the higher bracket (3.9): the source at the root is eliminated by momentum conservation, and the sources at the integrated vertices become derivatives of their surviving letters,

$$\lambda_{k,+\dot a}=-\sum_{\substack{v=1,\dots,k-1\\ \text{integrated vertices}}}\lambda_{v,+\dot a},\qquad \lambda_{v,+\dot a}\longmapsto\frac{\partial}{\partial x_v^{+\dot a}}=-\frac{1}{2}\partial_{v,+\dot a}\qquad (v<k),\tag{2.32}$$

with the Leibniz rule when several letters survive at v. A fully contracted integrated vertex has $\lambda_v=0$.

We now write the Wick fibre contraction explicitly so that the graph kernels determine higher brackets on words and their holomorphic derivatives.

Let F be the spacetime form in (3.9) and $e_a=(v_a,w_a)$ its ordered edges. For operators with no input holomorphic derivatives, the fibre contraction is

$$\begin{aligned}&W_\Gamma[F;\mathcal{O}_1,\dots,\mathcal{O}_k]\\ &\quad=F\prod_{a=1}^{|\Gamma_1|}{}^{\text{edge order}}\left(\begin{array}{c}\dfrac{\partial_L}{\partial \mathsf{b}_{A,v_a}}\dfrac{\partial_L}{\partial \mathsf{c}^A_{w_a}}+\dfrac{\partial_L}{\partial \mathsf{c}^A_{v_a}}\dfrac{\partial_L}{\partial \mathsf{b}_{A,w_a}}\\ -\dfrac{\partial_L}{\partial\beta^i_{v_a}}\dfrac{\partial_L}{\partial\gamma_{i,w_a}}-\dfrac{\partial_L}{\partial\gamma_{i,v_a}}\dfrac{\partial_L}{\partial\beta^i_{w_a}}\end{array}\right)\mathcal{O}_1\cdots\mathcal{O}_k.\end{aligned}\tag{2.33}$$

The fibre operators act on the ordered product from right to left. The displayed formula uses the same separation of spacetime and fibre factors as (3.5). For holomorphic jets, let X,Y be the paired fields in a summand and let r,s and p,q be their respective derivative orders. Each oriented edge $e=(v,w)$ replaces the two fibre derivatives in every summand of (2.33) by

$$\begin{aligned}&\frac{\partial_L}{\partial X_v}\frac{\partial_L}{\partial Y_w}\\ &\quad\longmapsto\sum_{r,s,p,q\ge0}(-2)^{r+s}2^{p+q}\frac{\partial^{r+p}}{\partial(z_e^{+\dot+})^{r+p}}\frac{\partial^{s+q}}{\partial(z_e^{+\dot-})^{s+q}}\frac{\partial_L}{\partial[(\partial^r_{++}\partial^s_{+-}X)(x_v)]}\frac{\partial_L}{\partial[(\partial^p_{++}\partial^q_{+-}Y)(x_w)]}.\end{aligned}\tag{2.34}$$

The z_e derivatives act only on the kernel of edge e, whose argument is $x_v-x_w+z_e$; set all $z_e=0$ afterwards. Indeed,

$$\partial_{v,+\dot a}[P_e]_\Omega=-2\frac{\partial}{\partial z_e^{+\dot a}}[P_e]_\Omega,\qquad \partial_{w,+\dot a}[P_e]_\Omega=2\frac{\partial}{\partial z_e^{+\dot a}}[P_e]_\Omega.$$

These identities also hold for the heat edge. The derivatives with respect to the input jets remove the contracted letters; all remaining letters are translated by (2.31). For example,

$$W_{(v,w)}[F;(\partial_{+\dot a}\mathsf{b}_A)(x_v),\mathsf{c}^B(x_w)]=-2\delta_A{}^B\left.\frac{\partial F}{\partial z_e^{+\dot a}}\right|_{z_e=0}.$$

The projection to $H(\bar\partial)$ is given in (2.30). Applying $\bar\partial$ to the ordered propagators gives the heat-edge signs $(-1)^{j-1}$ of (3.9).

For labelled vertices and a fixed edge order, every graph is included once. The derivatives supply the multiplicities of identical letters; the factor $1/(n+1)!$ in (8.1) accounts for the identical interaction insertions.

3 Holomorphic observables and graph operations

With the twisted theory of Section 2 in hand, we review its graph operations [3–6, 9]: the fields and the interaction, the higher brackets, and the current integrals that evaluate them.

3.1 Fields and interaction

The holomorphic twist of a four-dimensional $\mathcal{N}=1$ gauge theory organizes its fields and antifields into the Dolbeault fields $\mathsf{c}^A, \mathsf{b}_A, \gamma_i, \beta^i$; see details in Section 2.1.

Proposition 3.1 (Component dictionary)**.** *For canonical matter and quadratic gauge kinetic terms, the map between twisted theory and physical components is*

$$\begin{aligned}
\mathsf{c}^A &= c^A - \frac{1}{2} dx^-_{\dot a} V^{A\dot a}_- + \frac{i}{4} dx^{-\dot +} \wedge dx^{-\dot -} \lambda^A_-, \\
\gamma_i &= \bar\phi_i - \frac{1}{2\sqrt{2}} dx^-_{\dot a} \bar\psi^{\dot a}_i + \frac{1}{8} dx^{-\dot +} \wedge dx^{-\dot -} \bar{\mathcal{F}}_i, \\
\beta^i &= \bar{\mathcal{F}}^{\ddagger i} - \frac{i}{\sqrt{2}} \psi^i_+ + \frac{1}{2\sqrt{2}} dx^-_{\dot a} \bar\psi^{\ddagger \dot a i} + \frac{1}{8} dx^{-\dot +} \wedge dx^{-\dot -} \bar\phi^{\ddagger i}, \\
\mathsf{b}_A &= \frac{1}{4}\delta_{AB} f^B_{++} - \frac{i}{2}\lambda^{\ddagger -}_A + \frac{1}{4} dx^-_{\dot a} V^{\ddagger -\dot a}_A + \frac{1}{8} dx^{-\dot +} \wedge dx^{-\dot -} c^\ddagger_A, \\
\mathsf{w}^A_{\dot a} &= \bar W^A_{\dot a}\Big|_{\theta=0,\, \bar\theta^{\dot b} = dx^{-\dot b}/4} \\
&= \bar\lambda^A_{\dot a} + \frac{i}{4} dx^-_{\dot a} D^A + \frac{1}{8} dx^{-\dot b} \epsilon^{bc} \sigma^\mu_{b\dot a} \sigma^\nu_{c\dot b} f^A_{\mu\nu} \\
&\quad - \frac{i}{8} dx^{-\dot +} \wedge dx^{-\dot -} \nabla^b{}_{\dot a} \lambda^A_b .
\end{aligned} \tag{3.1}$$

where $\ddagger$ denotes the antifield.[9]

Proposition 3.2 (Twisted action and interaction density)**.** With $(t_A)^j{}_i$ the generators in the matter representation, the twisted action is

$$\begin{aligned}
S_{\mathrm{tw}} = \int_{\mathbb{C}^2} \Omega \wedge \Big[\mathsf{b}_A \left(\bar\partial \mathsf{c}^A - \frac{g}{2} C^A{}_{BC} \mathsf{c}^B \mathsf{c}^C \right) \\
+ \left(\bar\partial \gamma_i - i g \mathsf{c}^A (t_A)^j{}_i \gamma_j \right) \beta^i + \bar W(\gamma, \mathsf{w}) \Big]_{(0,2)} .
\end{aligned} \tag{3.3}$$

where the projection $[\cdot]_{(0,2)}$ selects antiholomorphic form degree two and the function $\bar W$ contains the antichiral superpotential and gauge kinetic terms expressed in γ and w. Ω is the holomorphic form $2\, dx^{+\dot +} \wedge dx^{+\dot -}$.

Throughout, $\bar W$ is a polynomial or, more generally, a formal power series in γ and w, read coefficientwise, as perturbation theory requires.

The interaction density entering the higher operations is

$$\mathcal{I} = -\frac{g}{2} C^A{}_{BC} \mathsf{b}_A \mathsf{c}^B \mathsf{c}^C - i g \mathsf{c}^A (t_A)^j{}_i \gamma_j \beta^i + \bar W(\gamma, \mathsf{w}). \tag{3.4}$$

[9] Homotopy transfer sends the antichiral field strength to

$$\begin{aligned}
\mathsf{w}^A_{\dot a} &\xmapsto{\text{homotopy transfer}} \frac{1}{2}\partial_{+\dot a} \mathsf{c}^A = -\frac{\partial \mathsf{c}^A}{\partial x^{+\dot a}} \\
&= \frac{1}{2}\partial_{+\dot a} c^A - \frac{1}{4} dx^-_{\dot b} \partial_{+\dot a} V^{A\dot b}_- + \frac{i}{8} dx^{-\dot +} \wedge dx^{-\dot -} \partial_{+\dot a} \lambda^A_- .
\end{aligned} \tag{3.2}$$

$$[\bar\lambda^A_{\dot a}] = [\mathsf{w}^{A(0)}_{\dot a}] = \frac{1}{2}[\partial_{+\dot a} \mathsf{c}^{A(0)}].$$

The doublet $(S_{\mathrm{BV}}, V^A_{+\dot a}) = -2\bar\lambda^A_{\dot a} + \nabla_{+\dot a} c^A$ identifies the gaugino with the zero-form part of $\frac{1}{2}\partial_{+\dot a}\mathsf{c}^A$ after transfer.

$\mathcal{I}$ in (3.4) is a polynomial in the full Dolbeault fields, retaining all antiholomorphic form degrees.

Its integrated interaction is $I = \int_{\mathbb{C}^2} \Omega \wedge [\mathcal{I}]_{(0,2)}$.

In (3.9), Wick contraction precedes the final $H(\bar\partial)$ projection: only the surviving fields are replaced by their holomorphic zero-form Taylor expansions.

The propagator forms remain in the integral and supply its antiholomorphic integration degree.

3.2 Propagators and higher brackets

The graph operations are built from the BV pairing, the heat kernel and the propagator. Define Δ_{vw} as the inverse BV pairing contracting the fields at vertices v, w.

Its fibre part is

$$\begin{aligned}\Delta_{vw} = &\frac{\partial_L}{\partial \mathsf{b}_{A,v}}\frac{\partial_L}{\partial \mathsf{c}^A_w} + \frac{\partial_L}{\partial \mathsf{c}^A_v}\frac{\partial_L}{\partial \mathsf{b}_{A,w}} \\ &- \frac{\partial_L}{\partial \beta^i_v}\frac{\partial_L}{\partial \gamma_{i,w}} - \frac{\partial_L}{\partial \gamma_{i,v}}\frac{\partial_L}{\partial \beta^i_w}.\end{aligned} \tag{3.5}$$

Dolbeault gauge fixing (a choice of Lagrangian submanifold) uses $Q^{\mathrm{GF}} = \bar\partial^*$ and $H = \{\bar\partial, Q^{\mathrm{GF}}\} = \{Q, Q^{\mathrm{GF}}\}$.

With ω_E^{-1} the inverse BV pairing, the heat kernel and, for $0 < \varepsilon < L$, the propagator are [5, 17]

$$K_t = (e^{-tH} \otimes 1)\omega_E^{-1}. \tag{3.6}$$

$$[K_t(x_1, x_2)]_\Omega = -\frac{e^{-|x_1 - x_2|_E^2/(2t)}}{32\pi^2 t^2} d(x_1 - x_2)^{-\dot+} \wedge d(x_1 - x_2)^{-\dot-}. \tag{3.7}$$

$$\begin{aligned}P_{\varepsilon,L} &= \int_\varepsilon^L (Q^{\mathrm{GF}} \otimes 1) K_t \, dt, \\ (\bar\partial \otimes 1 + 1 \otimes \bar\partial) P_{\varepsilon,L} &= -\int_\varepsilon^L \partial_t K_t \, dt = K_\varepsilon - K_L.\end{aligned} \tag{3.8}$$

The higher brackets are the basic objects in the computation of the anomalous supersymmetry transformation and of the quantum product.

Definition 3.3 (Brackets and the full quantum differential)**.** Let $\mathcal{O}_1, \ldots, \mathcal{O}_k$ be (composite or fundamental) operators with fermion numbers $|\mathcal{O}_i|$, with $\mathcal{O}_k$ at the origin.

The higher bracket is the sum of the Feynman graphs through these insertions, each summed over one cut edge, which sits at the ultraviolet boundary $t_e = \varepsilon$ of its Schwinger-time integral; the propagators are regularized by a Schwinger-time cutoff.

Let Γ run over connected contraction graphs, with edges $\Gamma_1 = (e_1, \ldots, e_{|\Gamma_1|})$ in their contraction order. Write Ω_v for the holomorphic volume form $\Omega = 2\, dx^{+\dot+} \wedge dx^{+\dot-}$ at the vertex v, $[P_{\varepsilon,L;e}]_\Omega$ for the propagator on the edge e, the Bochner–Martinelli form with Schwinger time integrated over $\varepsilon < t_e < L$, $[K_{\varepsilon;e}]_\Omega$ for the heat kernel at $t_e = \varepsilon$, and W_Γ for the ordered BV fibre contractions, which record how the Wick contractions act on the letters (Section 2). The final projection to

$H(\bar\partial)$ keeps the holomorphic local operators.

$$\begin{aligned}\{\mathcal{O}_1,\dots,\mathcal{O}_k\}_0 = (-1)^{\sum_{i=1}^k (k-i)|\mathcal{O}_i|}\sum_{\Gamma}\lim_{\varepsilon\downarrow 0}\sum_{j=1}^{|\Gamma_1|}(-1)^{j-1}\int_{(\mathbb{C}^2)^{k-1}}\left(\bigwedge_{v<k}\Omega_v\right)\\ \times W_\Gamma\left[\left(\bigwedge_{a<j}[P_{\varepsilon,L;e_a}]_\Omega\right)\wedge[K_{\varepsilon;e_j}]_\Omega\wedge\left(\bigwedge_{a>j}[P_{\varepsilon,L;e_a}]_\Omega\right);\right.\\ \left.\mathcal{O}_1(x_1),\dots,\mathcal{O}_k(0)\right]\Big|_{H(\bar\partial)}.\end{aligned}\tag{3.9}$$

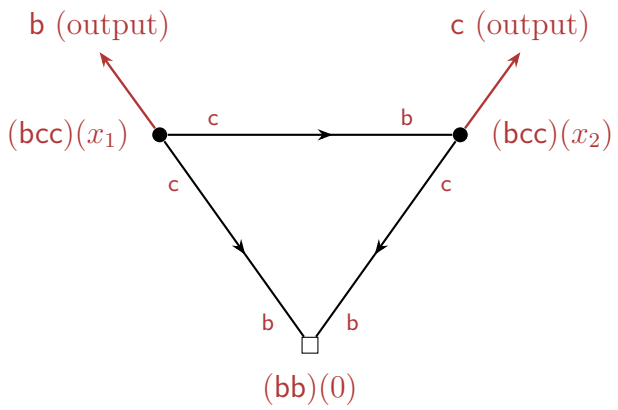


Figure 3.1: A contraction graph in $\{\mathsf{bcc},\mathsf{bcc},\mathsf{bb}\}_0$. The output letters are b at x_1 and c at x_2; internal arrows run from c to b.

For an interaction density $\mathcal{I}$ in the action and a holomorphic local operator $\mathcal{O}$, the full quantum BV differential is

$$\mathbf{Q}=\sum_{n\ge 0}\hbar^n Q_n,\qquad Q_n\mathcal{O}=\frac{1}{(n+1)!}\{\mathcal{I}^{\otimes(n+1)},\mathcal{O}\}_0.\tag{3.10}$$

The explicit derivative form of W_Γ and some other details are given in Section 2, equations (2.33)–(2.34).

3.3 Where the brackets come from

The contraction and cut prescription of (3.9) follows from the BV transformation of a renormalized insertion [3, 4, 6, 17]. We recall the argument; a reader who is familiar with the holomorphic twist and the higher brackets can skip this subsection.

Take the holomorphic interaction density $\mathcal{I}$ in (3.4) and a polynomial $\mathcal{O}$ in the zero-form fields $\mathsf{c},\mathsf{b},\gamma,\beta$ and their holomorphic jets.

$$I=\int_{\mathbb{C}^2}\Omega\wedge[\mathcal{I}]_{(0,2)},\qquad F=\mathcal{O}(0).$$

The operator $\partial_{P_{\varepsilon,L}}$ contracts one unordered pair using $P_{\varepsilon,L}$. At fixed $L>0$, evaluated on smooth compactly supported external fields, the effective interaction and insertion exist coefficientwise in perturbation theory [5, Theorem 5 and Corollary 4] as

$$\begin{aligned}e^{I[L]/\hbar}&=\lim_{\varepsilon\downarrow 0}e^{\hbar\partial_{P_{\varepsilon,L}}}e^{I/\hbar},\\ F[L]e^{I[L]/\hbar}&=\lim_{\varepsilon\downarrow 0}e^{\hbar\partial_{P_{\varepsilon,L}}}\left(\mathcal{O}(0)e^{I/\hbar}\right).\end{aligned}\tag{3.11}$$

Let $K_L = (e^{-LH} \otimes 1)\omega_E^{-1}$, let Δ_L contract one pair with K_L, and let $(\ ,\ )_L$ be the corresponding BV bracket. Write $\mathbf{Q}_L$ for the quantum BV differential acting on effective insertions.

$$\mathbf{Q}_L F[L] = Q_{\mathrm{BV}} F[L] + (I[L], F[L])_L + \hbar \Delta_L F[L].$$

This is the regulated meaning of the quantum transformation of the insertion.

The free differential annihilates the integrated holomorphic interaction and the zero-form holomorphic input. The free BV variation of the uncontracted fields therefore moves by descent and integration by parts onto the propagators [4, §2.3]. For the ordered edges $e_1, \dots, e_E$,

$$\begin{aligned} \bar\partial \bigwedge_{i=1}^{E} [P_{\varepsilon,L;e_i}]_\Omega = \sum_{\substack{j=1,\dots,E \\ \text{cut edges}}} (-1)^{j-1} \left(\bigwedge_{i<j} [P_{\varepsilon,L;e_i}]_\Omega \right) \\ \wedge [K_{\varepsilon;e_j} - K_{L;e_j}]_\Omega \wedge \left(\bigwedge_{i>j} [P_{\varepsilon,L;e_i}]_\Omega \right). \end{aligned} \tag{3.12}$$

The K_L contractions in $(I[L], F[L])_L + \hbar \Delta_L F[L]$ cancel the upper-end terms. The remaining K_ε is the lower endpoint of the propagator's Schwinger integral. This gives the cut at $t_e = \varepsilon$ and the sign $(-1)^{j-1}$ in (3.9).

A total-collision boundary is a finite, constant-coefficient holomorphic differential operator on the coincident vertex inputs [5, Theorem 6]. A cluster containing $\mathcal{O}$ and some of the interaction vertices gives a local word at the root, and summing its cut graphs gives exactly the higher operation (3.9). The uncollapsed vertices and propagators give the effective representative of that word, so that

$$\begin{aligned} &Q_{\mathrm{BV}} F[L] + (I[L], F[L])_L + \hbar \Delta_L F[L] \\ &\quad = (\mathbf{Q}\mathcal{O})[L] \\ &\quad = e^{-I[L]/\hbar} \lim_{\varepsilon \downarrow 0} e^{\hbar \partial_{P_{\varepsilon,L}}} ((\mathbf{Q}\mathcal{O})(0) e^{I/\hbar}). \end{aligned} \tag{3.13}$$

For m interaction vertices and the root $\mathcal{O}$, the surviving heat-edge graph has

$$|\Gamma_0| = m + 1, \qquad 2 + (|\Gamma_1| - 1) = 2m, \qquad n = |\Gamma_1| - |\Gamma_0| + 1 = m - 1. \tag{3.14}$$

Each propagator contraction contributes $\hbar$, and the marked contraction comes from $\hbar \Delta_\varepsilon$. The interaction exponential contributes $1/(m! \hbar^m)$. Hence

$$\frac{\hbar^{|\Gamma_1| - m}}{m!} = \frac{\hbar^{m-1}}{m!} = \frac{\hbar^n}{(n+1)!}. \tag{3.15}$$

which is the coefficient in (3.10).

At tree level the single heat edge gives $Q_0 \mathcal{O} = \{\mathcal{I}, \mathcal{O}\}_0$. At one loop, order the triangle edges as $(12, 1\mathcal{O}, 2\mathcal{O})$. The ultraviolet boundary is

$$\begin{aligned} &[K_{\varepsilon;12}]_\Omega \wedge [P_{\varepsilon,L;1\mathcal{O}}]_\Omega \wedge [P_{\varepsilon,L;2\mathcal{O}}]_\Omega \\ &- [P_{\varepsilon,L;12}]_\Omega \wedge [K_{\varepsilon;1\mathcal{O}}]_\Omega \wedge [P_{\varepsilon,L;2\mathcal{O}}]_\Omega \\ &+ [P_{\varepsilon,L;12}]_\Omega \wedge [P_{\varepsilon,L;1\mathcal{O}}]_\Omega \wedge [K_{\varepsilon;2\mathcal{O}}]_\Omega, \qquad \frac{\hbar^3}{2!\hbar^2} = \frac{\hbar}{2}. \end{aligned}$$

$$Q_1 \mathcal{O} = \frac{1}{2} \{\mathcal{I}, \mathcal{I}, \mathcal{O}\}_0.$$

The first cut contracts the two interactions.

The two remaining cuts contract an interaction with $\mathcal{O}$.

3.4 Positive Schwinger times and the operatope

Computing a higher bracket amounts to computing the kernels of its graphs, with the external momenta supplied by the uncontracted output letters.

Proposition 3.4 (Current representation; external input [9, Section 2.7, Equations (2.32)–(2.33)], [5, Lemma 2, Proposition 9])**.** *Let Γ have vertices $\Gamma_0 = \{1, \dots, k\}$, the insertion at $x_k = 0$, ordered edges $\Gamma_1 = (e_1, \dots, e_{2n+1})$ and Schwinger times $t_e > 0$. With the source λ_v of the output letters at v, and the edge displacement z_e generating the holomorphic derivatives of the input letters (2.34), and the edge currents $y_{e,+\dot a}$ of (3.22), the kernel $\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda; z)$ of (3.19) is*

$$\begin{aligned}\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda; z) = (-1)^n \int_{\Delta_\Gamma} e^{-\sum_e z_e^{+\dot a} y_{e,+\dot a}} \bigwedge_{e\in\Gamma_1} d^2 y_e \\ \times \prod_{v<k} \prod_{\dot a=\dot+,\dot-} \delta\left(\lambda_{v,+\dot a} - \sum_{e:e(0)=v} y_{e,+\dot a} + \sum_{e:e(1)=v} y_{e,+\dot a}\right).\end{aligned} \tag{3.16}$$

Here the currents obey the loop law around every oriented cycle C,

$$\sum_{\substack{e\in C\\ \text{along } C}} t_e y_{e,+\dot a} - \sum_{\substack{e\in C\\ \text{against } C}} t_e y_{e,+\dot a} = 0. \tag{3.17}$$

for one common set of positive Schwinger times, the same for both dotted components, and Δ_Γ carries the orientation induced by the UV faces (3.19):

$$\Delta_\Gamma = \bigcup_{(t_e)\in\mathbb{RP}^{|\Gamma_1|-1}_{>0}} \Big\{(y_{e,+\dot a})_{e\in\Gamma_1} : \sum_{\substack{e\in C\\ \text{along } C}} t_e y_{e,+\dot a} - \sum_{\substack{e\in C\\ \text{against } C}} t_e y_{e,+\dot a} = 0 \ \text{ for every cycle } C\Big\}. \tag{3.18}$$

We now derive this representation, following [5, 9].

The heat edge and the current form

Let $\Gamma_0 = \{1, \dots, n+2\}$ be the vertex set of a general n-loop Feynman graph satisfying the Laman condition, and order the $2n+1$ edges as $\Gamma_1 = (e_1, \dots, e_{2n+1})$.

Fix $x_{n+2} = 0$ and $\lambda_{n+2} = -\sum_{v<n+2} \lambda_v$.

We compute the kernel by applying $\bar\partial$, which places a heat kernel on each edge in turn [5, 9].

$$\begin{aligned}\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda; z) &= (-8\pi^2)^n \lim_{\varepsilon\downarrow 0} \sum_{j=1}^{2n+1} (-1)^{j-1} \int_{(\mathbb{C}^2)^{n+1}} \left(\bigwedge_{v<n+2} \Omega_v\right) e^{\sum_{v<n+2} \lambda_{v,+\dot a} x_v^{+\dot a}} \\ &\quad \times \left(\bigwedge_{i<j} [P_{\varepsilon,L;e_i}]_\Omega\right) \wedge [K_{\varepsilon;e_j}]_\Omega \wedge \left(\bigwedge_{i>j} [P_{\varepsilon,L;e_i}]_\Omega\right) \\ &= (-8\pi^2)^n \lim_{\varepsilon\downarrow 0} \int_{(\mathbb{C}^2)^{n+1}} \left(\bigwedge_{v<n+2} \Omega_v\right) e^{\sum_{v<n+2} \lambda_{v,+\dot a} x_v^{+\dot a}} \bar\partial \left(\bigwedge_{i=1}^{2n+1} [P_{\varepsilon,L;e_i}]_\Omega\right)\Big|_{\text{UV}} \\ &= (-1)^n (-8\pi^2)^n \lim_{\varepsilon\downarrow 0} \int_{(\mathbb{C}^2)^{n+1}} \left(\bigwedge_{v<n+2} \Omega_v\right) e^{\sum_{v<n+2} \lambda_{v,+\dot a} x_v^{+\dot a}} \\ &\quad \times \int_{\partial_{\text{UV}}[\varepsilon,L]^{2n+1}} \bigwedge_{i=1}^{2n+1} \Big([K_{t_{e_i};e_i}]_\Omega - dt_{e_i} \wedge [(Q^{\text{GF}} \otimes 1) K_{t_{e_i};e_i}]_\Omega\Big),\end{aligned} \tag{3.19}$$

$$\partial_{\text{UV}}[\varepsilon, L]^{2n+1} = \bigcup_{j=1}^{2n+1} \{t_{e_j} = \varepsilon, \ \ \varepsilon \le t_{e_i} \le L \ (i \ne j)\}, \tag{3.20}$$

Proof of the boundary signs.

The facet $t_{e_j} = \varepsilon$ carries $(-1)^{j-1} \bigwedge_{i \neq j} dt_{e_i}$.

Passing $\bar{\partial}$ through $j-1$ propagators gives the same sign.

On the facet $t_{e_j} = \varepsilon$, the other $2n$ factors each contribute $-dt_{e_i} \wedge [(Q^{\mathrm{GF}} \otimes 1) K_{t_{e_i};e_i}]_\Omega$, so their explicit minus signs multiply to 1. Moving these Schwinger differentials to the front crosses $0 + 1 + \cdots + (2n-1) = n(2n-1)$ propagator one-forms, giving $(-1)^{n(2n-1)} = (-1)^n$, while crossing the degree-two heat kernel contributes sign $+1$.

(The upper-boundary (IR boundary, or physical boundary) terms K_L combine with the scale-L BV operations $(I[L], \cdot\,)_L + \hbar\Delta_L$ [17].) $\square$

We can further simplify the last line of (3.19) by combining the terms into one two-form.

On $e = (v, w)$, the propagator and its current coordinate are

$$\begin{aligned}[P_{\varepsilon,L;e}]_\Omega = &- \frac{(x_v - x_w)^-_{\dot{a}}\, d(x_v - x_w)^{-\dot{a}}}{32\pi^2} \int_\varepsilon^L \frac{dt_e}{t_e^3} \\ &\times \exp\left[-\frac{(x_v - x_w)^-_{\dot{a}} (x_v - x_w + z_e)^{+\dot{a}}}{2t_e} \right].\end{aligned} \tag{3.21}$$

$$y_{e,+\dot{a}} = \frac{(x_{e(0)} - x_{e(1)})^-_{\dot{a}}}{2t_e}, \qquad d^2 y_e = dy_{e,++} \wedge dy_{e,+-}. \tag{3.22}$$

$$[K_{t_e;e}]_\Omega - dt_e \wedge [(Q^{\mathrm{GF}} \otimes 1) K_{t_e;e}]_\Omega = -\frac{e^{-y_{e,+\dot{a}}(x_{e(0)} - x_{e(1)} + z_e)^{+\dot{a}}}}{8\pi^2}\, d^2 y_e. \tag{3.23}$$

$$\begin{aligned}\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda; z) = &- \frac{1}{(8\pi^2)^{n+1}} \lim_{\varepsilon \downarrow 0} \int_{(\mathbb{C}^2)^{n+1}} \left(\bigwedge_{v<n+2} \Omega_v \right) e^{\sum_{v<n+2} \lambda_{v,+\dot{a}} x_v^{+\dot{a}}} \\ &\times \int_{\partial_{\mathrm{UV}}[\varepsilon, L]^{2n+1}} e^{-\sum_{e \in \Gamma_1} y_{e,+\dot{a}} (x_{e(0)} - x_{e(1)} + z_e)^{+\dot{a}}} \bigwedge_{e \in \Gamma_1} d^2 y_e.\end{aligned} \tag{3.24}$$

Now we can integrate out $x_v^{+\dot{a}}$ by analytically continuing these coordinates to purely imaginary values.

Proof of the current constraint and normalization.

For real λ_v and imaginary z_e, deform the Gaussian contour to imaginary x_v^+ and real x_v^-.

Grouping the exponent by vertex gives

$$\begin{aligned}&\sum_{v<n+2} \lambda_{v,+\dot{a}} x_v^{+\dot{a}} - \sum_e y_{e,+\dot{a}} (x_{e(0)} - x_{e(1)} + z_e)^{+\dot{a}} \\ &= -\sum_e y_{e,+\dot{a}} z_e^{+\dot{a}} + \sum_{v<n+2} \left(\lambda_{v,+\dot{a}} - \sum_{e:e(0)=v} y_{e,+\dot{a}} + \sum_{e:e(1)=v} y_{e,+\dot{a}} \right) x_v^{+\dot{a}}.\end{aligned} \tag{3.25}$$

$$\begin{aligned}&\int_{(i\mathbb{R})^2} \frac{dx_v^{++} \wedge dx_v^{+-}}{(2\pi i)^2} \exp\left[\left(\lambda_{v,+\dot{a}} - \sum_{e:e(0)=v} y_{e,+\dot{a}} + \sum_{e:e(1)=v} y_{e,+\dot{a}} \right) x_v^{+\dot{a}} \right] \\ &\quad = \prod_{\dot{a}=+,-} \delta\left(\lambda_{v,+\dot{a}} - \sum_{e:e(0)=v} y_{e,+\dot{a}} + \sum_{e:e(1)=v} y_{e,+\dot{a}} \right).\end{aligned} \tag{3.26}$$

$$\sum_{e:e(0)=v} y_{e,+\dot{a}} - \sum_{e:e(1)=v} y_{e,+\dot{a}} = \lambda_{v,+\dot{a}}. \tag{3.27}$$

Each vertex integral contributes $2(2\pi i)^2 = -8\pi^2$.

Thus $(-8\pi^2)^n(-1/8\pi^2)^{2n+1}(-8\pi^2)^{n+1} = 1$, leaving the sign $(-1)^n$ from the Schwinger differentials. The identity extends coefficient by coefficient to the formal series in λ, z. □

Independent currents

Choose a spanning tree T of the diagram.

Its grounded incidence matrix has determinant ± 1, so eliminating its currents by conservation has unit Jacobian for the two dotted components.

This gives

$$\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda; z) = \int_{\substack{y\in\Delta_\Gamma \\ \sum_{e:e(0)=v} y_e - \sum_{e:e(1)=v} y_e = \lambda_v}} e^{-\sum_e y_{e,+\dot{a}} z_e^{+\dot{a}}} \bigwedge_{e\notin T} (-dy_{e,+\dot{+}} \wedge dy_{e,+\dot{-}}). \tag{3.28}$$

The domain Δ_Γ consists of the current solutions for positive resistance ratios, with the heat-boundary orientation.

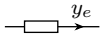
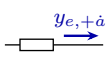
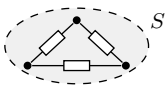
Remark 3.5 (Circuit dictionary)**.** Equation (3.16) describes an electrical network on Γ. The y_e are the currents on its edges and the λ_v the external currents injected at its vertices;

The delta functions in (3.16) impose current conservation at every vertex, and (3.17) is the voltage law around every loop, with resistances t_e;

The displacements z_e generate the moments of the edge currents: each $-\partial/\partial z_e^{+\dot{a}}$ inserts $y_{e,+\dot{a}}$. At $z = 0$ the kernel is therefore the oriented volume swept out by the independent currents as the ratios of the edge resistances range over $\mathbb{RP}^{|\Gamma_1|-1}_{>0}$, and z_e is the generating source that records the expectation values of the current y_e.

Table 3.1 collects this dictionary.

Table 3.1: Holomorphic graph kernels as electrical networks. $\Gamma_1[S]$ is the set of edges with both ends in the vertex set S; the Laman bound is [9, Equations (2.22)–(2.23)].

Graph kernel	Circuit	
Schwinger time t_e	resistance of the resistor e	
edge current $y_{e,+\dot{a}}$ (3.22), $\dot{a} = \dot{+}, \dot{-}$	current through e, two components	
output momentum λ_v (2.32)	current injected at the node v	
vertex delta functions (3.16), (3.27)	Kirchhoff's current law	
loop relation (3.17)	voltage law with resistances t_e	
weighted Laplacian [5, Remark 6]	conductance matrix	
$-\partial/\partial z_e^{+\dot{a}}$	insertion of the current $y_{e,+\dot{a}}$	
Laman bound $\lvert\Gamma_1[S]\rvert \le 2\lvert S\rvert - 3$ (3.31), S a vertex set with $\lvert S\rvert \ge 2$	local bound on independent current directions	
kernel at $z = 0$	oriented volume swept by the independent currents	

This analogy brings the series, parallel, and star–triangle transformations of circuits to the holomorphic kernels. One of our core observations is that they can be applied iteratively, transferring the sources at each step, as changes of variables on Schwinger space.

3.5 Restrictions on the graphs

Only a few graphs can have a nonzero contribution to (3.16); they must satisfy the so-called Laman condition. In addition, the diagrams in 4d separate further into a few classes.

Write $\Gamma_1[S]$ for the edges with both ends in a vertex set $S \subseteq \Gamma_0$.

Some of the proofs are easier to understand in terms of a circuit diagram, so we sometimes provide more than one proof to illustrate the idea.

Theorem 3.6 (Laman count)**.** *Every nonzero graph satisfies*

$$|\Gamma_1| = 2|\Gamma_0| - 3. \tag{3.29}$$

Proof (graph integral). The heat edge has antiholomorphic form degree two and every other propagator degree one; together they saturate the relative antiholomorphic integration directions:

$$2 + (|\Gamma_1| - 1) = 2(|\Gamma_0| - 1).$$

□

Proof (circuit). The resistance ratios sweep out the real components of the independent loop currents in (3.16):

$$\underbrace{|\Gamma_1| - 1}_{\text{independent resistance ratios}} = \underbrace{2}_{\text{holomorphic dimension}} \underbrace{(|\Gamma_1| - |\Gamma_0| + 1)}_{\text{loops}}.$$

□

Theorem 3.7 (Laman bound on subgraphs)**.** *Every $S \subseteq \Gamma_0$ with $\Gamma_1[S] \neq \emptyset$ satisfies*

$$|\Gamma_1[S]| \leq 2|S| - 3.$$

Proof (graph integral). Fix $v_0 \in S$; the coordinates $(x_v - x_{v_0})^{-\dot{a}}$, $v \in S \setminus \{v_0\}$, give $2(|S| - 1)$ one-form directions in the numerators of the propagators. The total form degree cannot exceed the number of one-form directions. If the heat kernel edge K lies in S,

$$|\Gamma_1[S]| + 1 \leq 2(|S| - 1).$$

If the heat edge lies outside S, the radial vector annihilates every internal propagator one-form [5, Proposition 5, Lemma 2], [9, Equation (2.15)]:

$$R = \sum_{u \in S \setminus \{v_0\}} (x_u - x_{v_0})^{-\dot{c}} \frac{\partial}{\partial x_u^{-\dot{c}}},$$
$$\iota_R\big[-(x_v - x_w)^-_{\dot{a}}\, d(x_v - x_w)^{-\dot{a}}\big] = -(x_v - x_w)^-_{\dot{a}}(x_v - x_w)^{-\dot{a}} = 0,$$

so the number of holomorphic directions drops by one.

$$|\Gamma_1[S]| \leq 2(|S| - 1) - 1.$$

□

Proof (circuit). Ohm's law is local: the $2|\Gamma_1[S]|$ current components inside S are fixed by the $2(|S| - 1)$ relative potentials and by $|\Gamma_1[S]| - 1$ independent resistances, a common rescaling of potentials and resistances, or the fixed resistance on the UV boundary, removing one. They vary in all directions only if

$$2|\Gamma_1[S]| \leq 2(|S| - 1) + |\Gamma_1[S]| - 1.$$

□

Theorem 3.7 immediately leads to:

Corollary 3.8 (No self-loops or parallel edges)**.** *A nonzero graph has no self-loop and at most one edge between any two vertices.*

Proof (propagators). A self-loop carries $d(x_v - x_v)^{-\dot{a}} = 0$ [5, Section 2.3]. Two edges between v and w, with holomorphic shifts z, z', share the one-form direction of $x_v - x_w$ [9, Section 2.1]:

$$\begin{aligned} [P_{\varepsilon,L;vw}(x_v - x_w + z)]_\Omega \wedge [P_{\varepsilon,L;vw}(x_v - x_w + z')]_\Omega = 0, \\ [K_{\varepsilon;vw}(x_v - x_w + z)]_\Omega \wedge [P_{\varepsilon,L;vw}(x_v - x_w + z')]_\Omega = 0. \end{aligned} \tag{3.30}$$

□

Proof (circuit). A self-loop is shorted: $t_e y_{e,+\dot{a}} = 0$ with $t_e > 0$ gives $y_e = 0$. Two resistors between the same nodes are in parallel: $t_e y_e = t_{e'} y_{e'}$, so their four current components vary in only three directions. □

Corollary 3.8 is the case $|S| \leq 2$ of Theorem 3.7.

These are the Laman constraints in two holomorphic dimensions [9, Equations (2.22)–(2.23)], [6, Section 3.3], [5, Definition 16, Proposition 5, Theorem 6].

Theorem 3.9 (Counts at n loops)**.** *A nonzero connected n-loop graph satisfies*

$$\begin{gathered} |\Gamma_0| = n + 2, \qquad |\Gamma_1| = 2n + 1, \qquad |\Gamma_1| - |\Gamma_0| + 1 = n, \\ |\Gamma_1[S]| \leq 2|S| - 3 \qquad (S \subseteq \Gamma_0,\ |S| \geq 2). \end{gathered} \tag{3.31}$$

Proof. Theorem 3.6 and $n = |\Gamma_1| - |\Gamma_0| + 1$ give the first line; Theorem 3.7 the second. □

We now turn to the condition special to $d = 4$.

By (3.10), the Feynman graphs of the loop-corrected supercharge Q_n have $n + 1$ interaction vertices and one insertion $\mathcal{O}$.

Proposition 3.10 (Only fan diagrams contribute to loop-corrected supercharges)**.** *For a translation-invariant interaction $\mathcal{I}$ whose vertices have degree at most three, every nonzero connected n-loop kernel with one local-operator insertion is a fan of cubic vertices (Figure 3.2), for $n \geq 1$. (The $n = 0$ case is a single edge connecting two vertices, representing the classical tree-level SUSY transformation.) The interaction vertices form a path; each is joined to the insertion, and one output field remains at each endpoint of the path.*

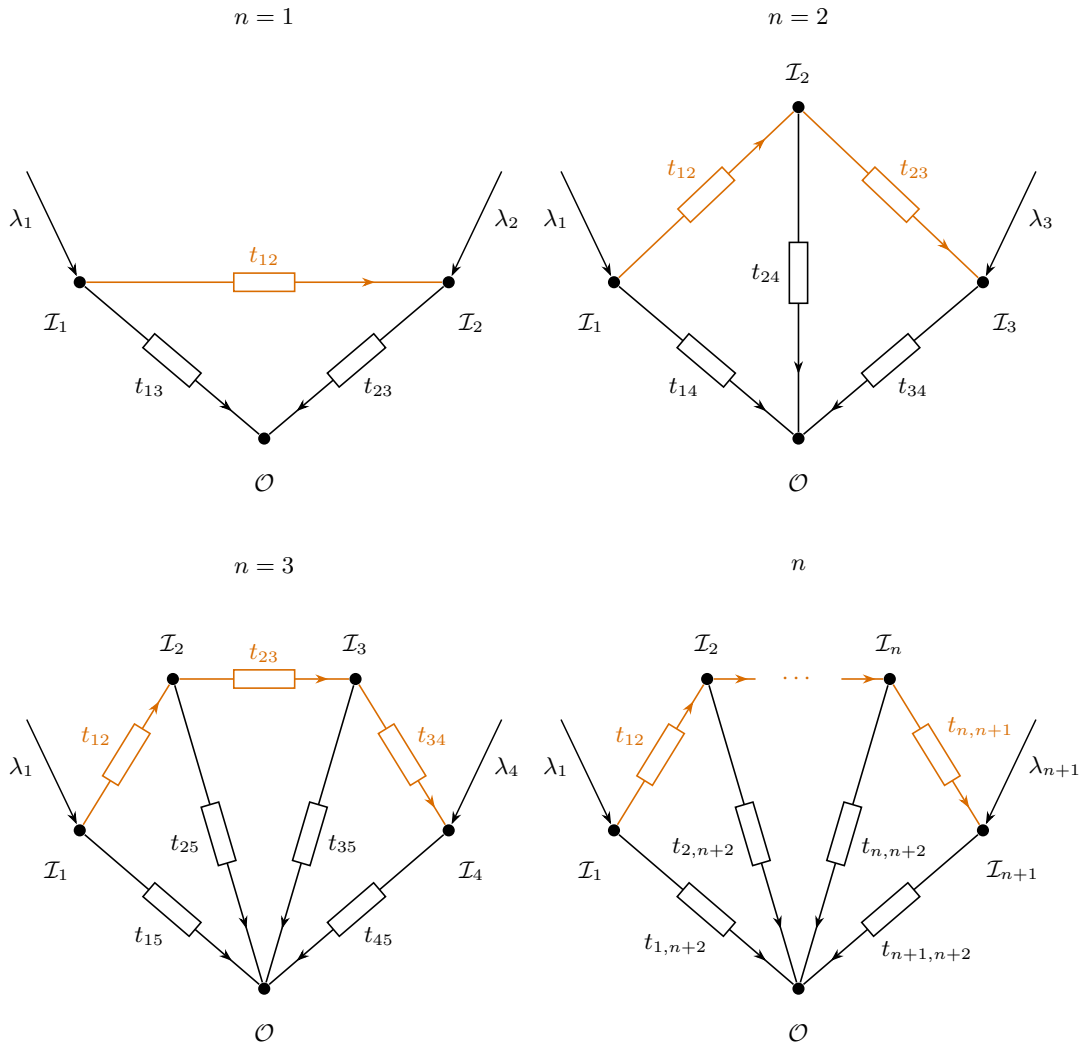


Figure 3.2: Fans at one, two, three and n loops as circuits, with the interaction path in orange, resistances t_{vw} and injected currents λ_1, λ_{n+1}.

By Theorem 3.9, there are $n+1$ interaction vertices besides the insertion $\mathcal{O}$.

The proof of Proposition 3.10 uses two lemmas:

Lemma 3.11 (Spokes)**.** *Let* $\deg_\Gamma \mathcal{O}$ *be the number of edges incident to* $\mathcal{O}$ *in a graph of Proposition 3.10. Then* $\deg_\Gamma \mathcal{O} = n+1$*: every interaction vertex is joined to* $\mathcal{O}$ *by one edge and exactly two interaction fields remain uncontracted.*

Proof (circuit). The uncontracted interaction fields inject the external currents. By Corollary 3.8 and Theorem 3.9, with equality in the first step only for cubic vertices,

$$\begin{aligned} \#(\text{sources}) &\le \underbrace{3}_{\text{degree bound}} \underbrace{(n+1)}_{\mathcal{I}\text{ vertices}} - \underbrace{2(2n+1-\deg_\Gamma \mathcal{O})}_{\substack{\text{letters contracted between} \\ \mathcal{I}\text{ vertices}}} - \underbrace{\deg_\Gamma \mathcal{O}}_{\substack{\text{edges from } \mathcal{O} \\ \text{to } \mathcal{I}\text{ vertices}}} \\ &= \underbrace{\deg_\Gamma \mathcal{O}}_{\substack{\le n+1 \\ \text{by Corollary 3.8}}} \quad -n+1 \le 2. \end{aligned} \tag{3.32}$$

If there is only one source, every current is a real multiple of the source current and sweeps at most $n < 2n$ dimensions; a nonzero kernel therefore has at least two sources, and both inequalities are then equalities. □

Lemma 3.12 (No interaction cycles)**.** *If every interaction vertex is joined to $\mathcal{O}$, no set S of interaction vertices carries a cycle.*

Proof. A loop of resistors on the interaction nodes S, together with their $|S|$ spokes, gives a subcircuit on $S \cup \{\mathcal{O}\}$ with more resistors than the local count of Ohm's law allows (Theorem 3.7):

$$|\Gamma_1[S \cup \{\mathcal{O}\}]| \geq 2|S| > 2(|S| + 1) - 3.$$

□

Proof of Proposition 3.10. By Lemma 3.11, each of the $n+1$ cubic interaction nodes is wired to $\mathcal{O}$ by one spoke, which leaves $2n + 1 - (n + 1) = n$ internal resistors; they close no loop (Lemma 3.12) and form a tree. Each node spends one terminal on its spoke and keeps at most two for internal resistors, so the tree is a chain: its interior nodes wire all three terminals, and each end keeps one free terminal, the two sources of Lemma 3.11. □

Corollary 3.13 (Renormalizable theories)**.** *For every renormalizable $\mathcal{N} = 1$ gauge theory, massless or massive, every nonzero connected graph of Q_n, $n \geq 1$, with one insertion $\mathcal{O}$ is a fan of cubic vertices.*

Proof. The superpotential has degree at most three, and Proposition 3.10 applies. □

The fan kernel is the simplest one to evaluate. If the theory is non-renormalizable (with $\mathcal{I}$ of degree greater than three), other kinds of kernels can also appear; they are, however, also highly constrained.

Proposition 3.14 (Superpotential vertices and antifield count)**.** *For the interaction (3.4), allow $\bar{W}(\gamma, \mathsf{w})$ to have arbitrary degree. Every nonzero connected n-loop graph of $Q_n\mathcal{O}$ with one insertion $\mathcal{O}$, $n \geq 1$, contains at most one $\bar{W}$ vertex. Each graph term lowers the number of b, β factors by n plus the number of $\bar{W}$ vertices.*

We prove Proposition 3.14 with the help of two lemmas.

Lemma 3.15 (Antifield terminals)**.**

Every contraction pairs a b or β factor with a c or γ factor. A gauge or matter vertex of (3.4) supplies one b or β factor, a $\bar{W}$ vertex none.

Proof. A gauge or matter node, bcc or $\mathsf{c}\gamma\beta$ in (3.4), has one antifield terminal, a node of $\bar{W}(\gamma, \mathsf{w})$ none. □

Lemma 3.16 (Internal edges)**.** *A nonzero connected n-loop graph of Q_n has at least n edges between interaction vertices.*

Proof. Two resistors from $\mathcal{O}$ to the same interaction node would be in parallel (Corollary 3.8), so at most $n + 1$ of the $2n + 1$ resistors (Theorem 3.9) end at $\mathcal{O}$, and $2n + 1 - \deg_\Gamma \mathcal{O} \geq 2n + 1 - (n + 1) = n$ join interaction nodes.

This is a weaker version of Lemma 3.11, which does not use the cubic condition. □

Proof of Proposition 3.14. By Lemmas 3.15 and 3.16,

$$\underbrace{n}_{\substack{\deg_\Gamma \mathcal{O} \leq n+1 \\ \text{at most } n+1 \text{ spokes}}} \leq \underbrace{2n + 1 - \deg_\Gamma \mathcal{O}}_{\text{internal resistors}} \leq \underbrace{n + 1 - \#(\bar{W} \text{ vertices})}_{\substack{\text{antifield terminals,} \\ \text{one per internal resistor}}} \tag{3.33}$$

$$\Longrightarrow \quad \#(\bar{W} \text{ vertices}) \leq 1,$$

and each of the $2n + 1$ resistors takes one antifield terminal, a b or β factor:

$$\begin{aligned} \#(\mathsf{b}, \beta \text{ in the output}) &= \#(\mathsf{b}, \beta \text{ in } \mathcal{O}) + [n + 1 - \#(\bar{W} \text{ vertices})] - (2n + 1) \\ &= \#(\mathsf{b}, \beta \text{ in } \mathcal{O}) - n - \#(\bar{W} \text{ vertices}). \end{aligned} \tag{3.34}$$

□

Corollary 3.17 (Finite loop order)**.** *For $\bar{W}$ of any degree, or a formal power series read coefficientwise, $Q_n\mathcal{O} = 0$ for $n > \#(\mathsf{b}, \beta \text{ in } \mathcal{O})$, and terms with a $\bar{W}$ vertex vanish already for $n + 1 > \#(\mathsf{b}, \beta \text{ in } \mathcal{O})$.*

Proof. The left side of (3.34) is nonnegative. □

Corollary 3.17 and dimensional counting make the perturbative expansion truncate at a finite order, so that the perturbative cohomology is well defined.

Proposition 3.18 (Graphs with one superpotential vertex)**.** *For the interaction (3.4), a nonzero connected n-loop graph of Q_n, $n \geq 1$, with one insertion $\mathcal{O}$ and one $\bar{W}$ vertex joins $\mathcal{O}$ to all $n+1$ interaction vertices, for any degree of $\bar{W}$. Removing $\mathcal{O}$ leaves a tree whose arms meet at $\bar{W}$.*

This type of kernel is called a spider.

Proof. With one $\bar{W}$ vertex the two bounds in (3.33) coincide:

$$n \leq 2n + 1 - \deg_\Gamma \mathcal{O} \leq n \quad \Longrightarrow \quad \deg_\Gamma \mathcal{O} = n + 1, \tag{3.35}$$

so every interaction node is wired to $\mathcal{O}$, and the n internal resistors close no loop (Lemma 3.12) and form a tree. A cubic node keeps at most two internal resistors besides its spoke, so the tree branches only at the $\bar{W}$ node; with a arms of ℓ_i cubic nodes,

$$\begin{aligned} &\Gamma \setminus \{\mathcal{O}\} = \text{arms of lengths } \ell_1, \dots, \ell_a \text{ meeting at } \bar{W}, \\ &\ell_1 + \cdots + \ell_a = n, \qquad \ell_i \geq 1, \qquad \deg \bar{W} \geq a + 1, \end{aligned} \tag{3.36}$$

where $\deg \bar{W}$, the number of field factors of the contributing monomial, covers the a internal resistors at the $\bar{W}$ node and its spoke. □

Remark 3.19. With at most two arms the graph of Proposition 3.18 is a fan; the bound $a + 1 \leq \deg \bar{W}$ gives this case for $\deg \bar{W} \leq 3$.

Corollary 3.20 (Graphs of Q_n)**.** *For the interaction (3.4), every nonzero connected graph of Q_n, $n \geq 1$, with one insertion $\mathcal{O}$ is a fan, or a spider whose arms meet at its $\bar{W}$ vertex, a book when all its arms have length one (Definition 7.4); for $\deg \bar{W} \leq 3$ it is a fan.*

Proof. Propositions 3.10, 3.14 and 3.18. □

This completes the classification of the diagrams of Q_n.

The calculation of the quantum product in Section 6 involves other kinds of kernels.

Proposition 3.21 (Superpotential vertices in quantum product diagrams)**.** *Let Γ be a nonzero connected marked graph of the quantum product (6.11) at order $\hbar^n$, $n \geq 1$, with insertions A, B, two arbitrary composite operators.*

(i) *Every connected component of $\Gamma \setminus \{A, B\}$ contains at most one $\bar{W}$ vertex.*

(ii) *Each graph term satisfies*

$$\#(\mathsf{b}, \beta \text{ in the output}) = \#(\mathsf{b}, \beta \text{ in } A) + \#(\mathsf{b}, \beta \text{ in } B) - n - \#(\bar{W} \text{ vertices}), \tag{3.37}$$

so the order-$\hbar^n$ term vanishes unless $\#(\mathsf{b}, \beta \text{ in } A) + \#(\mathsf{b}, \beta \text{ in } B) \geq n + \#(\bar{W} \text{ vertices})$.

Proof. (i) The $|T_j|$ nodes of a component T_j are connected through its internal resistors I_j, each of which takes its antifield terminal from a node of T_j (Lemma 3.15):

$$|T_j| - 1 \leq |I_j| \leq |T_j| - \#(\bar{W} \text{ vertices in } T_j).$$

(ii) Each of the $2n$ physical resistors takes one antifield terminal; the n interaction nodes have $n - \#(\bar{W} \text{ vertices})$ of them. □

The graphs of the quantum product are forests of interaction trees attached to both insertions, classified in Proposition 7.2.

4 Summary of results

4.1 Graph kernels as circuits

In the holomorphically twisted theory in four dimensions, the raw-cut kernel of a graph is the current integral of [5, 9], with oriented edges $e : e(0) \to e(1)$, an arbitrary spanning tree T, edge currents $y_{e,+\dot{a}}$, sources λ_v, and displacements z_e generating holomorphic derivatives:

$$\begin{aligned} \mathcal{I}_\Gamma^{\text{raw cut}}(\lambda; z) = \int_{y \in \Delta_\Gamma} e^{-\sum_{e \in \Gamma_1}^{\text{edges}} z_e^{+\dot{a}} y_{e,+\dot{a}}} \bigwedge_{e \notin T} (-dy_{e,+\dot{+}} \wedge dy_{e,+\dot{-}}), \\ \sum_{\substack{e : e(0) = v \\ \text{outgoing edges}}} y_e - \sum_{\substack{e : e(1) = v \\ \text{incoming edges}}} y_e = \lambda_v. \end{aligned} \tag{4.1}$$

The domain Δ_Γ consists of currents obeying Kirchhoff's voltage law for a common set of positive resistances t_e; the constraint in (4.1) is Kirchhoff's current law; the voltage law is (3.17). This electrical-network reading of the operatope equations [9, Equations (2.26), (2.31), (2.33)] gives the dictionary in Table 3.1.

The contributing graphs are strongly constrained.

Proposition 4.1 (Graphs of the supercharge and of the product)**.** *For the interaction* (3.4), $\mathcal{I} = -\frac{g}{2} C^A{}_{BC} \mathsf{b}_A \mathsf{c}^B \mathsf{c}^C - ig \mathsf{c}^A (t_A)^j{}_i \gamma_j \beta^i + \bar{W}(\gamma, \mathsf{w})$, *let* $\bar{W}$ *be a gauge-invariant polynomial of arbitrary degree.*

(i) *Every nonzero connected graph of* Q_n, $n \geq 1$, *is a fan or a spider whose arms meet at its single* $\bar{W}$ *vertex; the number of arms is at most* $\deg \bar{W} - 1$ *(Figure 4.1). For interactions of degree at most three, including renormalizable* $\mathcal{N} = 1$ *gauge theories with mass terms, only fans contribute.*

(a) Fan (b) Spider

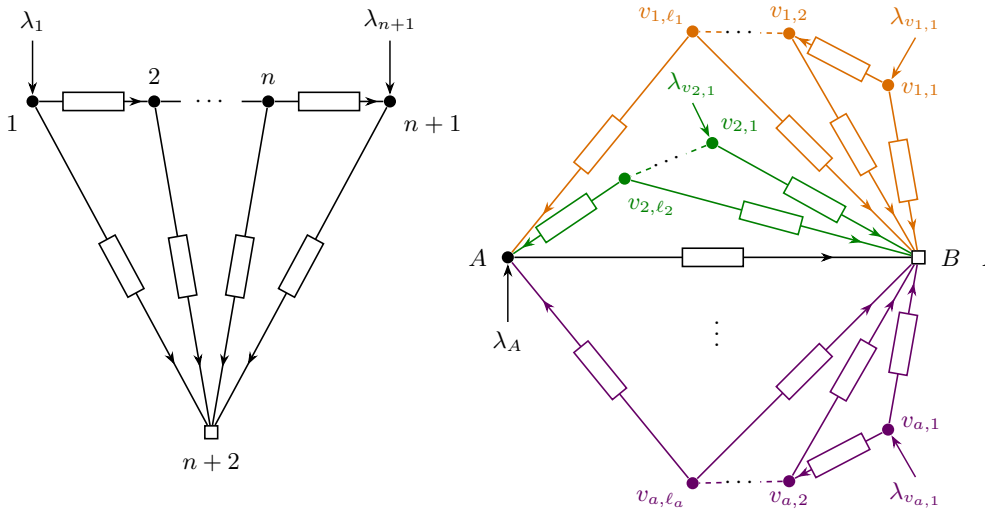


Figure 4.1: (a) A *fan*: the interaction vertices $1, \ldots, n+1$ form a path, and each of them is joined to the insertion $n+2$ by a spoke. (b) A *spider*: several such paths, the arms, meet at a vertex A, and every vertex, A included, is joined to a second vertex B; in Q_n, A is the $\bar{W}$ vertex and $B = \mathcal{O}$ (Definition 7.4).

(ii) *A nonzero connected graph of the quantum product at order* $\hbar^n$ *consists of a marked edge* AB *between the two insertions and a forest of interaction trees* T_j*; each tree has at most one*

$\bar{W}$ vertex and connects to both insertions by $|T_j|+1$ spokes (Figure 4.2). For cubic interactions each tree carries exactly one source, at its unique vertex of degree two.

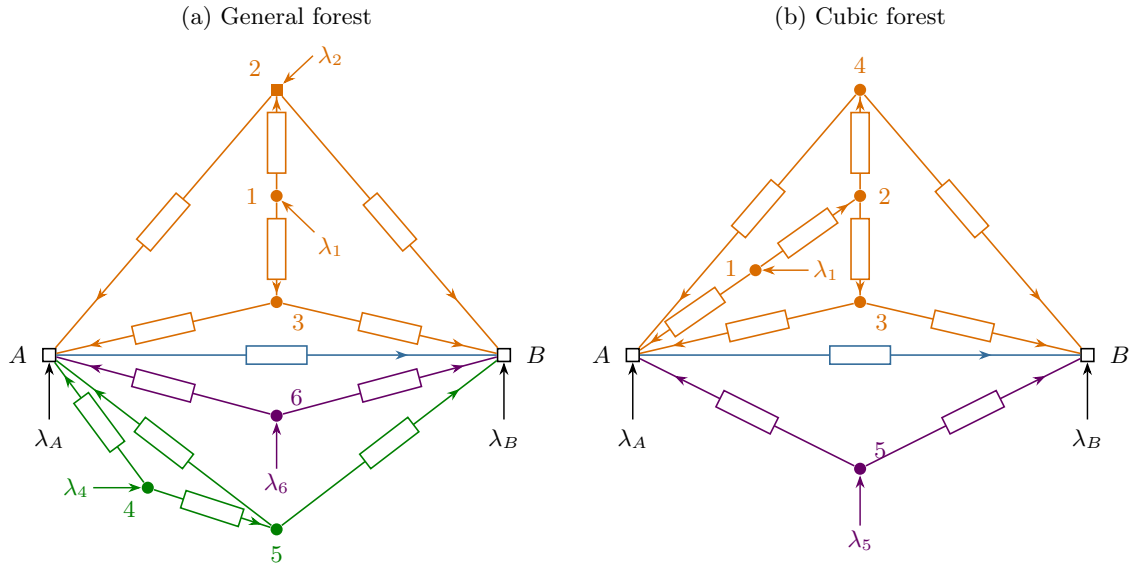


Figure 4.2: Graphs of the quantum product: the marked edge AB (blue) between the insertions and a forest of interaction trees, each in its own colour and joined to A and B by one more spoke than it has vertices. (a) For general $\bar{W}$, each tree contains at most one $\bar{W}$ vertex (square). (b) For cubic interactions, each tree carries one source, at its vertex of degree two.

Part (i) combines Propositions 3.10, 3.14 and 3.18; part (ii) is Propositions 7.2 and 7.3.

4.2 Iterated reduction and closed kernels at every loop order

Series and parallel moves become changes of variables for the kernel when the eliminated source is transferred to its neighbouring nodes. Figure 4.3 shows the three reduction moves: the series–parallel move, the star–triangle move at a vertex, and the simultaneous removal of two source-free vertices.

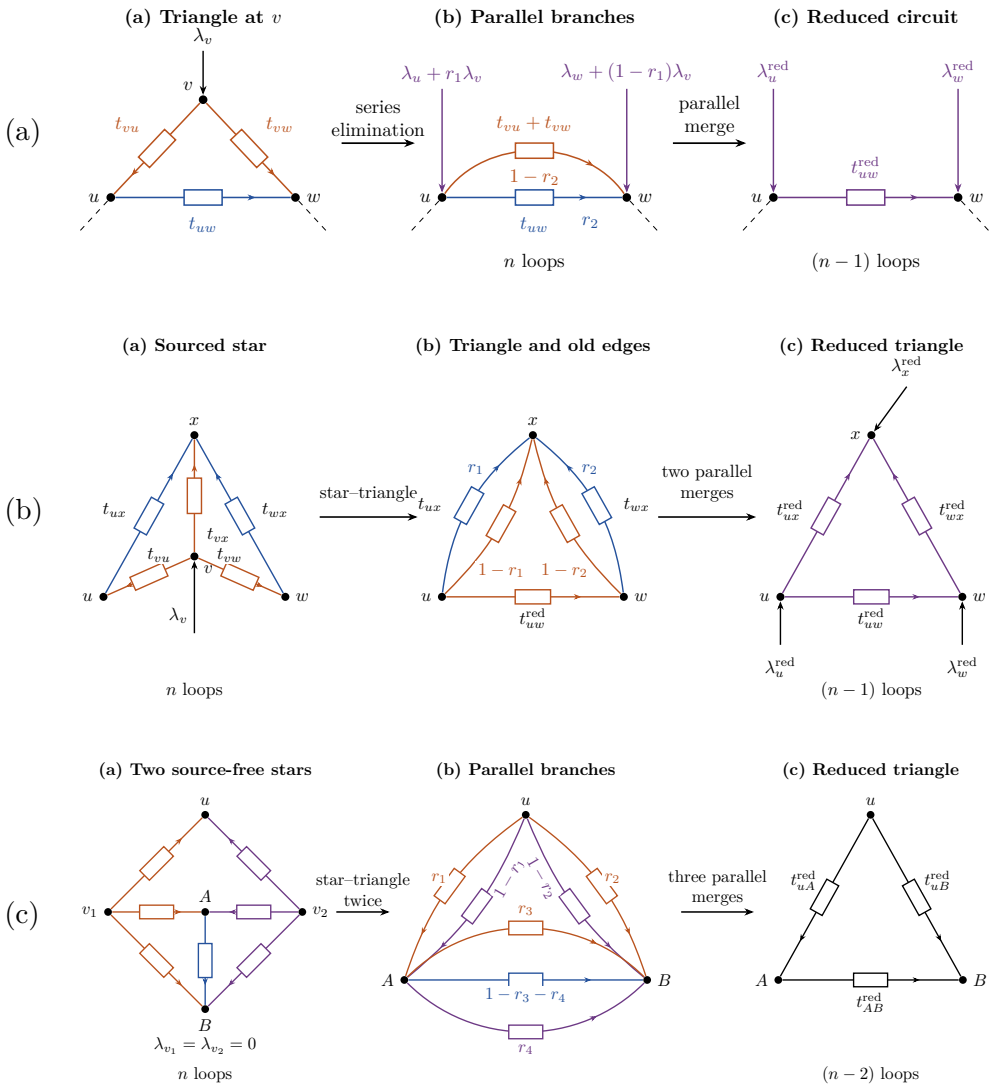


Figure 4.3: The three reduction moves of Appendix A.3. (a) Series–parallel reduction (Theorem A.1): the source λ_v is transferred to u and w, and the series branch (orange) merges with the original uw edge (blue), removing one loop. (b) Sourced star–triangle reduction (Theorem A.2): the source at v is transferred to u, w, x; the blue and orange labels are fractions of the final merged currents. (c) Two source-free stars reduce to a triangle by two star–triangle moves and three parallel merges (Theorem A.3).

For the series and parallel steps, the resistance, source and current transformations are (A.8) and (A.10):

$$r = \frac{t_{vu}}{t_{vu}+t_{vw}}, \qquad t^{(1)}_{uw} = t_{vu}+t_{vw}, \qquad \lambda^{\mathrm{red}}_u = \lambda_u + (1-r)\lambda_v, \qquad \lambda^{\mathrm{red}}_w = \lambda_w + r\lambda_v,$$
$$\frac{1}{t^{\mathrm{red}}_{uw}} = \sum_{\substack{j \\ \text{parallel resistors}}} \frac{1}{t^{(j)}_{uw}}, \qquad r_j = \frac{t^{\mathrm{red}}_{uw}}{t^{(j)}_{uw}}, \qquad \sum_{\substack{j \\ \text{parallel resistors}}} r_j = 1, \tag{4.2}$$
$$y_{vu} = (1-r)\lambda_v - y^{(1)}_{uw}, \qquad y_{vw} = r\lambda_v + y^{(1)}_{uw}, \qquad y^{(j)}_{uw} = r_j\, y^{\mathrm{red}}_{uw}.$$

The transformations of the two star–triangle moves are given in Theorems A.2 and A.3.

Theorem 4.2 (Kernels at every loop order)**.**

(i) *For the n-loop fan with path $1-2-\cdots-(n+1)$, insertion $n+2$, sources λ_1, λ_{n+1} at the endpoints of the path and displacements (holomorphic derivatives) $z_{i,n+2} = -z_i$ on the spokes, in*

the coordinates $r_1,\dots,r_{2n}$ *of* (5.6),

$$\begin{aligned}&\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda;z)\big|_{\lambda_2=\cdots=\lambda_n=0,\ z_{i,i+1}=0,\ z_{i,n+2}=-z_i}\\&=(\lambda_{1,+\dot a}\lambda_{n+1,+}{}^{\dot a})^n\int d^{2n}r\,\left[\prod_{j=1}^{n-1}r_{2j-1}^{n-j}r_{2j}(1-r_{2j})^{j-1}\right]r_{2n}^{n-1}e^{\sum_{j=1}^{n+1}\ \ y_{j,n+2,+\dot a}z_j^{+\dot a}}\\&\qquad\qquad\qquad

(iii) *Every product forest with cubic vertices is reduced to one resistor by star–triangle moves at its source-free trivalent nodes, series steps and parallel merges (Figure 4.5). The reversed coordinates make the currents and density polynomial; each Taylor coefficient is a finite sum of the interval and simplex moments* (A.42).

Every coefficient of the complete displacement expansions of (i)–(ii) is an explicit finite factorial sum in (5.14) *and* (7.15).

The proofs are in Sections 5 and 7 and Appendix A.[10]

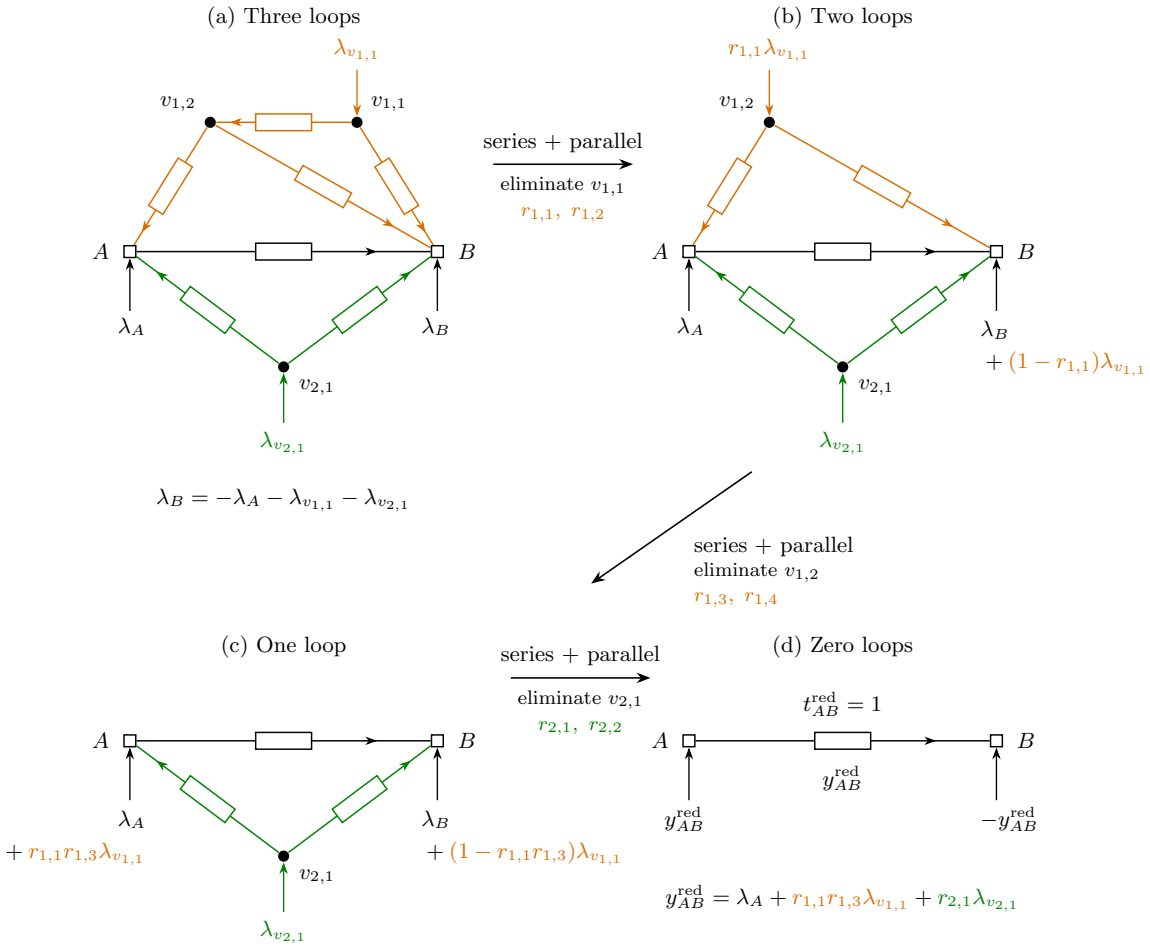


Figure 4.4: Iterated reduction of the three-loop spider with arms $\ell_1 = 2$ and $\ell_2 = 1$ (Section 7.2); for Q_n, A is the $\bar{W}$ vertex and $B = \mathcal{O}$. Each step eliminates an arm vertex by a series step, which passes the fraction $r_{i,2j-1}$ of its source along the arm and the rest to B, and a parallel merge; each step lowers the loop number by one, down to the unit resistor carrying y_{AB}^{red}.

[10] For $n = 1, 2, 3$, (4.5) agrees with the operatope values [9, Equations (4.20), (4.44), (4.67)], for which the code [18] is public, and with the fans of Gui, Wang and Williams [13, Sections 4.4–4.6]; at $n = 1$ it is the coefficient of [4, Equation (2.48)].

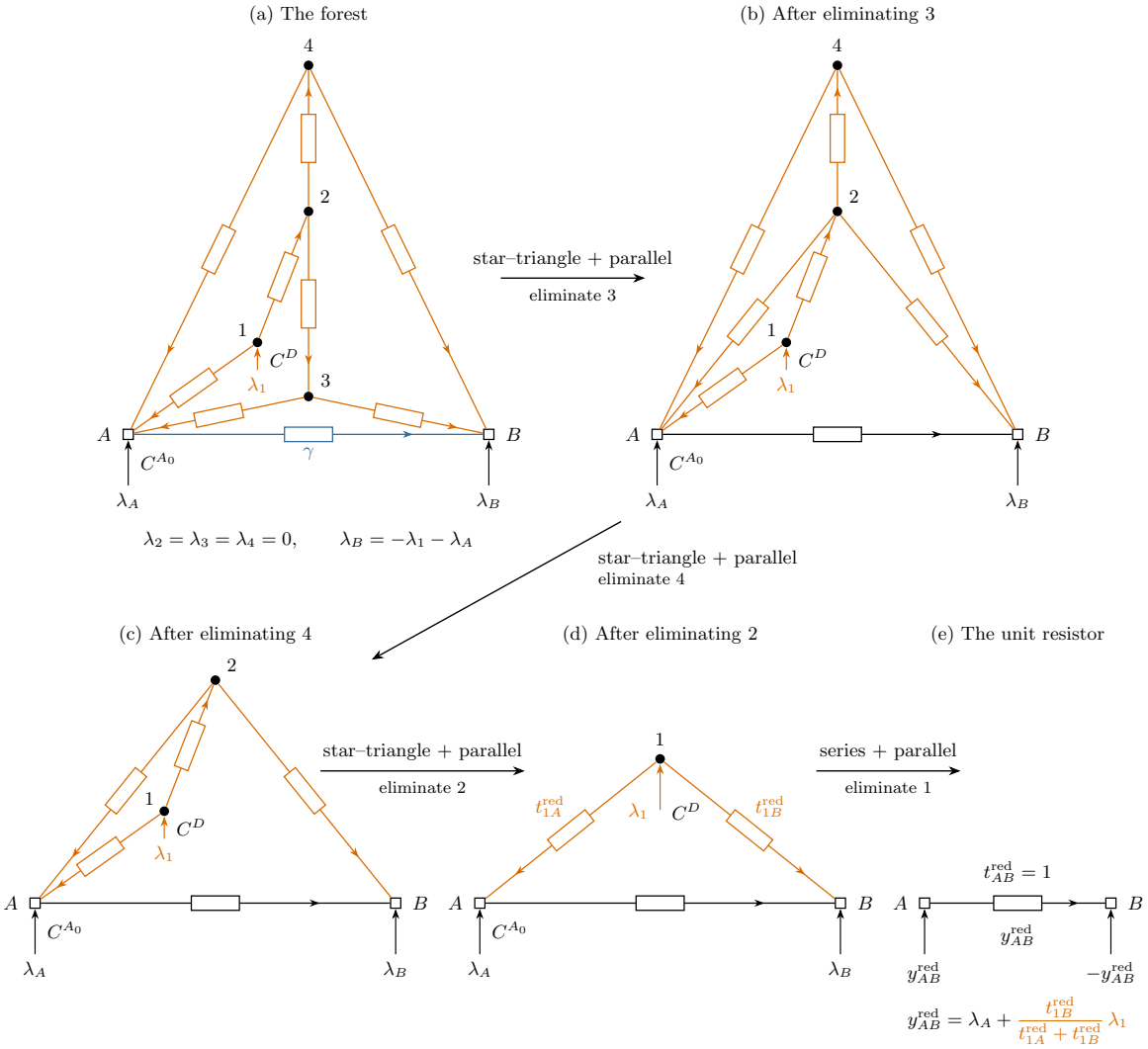


Figure 4.5: Reduction of a four-loop product forest with tree edges $12, 23, 24$, spokes $1A, 3A, 3B, 4A, 4B$ and the marked edge AB in blue (Section 7.3). Star–triangle moves remove the source-free vertices 3, 4 and 2, one per step, and the new resistors merge in parallel with existing ones; the root 1, carrying λ_1, is removed last by a series step and a parallel merge with AB.

4.3 The all-order supercharge of $\mathcal{N} = 4$ Yang–Mills theory

For $\mathcal{N} = 4$ Yang–Mills theory, the fan kernel evaluates every Q_n, including its action on all holomorphic input derivatives.

Theorem 4.3 (All-loop supercharge of $\mathcal{N} = 4$ Yang–Mills theory)**.** *With the superfield action* (A.3)*, superfield C, structure constants C_{ABC}, odd coordinates θ, and integration domain* (5.6)*,*

the n-loop supercharge for $n \geq 1$ and an insertion at x is

$$
\begin{aligned}
Q_n = \frac{(-1)^n i^{n+1}}{2(32\pi^2)^n} \int_{(5.6)} dr_1 \cdots dr_{2n} \left[\prod_{j=1}^{n-1} r_{2j-1}^{n-j} r_{2j} (1-r_{2j})^{j-1} \right] r_{2n}^{n-1} \int d^3\theta \int \prod_{j=1}^{n+1} d^4 z_j \\
\times C_{A_1 D R_1} \left(\prod_{j=2}^{n} C_{A_j R_{j-1} R_j} \right) C_{A_{n+1} R_n E} \\
\times (\partial_{+\dot{a}_1} \cdots \partial_{+\dot{a}_n} C^D)(x + x_1(r,z), \theta) (\partial_+{}^{\dot{a}_1} \cdots \partial_+{}^{\dot{a}_n} C^E)(x + x_{n+1}(r,z), \theta) \\
\times \frac{\delta}{\delta C^{A_1}(x+z_1,\theta)} \cdots \frac{\delta}{\delta C^{A_{n+1}}(x+z_{n+1},\theta)},
\end{aligned}
\tag{4.8}
$$

The displayed derivatives act on the output fields before evaluation at the translated positions

$$
\begin{aligned}
x_1^{+\dot{a}}(r,z) = \sum_{\substack{j=1 \\ \text{path vertices}}}^{n} (1-r_{2j-1}) \left(\prod_{k=1}^{j-1} r_{2k-1} \right) \left[\left(\prod_{k=1}^{j-1} (1-r_{2k}) \right) z_1^{+\dot{a}} + \sum_{\substack{i=2 \\ \text{path vertices}}}^{j} r_{2i-2} \left(\prod_{k=i}^{j-1} (1-r_{2k}) \right) z_i^{+\dot{a}} \right] \\
+ \left(\prod_{k=1}^{n} r_{2k-1} \right) z_{n+1}^{+\dot{a}}, \\
x_{n+1}^{+\dot{a}}(r,z) = r_{2n} \left[\left(\prod_{k=1}^{n-1} (1-r_{2k}) \right) z_1^{+\dot{a}} + \sum_{\substack{i=2 \\ \text{path vertices}}}^{n} r_{2i-2} \left(\prod_{k=i}^{n-1} (1-r_{2k}) \right) z_i^{+\dot{a}} \right] + (1-r_{2n}) z_{n+1}^{+\dot{a}}.
\end{aligned}
\tag{4.9}
$$

These are (8.3) and (5.12); the functional derivatives act at the input positions $x + z_j$. Each r_j is a parallel current-division ratio (Figure 4.3): for $i < n$, r_{2i-1} is the fraction of the source at i passed to $i+1$ and r_{2i} the fraction of the merged spoke current carried by the original spoke $(i+1, n+2)$, and in the final triangle r_{2n-1} and r_{2n} are the fractions of the sources at n and $n+1$ that flow through the edge $(n, n+1)$. Q_n replaces $n+1$ input letters by two output letters.[11]

Lemma 4.4 (Derivative grading)**.** *For an interaction without holomorphic derivatives, Q_n raises each holomorphic derivative total—the numbers of ∂_{++} and $\partial_{+\dot{-}}$—by n. With homotopy contractions homogeneous in these totals, the successive obstruction differential d_r of Section 9.3 raises each by r; a nonzero class can therefore be d_r-exact only if both totals of its leading representative are at least r. In particular, a class whose leading representative has no holomorphic derivatives is d_r-exact for no $r \geq 1$.*

The $\mathcal{N} = 4$ grading is Corollary B.2; the same derivative count applies to the massive SQED example of Section 10.9.

Theorem 4.5 (Finite loop order)**.** *For the interaction (3.4) with $\bar{W}$ of any degree, and every monomial $\mathcal{O}$,*

$$
Q_n \mathcal{O} = 0 \qquad \text{for } n > \#(\mathsf{b}, \beta \text{ in } \mathcal{O}). \tag{4.10}
$$

This is Corollary 3.17.

The quadratic identities imply $\mathbf{Q}^2 = 0$ when the interaction satisfies the quantum master equation (6.4) (Lemma 6.1). For the off-shell $\mathcal{N} = 1$ gauge theory with gauge-invariant $\bar{W}$, with matter representation satisfying $\mathrm{Tr}_R(t_A\{t_B, t_C\}) = 0$, the interaction transforms as:

$$
Q_0 \mathcal{I} = 0, \qquad Q_1 \mathcal{I} = \frac{ig^3}{128\pi^2} \mathrm{Tr}_R(t_A\{t_B, t_C\})\, \mathsf{c}^A (\partial_{+\dot{a}} \mathsf{c}^B)(\partial_+{}^{\dot{a}} \mathsf{c}^C) = 0, \qquad Q_{n \geq 2} \mathcal{I} = 0, \tag{4.11}
$$

Thus $\mathrm{Tr}_R(t_A\{t_B, t_C\}) = 0$ makes the interaction satisfy the quantum master equation at every order (Theorem 6.2).

[11] At one loop, Q_1 agrees with the operator of Budzik and Kulp [4, Equations (3.64)–(3.70)] up to normalization; its component (8.6) is the Konishi anomaly.

4.4 The quantum product and the quantum BPS ring

The higher-loop supercharge acts on several input fields at once, so ordinary multiplication must be replaced by a product compatible with $\mathbf{Q}$. The marked operation of Gaiotto, Kulp and Wu [6, Sections 4.3 and 4.5] defines

$$A \star B = \sum_{\substack{n \geq 0 \\ \text{loop orders}}} \frac{\hbar^n}{n!} \{\mathcal{I}^{\otimes n}, A, B\}^{\gamma,0}, \tag{4.12}$$

Here $\gamma = \{(A, B)\}$ marks the edge joining the insertions, contracted last with $z_{AB} = 0$ (6.26). For quantum cocycles A, B, the coefficient equations of $\mathbf{Q}(A \star B) = 0$ give a closed completion with leading term $A_0 B_0$ (6.35). For $\hbar$-independent polynomial inputs, the coefficient of $\hbar^n$ vanishes when n exceeds their combined number of b, β factors (finiteness bound in Section (6.3)).

Theorem 4.6 (Quantum BPS ring)**.** *For the holomorphic graph operations of Sections 3 and 6, suppose that the interaction satisfies the quantum master equation* (6.4).

(i) $\mathbf{Q}(A \star B) = (\mathbf{Q}A) \star B + (-1)^{|A|} A \star (\mathbf{Q}B)$, *with* $|A|$ *the fermion number of* A.

(ii) *The quantum product induces an associative, graded-commutative, unital multiplication on* $H(\mathbf{Q})$.[12]

(iii) *On the pages* $E_1 = H(Q_0)$, $E_{r+1} = H(E_r, d_r)$ *of the successive homotopy transfer (Section 9.3), it induces the ordinary product of the* $\hbar^0$ *terms of representatives, and every differential* d_r *is a derivation of it.*

(iv) *For homogeneous quantum cocycles* A, B *and every* $L > 0$, $A \star B$ *is cohomologous, in the effective-observable complex at scale* L*, to the quantum normal-ordering product of [3, Section 2.3], the contour integral* (6.41) *of the scale-*L *two-insertion expansion of* A *around* B.

These are Theorems 6.3, 6.4 and 6.6.[13] Consequently the quantum BPS ring is reached from the classical BPS ring $E_1 = H(Q_0)$ by successive contractions: each page $E_{r+1} = H(E_r, d_r)$ is the cohomology of the ring E_r with respect to the derivation d_r, a subquotient of E_r that inherits its product. The quantum ring is thus a subquotient of the classical ring, in general not a subring.

The expansion (4.12) begins with the ordinary product,

$$A \star B = AB + \hbar\, \{\mathcal{I}, A, B\}^{\{(2,3)\},0} + \frac{\hbar^2}{2} \{\mathcal{I}, \mathcal{I}, A, B\}^{\{(3,4)\},0} + O(\hbar^3), \tag{4.13}$$

and in $\mathcal{N} = 4$ Yang–Mills theory its one-loop term is

$$\begin{aligned} \{\mathcal{I}, A, B\}^{\{(2,3)\},0} = \frac{i}{32\pi^2} \int_0^1 dr_1 \int_0^{1-r_1} dr_2 \int d^3\theta\, C_{XYZ} \int d^4 z_1\, d^4 z_2 \\ \times (\partial_{+\dot{a}} C^X)(r_1 z_1 + (1-r_1) z_2, \theta) \\ \times \partial_{2,+}{}^{\dot{a}} \left(\frac{\delta A(x_2)}{\delta C^Y(x_2 + z_1, \theta)} \right) \Big|_{x_2 = r_2(z_2 - z_1)} \frac{\delta B(0)}{\delta C^Z(z_2, \theta)}, \end{aligned} \tag{4.14}$$

The all-order graph construction uses the spider kernels (4.6) and, from three loops onwards, the remaining cubic product forests of Theorem 4.2(iii).

[12] The product is associative only up to exact terms.

[13] The quantum product is the marked operation of Gaiotto, Kulp and Wu [6, Sections 4.3 and 4.5]. Part (i) follows from the quadratic identities of the higher brackets [9, Section 3.2], [5, Theorem 7], [6, Equations (4.19), (4.42)] with one marked edge; the Leibniz rule and a ring structure for the quantum normal-ordering product were anticipated in [3, Section 2.2 and footnote 28], and the two-mark associator graph is drawn in [6, Figure 7]. The all-order homotopy and strict unit, the page product, and the normal-ordering comparison are established in Section 6. The kernel formulas evaluate these operations graph by graph, with the complete one- and two-loop products in Section 6.4.

Uses of the quantum product. The chain identity supplies quantum completions and determines how multiplication acts on successive cohomology pages.

(a) *Multi-gravitons remain quantum cocycles.* The single gravitons G, the components of $\mathrm{Tr}(dC)^p$ [1, Section 4], satisfy $Q_n G = 0$ for every n (9.5). Theorem 9.1 completes any classical product of single gravitons $G_1, \dots, G_m$ by an iterated quantum product $\mathcal{O}$, with any bracketing:

$$\mathbf{Q}\mathcal{O} = 0, \qquad \mathcal{O}|_{\hbar=0} = G_1 \cdots G_m, \tag{4.15}$$

and each fixed component has finitely many corrections: with $\deg_\theta$ equal to $0, 1, 2, 3$ on $\mathsf{c}, \gamma, \beta, \mathsf{b}$ (B.2), the coefficient of $\hbar^n$ vanishes once $3n > \deg_\theta(G_1 \cdots G_m)$ (Theorem 9.1).

(b) *Dressed classes are completed by one more product.* A closed completion $\mathcal{O}[\mathcal{O}_0]$ gives the closed completion $G \star \mathcal{O}[\mathcal{O}_0]$ of $G\mathcal{O}_0$ [14, supplement, Section S8.3].

(c) *Higher differentials vanish on completed classes.* A completed class is annihilated by d_r on every page on which it is defined (9.47).[14] For a finite-dimensional charge sector of $\mathcal{N} = 4$ theory, cohomological degree k, and any finite number R of pages,

$$\sum_{\substack{r=1 \\ \text{pages}}}^{R} \operatorname{rank} d_r^k \le \dim E_1^k - \dim \operatorname{span}\left\{ [\mathcal{O}_0]_1 \in E_1^k : \mathcal{O}_0 \text{ is a multi-graviton} \right\}. \tag{4.16}$$

The right-hand side is the dimension of the classical cohomology modulo the span of the multi-gravitons, that is, of the space of fortuitous classes (9.51).

(d) *The ring is explicit.* Theorem 4.6 gives the multiplication on $H(\mathbf{Q})$, the perturbative quantum BPS ring of $\mathcal{N} = 4$ Yang–Mills theory, and (4.12) computes it graph by graph.

(e) *Closure tests reduce to generators.* Every d_r is a derivation of the product on E_r, so its values on algebra generators of that page determine its action on all products (9.43). The multi-graviton generators already have zero obstruction; the remaining generators carry the new closure tests.

4.5 Loop obstructions and lifting beyond one loop

The two-loop lifting results are listed in Sections 10.4 (Leigh–Strassler), 10.5 (the β-deformation), and 10.9 (massive SQED).

4.6 Non-Cartan classes and the theory-by-theory results

For class representatives, lifting results and certificates, see Section 10; the non-Cartan G_2 class is in Section 10.3.

5 Fan kernels

In Section 2, we show that only the superpotential $\bar{W}$ of the original physical action enters the holomorphic theory; all other terms in the action of the holomorphic twist come from the kinetic BV terms.

In four dimensions, a renormalizable $\mathcal{N} = 1$ supersymmetric gauge theory has a superpotential of degree at most three, and its holomorphic gauge and matter vertices are cubic. The fan kernels therefore determine every loop correction to the supersymmetry transformation in these theories (Corollary 3.13), and now we evaluate them at arbitrary loop order. Appendix A gives the detailed reductions; here we focus on the physical intuition and the final result.

[14] A closed completion may still be removed by a d_r-exact leading class. If $\mathbf{Q}X = \hbar^r \mathcal{O}$, then $[\mathcal{O}]$ is annihilated by $\hbar^r$ over formal power series, while $\mathcal{O} = \mathbf{Q}(\hbar^{-r} X)$ after inverting $\hbar$ (9.52). Moreover, the product of a closed class with a class that is not closed can be closed; such products are quantum analogues of fortuitous classes.

5.1 The one-loop triangle diagram

We first review the well-known one-loop result in the circuit picture.

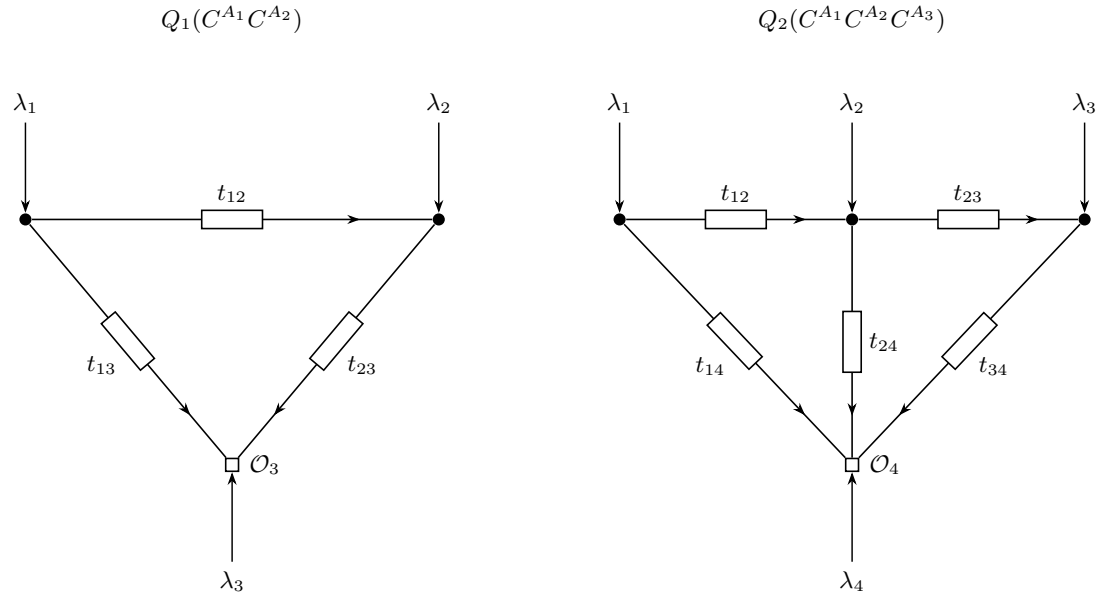


Figure 5.1: The one- and two-loop fan circuits. Dots are interaction vertices and the square is the local insertion. Each edge carries a resistor t_{vw}; λ_v is the current injected at vertex v.

The vertices of the triangle are $1, 2, 3$, with 3 at the origin, the insertion. With the gauge $t_{12} + t_{13} + t_{23} = 1$, current conservation and the cycle equation give[15]

$$\begin{aligned} y_{12,+\dot a} &= t_{13}\lambda_{1,+\dot a} - t_{23}\lambda_{2,+\dot a}, \\ y_{13,+\dot a} &= \lambda_{1,+\dot a} - y_{12,+\dot a}, \\ y_{23,+\dot a} &= \lambda_{2,+\dot a} + y_{12,+\dot a}. \end{aligned} \tag{5.1}$$

Use simplex coordinates r_1, r_2 with $(t_{12}, t_{13}, t_{23}) = (1 - r_1 - r_2, r_1, r_2)$.

The Jacobian is

$$-dy_{12,++} \wedge dy_{12,+-} = (\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a})\, dr_1 \wedge dr_2. \tag{5.2}$$

Then

$$\mathcal{I}_{\triangle}^{\text{raw cut}}(\lambda; z) = (\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a}) \int_0^1 dr_1 \int_0^{1-r_1} dr_2\, e^{-\sum_e y_{e,+\dot b} z_e^{+\dot b}}. \tag{5.3}$$

Expanding the exponential and integrating each monomial yields the entire displacement dependence, as in [4, Equation (2.47)]:[16]

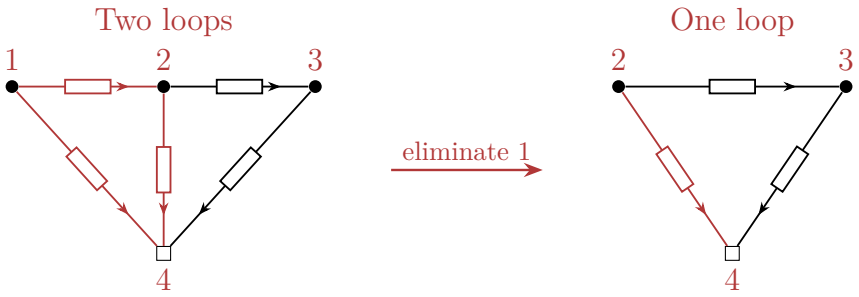
$$\begin{aligned} \mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda; z) = (\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a})\, & e^{-\lambda_{1,+\dot a} z_{13}^{+\dot a} - \lambda_{2,+\dot a} z_{23}^{+\dot a}} \\ & \times \sum_{i,j \geq 0} \frac{[-\lambda_{1,+\dot b}(z_{12} + z_{23} - z_{13})^{+\dot b}]^i [\lambda_{2,+\dot c}(z_{12} + z_{23} - z_{13})^{+\dot c}]^j}{(i + j + 2)!}. \end{aligned} \tag{5.4}$$

[15] The first line of (5.1) is read off from $\partial y_{12}/\partial\lambda_1 = t_{13}$ and $\partial y_{12}/\partial\lambda_2 = -t_{23}$. With λ_2 switched off, the current λ_1 enters at vertex 1 and leaves at vertex 3 through the two parallel paths $1 \to 3$ and $1 \to 2 \to 3$; the path through edge 12, of resistance $t_{12} + t_{23}$, carries the fraction $t_{13}/(t_{12} + t_{13} + t_{23}) = t_{13}$. The other two lines are just current conservation at vertices 1 and 2.

[16] As in (2.34), the displacements generate the holomorphic derivatives of the letters of $\mathcal{O}$.

5.2 One elimination step

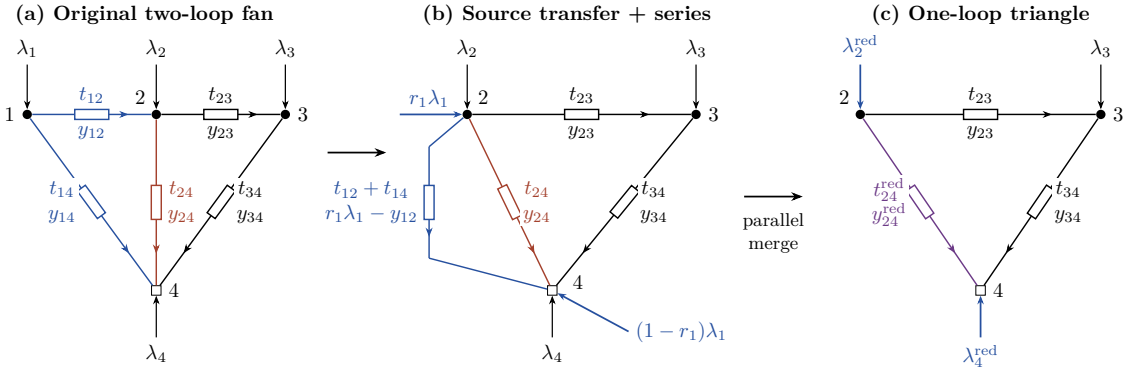
The fan is reduced recursively: each step eliminates the leftmost vertex and redistributes its source between the two remaining endpoints. For the n-loop fan of Figure 3.2, with path $1 \to 2 \to \cdots \to n+1$, spokes oriented towards the insertion $n+2$, and $n \geq 2$, the first step is the following (Figure 5.2).



$$\mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda; z) = \int_0^1 dr_1 \int_0^1 dr_2 \, r_2 \, e^{-\lambda_{1,+\dot{a}}[r_1 z_{12} + (1-r_1) z_{14}]^{+\dot{a}}} \times \lambda_{1,+}^{\dot{b}} \frac{\partial}{\partial z_{24}^{\text{red},+\dot{b}}} \mathcal{I}_{\triangle}^{\text{raw cut}}\Big(\lambda_2 + r_1\lambda_1, \lambda_3, \lambda_4 + (1-r_1)\lambda_1; z^{\text{red}}\Big).$$

$$z_{24}^{\text{red}} = r_2 z_{24} + (1-r_2)(z_{14} - z_{12}), \qquad z_{23}^{\text{red}} = z_{23}, \quad z_{34}^{\text{red}} = z_{34}.$$

Figure 5.2: The two-loop fan on vertices $1, 2, 3, 4$ reduces to the triangle on $2, 3, 4$ (Proposition 5.1); vertex 4 is the insertion. The boxed formula integrates the one-loop kernel over r_1, r_2.



$$r_1 = \frac{t_{14}}{t_{12}+t_{14}}, \qquad r_2 = \frac{t_{12}+t_{14}}{t_{12}+t_{14}+t_{24}}, \qquad \lambda_2^{\text{red}} = \lambda_2 + r_1\lambda_1, \quad \lambda_4^{\text{red}} = \lambda_4 + (1-r_1)\lambda_1.$$

$$-[y_{12} - r_1\lambda_1] = y_{14} - (1-r_1)\lambda_1 = (1-r_2)y_{24}^{\text{red}}, \qquad y_{24} = r_2 y_{24}^{\text{red}}.$$

$$t_{24}^{\text{red}} = \left[\frac{1}{t_{12}+t_{14}} + \frac{1}{t_{24}}\right]^{-1}, \qquad y_{24}^{\text{red}} = \underbrace{r_1\lambda_1 - y_{12}}_{\text{series branch}} + \underbrace{y_{24}}_{\text{original spoke}} = \lambda_2 + r_1\lambda_1 - y_{23}.$$

$$y_{12} = r_1\lambda_1 - (1-r_2)y_{24}^{\text{red}}, \qquad y_{14} = \lambda_1 - y_{12}, \qquad y_{24} = r_2 y_{24}^{\text{red}}.$$

Figure 5.3: Two-loop series–parallel reduction, supplementing Figure 5.2. Blue source arrows inject $r_1\lambda_1$ at 2 and $(1-r_1)\lambda_1$ at insertion 4; the series branch carries $r_1\lambda_1 - y_{12}$. The parallel merge gives the triangle on $2, 3, 4$; the formulas beneath the circuits reconstruct the original currents.

Proposition 5.1 (One elimination step)**.**

Set

$$r_1 = \frac{t_{1,n+2}}{t_{12}+t_{1,n+2}}, \qquad r_2 = \frac{t_{12}+t_{1,n+2}}{t_{12}+t_{1,n+2}+t_{2,n+2}},$$

where r_1 will be the fraction of λ_1 passed to vertex 2 and r_2 will be the fraction of the merged spoke current carried by the original spoke $(2, n+2)$. Eliminating vertex 1 replaces the resistors $t_{12}, t_{1,n+2}, t_{2,n+2}$ by one resistor between 2 and $n+2$ and moves the source λ_1 to the remaining endpoints:

$$t^{\mathrm{red}}_{2,n+2} = \left[\frac{1}{t_{12}+t_{1,n+2}} + \frac{1}{t_{2,n+2}}\right]^{-1}, \qquad \lambda^{\mathrm{red}}_2 = \lambda_2 + r_1\lambda_1, \qquad \lambda^{\mathrm{red}}_{n+2} = \lambda_{n+2} + (1-r_1)\lambda_1.$$

The removed currents follow from the reduced spoke current $y^{\mathrm{red}}_{2,n+2} = \lambda_2 + r_1\lambda_1 - y_{23}$:

$$y_{2,n+2} = r_2 y^{\mathrm{red}}_{2,n+2}, \qquad y_{12} = r_1\lambda_1 - (1-r_2)y^{\mathrm{red}}_{2,n+2} = r_1 r_2\lambda_1 - (1-r_2)\lambda_2 + (1-r_2)y_{23}, \qquad y_{1,n+2} = \lambda_1 - y_{12}.$$

With Γ^{red} the reduced fan on vertices $2, \dots, n+2$, the kernel satisfies

$$\begin{aligned}
&\mathcal{I}^{\text{raw cut}}_{\Gamma}(\lambda; z) \\
&= \int_0^1 dr_1 \int_0^1 dr_2 \; e^{-\lambda_{1,+\dot a}(r_1 z_{12} + (1-r_1) z_{1,n+2})^{+\dot a}} \\
&\quad \times \underbrace{r_2 \lambda_{1,+}{}^{\dot b} \frac{\partial}{\partial z^{\mathrm{red},+\dot b}_{2,n+2}}}_{\substack{\text{inserts the current-area Jacobian} \\ r_2(\lambda_{1,+\dot b} y^{\mathrm{red}}_{2,n+2,+}{}^{\dot b})}} \mathcal{I}^{\text{raw cut}}_{\Gamma^{\mathrm{red}}}(\lambda^{\mathrm{red}}; z^{\mathrm{red}}).
\end{aligned} \tag{5.5}$$

The derivative is taken before substituting

$$z^{\mathrm{red}}_{2,n+2} = r_2 z_{2,n+2} + (1-r_2)(z_{1,n+2} - z_{12}), \qquad z^{\mathrm{red}}_e = z_e \quad (e \neq (2, n+2)).$$

Proof.

Conservation at 1 and the cycle $1 \to 2 \to n+2 \to 1$ give:

$$y_{1,n+2} = \lambda_1 - y_{12}, \qquad t_{2,n+2} y_{2,n+2} = t_{1,n+2}\lambda_1 - (t_{12}+t_{1,n+2}) y_{12}.$$

The resistors on $2 \to 1 \to n+2$ combine in series, with resistance $t_{12}+t_{1,n+2}$, and carry

$$y_{eff\,2,n+2} = r_1\lambda_1 - y_{12} = \frac{t_{2,n+2} y_{2,n+2}}{t_{12}+t_{1,n+2}}.$$

In parallel with $t_{2,n+2}$ the conductances add, and the merged current $y^{\mathrm{red}}_{2,n+2} = (r_1\lambda_1 - y_{12}) + y_{2,n+2}$ divides as

$$r_1\lambda_1 - y_{12} = (1-r_2)\, y^{\mathrm{red}}_{2,n+2}, \qquad y_{2,n+2} = r_2\, y^{\mathrm{red}}_{2,n+2},$$

while conservation at 2 in the reduced network gives $y^{\mathrm{red}}_{2,n+2} = \lambda^{\mathrm{red}}_2 - y_{23}$. Solving for y_{12} gives the stated formula.

For the kernel, substituting the removed currents gives

$$\begin{aligned}
\sum_{e \in \Gamma_1} y_{e,+\dot a} z_e^{+\dot a} &= \lambda_{1,+\dot a}(r_1 z_{12} + (1-r_1) z_{1,n+2})^{+\dot a} \\
&\quad + \sum_{e \in \Gamma^{\mathrm{red}}_1} y^{\mathrm{red}}_{e,+\dot a} z^{\mathrm{red},+\dot a}_e.
\end{aligned}$$

Modulo the differentials of the remaining loop currents,

$$\begin{aligned} dy^{\mathrm{red}}_{2,n+2,+\dot a} &\equiv \lambda_{1,+\dot a}\, dr_1, \\ dy_{12,+\dot a} &\equiv r_2\lambda_{1,+\dot a}\, dr_1 + y^{\mathrm{red}}_{2,n+2,+\dot a}\, dr_2, \\ -dy_{12,+\dot +} \wedge dy_{12,+\dot -} &\equiv - \underbrace{r_2(\lambda_{1,+\dot a} y^{\mathrm{red}}_{2,n+2,+}{}^{\dot a})}_{\text{current-area Jacobian with induced orientation}} dr_1 \wedge dr_2, \\ \det \frac{\partial(t_{12}, t_{1,n+2}, t_{2,n+2})}{\partial(r_1, r_2, t^{\mathrm{red}}_{2,n+2})} &= -\frac{(t^{\mathrm{red}}_{2,n+2})^2}{r_2^2(1-r_2)^3}. \end{aligned}$$

The two orientation signs cancel as in (A.11).

Differentiation under the remaining current integral inserts the required pairing:

$$\begin{aligned} & r_2\lambda_{1,+}{}^{\dot a} \frac{\partial}{\partial z^{\mathrm{red},+\dot a}_{2,n+2}} e^{-\sum_{e\in\Gamma_1^{\mathrm{red}}} y^{\mathrm{red}}_{e,+\dot b} z_e^{\mathrm{red},+\dot b}} \\ &\quad = \underbrace{r_2(\lambda_{1,+\dot a} y^{\mathrm{red}}_{2,n+2,+}{}^{\dot a})}_{\text{current-area Jacobian insertion}} \quad e^{-\sum_{e\in\Gamma_1^{\mathrm{red}}} y^{\mathrm{red}}_{e,+\dot b} z_e^{\mathrm{red},+\dot b}}. \end{aligned}$$

□

The next step acts on vertex 2 with the reduced source $\lambda_2 + r_1\lambda_1$ and the reduced spoke $t^{\mathrm{red}}_{2,n+2}$ (Appendix A.4). The iteration eventually reaches one loop.

5.3 Final result at arbitrary loop order

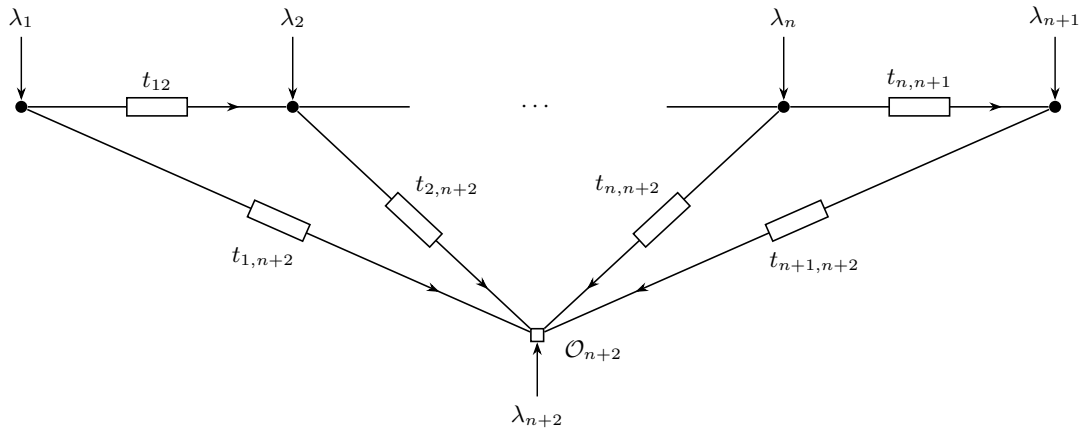


Figure 5.4: The n-loop fan circuit: a path of $n+1$ cubic interactions joined to the insertion. The resistors have resistances t_{vw}, and the external arrows carry the currents λ_v. The short stubs and central ellipsis indicate omitted intermediate vertices.

We keep the sources $\lambda_2, \dots, \lambda_n$ arbitrary. In a renormalizable theory, however, these λ's are zero in Q_n because there are no uncontracted letters at these vertices; only λ_1 and λ_{n+1} (and $\lambda_{n+2} = -\sum_{v<n+2}\lambda_v$) are nonzero. (By Propositions 3.14 and 3.18, even in a non-renormalizable theory, at most one of $\lambda_2, \dots, \lambda_n$ is nonzero in every fan kernel considered here.)

For $n \geq 2$, iterating Proposition 5.1 along the interaction path, at vertex 1, then vertex 2, and so on, reduces the fan to the triangle.

The independent coordinates range over

$$0 < r_j < 1 \quad (1 \leq j \leq 2n-2), \qquad r_{2n-1} > 0, \quad r_{2n} > 0, \quad r_{2n-1} + r_{2n} < 1. \tag{5.6}$$

The final two coordinates form a triangle; the earlier coordinates range independently over intervals. **Each r_j with $j \le 2n-2$ is a current-division ratio of the iteration: r_{2i-1} is the fraction of the source at the eliminated vertex i that the series step passes to vertex $i+1$, and r_{2i} is the fraction of the merged current carried by the original spoke $(i+1, n+2)$ after the parallel merge.** The last two, r_{2n-1} and r_{2n}, are the simplex coordinates of the final triangle, as in (5.3).

Before eliminating vertex i, the reduced source is

$$\lambda_i^{\mathrm{red}} = \sum_{\substack{1\le j\le i\\ \text{vertices}}} \left(\prod_{k=j}^{i-1} r_{2k-1}\right)\lambda_j, \qquad 1 \le i \le n,$$

$$\lambda_1^{\mathrm{red}} = \lambda_1, \qquad \lambda_2^{\mathrm{red}} = \lambda_2 + r_1\lambda_1, \qquad \lambda_3^{\mathrm{red}} = \lambda_3 + r_3(\lambda_2 + r_1\lambda_1) \quad (n \ge 3).$$

The first step is (5.5).

For $n \ge 3$, eliminating vertex 2 from Γ^{red} gives $(\Gamma^{\mathrm{red}})^{\mathrm{red}}$:

$$\begin{aligned}
&\mathcal{I}^{\text{raw cut}}_{\Gamma^{\mathrm{red}}}(\lambda^{\mathrm{red}}; z^{\mathrm{red}})\\
&= \int_0^1 dr_3 \int_0^1 dr_4\, e^{-\lambda^{\mathrm{red}}_{2,+\dot a}(r_3 z_{23} + (1-r_3) z^{\mathrm{red}}_{2,n+2})^{+\dot a}}\\
&\quad\times \underbrace{r_4 \lambda^{\mathrm{red}}_{2,+}{}^{\dot c} \frac{\partial}{\partial z^{\mathrm{red},+\dot c}_{3,n+2}}}_{\substack{\text{inserts the current-area Jacobian}\\ r_4(\lambda^{\mathrm{red}}_{2,+\dot c} y^{\mathrm{red}}_{3,n+2,+}{}^{\dot c})}} \mathcal{I}^{\text{raw cut}}_{(\Gamma^{\mathrm{red}})^{\mathrm{red}}}\begin{pmatrix} \lambda_3 + r_3\lambda_2^{\mathrm{red}}, \lambda_4, \dots, \lambda_{n+1},\\ \lambda_{n+2} + (1-r_1)\lambda_1\\ +(1-r_3)\lambda_2^{\mathrm{red}}; z^{\mathrm{red}}\end{pmatrix},\\
&z^{\mathrm{red}}_{3,n+2} = r_4 z_{3,n+2} + (1-r_4)(z^{\mathrm{red}}_{2,n+2} - z_{23}).
\end{aligned}$$

In each kernel on the right, red denotes the network produced by that step; the derivative precedes the displayed displacement substitution.

The first two inserted-current factors are

$$\begin{aligned}
r_2\lambda_{1,+\dot a} y^{\mathrm{red}}_{2,n+2,+}{}^{\dot a} &= r_2\lambda_{1,+\dot a}(\lambda_{2,+}{}^{\dot a} - y_{23,+}{}^{\dot a}(r)),\\
r_4\lambda^{\mathrm{red}}_{2,+\dot a} y^{\mathrm{red}}_{3,n+2,+}{}^{\dot a} &= r_4(\lambda_{2,+\dot a} + r_1\lambda_{1,+\dot a})(\lambda_{3,+}{}^{\dot a} - y_{34,+}{}^{\dot a}(r)).
\end{aligned}$$

For $n \ge 3$, the first two reductions and their iteration give the equality chain below.

In each term, take the displayed derivatives first, then apply the displacement substitutions from the last reduction to the first.

For $n=2$, the first reduced kernel is the triangle (5.3), so the second reduction is omitted:

$$
\begin{aligned}
&\mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda;z)\\
&=\int_0^1 dr_1\int_0^1 dr_2\; e^{-\lambda_{1,+\dot a}(r_1z_{12}+(1-r_1)z_{1,n+2})^{+\dot a}}\\
&\quad\times\underbrace{r_2\lambda_{1,+}{}^{\dot b}\frac{\partial}{\partial z^{\text{red},+\dot b}_{2,n+2}}}_{\substack{\text{inserts the current-area Jacobian}\\ r_2(\lambda_{1,+\dot b}y^{\text{red}}_{2,n+2,+}{}^{\dot b})}}\qquad\underbrace{\mathcal{I}^{\text{raw cut}}_{\Gamma^{\text{red}}}(\lambda^{\text{red}};z^{\text{red}})}_{(n-1)\text{-loop kernel}}\\
&=\int_0^1 dr_1\int_0^1 dr_2\int_0^1 dr_3\int_0^1 dr_4\; e^{-\lambda_{1,+\dot a}(r_1z_{12}+(1-r_1)z_{1,n+2})^{+\dot a}}\\
&\quad\times e^{-\lambda^{\text{red}}_{2,+\dot a}(r_3z_{23}+(1-r_3)z^{\text{red}}_{2,n+2})^{+\dot a}}\\
&\quad\times\underbrace{r_2\left[(1-r_3)\lambda_{1,+\dot a}\lambda^{\text{red}\,\dot a}_{2,+}+(1-r_4)\lambda_{1,+}{}^{\dot b}\frac{\partial}{\partial z^{\text{red},+\dot b}_{3,n+2}}\right]}_{\substack{\text{inserts the first current-area Jacobian}\\ r_2(\lambda_{1,+\dot b}y^{\text{red}}_{2,n+2,+}{}^{\dot b})}}\\
&\quad\times\underbrace{r_4\lambda^{\text{red}\,\dot c}_{2,+}\frac{\partial}{\partial z^{\text{red},+\dot c}_{3,n+2}}}_{\substack{\text{inserts the current-area Jacobian}\\ r_4(\lambda^{\text{red}}_{2,+\dot c}y^{\text{red}}_{3,n+2,+}{}^{\dot c})}}\underbrace{\mathcal{I}^{\text{raw cut}}_{(\Gamma^{\text{red}})^{\text{red}}}\begin{pmatrix}\lambda_3+r_3\lambda^{\text{red}}_2,\lambda_4,\dots,\lambda_{n+1},\\ \lambda_{n+2}+(1-r_1)\lambda_1\\ +(1-r_3)\lambda^{\text{red}}_2;z^{\text{red}}\end{pmatrix}}_{(n-2)\text{-loop kernel}}\\
&=\cdots\\
&=\int_{(5.6)}d^{2n}r\prod_{i=1}^{n-1}\underbrace{r_{2i}\left[\sum_{j=1}^{i}\left(\prod_{k=j}^{i-1}r_{2k-1}\right)\lambda_{j,+\dot a_i}\right](\lambda_{i+1,+}{}^{\dot a_i}-y_{i+1,i+2,+}{}^{\dot a_i}(r))}_{\text{kernel factor from eliminating vertex } i}\\
&\quad\times\underbrace{\left[\sum_{j=1}^{n}\left(\prod_{k=j}^{n-1}r_{2k-1}\right)\lambda_{j,+\dot b}\right]\lambda_{n+1,+}{}^{\dot b}}_{\text{final triangle factor}}\;e^{-\sum_e y_{e,+\dot a}(r)z_e^{+\dot a}}.
\end{aligned}
\tag{5.7}
$$

We now recover the original y's in the new coordinates.

We invert the reduction to obtain $y_{i,i+1}$; $y_{i,n+2}$ then follows immediately from current conservation.

Along the path,[17]

$$
\begin{aligned}
y_{n,n+1,+\dot a}&=r_{2n-1}\sum_{j=1}^{n}\left(\prod_{k=j}^{n-1}r_{2k-1}\right)\lambda_{j,+\dot a}-r_{2n}\lambda_{n+1,+\dot a},\\
y_{i,i+1,+\dot a}&=r_{2i-1}r_{2i}\sum_{j=1}^{i}\left(\prod_{k=j}^{i-1}r_{2k-1}\right)\lambda_{j,+\dot a}\\
&\qquad-(1-r_{2i})\lambda_{i+1,+\dot a}+(1-r_{2i})y_{i+1,i+2,+\dot a},\qquad 1\le i<n.
\end{aligned}
\tag{5.8}
$$

[17]Equation (5.8) is Proposition 5.1 applied at vertex i of the reduced network,

$$y_{i,i+1}=r_{2i-1}r_{2i}\lambda^{\text{red}}_i-(1-r_{2i})\lambda_{i+1}+(1-r_{2i})y_{i+1,i+2},\qquad \lambda^{\text{red}}_i=\lambda_i+r_{2i-3}\lambda^{\text{red}}_{i-1},\quad \lambda^{\text{red}}_1=\lambda_1,$$

with the resistance ratios $r_{2i-1}=t^{\text{red}}_{i,n+2}/(t_{i,i+1}+t^{\text{red}}_{i,n+2})$ and $r_{2i}=(t_{i,i+1}+t^{\text{red}}_{i,n+2})/(t_{i,i+1}+t^{\text{red}}_{i,n+2}+t_{i+1,n+2})$ of Theorem A.1; the first line is the triangle (5.1) with t_{13},t_{23} replaced by $t^{\text{red}}_{n,n+2}=r_{2n-1}$ and $t_{n+1,n+2}=r_{2n}$.

The spoke currents are

$$\begin{aligned} y_{1,n+2,+\dot{a}} &= \lambda_{1,+\dot{a}} - y_{12,+\dot{a}}, \\ y_{i,n+2,+\dot{a}} &= \lambda_{i,+\dot{a}} + y_{i-1,i,+\dot{a}} - y_{i,i+1,+\dot{a}}, \qquad 2 \le i \le n, \\ y_{n+1,n+2,+\dot{a}} &= \lambda_{n+1,+\dot{a}} + y_{n,n+1,+\dot{a}}. \end{aligned} \tag{5.9}$$

This reflects the order of our iteration: the path currents are fixed first, backward from $y_{n,n+1}$ to y_{12}, and current conservation at each path vertex then determines every spoke current.

All currents are polynomial in the new coordinates, without the spanning-tree denominators that appeared in previous work, so every fixed Taylor coefficient is a polynomial integral over the explicit domain (5.6), for arbitrary sources on the path.

5.4 Endpoint sources and final Taylor expansion

As noted above, in the computation of Q_n in renormalizable theories, the middle λ's are all zero.

Moreover, the interaction vertices carry no derivatives, so only the holomorphic derivatives of the letters of $\mathcal{O}$ generate displacements. These sit on the spokes: $z_{i,i+1} = 0$ and $z_{i,n+2} = -z_i$, where z_i displaces the letter contracted on spoke i. The density then factorizes:

$$\begin{aligned} &\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda; z)\big|_{\lambda_2=\cdots=\lambda_n=0,\ z_{i,i+1}=0,\ z_{i,n+2}=-z_i} \\ &= (\lambda_{1,+\dot{a}}\lambda_{n+1,+}{}^{\dot{a}})^n \int_{(5.6)} d^{2n}r \left[\prod_{j=1}^{n-1} r_{2j-1}^{n-j} r_{2j}(1-r_{2j})^{j-1}\right] r_{2n}^{n-1} e^{\sum_j y_{j,n+2,+\dot{a}} z_j^{+\dot{a}}}. \end{aligned} \tag{5.10}$$

The exponential in (5.10) translates the two endpoint fields. Substituting the currents (5.8)–(5.9) gives

$$\sum_{j=1}^{n+1} y_{j,n+2,+\dot{a}}(r) z_j^{+\dot{a}} = \lambda_{1,+\dot{a}} x_1^{+\dot{a}}(r,z) + \lambda_{n+1,+\dot{a}} x_{n+1}^{+\dot{a}}(r,z), \tag{5.11}$$

where the translated positions are

$$\begin{aligned} x_1^{+\dot{a}}(r,z) = &\sum_{j=1}^{n}(1-r_{2j-1})\left(\prod_{k=1}^{j-1} r_{2k-1}\right) \\ &\times \left[\left(\prod_{k=1}^{j-1}(1-r_{2k})\right) z_1^{+\dot{a}} + \sum_{i=2}^{j} r_{2i-2}\left(\prod_{k=i}^{j-1}(1-r_{2k})\right) z_i^{+\dot{a}}\right] \\ &+ \left(\prod_{k=1}^{n} r_{2k-1}\right) z_{n+1}^{+\dot{a}}, \\ x_{n+1}^{+\dot{a}}(r,z) = &\, r_{2n}\left[\left(\prod_{k=1}^{n-1}(1-r_{2k})\right) z_1^{+\dot{a}} + \sum_{i=2}^{n} r_{2i-2}\left(\prod_{k=i}^{n-1}(1-r_{2k})\right) z_i^{+\dot{a}}\right] \\ &+ (1-r_{2n}) z_{n+1}^{+\dot{a}}. \end{aligned} \tag{5.12}$$

We now give the Taylor expansion in explicit form, ready for direct use in the machine enumeration of the later sections.

The monomial moments on the parameter domain are

$$\int_{(5.6)} dr_1 \cdots dr_{2n} \prod_{j=1}^{2n} r_j^{m_j} = \frac{m_{2n-1}!\, m_{2n}!}{(m_{2n-1}+m_{2n}+2)!} \prod_{j=1}^{2n-2} \frac{1}{m_j+1}. \tag{5.13}$$

Grouping by the total displacement degree m, the full Taylor expansion is

$$\begin{aligned}
&\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda;z)\Big|_{\lambda_2=\cdots=\lambda_n=0}\\
&=(\lambda_{1,+\dot a}\lambda_{n+1,+}{}^{\dot a})^n\sum_{m=0}^{\infty}\sum_{\substack{r_0,s_0,r_{ik},s_i\geq 0\\ r_0+s_0+\sum_{i=1}^n s_i+\sum_{k=1}^n\sum_{i=1}^k r_{ik}=m}}\\
&\quad\times\frac{[-\lambda_{1,+\dot b}z_{1,n+2}^{+\dot b}]^{r_0}[-\lambda_{n+1,+\dot c}z_{n+1,n+2}^{+\dot c}]^{s_0}}{r_0!s_0!}\\
&\quad\times\prod_{1\leq i\leq k\leq n}\frac{\left[-\lambda_{1,+\dot d}(z_{i,i+1}+z_{i+1,n+2}-z_{i,n+2})^{+\dot d}\right]^{r_{ik}}}{r_{ik}!}\\
&\quad\times\prod_{i=1}^n\frac{[\lambda_{n+1,+\dot e}(z_{i,i+1}+z_{i+1,n+2}-z_{i,n+2})^{+\dot e}]^{s_i}}{s_i!}\\
&\quad\times\frac{(\sum_{i=1}^n r_{in})!\,(n-1+\sum_{i=1}^n s_i)!}{(n+1+\sum_{i=1}^n r_{in}+\sum_{i=1}^n s_i)!}\\
&\quad\times\prod_{j=1}^{n-1}\frac{1}{n-j+1+\sum_{k=j}^n\sum_{i=1}^k r_{ik}}\\
&\quad\times\prod_{j=1}^{n-1}\frac{\left(1+\sum_{i=1}^j r_{ij}\right)!\left(j-1+\sum_{i=1}^j\left[s_i+\sum_{k=j+1}^n r_{ik}\right]\right)!}{\left(j+1+\sum_{i=1}^j\left[s_i+\sum_{k=j}^n r_{ik}\right]\right)!}.
\end{aligned}\tag{5.14}$$

For arbitrary edge displacements, sum over nonnegative integers r_0, s_0, r_{ik}, s_i, with $1\leq i\leq k\leq n$.

Its simplest coefficient, without displacements, is

$$\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda;0)\Big|_{\lambda_{2,+\dot a}=\cdots=\lambda_{n,+\dot a}=0}=\frac{(\lambda_{1,+\dot a}\lambda_{n+1,+}{}^{\dot a})^n}{(n!)^3(n+1)}.\tag{5.15}$$

For $n=1,2,3$, (5.15) reproduces [9, Equations (4.20), (4.44), (4.67)], [4, Equation (2.48)] and [13, Sections 4.4–4.6].

6 Quantum multiplication and the Leibniz rule

As we have seen, the differential Q_1 contains two input derivatives [4].

$$\begin{aligned}
Q_1=&-\frac{1}{64\pi^2}\int_0^1 dr_2\int_0^{1-r_2}dr_1\int d^3\theta\int d^4z_1\,d^4z_2\,C_{ARD}C_{BRE}\\
&\times(\partial_{+\dot a}C^D)(x+(1-r_1)z_1+r_1z_2,\theta)\,(\partial_+{}^{\dot a}C^E)(x+r_2z_1+(1-r_2)z_2,\theta)\\
&\times\frac{\delta}{\delta C^A(x+z_1,\theta)}\frac{\delta}{\delta C^B(x+z_2,\theta)}.
\end{aligned}\tag{6.1}$$

Its two functional derivatives contract two input letters, which can belong to different factors (equation (6.2) below).

Acting on a product of two composite operators $A\times B$, these derivatives can land on different factors, producing a cross-contraction that explicitly violates the Leibniz rule. For example, in the zero-input-jet part of $\mathcal{N}=4$ SYM, the Leibniz rule defect reads:

$$\begin{aligned}
&Q_1(AB)-(Q_1A)B-(-1)^{|A|}A(Q_1B)\\
&\quad=-\frac{1}{64\pi^2}\int d^3\theta\,C_{XRD}C_{YRE}(\partial_{+\dot a}C^D)(\partial_+{}^{\dot a}C^E)\frac{\delta A}{\delta C^X}\frac{\delta B}{\delta C^Y}.
\end{aligned}\tag{6.2}$$

Here $C(z^{+\dot{a}}, \theta_i)$ is the superfield of Appendix A.1.

Even when both A and B are annihilated by Q_0 and Q_1 (so that they remain quantum cocycles), this cross-contraction can be nonzero; in fact, it is generically nonzero.

This raises an apparent tension with the ring structure of supercharge cohomology: supersymmetry acts as a graded derivation on separated operator products, the product of two closed operators should again define a closed class, and the Q-cohomology, or BPS ring, is expected to have a ring structure.

This raises a question: given two classes $[A]$ and $[B]$ of Q-closed operators, we expect a class, denoted $[AB]$, with the physical meaning of the product of A and B.

The failure of the Leibniz rule for Q_1, Q_2, ...[18] shows that this is not trivial: at the least, $[AB]$ cannot be the naive Wick product of A and B.

The quantum multiplication, defined in previous work [6, Sections 4.3 and 4.5], resolves this tension at every perturbative order [3, footnote 28]: $\mathbf{Q}$ obeys the Leibniz rule for the quantum "normal-ordering product".

The definition of the quantum product also uses the higher brackets, together with two auxiliary free fields.

The quadratic identities of the higher brackets, together with the quantum master equation, give the corrected Leibniz identity (6.13). Thus $\mathbf{Q}A = \mathbf{Q}B = 0$ implies $\mathbf{Q}(A \star B) = 0$. We first prove this identity and, from the marked quadratic identities with three insertions, the ring laws on cohomology; we then give the graph formula and, in Section 7, the way to calculate these graphs.

For the enumeration of $\mathbf{Q}$-cohomology representatives, the quantum product usually need not be computed explicitly. It is an explicit construction showing that the product of two quantum cocycles is a quantum cocycle: for quantum cocycles A and B, $A \star B$ is a quantum cocycle completing $A_0 B_0$,[19] and it is lifted exactly when it becomes a quantum coboundary. In the successive homotopy transfer of Section 9.3, which is the algorithm used for the lifting tests, the quantum product induces the ordinary product of leading representatives on every page (Theorem 6.4).

6.1 The Leibniz rule of the quantum supercharge

Here $\{\cdot\}_0$ denotes the brackets of Section 3. The quantum differential and the interacting bracket of two operators are

$$\mathbf{Q}\mathcal{O} = \sum_{s\geq 1} \frac{\hbar^{s-1}}{s!} \{\mathcal{I}^{\otimes s}, \mathcal{O}\}_0, \qquad \{X, Y\}_{\mathcal{I}} = \sum_{r\geq 0} \frac{\hbar^r}{r!} \{\mathcal{I}^{\otimes r}, X, Y\}_0. \tag{6.3}$$

Lemma 6.1. *Suppose that the interaction satisfies*

$$\sum_{k\geq 2} \frac{\hbar^{k-2}}{k!} \{\mathcal{I}^{\otimes k}\}_0 = \sum_{n\geq 0} \frac{\hbar^n}{n+2} Q_n \mathcal{I} = 0. \tag{6.4}$$

Then $\mathbf{Q}^2 = 0$; at the first orders,

$$\begin{aligned} Q_0^2 = 0, \qquad & Q_0 Q_1 + Q_1 Q_0 = 0, \\ Q_0 Q_2 + Q_1^2 + Q_2 Q_0 = 0, \quad & Q_0 Q_3 + Q_1 Q_2 + Q_2 Q_1 + Q_3 Q_0 = 0. \end{aligned} \tag{6.5}$$

[18] The higher corrections Q_n likewise act on $n+1$ input fields and produce contractions shared between at most $n+1$ factors, violating the ordinary Leibniz rule more severely.

[19] Here, the completion of a Q_0 cocycle O_0 is a series $O = \sum_{\substack{i\geq 0 \\ \text{loop orders}}} \hbar^i O_i = O_0 + \hbar O_1 + \hbar^2 O_2 + \cdots$ (a finite sum) satisfying $\mathbf{Q} \sum_{\substack{i\geq 0 \\ \text{loop orders}}} \hbar^i O_i = 0$. Not all Q_0 cocycles have a completion; those without one are the "lifted" operators.

Proof. The quadratic identity of [6], with N interaction entries and one operator, reads

$$\begin{aligned} 0 = &\sum_{r=1}^{N-1} \binom{N}{r} \{\mathcal{I}^{\otimes(N-r)}, \{\mathcal{I}^{\otimes r}, \mathcal{O}\}_0\}_0 \\ &- \sum_{k=2}^{N} \binom{N}{k} \{\mathcal{I}^{\otimes(N-k)}, \{\mathcal{I}^{\otimes k}\}_0, \mathcal{O}\}_0. \end{aligned} \tag{6.6}$$

Multiplying by $\hbar^{N-2}/N!$ and summing over N gives

$$\begin{aligned} 0 = &\sum_{a,b\geq 0} \frac{\hbar^{a+b}}{(a+1)!\,(b+1)!} \{\mathcal{I}^{\otimes(a+1)}, \{\mathcal{I}^{\otimes(b+1)}, \mathcal{O}\}_0\}_0 \\ &- \sum_{m\geq 0,\ k\geq 2} \frac{\hbar^{m+k-2}}{m!\,k!} \{\mathcal{I}^{\otimes m}, \{\mathcal{I}^{\otimes k}\}_0, \mathcal{O}\}_0. \end{aligned} \tag{6.7}$$

The first sum is $\mathbf{Q}^2\mathcal{O}$ and the second vanishes by (6.4). The first coefficients agree with [4, Equations (1.12) and (2.25)–(2.26)]. □

Theorem 6.2. *Suppose that $\bar{W}$ is gauge invariant and the matter representation obeys*

$$\mathrm{Tr}_R(t_A\{t_B, t_C\}) = 0. \tag{6.8}$$

For the interaction (3.4), the integrated interaction satisfies

$$\bar{\partial}\mathcal{I} = 0, \qquad Q_0\mathcal{I} = 0, \qquad Q_1\mathcal{I} = 0, \qquad Q_2\mathcal{I} = Q_3\mathcal{I} = \cdots = 0. \tag{6.9}$$

Consequently, $\mathcal{I}$ satisfies (6.4), $(\bar{\partial} + Q_0 + \sum_{n\geq 1}\hbar^n Q_n)\mathcal{I} = 0$, and $\mathbf{Q}^2 = 0$ by Lemma 6.1.

Proof. The free differential $\bar{\partial}$ maps the integrated Dolbeault density of $\mathcal{I}$ to a total derivative. The gauge and matter vertices satisfy the classical master equation by

$$C^A{}_{B[C}C^B{}_{DE]} = 0, \qquad [t_A, t_B] = iC^C{}_{AB}t_C.$$

Since $\bar{W}$ contains no b or β, its self-bracket vanishes, and, up to holomorphic total derivatives,

$$\begin{aligned} Q_0\mathcal{I} = \{\mathcal{I}, \mathcal{I}\}_0 = &-2ig\mathsf{c}^A(t_A)^j{}_i\gamma_j\frac{\partial\bar{W}}{\partial\gamma_i} \\ &- 2g\mathsf{c}^AC^B{}_{AC}\mathsf{w}^C_{\dot{a}}\frac{\partial_L\bar{W}}{\partial\mathsf{w}^B_{\dot{a}}} = 0. \end{aligned}$$

The last equality is gauge invariance of $\bar{W}$.

At one loop, the triangle graphs give

$$Q_1\mathcal{I} = \frac{ig^3}{128\pi^2}\,\mathrm{Tr}_R(t_A\{t_B, t_C\})\,\mathsf{c}^A(\partial_{+\dot{a}}\mathsf{c}^B)(\partial_+{}^{\dot{a}}\mathsf{c}^C) = 0. \tag{6.10}$$

A circuit of $Q_n\mathcal{I}$ has $n+2$ interaction nodes and $2n+1$ resistors, and each resistor takes its own antifield terminal (Lemma 3.15): a gauge or matter node has one, a superpotential node none. Thus

$$2n + 1 \leq n + 2 - \#(\bar{W}\text{ nodes}),$$

which is impossible for $n \geq 2$. Finally, $Q_n\mathcal{I} = \{\mathcal{I}^{\otimes(n+2)}\}_0/(n+1)!$ gives (6.4). □

Introduce a decoupled auxiliary pair, with action $\int \Omega \wedge [a\bar\partial b_{\mathrm{aux}}]_{(0,2)}$ and $|a| = 0$, $|b_{\mathrm{aux}}| = 1$. For every $n \geq 0$, $Q_n a = Q_n b_{\mathrm{aux}} = 0$; the free differential is still $Q_{\mathrm{BV}} = \bar\partial$, which vanishes on the holomorphic auxiliary jets. Thus

$$\mathbf{Q}(Aa) = (\mathbf{Q}A)a, \qquad \mathbf{Q}(b_{\mathrm{aux}}B) = -b_{\mathrm{aux}}\mathbf{Q}B.$$

The quantum product is defined by the unique auxiliary contraction; $\gamma = \{(A, B)\}$ marks this edge, and $\mathrm{aux} = 0$ sets the remaining auxiliary fields to zero after contraction:

$$\begin{aligned} A \star B &= \sum_{n\geq 0} \frac{\hbar^n}{n!} \{\mathcal{I}^{\otimes n}, A, B\}^{\gamma,0} = \{Aa, b_{\mathrm{aux}}B\}_{\mathcal{I}}\big|_{\mathrm{aux}=0}\,, \\ \{A, B\}^{\gamma,0} &= (-1)^{|A|} \frac{\partial_L}{\partial a} \frac{\partial_L}{\partial b_{\mathrm{aux}}} (Aa\, b_{\mathrm{aux}} B) = AB. \end{aligned} \tag{6.11}$$

We will show later that these auxiliary fields serve precisely to define an integration contour, analogous to normal ordering in 2d CFT.

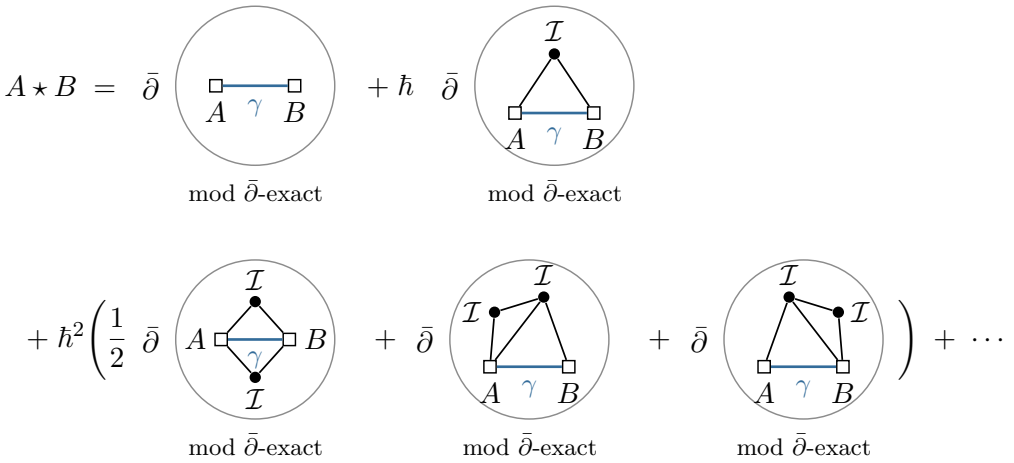


Figure 6.1: The quantum product (6.11) through order $\hbar^2$: the free differential $\bar\partial$ acting on the marked graphs, as in (3.19). Squares are the insertions A and B, dots the interaction vertices $\mathcal{I}$. The blue edge γ is the propagator of the pair a, b_{aux} in (6.26), which acts as the integration contour of (6.41); at order zero $\bar\partial$ turns it into the heat kernel and gives AB (6.11). Black edges contract fields of $\mathcal{I}$, A and B. Each coefficient is $\hbar^n/n!$ times the number of labellings of the interaction vertices.

Theorem 6.3 (Leibniz rule)**.**

Suppose that the interaction satisfies (6.4)*, which is the quantum master equation. (Explicitly, it is the gauge anomaly cancellation condition.) For operators X, Y, with $|X|$ the fermion number of X,*[20]

$$\mathbf{Q}\{X, Y\}_{\mathcal{I}} - \{\mathbf{Q}X, Y\}_{\mathcal{I}} + (-1)^{|X|}\{X, \mathbf{Q}Y\}_{\mathcal{I}} = 0, \tag{6.12}$$

and the quantum product obeys

$$\mathbf{Q}(A \star B) = (\mathbf{Q}A) \star B + (-1)^{|A|} A \star (\mathbf{Q}B). \tag{6.13}$$

In particular, quantum multiplication takes quantum cocycles to quantum cocycles and descends to $\mathbf{Q}$-cohomology.

[20] Equation (6.14) is the quadratic identity of the higher brackets [9, Section 3.2], [5, Theorem 7], [6, Equation (4.19)]: in its first three terms the inner bracket contains both X and Y, only X, and only Y, and the terms whose inner bracket contains interactions alone cancel by (6.4).

Proof. For $N \geq 1$ interaction entries and the entries X, Y, the quadratic identity of the higher brackets [9, Section 3.2], [5, Theorem 7], [6, Equation (4.19)], in the conventions of (6.6), sorts its terms by the content of the inner bracket:

$$\begin{aligned} 0 = {} & \underbrace{\sum_{r=0}^{N-1} \binom{N}{r} \{\mathcal{I}^{\otimes(N-r)}, \{\mathcal{I}^{\otimes r}, X, Y\}_0\}_0}_{\text{inner bracket contains } X \text{ and } Y} - \underbrace{\sum_{r=1}^{N} \binom{N}{r} \{\mathcal{I}^{\otimes(N-r)}, \{\mathcal{I}^{\otimes r}, X\}_0, Y\}_0}_{\text{inner bracket contains } X} \\ & + (-1)^{|X|} \underbrace{\sum_{r=1}^{N} \binom{N}{r} \{\mathcal{I}^{\otimes(N-r)}, X, \{\mathcal{I}^{\otimes r}, Y\}_0\}_0}_{\text{inner bracket contains } Y} + \underbrace{\sum_{k=2}^{N} \binom{N}{k} \{\mathcal{I}^{\otimes(N-k)}, \{\mathcal{I}^{\otimes k}\}_0, X, Y\}_0}_{\text{inner bracket contains only interactions}} . \end{aligned} \tag{6.14}$$

Multiply by $\hbar^{N-1}/N!$ and sum over $N \geq 1$. Since $\binom{N}{r}/N! = 1/(r!\,(N-r)!)$, each double sum factorizes into an outer series in $s = N - r$ and an inner series:

$$\begin{aligned} 0 = {} & \underbrace{\sum_{s\geq 1} \frac{\hbar^{s-1}}{s!} \Big\{\mathcal{I}^{\otimes s}, \sum_{r\geq 0} \frac{\hbar^r}{r!} \{\mathcal{I}^{\otimes r}, X, Y\}_0\Big\}_0}_{\mathbf{Q}\{X,Y\}_{\mathcal{I}}} - \underbrace{\sum_{s\geq 0} \frac{\hbar^s}{s!} \Big\{\mathcal{I}^{\otimes s}, \sum_{r\geq 1} \frac{\hbar^{r-1}}{r!} \{\mathcal{I}^{\otimes r}, X\}_0, Y\Big\}_0}_{\{\mathbf{Q}X,Y\}_{\mathcal{I}}} \\ & + (-1)^{|X|} \underbrace{\sum_{s\geq 0} \frac{\hbar^s}{s!} \Big\{\mathcal{I}^{\otimes s}, X, \sum_{r\geq 1} \frac{\hbar^{r-1}}{r!} \{\mathcal{I}^{\otimes r}, Y\}_0\Big\}_0}_{\{X,\mathbf{Q}Y\}_{\mathcal{I}}} \\ & + \underbrace{\sum_{s\geq 0} \frac{\hbar^{s+1}}{s!} \Big\{\mathcal{I}^{\otimes s}, \sum_{k\geq 2} \frac{\hbar^{k-2}}{k!} \{\mathcal{I}^{\otimes k}\}_0, X, Y\Big\}_0}_{0 \text{ by (6.4)}} . \end{aligned} \tag{6.15}$$

The inner series are (6.3) and the left side of (6.4); this is (6.12). The Jacobi identity expresses the Leibniz rule for an adjoint action on a bracket, and (6.12) is its higher L_∞ form, with the interaction insertions supplying every perturbative order [6, Sections 4.3–4.5].

Set $X = Aa$ and $Y = b_{\mathrm{aux}} B$ in (6.12), and extract the auxiliary contraction:

$$0 = \mathbf{Q}(A \star B) - (\mathbf{Q}A) \star B - (-1)^{|A|} A \star (\mathbf{Q}B).$$

Consequently, $\mathbf{Q}A = \mathbf{Q}B = 0$ gives $\mathbf{Q}(A \star B) = 0$. For homogeneous X, Y with $|X| = |A| + 1$ and $|Y| = |B| + 1$, the same identity and $\mathbf{Q}^2 = 0$ (Lemma 6.1) give

$$\begin{aligned} & (A + \mathbf{Q}X) \star (B + \mathbf{Q}Y) - A \star B \\ & \qquad = \mathbf{Q}(X \star B + (-1)^{|A|} A \star Y + X \star \mathbf{Q}Y). \end{aligned} \tag{6.16}$$

The resulting cohomology class is therefore independent of representatives. □

6.2 The quantum BPS ring

Write $[\cdot]_{\mathbf{Q}}$ for the class of a quantum cocycle.

Theorem 6.4. *Suppose the interaction satisfies (6.4), the quantum master equation. The quantum product (6.11) induces an associative, graded-commutative, unital multiplication on $H(\mathbf{Q})$. For quantum cocycles A, B, D,*

$$[(A \star B) \star D]_{\mathbf{Q}} = [A \star (B \star D)]_{\mathbf{Q}}, \tag{6.17}$$

$$[A \star B]_{\mathbf{Q}} = (-1)^{|A||B|} [B \star A]_{\mathbf{Q}}. \tag{6.18}$$

The unit is strict:

$$A \star 1 = 1 \star A = A. \tag{6.19}$$

In the successive homotopy transfer of Section 9.3, the quantum product induces on every page E_r the ordinary product of leading representatives, and every d_r is a derivation of it.

Proof. The chain identity (6.13) makes the product well-defined on cohomology. A class of E_r has a representative $a + \hbar a_1 + \cdots$ whose $\mathbf{Q}$-variation starts at order $\hbar^r$ with $d_r a$. Since $A \star B = AB + O(\hbar)$, the product of two such representatives has leading term ab, and by (6.13) its $\mathbf{Q}$-variation starts at order $\hbar^r$ with $(d_r a)b + (-1)^{|a|} a\, d_r b$. For the unit, the identity insertion has no fields, so in a marked circuit with $n \geq 1$ interaction nodes it is wired only by the marked resistor. It is a dead end: the marked resistor carries the current injected at the identity insertion, whatever its resistance, so this resistance moves no current and the current form vanishes. So only the zero-interaction marked segment contributes.

For three closed insertions X, Y, Z, sum the quadratic identities with all interaction entries, as in [6, Sections 4.3 and 4.5]. The pure-interaction terms vanish by (6.4), and the single-input terms vanish by closure. With the ordered-bracket sign of (3.9), the remaining identity is

$$\begin{aligned} 0 = {} & \{X, \{Y, Z\}_{\mathcal{I}}\}_{\mathcal{I}} - \{\{X, Y\}_{\mathcal{I}}, Z\}_{\mathcal{I}} \\ & - (-1)^{(|X|+1)(|Y|+1)} \{Y, \{X, Z\}_{\mathcal{I}}\}_{\mathcal{I}} \\ & + \mathbf{Q} \sum_{n \geq 0} \frac{\hbar^{n+1}}{n!} \{\mathcal{I}^{\otimes n}, X, Y, Z\}_0. \end{aligned} \tag{6.20}$$

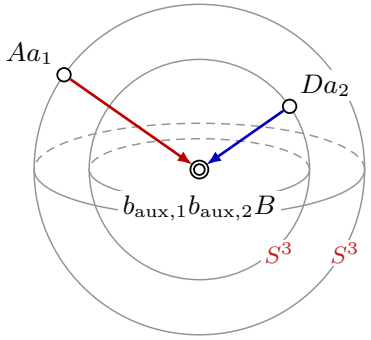
A ternary graph with n interactions has $n + 3$ vertices and $2n + 3$ edges, hence loop number $n + 1$. This fixes the power of $\hbar$ in the last line.

Take two independent copies $(a_1, b_{\mathrm{aux},1})$ and $(a_2, b_{\mathrm{aux},2})$ of the free auxiliary pair defined above. Each a_i is even and each $b_{\mathrm{aux},i}$ is odd. Substitute

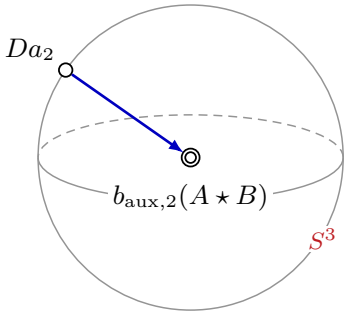
$$X = Aa_1, \qquad Y = Da_2, \qquad Z = b_{\mathrm{aux},1} b_{\mathrm{aux},2} B.$$

The last insertion is the root. This marks the edges AB and DB, as in Figure 6.2.

(a) Both marks end at the root

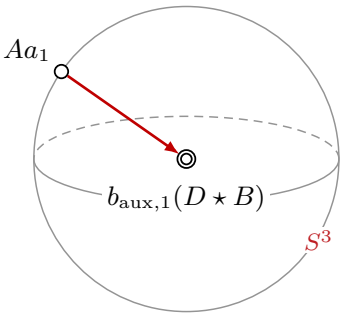


(b) Collapse A, B at B



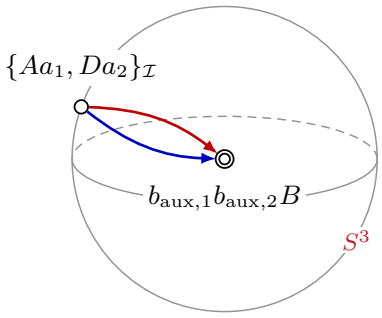
The second auxiliary field stays at the root.

(c) Collapse D, B at B



The first auxiliary field stays at the root.

(d) Collapse A, D



$P \wedge P = 0, \qquad K \wedge P = 0$

Both marks now join the same two vertices.

Figure 6.2: The common-root contraction and its three pair-collapse channels. Red and blue arrows mark the first and second auxiliary pairs; double circles mark the root. Each non-root insertion lies on its own integration sphere S^3 centered at the root, shown schematically in gray. Only auxiliary contractions are shown. Each component may also contain interaction vertices and physical contractions. The labels in (b) and (c) display the surviving fields; their Koszul signs are given in the proof.

The ordered-bracket sign gives

$$\begin{aligned} \{Aa_1, b_{\mathrm{aux},1}b_{\mathrm{aux},2}B\}_{\mathcal{I}}|_{a_1=b_{\mathrm{aux},1}=0} &= (-1)^{|A|}b_{\mathrm{aux},2}(A \star B), \\ \{Da_2, b_{\mathrm{aux},1}b_{\mathrm{aux},2}B\}_{\mathcal{I}}|_{a_2=b_{\mathrm{aux},2}=0} &= (-1)^{|D|+1}b_{\mathrm{aux},1}(D \star B). \end{aligned} \tag{6.21}$$

In the second line the odd derivative passes through $b_{\mathrm{aux},1}$, contributing the extra minus sign.

The third binary channel first collapses A, D. Both marked resistors then join that collapsed node to B: they are in parallel, and their proportional currents kill the measure for every holomorphic derivative (Corollary 3.8, (3.30)). Consequently,

$$\{\{Aa_1, Da_2\}_{\mathcal{I}}, b_{\mathrm{aux},1}b_{\mathrm{aux},2}B\}_{\mathcal{I}}|_{\mathrm{aux}=0} = 0. \tag{6.22}$$

On substituting (6.21) and (6.22) into (6.20), each of its three nested brackets becomes a product of three operators:

$$0 = \underbrace{(-1)^{|D|+1}A \star (D \star B)}_{\{X,\{Y,Z\}_{\mathcal{I}}\}_{\mathcal{I}}} - \underbrace{0}_{\substack{\{\{X,Y\}_{\mathcal{I}},Z\}_{\mathcal{I}}: \\ \text{parallel marks}}} \underbrace{-(-1)^{(|A|+1)(|D|+1)+|A|}D \star (A \star B)}_{-(-1)^{(|X|+1)(|Y|+1)}\{Y,\{X,Z\}_{\mathcal{I}}\}_{\mathcal{I}}} + \underbrace{\mathbf{Q}\,H}_{\mathbf{Q}\text{-exact}}, \tag{6.23}$$

with H the bracket in the last line of (6.20) at these inputs. The middle term vanishes, so only the two bracketings $A \star (D \star B)$ and $D \star (A \star B)$ remain. Multiplying by $(-1)^{|D|+1}$ gives the local homotopy

$$\begin{aligned} &A \star (D \star B) - (-1)^{|A||D|} D \star (A \star B) \\ &\quad = (-1)^{|D|} \mathbf{Q} \left[\sum_{n \geq 0} \frac{\hbar^{n+1}}{n!} \{\mathcal{I}^{\otimes n}, Aa_1, Da_2, b_{\mathrm{aux},1} b_{\mathrm{aux},2} B\}_0 \Big|_{\mathrm{aux}=0} \right]. \end{aligned} \tag{6.24}$$

Moreover, its n-interaction term has $2n+1$ physical contractions. The interactions supply at most n fields of type b or β, so at least $n+1$ must come from the inputs. The homotopy is therefore a finite sum of local operators for finite polynomial inputs.

Setting $B = 1$ proves graded commutativity. For three closed inputs, (6.13), graded commutativity, and (6.24) give

$$\begin{aligned} [(A \star B) \star D]_{\mathbf{Q}} &\underset{(6.18)}{=} (-1)^{(|A|+|B|)|D|} [D \star (A \star B)]_{\mathbf{Q}} \\ &\underset{(6.24)}{=} (-1)^{|B||D|} [A \star (D \star B)]_{\mathbf{Q}} \\ &\underset{(6.18)}{=} [A \star (B \star D)]_{\mathbf{Q}}. \end{aligned} \tag{6.25}$$

The $\mathbf{Q}$-exact term of (6.24) drops out of the classes; this is the potential $\mathbf{Q}$-exact difference between the two bracketings in (6.17). This proves the stated ring laws. □

6.3 The graph formula of the quantum product

For a physical edge (v, w), the operator Δ_{vw} contracts one field at vertex v with its BV conjugate at vertex w, using the pairing (3.5). For example, in the $\mathcal{N} = 4$ superfield variables, at fixed vertex positions,

$$\Delta_{vw} = -\delta^{DE} \int d^3\theta \, \frac{\delta}{\delta C_v^D(\theta)} \frac{\delta}{\delta C_w^E(\theta)}.$$

Here C_v^D is the field at vertex v, and the fibre derivatives are given by (A.4). Composing these contractions in edge order gives the graph coefficient

$$\begin{aligned} \frac{1}{n!} \{\mathcal{I}^{\otimes n}, A, B\}^{\gamma,0} &= \frac{1}{n!} \{\mathcal{I}^{\otimes n}, Aa, b_{\mathrm{aux}} B\}_0 \Big|_{\mathrm{aux}=0} \\ &= \frac{(-1)^n}{n! (8\pi^2)^n} \sum_{\substack{\Gamma \text{ labelled, Laman} \\ |\Gamma_0|=n+2,\ |\Gamma_1|=2n+1 \\ (n+1,n+2) \in \Gamma_1}} \Bigg[\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda; z) \\ &\qquad \times \prod_{(v,w) \in \Gamma_1 \setminus \{(n+1,n+2)\}}^{\text{edge order}} \Delta_{vw} (\mathcal{I}_1 \cdots \mathcal{I}_n A_{n+1} B_{n+2}) \Bigg]_{\text{jets},\, x_v = x,\, z = 0}. \end{aligned} \tag{6.26}$$

Here $\gamma = \{(n+1, n+2)\}$, with aux $= 0$ imposed after contraction.

In the subscript jets, each derivative on a contracted input field differentiates the kernel through its edge displacement, while λ_v differentiates the surviving fields at vertex v.

The displacements are then set to zero and all vertex positions to the common insertion point x. The graph restrictions of Section 7 reduce this sum to product forests.

Corollary 6.5 (Finiteness of the quantum product)**.**

For polynomial local inputs and a polynomial superpotential in the interaction (3.4), the quantum product has finitely many powers of $\hbar$.

Proof. Each order-n product graph has n interaction vertices and $2n$ physical contractions, so Proposition 3.21 gives

$$\begin{aligned}\#(\mathsf{b},\beta \text{ in output}) &= \#(\mathsf{b},\beta \text{ in } A)+\#(\mathsf{b},\beta \text{ in } B)\\ &\quad+\underbrace{n-\#(\bar{W} \text{ vertices})}_{\text{interaction terminals}}-\underbrace{2n}_{\text{physical contractions}}\\ &= \#(\mathsf{b},\beta \text{ in } A)+\#(\mathsf{b},\beta \text{ in } B)-n-\#(\bar{W} \text{ vertices})\geq 0.\end{aligned}$$

Writing $[\hbar^n]$ for the coefficient of $\hbar^n$, we obtain

$$[\hbar^n](A\star B)=0 \quad \text{if} \quad n>\#(\mathsf{b},\beta \text{ in } A)+\#(\mathsf{b},\beta \text{ in } B). \tag{6.27}$$

For polynomial inputs the counts are the maxima over their monomials; the same loss adds at the binary vertices of any fixed bracketing of finitely many inputs.

For a formal superpotential the bound holds coefficientwise in its coefficients. □

6.4 One- and two-loop examples in $\mathcal{N}=4$ Yang–Mills theory

The one-loop triangle has vertices $1:\mathcal{I}$, $2:A$ and $3:B$, and marked edge $\gamma=\{(2,3)\}$. Its kernel is exactly the same as the fan kernel. Pairing its displacement exponential with the spatial functional derivatives of (8.2) gives

$$\begin{aligned}\{\mathcal{I},A,B\}^{\{(2,3)\},0}(x) &= \frac{i}{32\pi^2}\int_0^1 dr_1\int_0^{1-r_1} dr_2\int d^3\theta\, C_{XYZ}\\ &\quad\times\int d^4z_1\, d^4z_2\\ &\quad\times(\partial_{+\dot{a}}C^X)\big(x+r_1z_1+(1-r_1)z_2,\theta\big)\\ &\quad\times\partial_{2,+}{}^{\dot{a}}\left(\frac{\delta A(x_2)}{\delta C^Y(x_2+z_1,\theta)}\right)\Big|_{x_2=x+r_2(z_2-z_1)}\frac{\delta B(x)}{\delta C^Z(x+z_2,\theta)}.\end{aligned} \tag{6.28}$$

Here x is the arbitrary common position of the local product.

At two loops the physical graphs have four edges (plus, as always, one edge for the auxiliary fields). They consist of a book and two triangles attached to opposite inputs (Figure 6.3). Their ordered edges, for $3=A$ and $4=B$, are

$$\begin{aligned}\Gamma_{\text{book}}&:(13,14,23,24,34),\\ \Gamma_A&:(12,13,23,24,34),\\ \Gamma_B&:(12,14,23,24,34).\end{aligned} \tag{6.29}$$

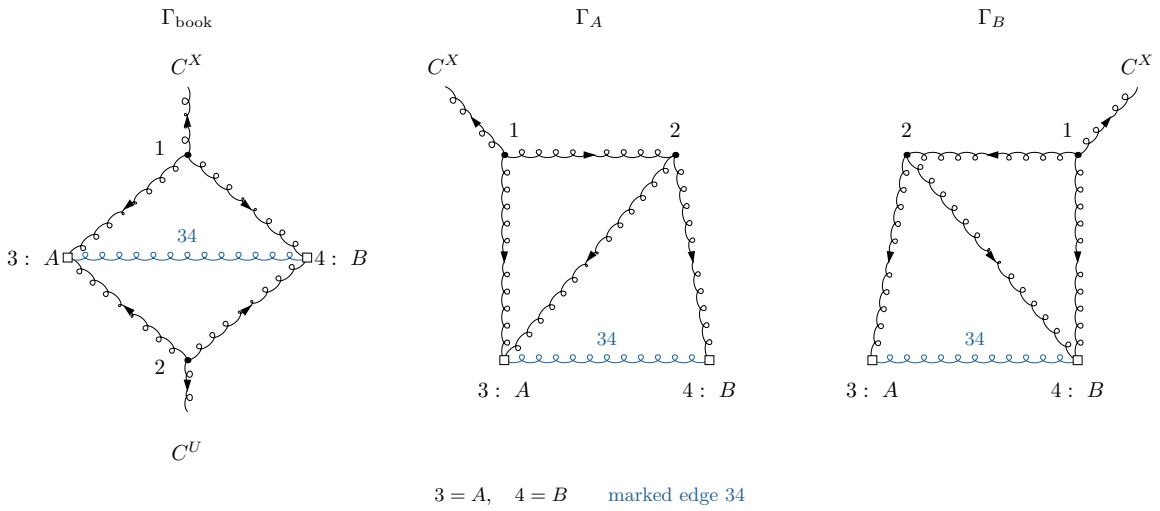


Figure 6.3: The two-loop quantum-product graphs Γ_{book}, Γ_A and Γ_B. Squares are the insertions $3 : A$ and $4 : B$; dots are cubic interactions. Black arrows orient the physical contractions, and the blue edge $(3, 4)$ carries the auxiliary pairing. The book leaves C^X, C^U uncontracted; each triangle leaves C^X.

The last edge is the auxiliary edge. These are still fan kernels (but with more free injected momenta).

$$\begin{aligned}
\mathcal{I}^{\text{raw cut}}_{\Gamma_{\text{book}}}(\lambda; 0) &= \frac{1}{24}\big[\lambda_{1,+\dot{a}}(\lambda_{2,+}{}^{\dot{a}} + 2\lambda_{3,+}{}^{\dot{a}})\big]\big[\lambda_{2,+\dot{b}}(\lambda_{1,+}{}^{\dot{b}} + 2\lambda_{3,+}{}^{\dot{b}})\big], \\
\mathcal{I}^{\text{raw cut}}_{\Gamma_A}(\lambda; 0) &= -\frac{1}{24}(\lambda_{1,+\dot{a}}\lambda_{3,+}{}^{\dot{a}})^2, \\
\mathcal{I}^{\text{raw cut}}_{\Gamma_B}(\lambda; 0) &= \frac{1}{24}(\lambda_{1,+\dot{a}}\lambda_{3,+}{}^{\dot{a}})^2.
\end{aligned} \tag{6.30}$$

For the full result, the book kernel gives

$$\begin{aligned}
&\frac{1}{2}\{\mathcal{I}, \mathcal{I}, A, B\}^{\gamma,0}(x)\Big|_{\Gamma_{\text{book}}} \\
&\quad = \frac{C_{XYZ}C_{UVW}}{2(32\pi^2)^2}\int_0^1 dr_1 \int_0^1 dr_3 \int_0^1 dr_2 \int_0^{1-r_2} dr_4\,(1 - r_2 - r_4) \\
&\times \int d^3\theta^{(1)} d^3\theta^{(2)} \int \prod_{e\in\{13,14,23,24\}} d^4 z_e \\
&\times \partial_{1,+\dot{a}}(\partial_{3,+}{}^{\dot{a}} + r_3\partial_{2,+}{}^{\dot{a}})\partial_{2,+\dot{b}}(\partial_{3,+}{}^{\dot{b}} + r_1\partial_{1,+}{}^{\dot{b}}) \\
&\times \Bigg[C^X(x_1, \theta^{(1)})C^U(x_2, \theta^{(2)}) \\
&\qquad \times \frac{\delta^2 A(x_3)}{\delta C^Y(x_3 + z_{13}, \theta^{(1)})\delta C^V(x_3 + z_{23}, \theta^{(2)})}\frac{\delta^2 B(x)}{\delta C^Z(x + z_{14}, \theta^{(1)})\delta C^W(x + z_{24}, \theta^{(2)})}\Bigg]_{x_v = x_v(x;r,z),\, v=1,2,3}.
\end{aligned} \tag{6.31}$$

The translated positions are

$$\begin{aligned}
x_1(x; r, z) &= x + r_1 z_{13} + (1 - r_1)z_{14} + r_1\big[r_2(z_{14} - z_{13}) + r_4(z_{24} - z_{23})\big], \\
x_2(x; r, z) &= x + r_3 z_{23} + (1 - r_3)z_{24} + r_3\big[r_2(z_{14} - z_{13}) + r_4(z_{24} - z_{23})\big], \\
x_3(x; r, z) &= x + r_2(z_{14} - z_{13}) + r_4(z_{24} - z_{23}).
\end{aligned}$$

For Γ_A, the translated positions are

$$\begin{aligned}
x_1^{+\dot a}(x;r,z) &= x^{+\dot a} + \big[r_1r_2 + (1-r_2)(r_1r_3+r_4)\big]z_{13}^{+\dot a} - r_2\big[r_1(1-r_3)-r_4\big]z_{23}^{+\dot a} \\
&\quad + (1-r_1r_3-r_4)z_{24}^{+\dot a}, \\
x_3^{+\dot a}(x;r,z) &= x^{+\dot a} + (1-r_3-r_4)\big[z_{24}^{+\dot a} - (1-r_2)z_{13}^{+\dot a} - r_2 z_{23}^{+\dot a}\big].
\end{aligned}$$

The full contribution is

$$\begin{aligned}
&\frac{1}{2}\{\mathcal{I},\mathcal{I},A,B\}^{\gamma,0}(x)\Big|_{\Gamma_A} \\
&\quad = \frac{C_{XYR}C_{RZW}}{(32\pi^2)^2}\int_0^1 dr_1\int_0^1 dr_2\int_0^1 dr_3\int_0^{1-r_3} dr_4 \\
&\times r_2(1-r_1)(1-r_3-r_4)\int d^3\theta\int\prod_{e\in\{13,23,24\}} d^4z_e \\
&\times (\partial_{+\dot a}\partial_{+\dot b}C^X)(x_1(x;r,z),\theta) \\
&\times \partial_{3,+}{}^{\dot a}\partial_{3,+}{}^{\dot b}\left(\frac{\delta^2 A(x_3)}{\delta C^Y(x_3+z_{13},\theta)\,\delta C^Z(x_3+z_{23},\theta)}\right)\Big|_{x_3=x_3(x;r,z)}\frac{\delta B(x)}{\delta C^W(x+z_{24},\theta)}.
\end{aligned} \tag{6.32}$$

For Γ_B, the translated positions are

$$\begin{aligned}
x_1^{+\dot a}(x;r,z) &= x^{+\dot a} + \big[1-r_1r_2-r_1(1-r_2)r_3\big]z_{14}^{+\dot a} + r_1r_3z_{23}^{+\dot a} + r_1r_2(1-r_3)z_{24}^{+\dot a}, \\
x_3^{+\dot a}(x;r,z) &= x^{+\dot a} + r_4\big[(1-r_2)z_{14}^{+\dot a} - z_{23}^{+\dot a} + r_2 z_{24}^{+\dot a}\big].
\end{aligned}$$

Thus

$$\begin{aligned}
&\frac{1}{2}\{\mathcal{I},\mathcal{I},A,B\}^{\gamma,0}(x)\Big|_{\Gamma_B} \\
&\quad = -\frac{C_{XZR}C_{RYW}}{(32\pi^2)^2}\int_0^1 dr_1\int_0^1 dr_2\int_0^1 dr_3\int_0^{1-r_3} dr_4 \\
&\times r_1r_2r_4\int d^3\theta\int\prod_{e\in\{14,23,24\}} d^4z_e \\
&\times (\partial_{+\dot a}\partial_{+\dot b}C^X)(x_1(x;r,z),\theta) \\
&\times \partial_{3,+}{}^{\dot a}\partial_{3,+}{}^{\dot b}\left(\frac{\delta A(x_3)}{\delta C^Y(x_3+z_{23},\theta)}\right)\Big|_{x_3=x_3(x;r,z)}\frac{\delta^2 B(x)}{\delta C^Z(x+z_{14},\theta)\,\delta C^W(x+z_{24},\theta)}.
\end{aligned} \tag{6.33}$$

The exchange $3\leftrightarrow 4$ maps Γ_A to Γ_B: after restoring the displayed edge order, $\mathcal{I}_{\Gamma_B}^{\text{raw cut}}(\lambda;0) = -\mathcal{I}_{\Gamma_A}^{\text{raw cut}}(\lambda;0)$ in (6.30); the full displaced formulas also require exchanging the inputs, their edge displacements, and the root.

6.5 Order-by-order representatives

For $Q_0A = Q_0B = 0$, the first two orders of (6.13) are

$$\begin{aligned}
Q_0\{\mathcal{I},A,B\}^{\gamma,0} + Q_1(AB) &= (Q_1A)B + (-1)^{|A|}A(Q_1B), \\
\frac{1}{2}Q_0\{\mathcal{I},\mathcal{I},A,B\}^{\gamma,0} + Q_1\{\mathcal{I},A,B\}^{\gamma,0} + Q_2(AB) &= (Q_2A)B + (-1)^{|A|}A(Q_2B) \\
&\quad + \{\mathcal{I},Q_1A,B\}^{\gamma,0} + (-1)^{|A|}\{\mathcal{I},A,Q_1B\}^{\gamma,0}.
\end{aligned} \tag{6.34}$$

Thus, the classical variation of the one-loop term of the quantum product cancels the one-loop cross-contraction. Its two-loop term absorbs the remaining two-loop defect together with the action of Q_1 on the one-loop term.

If $A = \sum_i \hbar^i A_i$ and $B = \sum_j \hbar^j B_j$ are quantum cocycles, their product has coefficients

$$\mathcal{O}_r = \sum_{i+j+n=r} \frac{1}{n!}\{\mathcal{I}^{\otimes n}, A_i, B_j\}^{\gamma,0}, \qquad \mathcal{O}_0 = A_0 B_0 \text{ (6.11)}, \tag{6.35}$$

$$\begin{aligned} \mathcal{O}_1 &= A_1 B_0 + A_0 B_1 + \{\mathcal{I}, A_0, B_0\}^{\gamma,0}, \\ \mathcal{O}_2 &= A_2 B_0 + A_1 B_1 + A_0 B_2 \\ &\quad + \{\mathcal{I}, A_1, B_0\}^{\gamma,0} + \{\mathcal{I}, A_0, B_1\}^{\gamma,0} + \frac{1}{2}\{\mathcal{I}, \mathcal{I}, A_0, B_0\}^{\gamma,0}. \end{aligned} \tag{6.36}$$

Equation (6.13) gives $\sum_{k=0}^{r} Q_k \mathcal{O}_{r-k} = 0$ at every order. These are the lifting equations (9.17) for the product $A_0 B_0$: the quantum product gives in closed form the completion that the homotopy transfer of Section 9.3 constructs order by order.

6.6 Normal ordering and an explicit associator

Normal ordering. The quantum normal-ordering product is represented by the Bochner–Martinelli contour (6.41) [3, Section 2.3].

We now prove that it agrees, up to exact terms, with the quantum product of Sections 6.1–6.5.

Here $(A(x)B(0))[L]$ denotes the joint renormalized two-insertion expansion at scale L, defined for local insertions X by $X[L]\, e^{I[L]/\hbar} = \lim_{\varepsilon \downarrow 0} e^{\hbar \partial_{P_{\varepsilon,L}}} (X\, e^{I/\hbar})$ with $I = \int_{\mathbb{C}^2} \Omega \wedge [\mathcal{I}]_{(0,2)}$ [17]. (The scale L belongs to Costello's formalism of BV renormalization theory; it never affects our calculations.)

The contour argument uses renormalized separated insertions satisfying the supersymmetry descent Ward identity [3, Sections 2.2–2.3].

Let $A(x, \bar\theta)[L]$ be the renormalized supersymmetry multiplet of the effective insertion $A[L]$. In the coordinates of (2.12),

$$\begin{aligned} A^{(1)}(x)[L] &= \frac{1}{4} dx^{-\dot a}\, \frac{\partial_L A(x,\bar\theta)[L]}{\partial \bar\theta^{\dot a}}\bigg|_{\bar\theta=0}, \\ (\bar\partial + (I[L], \cdot)_L + \hbar\Delta_L) A^{(1)}[L] + \bar\partial_x A[L] &= (\mathbf{Q}A)^{(1)}[L]. \end{aligned} \tag{6.37}$$

Taking the one-form component of the full composite multiplet gives its descendant [3, Section 2.2].

For $\mathbf{Q}A = 0$, write the form to the left of its coefficient:

$$\begin{aligned} A^{(1)}[L] &= dx^{-\dot a} A^{(1)}_{-\dot a}[L], \\ \frac{\partial A[L]}{\partial x^{-\dot a}} &= (\bar\partial + (I[L], \cdot)_L + \hbar\Delta_L) A^{(1)}_{-\dot a}[L]. \end{aligned} \tag{6.38}$$

This gives the primitive of an antiholomorphic translation in the effective-observable complex.

At zero displacement, (3.21) gives

$$\begin{gathered} [P_{0,\infty}(x)]_\Omega = -\frac{x^-_{\dot a}\, dx^{-\dot a}}{8\pi^2 |x|^4_E}, \qquad d(\Omega \wedge [P_{0,\infty}]_\Omega) = \delta^{(4)}_E(x) d^4 x_E, \\ \int_{S^3} \Omega \wedge [P_{0,\infty}]_\Omega = 1. \end{gathered} \tag{6.39}$$

The last integral holds for every positively oriented embedded three-sphere enclosing the origin once; its normalization is that of the heat kernel,

$$\int_{|x|_E\le r}\Omega\wedge[K_\varepsilon]_\Omega = 1-\left(1+\frac{r^2}{2\varepsilon}\right)e^{-r^2/(2\varepsilon)}\xrightarrow{\varepsilon\downarrow 0}1.$$

Let $(A(x)B(0))^{(1)}[L]$ be the first descendant of the joint insertion with respect to the position x.

For closed inputs and $S_1^3-S_0^3=\partial V$, with V disjoint from all insertions,

$$\begin{aligned}&\int_{S_1^3}\Omega\wedge[P_{0,\infty}]_\Omega(A(x)B(0))[L]-\int_{S_0^3}\Omega\wedge[P_{0,\infty}]_\Omega(A(x)B(0))[L]\\&\quad=-(\bar\partial+(I[L],\cdot)_L+\hbar\Delta_L)\int_V\Omega\wedge[P_{0,\infty}]_\Omega\wedge(A(x)B(0))^{(1)}[L].\end{aligned}\tag{6.40}$$

For $\mathbf{Q}A=\mathbf{Q}B=0$, descent gives $\bar\partial_x(A(x)B(0))[L]=-(\bar\partial+(I[L],\cdot)_L+\hbar\Delta_L)(A(x)B(0))^{(1)}[L]$; the propagator is closed on V, so

$$\begin{aligned}\int_{\partial V}\Omega\wedge[P_{0,\infty}]_\Omega(A(x)B(0))[L]&=-\int_V\Omega\wedge[P_{0,\infty}]_\Omega\wedge\bar\partial_x(A(x)B(0))[L]\\&=-(\bar\partial+(I[L],\cdot)_L+\hbar\Delta_L)\int_V\Omega\wedge[P_{0,\infty}]_\Omega\wedge(A(x)B(0))^{(1)}[L].\end{aligned}$$

Equation (6.40) proves equality of the contour classes in the effective-observable complex.

Theorem 6.6 (Normal-ordering comparison)**.** *Let A and B be homogeneous quantum cocycles and $L>0$. In the effective-observable complex at scale L, with differential $\mathbf{Q}_L=Q_{\mathrm{BV}}+(I[L],\cdot)_L+\hbar\Delta_L$, let B_R be the ball of radius R centred at the insertion of B and $S_R^3=\partial B_R$. The contour product and the effective representative of (6.11) satisfy*

$$\begin{aligned}&\int_{S_R^3}\Omega\wedge[P_{0,\infty}(x)]_\Omega\,(A(x)B(0))[L]\\&\quad=(A\star B)[L]-\mathbf{Q}_L\int_{B_R}\Omega\wedge[P_{0,\infty}(x)]_\Omega\wedge(A(x)B(0))^{(1)}[L].\end{aligned}\tag{6.41}$$

Proof.

Write $F(x)=(A(x)B(0))[L]$ and $F^{(1)}(x)=(A(x)B(0))^{(1)}[L]$. The volume integral uses the full joint descendant and has type $(2,2)$. Evaluate it as one coefficientwise graph limit in the decoupled auxiliary theory: physical edges carry $P_{\varepsilon,L}$ and the marked edge carries $P_{\varepsilon,\infty}$. Take the limit only after combining the ordered graphs. The marked edge is last, has zero displacement, and its single Wick factor $\hbar$ is divided out.

$$\begin{aligned}\bar\partial P_{\varepsilon,L}&=K_\varepsilon-K_L,\qquad&\bar\partial P_{\varepsilon,\infty}&=K_\varepsilon,\\|\Gamma_0|&=n+2,\qquad&|\Gamma_1|&=2n+1,\qquad&\frac{1}{\hbar}\frac{\hbar^{2n+1}}{n!\hbar^n}&=\frac{\hbar^n}{n!}.\end{aligned}\tag{6.42}$$

Apply (6.42) to every edge, with the signs $(-1)^{j-1}$ of (3.9). The upper physical-edge faces assemble into $\mathbf{Q}_L$; pure-interaction collision faces cancel by the quantum master equation, and faces with only one insertion vanish by $\mathbf{Q}A=\mathbf{Q}B=0$. A remaining cluster contains both insertions and n interactions. The rescaling $t_e\mapsto\varepsilon t_e$, $x_v\mapsto\sqrt{\varepsilon}\,x_v$ sends the physical upper endpoint to $L/\varepsilon\to\infty$, yielding exactly the projective marked kernel (6.26). The ordered Wick signs are those of (3.9); the zero-interaction identity in (6.11) fixes the collision sign to $+AB$.

$$\left[\sum_{n\ge0}\frac{\hbar^n}{n!}\{\mathcal{I}^{\otimes n},Aa,b_{\mathrm{aux}}B\}_0\Big|_{\mathrm{aux}=0}\right][L]=(A\star B)[L].\tag{6.43}$$

The graphs outside each collapsed cluster reconstruct its effective representative by the Ward identity $\mathbf{Q}_L(\mathcal{O}[L]) = (\mathbf{Q}\mathcal{O})[L]$ of a local insertion, the zero-form counterpart of (6.37). The same ultraviolet-finite holomorphic graph prescription applies to the decoupled auxiliary extension. Away from the collision faces, the closed-input descent identity is $\bar\partial_x F = -\mathbf{Q}_L F^{(1)}$. The remaining boundary is S^3_R. Since $\Omega \wedge [P_{0,\infty}]_\Omega$ has odd degree, the regulated Stokes identity gives

$$\begin{aligned}\mathbf{Q}_L \int_{B_R} \Omega \wedge [P_{0,\infty}]_\Omega \wedge F^{(1)} &= -\int_{B_R} \Omega \wedge [P_{0,\infty}]_\Omega \wedge \mathbf{Q}_L F^{(1)} \\ &= \int_{B_R} \Omega \wedge [P_{0,\infty}]_\Omega \wedge \bar\partial_x F \\ &= (A \star B)[L] - \int_{S^3_R} \Omega \wedge [P_{0,\infty}]_\Omega F.\end{aligned} \tag{6.44}$$

This is (6.41). The collision coefficient is independent of L by this rescaling [5, Theorem 6]; L remains in the effective representatives and their differential. □

An explicit associator example. **Example 6.7.** In $\mathcal{N} = 4$ Yang–Mills theory, let A and B be gauge-invariant polynomials in the scalar fields $(\gamma^I)_X$ (a subsector of the BMN sector), without holomorphic derivatives, and set $D = (\gamma^3)_X(\beta_2)^X$. Then

$$\begin{aligned}&(A \star B) \star D - A \star (B \star D) \\ &\quad = \mathbf{Q}\left\{\frac{\hbar}{64\pi^2}\left[(\partial_{+\dot a}B)\partial_+{}^{\dot a}\frac{\partial A}{\partial(\gamma^2)_X} + (\partial_{+\dot a}A)\partial_+{}^{\dot a}\frac{\partial B}{\partial(\gamma^2)_X}\right](\gamma^3)_X\right\}.\end{aligned} \tag{6.45}$$

Proof. Gauge invariance gives $Q_0 A = Q_0 B = 0$. Since $C^{XYZ}(\gamma^3)_X(\gamma^3)_Y(\gamma^1)_Z$ is zero, $Q_0 D$ is also zero. The possible one-loop action on D contains $(\theta_1\theta_2)\theta_2 = 0$, and $Q_{n\geq 2}$ requires at least three input fields. Thus $\mathbf{Q}D = 0$. Equation (6.28) gives

$$\{\mathcal{I}, A, D\}^{\gamma,0} = \frac{g}{64\pi^2} C_{EX}{}^Y(\partial_{+\dot a}\mathsf{c}^E)\partial_+{}^{\dot a}\left(\frac{\partial A}{\partial(\gamma^2)_X}\right)(\gamma^3)_Y.$$

Each physical contraction uses one b or β. For an order-n product, the n interactions supply at most n such factors, and D supplies one; its $2n$ physical contractions therefore require $n \leq 1$ (one-loop exact). Products between operators containing only c and γ have no quantum correction. Expanding the two bracketings consequently gives, exactly,

$$\begin{aligned}&(A \star B) \star D - A \star (B \star D) \\ &\quad = \frac{\hbar g}{64\pi^2} C_{EX}{}^Y(\partial_{+\dot a}\mathsf{c}^E)\left[\frac{\partial A}{\partial(\gamma^2)_X}\partial_+{}^{\dot a}B + (\partial_+{}^{\dot a}A)\frac{\partial B}{\partial(\gamma^2)_X}\right](\gamma^3)_Y.\end{aligned}$$

We have

$$Q_0\left[\partial_+{}^{\dot a}\left(\frac{\partial A}{\partial(\gamma^2)_X}\right)(\gamma^3)_X\right] = -g C_{EX}{}^Y(\partial_+{}^{\dot a}\mathsf{c}^E)\frac{\partial A}{\partial(\gamma^2)_X}(\gamma^3)_Y.$$

The same identity holds for B. Together with the antisymmetry of the dotted-index contraction, it evaluates the displayed $\mathbf{Q}$-variation to the associator. For an operator containing only γ and its derivatives, an n-loop supercharge graph would require $2n + 1 \leq n + 1$, so all its positive-loop variations vanish. This also gives $\mathbf{Q}A = \mathbf{Q}B = 0$ and shows that no further terms enter the primitive. □

For example, take $A = B = \delta^{XY}(\gamma^2)_X(\gamma^2)_Y$. The two gradient terms coincide, and the formula becomes

$$\begin{aligned}&(A \star B) \star D - A \star (B \star D) \\ &\quad = \frac{\hbar g}{16\pi^2} C_E{}^{XY}(\partial_{+\dot a}\mathsf{c}^E)(\partial_+{}^{\dot a}A)(\gamma^2)_X(\gamma^3)_Y \\ &\quad = \mathbf{Q}\left[\frac{\hbar}{16\pi^2}\delta^{XY}(\partial_{+\dot a}A)(\partial_+{}^{\dot a}(\gamma^2)_X)(\gamma^3)_Y\right].\end{aligned} \tag{6.46}$$

The anomaly brackets, with their holomorphic spectral parameters, form a homotopical Lie conformal algebra: their Jacobi identities hold through the higher brackets [6, Sections 4.2–4.5]. The normal-ordering product and its associator homotopies are additional operations [3, Section 2.3]. The corrected Leibniz rule also permits a normal-ordering calculation of the supercharge on composites; [3, Section 3.2] illustrates this with the one-loop supersymmetry anomaly of the stress tensor.

7 Other kernels and product forests

As derived in Section 3, kernels beyond the fans arise in two cases:

1. The quantum product has general product forests: the marked edge AB and interaction trees, each joined to both insertions by $|T_j|+1$ spokes. Marked books are the special case in which every interaction tree is a single vertex.

1. For a superpotential of degree greater than three (non-renormalizable), the loop corrections to the supersymmetry transformation themselves contain spiders (Proposition 3.18).

7.1 Restrictions on quantum-product graphs

Proposition 7.1 (Edge counts of the product graphs)**.** *Let Γ be a nonzero connected marked graph of (6.11) at order $\hbar^n$, $n \geq 1$. It has n interaction vertices, the two insertions A, B, and $2n+1$ edges: the auxiliary edge AB, which contributes its kernel, and $2n$ edges contracting fields in $\mathcal{I}$, A and B. Removing A and B leaves the interaction subgraph (Figure 7.1). For one of its components T_j, with $|T_j|$ vertices, I_j denotes its internal edges and E_j its edges to A or B, the spokes. Every edge other than AB is an internal edge or a spoke of exactly one component, and*

$$|T_j| - 1 \;\leq\; |I_j| \;\leq\; |T_j| - \#(\bar{W} \text{ vertices in } T_j), \qquad |I_j| + |E_j| = 2|T_j|. \tag{7.1}$$

Proof. In the circuit of Γ (Remark 3.5) every edge is a resistor, and the marked resistor joins the insertions A and B.

Internal resistors.

$$\underbrace{|T_j| - 1}_{T_j \text{ connected}} \;\leq\; |I_j| \;\leq\; \underbrace{|T_j| - \#(\bar{W} \text{ vertices in } T_j)}_{\text{antifield terminals in } T_j}.$$

It is the antifield-terminal count of Lemma 3.15.

Resistors at the insertions. A physical resistor between A and B would be in parallel with the marked resistor (Corollary 3.8). Every physical resistor therefore ends at an interaction node, and it is an internal resistor or a spoke of the component of that node. Summing over the components,

$$\sum_j (|I_j| + |E_j|) = 2n = 2\sum_j |T_j|. \tag{7.2}$$

Local Ohm's law on $T_j \cup \{A, B\}$. The subcircuit formed by the nodes of T_j, the two insertions and the marked resistor obeys the local count of Local Ohm's law (Theorem 3.7):

$$\begin{aligned} & |I_j| + |E_j| + \underbrace{1}_{AB} \;\leq\; 2(|T_j| + 2) - 3 \\ \Longrightarrow \quad & |I_j| + |E_j| \;\leq\; 2|T_j|. \end{aligned}$$

By (7.2), equality holds for every component. □

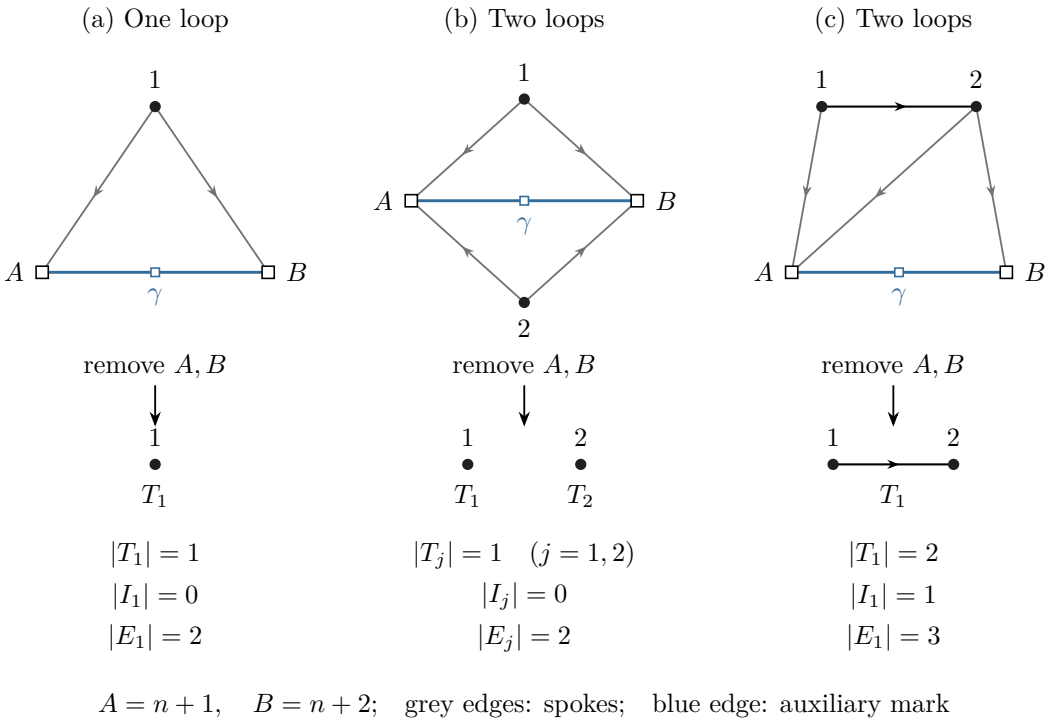


Figure 7.1: Marked graph topologies through two loops and their internal forests. The squares $A = n+1$ and $B = n+2$ are the two operator insertions; the blue edge carries the auxiliary mark γ. Deleting A, B and all incident edges leaves (a) one isolated vertex, (b) two isolated vertices, and (c) a two-vertex tree. The black edge is internal; the grey edges are spokes to the insertions. The displayed spoke counts refer to the full graphs in the top row.

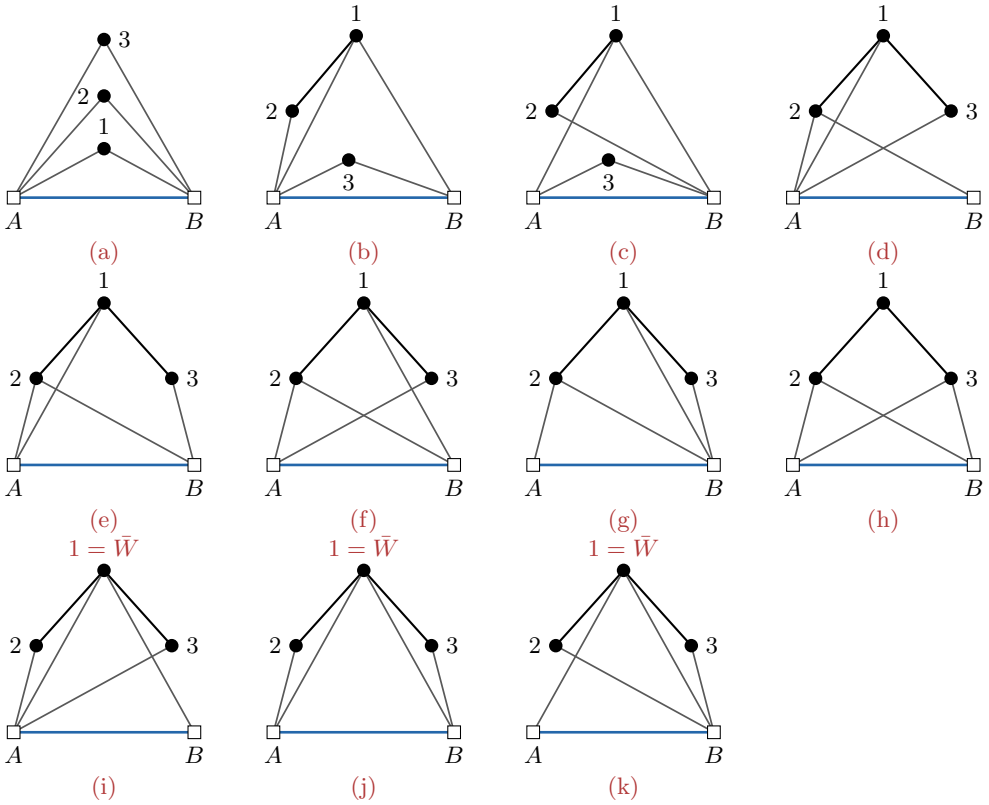


Figure 7.2: All eleven marked three-loop topologies up to permutations of the interaction vertices, fixing A and B. Panels (a)–(h) allow cubic interactions; in (i)–(k), vertex 1 is a superpotential vertex with four resistors. Blue: the marked edge AB; black: internal edges; gray: spokes.

Proposition 7.2 (Graphs of the quantum product)**.** *For the interaction* (3.4)*, with* $\bar{W}$ *of any degree, let* Γ *be a nonzero connected marked graph of* (6.11) *at order* $\hbar^n$*,* $n \geq 1$*. Then*

$$\Gamma \setminus \{A, B\} = \coprod_j T_j, \qquad T_j \text{ a tree}, \qquad \#(\bar{W} \text{ vertices in } T_j) \leq 1, \qquad |E_j| = |T_j| + 1. \tag{7.3}$$

Every T_j *is joined to both* A *and* B*, and loop diagrams contain no quadratic term of the superpotential* $\bar{W}$ *(only terms of at least cubic degree enter the theory). Each tree has one uncontracted field if all its vertices are cubic, and* $\deg \bar{W} - 2$ *if it contains a superpotential monomial; its number of source-bearing vertices is bounded by that number.*

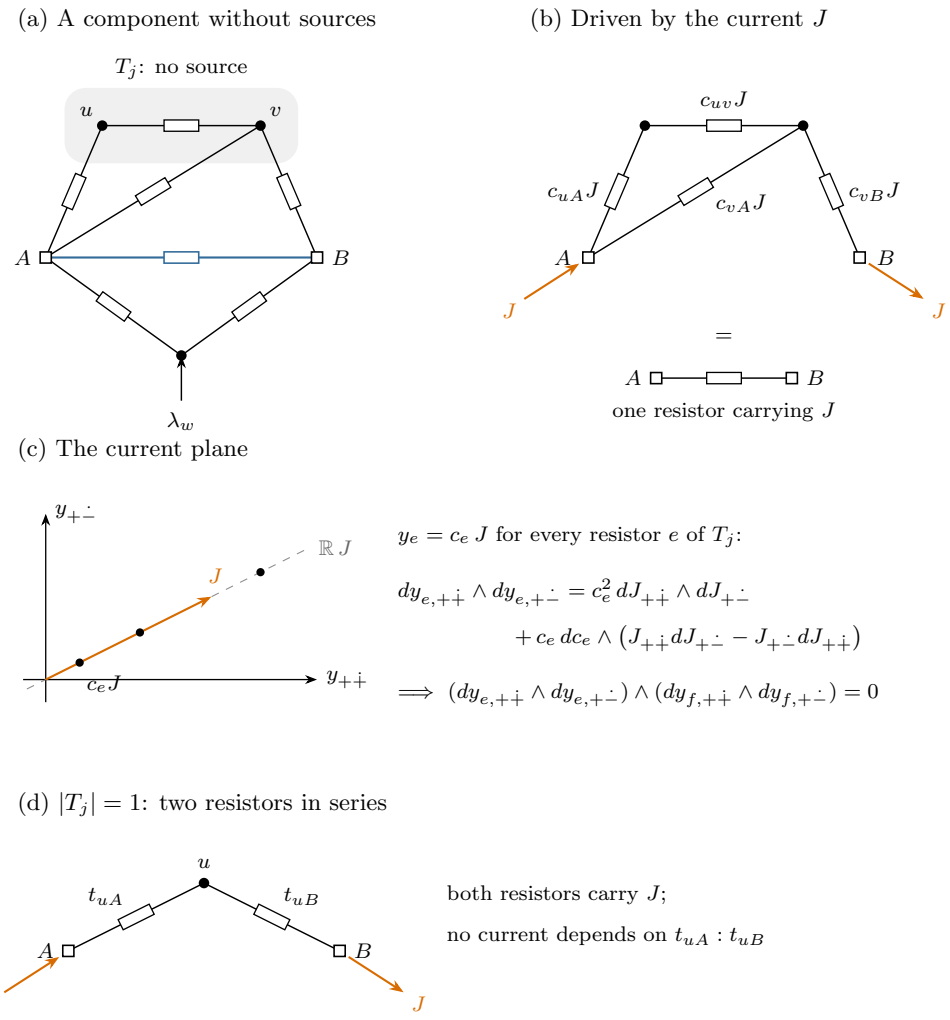


Figure 7.3: A component without sources. (a) A component T_j with no injected current meets the rest of the circuit only at the insertions A and B; here the rest is the marked resistor (blue) and a node fed by λ_w. (b) The current J entering T_j at A leaves it at B, and every current of T_j is c_eJ with a real current-division ratio c_e: the component acts as one resistor carrying J. (c) All currents of T_j lie on the line $\mathbb{R}J$ of the current plane, so each of their area forms is (7.4), and the product of two of them vanishes. (d) For $|T_j| = 1$ the two resistors are in series, and no current depends on the ratio of their resistances.

Proof.

The proof parallels that of Proposition 3.10.

Components without sources. Suppose that no current is injected at the nodes of T_j (Figure 7.3(a)). The component touches the rest of the circuit only at the insertions A and B, so the current J that enters it at A leaves it at B, and inside it Kirchhoff's laws are those of a passive network driven by J. Every current of the component is therefore a real multiple c_eJ of J, with a current-division ratio c_e fixed by its resistances: the component acts as one resistor carrying J (Figure 7.3(b)). As the resistances vary, its currents move along the single direction of J in the current plane (Figure 7.3(c)):

$$dy_{e,+\dot{+}} \wedge dy_{e,+\dot{-}} = c_e^2\,dJ_{+\dot{+}} \wedge dJ_{+\dot{-}} + c_e\,dc_e \wedge (J_{+\dot{+}}dJ_{+\dot{-}} - J_{+\dot{-}}dJ_{+\dot{+}}). \tag{7.4}$$

Replace the source-free component by its effective resistor between A, B and merge it with the marked resistor; call the resulting circuit Γ^{red}. If $|T_j| = 1$, its vertex v has $\lambda_v = 0$, and Theorem A.1 gives

$$\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda;z)=\int_0^1 dr_1\int_0^1 dr_2\ \underbrace{r_2\lambda_{v,+}{}^{\dot a}\frac{\partial}{\partial z_{AB}^{\text{red},+\dot a}}\Big|_{\lambda_v=0}}_{\text{zero current-area operator}}\mathcal{I}_{\Gamma^{\text{red}}}^{\text{raw cut}}(\lambda;z^{\text{red}})=0.$$

If $|T_j|\geq 2$, choose a spanning tree containing AB and $|T_j|$ resistors of the component. There are $|T_j|$ complementary component resistors; for any two, e,f, (7.4) gives

$$\begin{aligned}
&(dy_{e,++}\wedge dy_{e,+-})\wedge(dy_{f,++}\wedge dy_{f,+-})\\
&=\Big[c_e^2 dJ_{++}\wedge dJ_{+-}+c_e\,dc_e\wedge(J_{++}dJ_{+-}-J_{+-}dJ_{++})\Big]\\
&\quad\wedge\Big[c_f^2 dJ_{++}\wedge dJ_{+-}+c_f\,dc_f\wedge(J_{++}dJ_{+-}-J_{+-}dJ_{++})\Big]=0,\\
&\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda;z)=\underbrace{0}_{\text{current-area Jacobian of the component}}\qquad\mathcal{I}_{\Gamma^{\text{red}}}^{\text{raw cut}}(\lambda;z^{\text{red}})=0.
\end{aligned}$$

Trees. If the internal resistors of T_j close a loop, they number at least $|T_j|$; by (7.1) they number exactly $|T_j|$, T_j has no $\bar W$ node, and $|E_j|=|T_j|$. The $2|T_j|$ ends of the internal resistors and the $|T_j|$ spokes then wire all $3|T_j|$ terminals of its cubic nodes. No current is injected in T_j, and it becomes a component without sources, which gives zero. Every T_j is therefore a tree, $|I_j|=|T_j|-1$, and (7.1) gives at most one $\bar W$ node and $|E_j|=|T_j|+1$, which is (7.3).

Spokes and degrees. Since $|E_j|=|T_j|+1$ and each node can be joined to each of A and B at most once, each component must have spokes to both A and B.

Quadratic terms of $\bar W$. A quadratic term of $\bar W$ is a node with two terminals, both wired by the previous step, so it injects no current. It is the only $\bar W$ node of its tree, whose $3(|T_j|-1)+2=3|T_j|-1$ terminals are then all wired by the $|T_j|-1$ internal resistors and the $|T_j|+1$ spokes. The tree carries no source, and the first step gives zero.

A source at a vertex is supplied by its uncontracted fields. For cubic vertices the count is one; if T_j contains a superpotential monomial,

$$\begin{aligned}
\#(\text{uncontracted fields in } T_j)&=3(|T_j|-1)+\deg\bar W-2(|T_j|-1)-(|T_j|+1)\\
&=\deg\bar W-2,\\
\#\{v\in T_j:\lambda_v\neq 0\}&\leq\#(\text{uncontracted fields in } T_j)=\deg\bar W-2.
\end{aligned}$$

Thus the bound is one for cubic interactions, and is not uniformly one for arbitrary $\bar W$. □

Proposition 7.3 (Cubic interactions have a starting point for the iteration: a vertex of degree two)**.** *Let every vertex of $\mathcal{I}$ be cubic, as are the gauge and matter vertices of (3.4) together with a cubic $\bar W$. In every tree T_j of (7.3), exactly one vertex has degree two, and its uncontracted field is the only output field of T_j; all other vertices have degree three. The tree degree of a vertex is its number of internal edges, and the number of spokes at a vertex of given tree degree is*

tree degree	0	1	2	3
spokes if $\deg_\Gamma v=3$	–	2	1	0
spokes if $\deg_\Gamma v=2$	2	1	0	–

(7.5)

and a vertex with two spokes is joined to both A and B (Figure 7.4).

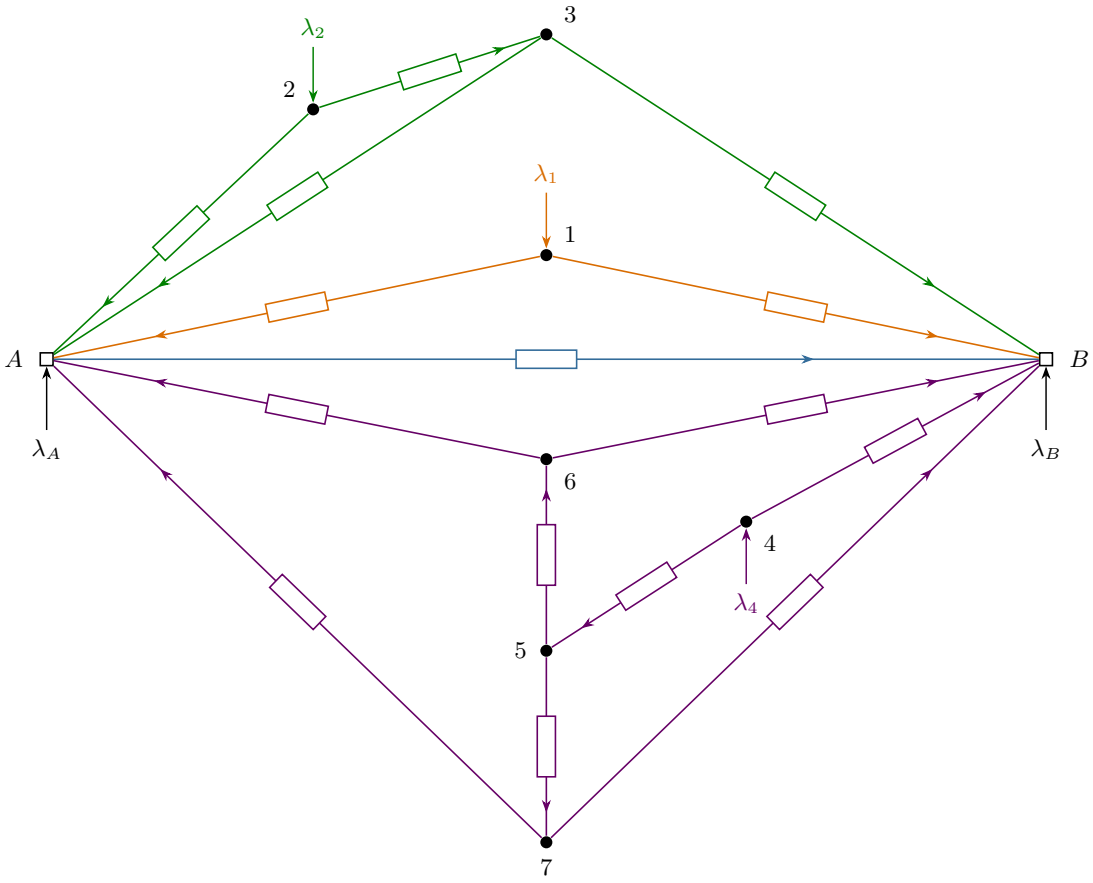


Figure 7.4: A seven-loop product graph with three trees, each in its own colour, between the insertions A and B; the marked edge AB is blue. Each tree has one more spoke than vertices, and its source enters at its only vertex of degree two: 1, 2 and 4. Leaves of degree three are joined to both insertions, and the branching vertex 5 has no spoke.

Proof. A cubic node has three terminals. In T_j the $|T_j|-1$ internal resistors wire $2(|T_j|-1)$ terminals and the $|T_j|+1$ spokes wire $|T_j|+1$ more, so the numbers of resistors at its nodes add up to

$$\sum_{v\in T_j} \deg_\Gamma v = 2(|T_j|-1) + (|T_j|+1) = 3|T_j| - 1. \tag{7.6}$$

Every node has two or three resistors by Proposition 7.2, so exactly one node has two, and

$$\underbrace{3|T_j|}_{\text{terminals}} - \underbrace{2(|T_j|-1)}_{\text{internal resistor ends}} - \underbrace{(|T_j|+1)}_{\text{spokes}} = 1 \tag{7.7}$$

terminal stays free, at that node: each tree carries a single source, injected at its node with two resistors. □

The final iteration of these forest diagrams requires star–triangle transformations and is more involved; the details are given in Appendix A. Since the graphs have many different topologies, there is no unique way to derive the final result in one step. At low loop orders ($n \leq 2$), however, all these forest diagrams belong to the fan and spider kernels, which can be computed exactly, and we have not yet needed the explicit form of these results.

As shown in Section 3, the calculation of Q_n in non-renormalizable theories involves the spider kernel.

7.2 Spiders

A superpotential vertex of degree greater than three can join several arms (Proposition 3.18).

Definition 7.4 (Spiders)**.** A *spider* consists of paths meeting only at a vertex A, together with a second vertex B joined to every vertex of these paths, including A. In the supercharge Q_n graphs, A carries $\bar{W}$ and $B = \mathcal{O}$. The ith arm contains ℓ_i cubic vertices, labelled $v_{i,1}, \dots, v_{i,\ell_i}$ from its outer end towards A; its length is ℓ_i and the loop number is $n = \sum_i \ell_i$.

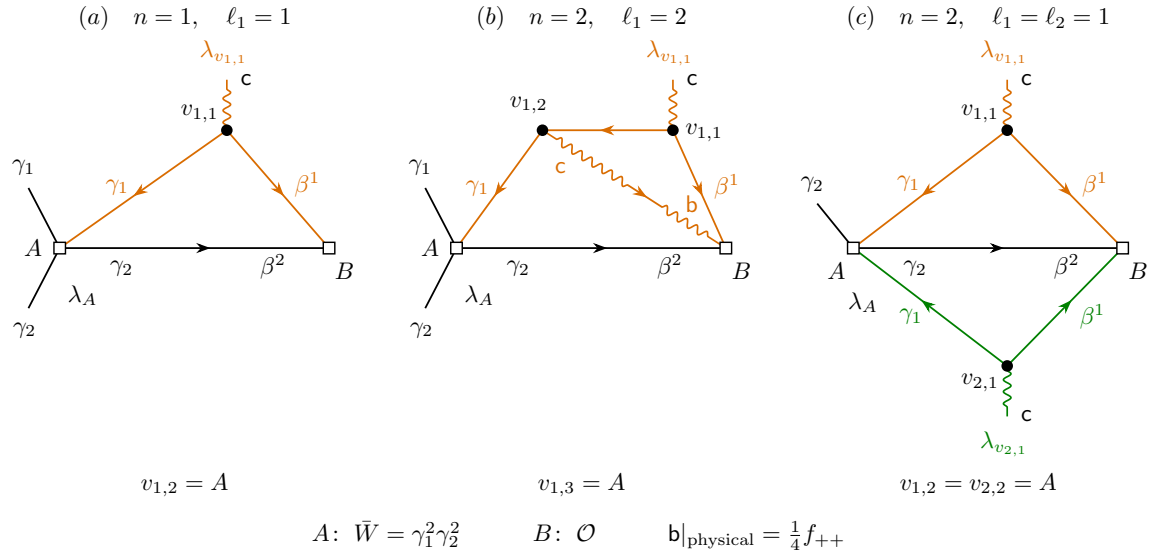


Figure 7.5: One- and two-loop spiders for $\bar{W} = \gamma_1^2\gamma_2^2$: (a) $\ell_1 = 1$; (b) $\ell_1 = 2$; (c) $\ell_1 = \ell_2 = 1$. The superpotential vertex is A and the operator insertion is B; each arm, with its spokes, has its own colour. Arm indices increase towards $A = v_{i,\ell_i+1}$. Each $v_{i,1}$ carries an output c; the output matter at A supplies λ_A.

On arm i, $v_{i,1}$ carries the source, v_{i,ℓ_i} is adjacent to A, and $v_{i,\ell_i+1} = A$. The remaining arm vertices are fully contracted:

$$\begin{aligned} \Gamma_1 = \big((v_{i,j}, v_{i,j+1}), (v_{i,j}, B)\big)_{i=1,\dots,a;\ j=1,\dots,\ell_i} \cup \{(A,B)\}, \\ \ell_1 + \cdots + \ell_a = n, \qquad \lambda_{v_{i,j},+\dot{a}} = \delta_{j1}\lambda_{v_{i,1},+\dot{a}}. \end{aligned} \tag{7.8}$$

All computations of Q_n involve only the spider kernel.

The computation of the spider kernel is similar to repeating the computation of the fan kernel on each path.

Each arm is reduced as a fan is, from its outer end (Figure 7.6).

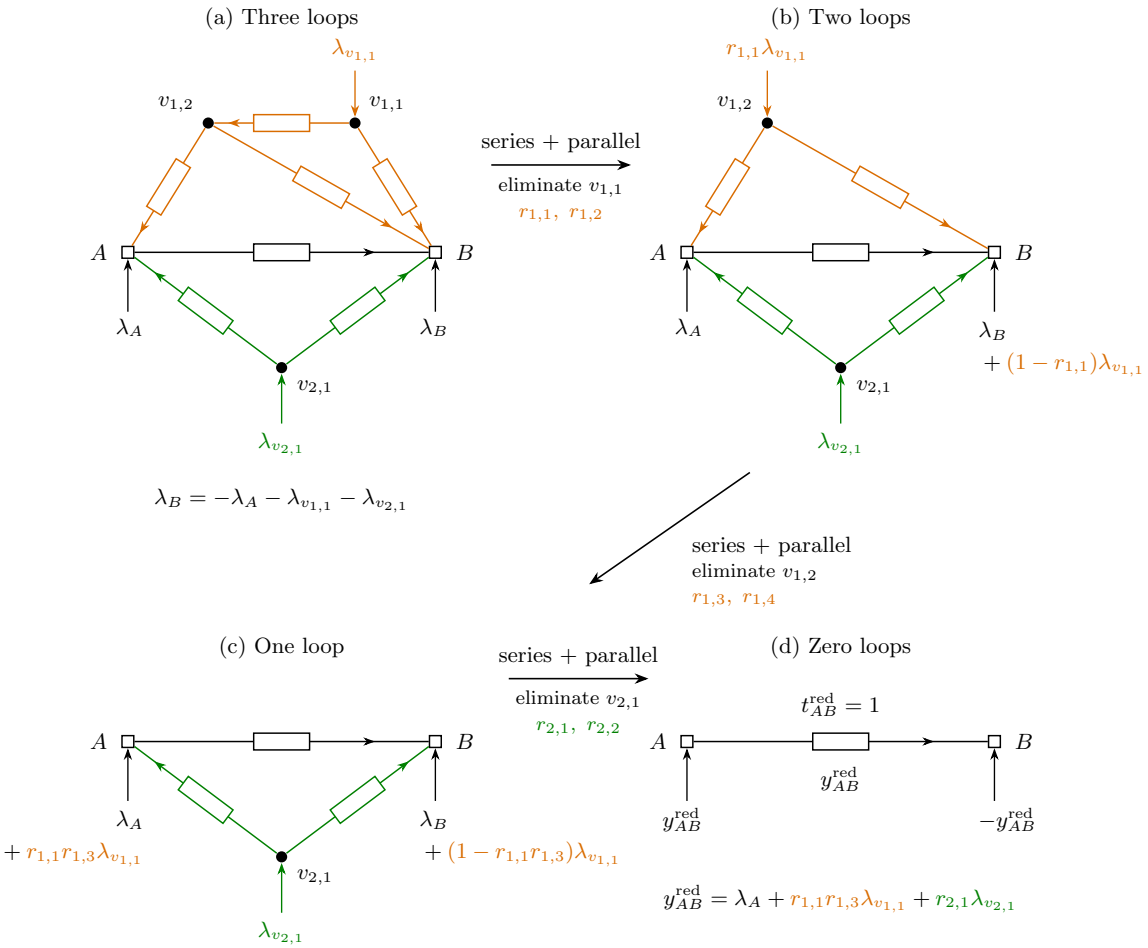


Figure 7.6: Reduction of the three-loop spider with arms $\ell_1 = 2$ (orange) and $\ell_2 = 1$ (green). Each step eliminates an arm vertex by a series step, which passes the fraction $r_{i,2j-1}$ of its source along the arm and the rest to B, and a parallel merge with the next spoke or with AB; each step lowers the loop number by one. The last panel is the unit resistor carrying (7.10).

Each merge lowers the loop number by one. When every arm is reduced, its branch and the original edge AB merge into one resistor, whose Schwinger time is fixed to one. Appendix A.5 evaluates the iterated kernel operators, using the coordinates

$$0 < r_{i,j} < 1 \quad (j < 2\ell_i), \qquad r_{i,2\ell_i} > 0, \qquad \sum_i r_{i,2\ell_i} < 1. \tag{7.9}$$

Here $r_{i,2j-1}$ is the fraction of the source at $v_{i,j}$, after the earlier eliminations, that the series step passes to $v_{i,j+1}$, the rest going to B;

for $j < \ell_i$, $r_{i,2j}$ is the fraction of the merged spoke current at $v_{i,j+1}$ carried by the new branch;

and $r_{i,2\ell_i}$ is the fraction of the final current carried by the branch of arm i, the original edge AB carrying $1 - \sum_i r_{i,2\ell_i}$.

The final resistor carries the reduced current y^{red}_{AB}, which by conservation at A is the source collected at A: λ_A and, from each arm, the fraction of its source passed along the whole arm,

$$y^{\mathrm{red}}_{AB,+\dot{a}} = \lambda_{A,+\dot{a}} + \sum_i \left(\prod_{k=1}^{\ell_i} r_{i,2k-1} \right) \lambda_{v_{i,1},+\dot{a}}. \tag{7.10}$$

Reversing the reductions splits y^{red}_{AB} back into the original currents (A.37)–(A.38).

Deleting the arms one at a time gives the kernel:

$$
\begin{aligned}
&\mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda;z)\\
&= \left[\prod_{j=1}^{\ell_1,\longrightarrow}\left\{\begin{array}{ll}\displaystyle\int_0^1 dr_1\int_0^1 dr_2\, e^{-\lambda_{v_{1,j},+\dot a}[r_1 z_{v_{1,j}v_{1,j+1}}+(1-r_1)z_{v_{1,j}B}]^{+\dot a}} & \\ \quad\times \quad \underbrace{r_2\lambda_{v_{1,j},+^{\dot b}}\dfrac{\partial}{\partial z^{\text{red},+\dot b}_{v_{1,j+1}B}}}_{\text{current-area Jacobian insertion}} & \end{array}\right\}\right]\\
&\quad\times \underbrace{\mathcal{I}^{\text{raw cut}}_{\Gamma\setminus\{v_{1,1},\dots,v_{1,\ell_1}\}}(\lambda^{\text{red}};z^{\text{red}})}_{(n-\ell_1)\text{-loop kernel}}\\
&= \left[\prod_{i=1}^{2,\longrightarrow}\prod_{j=1}^{\ell_i,\longrightarrow}\left\{\begin{array}{ll}\displaystyle\int_0^1 dr_1\int_0^1 dr_2\, e^{-\lambda_{v_{i,j},+\dot a}[r_1 z_{v_{i,j}v_{i,j+1}}+(1-r_1)z_{v_{i,j}B}]^{+\dot a}} & \\ \quad\times r_2\lambda_{v_{i,j},+^{\dot b}}\dfrac{\partial}{\partial z^{\text{red},+\dot b}_{v_{i,j+1}B}} & \end{array}\right\}\right]\\
&\quad\times \underbrace{\mathcal{I}^{\text{raw cut}}_{\Gamma\setminus\{v_{1,1},\dots,v_{1,\ell_1},v_{2,1},\dots,v_{2,\ell_2}\}}(\lambda^{\text{red}};z^{\text{red}})}_{(n-\ell_1-\ell_2)\text{-loop kernel}}\\
&= \cdots\\
&= \int_{(7.9)}\prod_{i=1}^{a}\prod_{j=1}^{2\ell_i} dr_{i,j}\ \underbrace{\left(1-\sum_i r_{i,2\ell_i}\right)}_{\text{final parallel merge}}\\
&\quad\times \prod_{i=1}^{a}\underbrace{r_{i,2\ell_i}^{\ell_i-1}\prod_{j=1}^{\ell_i-1}\left[r_{i,2j-1}^{\ell_i-j}\,r_{i,2j}^{j-1}\,(1-r_{i,2j})\right]\left[\lambda_{v_{i,1},+\dot a}y^{\text{red}}_{AB,+^{\dot a}}\right]^{\ell_i}}_{\text{diagonal blocks of arm } i}\ e^{-\sum_e y_{e,+\dot a}(r)z_e^{+\dot a}}.
\end{aligned}
\tag{7.11}
$$

In the first two lines, the ordered products are compositions of the displayed integral operators, with the first vertex of each arm outermost. Each factor uses the current graph and its own local r_1, r_2 of Theorem A.1, with $v = v_{i,j}$, $u = v_{i,j+1}$ and $w = B$; each derivative acts on the entire expression to its right, including the later exponential factors and displacement substitutions,

$$
\begin{aligned}
\lambda^{\text{red}}_u &= \lambda_u + r_1\lambda_v, \qquad \lambda^{\text{red}}_B = \lambda_B + (1-r_1)\lambda_v,\\
z^{\text{red}}_{uB} &= r_2 z_{uB} + (1-r_2)(z_{vB}-z_{vu}), \qquad z^{\text{red}}_e = z_e \quad (e \neq uB).
\end{aligned}
$$

Deleting an arm includes these source transfers and parallel merges; the second line is omitted if there is only one arm. The last line uses the joint coordinates (7.9), whose accumulated operator factors give (A.34).

The cubic arm vertices carry no derivatives, so $z_{v_{i,j}v_{i,j+1}} = 0$ for $j < \ell_i$; the spokes, the last arm edges and AB keep general displacements if we want to compute results with holomorphic derivatives.

The exponential then translates the outputs at $v_{i,1}$ and A:

$$
-\sum_e y_{e,+\dot a}(r)z_e^{+\dot a} = \lambda_{A,+\dot a}x_A^{+\dot a}(r,z) + \sum_{i=1}^{a}\lambda_{v_{i,1},+\dot a}x^{+\dot a}_{v_{i,1}}(r,z),
\tag{7.12}
$$

with translated positions

$$\begin{aligned}
x_A^{+\dot a}(r,z) =& -\left(1-\sum_i r_{i,2\ell_i}\right) z_{AB}^{+\dot a} \\
&- \sum_{i=1}^{a} r_{i,2\ell_i}\left[\sum_{j=1}^{\ell_i}(1-r_{i,2j-2})\left(\prod_{k=j}^{\ell_i-1} r_{i,2k}\right) z_{v_{i,j}B}^{+\dot a} - z_{v_{i,\ell_i}A}^{+\dot a}\right], \\
x_{v_{i,1}}^{+\dot a}(r,z) =& -\sum_{s=1}^{\ell_i}(1-r_{i,2s-1})\left(\prod_{k=1}^{s-1} r_{i,2k-1}\right)\sum_{j=1}^{s}(1-r_{i,2j-2})\left(\prod_{k=j}^{s-1} r_{i,2k}\right) z_{v_{i,j}B}^{+\dot a} \\
&+ \left(\prod_{k=1}^{\ell_i} r_{i,2k-1}\right)(x_A^{+\dot a}(r,z) - z_{v_{i,\ell_i}A}^{+\dot a}), \qquad r_{i,0}=0.
\end{aligned} \tag{7.13}$$

For nonnegative integers $\kappa_{i,j}$ and $\bar\kappa_{i,2j}$, the weighted monomial moments on (7.9) are

$$\begin{aligned}
&\int_{(7.9)} \prod_{i,j} dr_{i,j}\left(1-\sum_i r_{i,2\ell_i}\right)\prod_{i=1}^{a}\left[r_{i,2\ell_i}^{\ell_i-1+\kappa_{i,2\ell_i}}\prod_{j=1}^{\ell_i} r_{i,2j-1}^{\ell_i-j+\kappa_{i,2j-1}}\prod_{j=1}^{\ell_i-1} r_{i,2j}^{j-1+\kappa_{i,2j}}(1-r_{i,2j})^{1+\bar\kappa_{i,2j}}\right] \\
&= \frac{\prod_i(\ell_i-1+\kappa_{i,2\ell_i})!}{(n+1+\sum_i\kappa_{i,2\ell_i})!}\prod_{i=1}^{a}\left[\prod_{j=1}^{\ell_i}\frac{1}{\ell_i-j+1+\kappa_{i,2j-1}}\prod_{j=1}^{\ell_i-1}\frac{(j-1+\kappa_{i,2j})!\,(1+\bar\kappa_{i,2j})!}{(j+1+\kappa_{i,2j}+\bar\kappa_{i,2j})!}\right].
\end{aligned} \tag{7.14}$$

We organize the final result: Sum over nonnegative integers m_{iA}, m_{ik} $(k\neq i)$ with $m_{iA}+\sum_{k\neq i} m_{ik}=\ell_i$, p_{ijs} $(1\le j\le s\le \ell_i)$, and q_{ijA}, q_{ijk} $(1\le j\le \ell_i,\ 1\le k\le a)$, and write $q_{ij}=q_{ijA}+\sum_k q_{ijk}$ and $M_i=\sum_{k\neq i} m_{ki}+\sum_{k=1}^{a}\sum_{j=1}^{\ell_k} q_{kji}$.

The multinomial exponents m_{iA} and m_{ik} choose the λ_A and $\lambda_{v_{k,1}}$ terms in $[\lambda_{v_{i,1}} y_{AB}^{\mathrm{red}}]^{\ell_i}$. In the displacement exponential, p_{ijs} counts the sth $\lambda_{v_{i,1}}$ term of (A.39), including $s=\ell_i$, while q_{ijA} and q_{ijk} count the λ_A and $(\prod_{h=1}^{\ell_k} r_{k,2h-1})\lambda_{v_{k,1}}$ terms of y_{AB}^{red} in its last term.

The full Taylor expansion is

$$\begin{aligned}
\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda;z) = e^{-\lambda_{A,+\dot a} z_{AB}^{+\dot a} - \sum_i \lambda_{v_{i,1},+\dot a} z_{v_{i,1}B}^{+\dot a}} \sum_{m,p,q}\prod_{i=1}^{a}\Bigg\{ & \\
\ell_i!\,\frac{(\lambda_{v_{i,1},+\dot b}\lambda_{A,+}{}^{\dot b})^{m_{iA}}}{m_{iA}!}\prod_{k\neq i}\frac{(\lambda_{v_{i,1},+\dot c}\lambda_{v_{k,1},+}{}^{\dot c})^{m_{ik}}}{m_{ik}!} & \\
\times\prod_{j=1}^{\ell_i}\Bigg[\prod_{s=j}^{\ell_i}\frac{\left[-\lambda_{v_{i,1},+\dot d}(z_{v_{i,j}v_{i,j+1}}+z_{v_{i,j+1}B}-z_{v_{i,j}B})^{+\dot d}\right]^{p_{ijs}}}{p_{ijs}!} & \\
\times\frac{\left[\lambda_{A,+\dot e}(z_{v_{i,j}v_{i,j+1}}+z_{v_{i,j+1}B}-z_{v_{i,j}B})^{+\dot e}\right]^{q_{ijA}}}{q_{ijA}!} & \\
\times\prod_{k=1}^{a}\frac{\left[\lambda_{v_{k,1},+\dot f}(z_{v_{i,j}v_{i,j+1}}+z_{v_{i,j+1}B}-z_{v_{i,j}B})^{+\dot f}\right]^{q_{ijk}}}{q_{ijk}!}\Bigg]\Bigg\} & \\
\times\frac{\prod_i\left(\ell_i-1+\sum_j q_{ij}\right)!}{\left(n+1+\sum_{i,j} q_{ij}\right)!}\prod_{i=1}^{a}\prod_{t=1}^{\ell_i}\frac{1}{\ell_i-t+1+M_i+\sum_{s=t}^{\ell_i}\sum_{j=1}^{s} p_{ijs}} & \\
\times\prod_{i=1}^{a}\prod_{t=1}^{\ell_i-1}\frac{\left(t-1+\sum_{j=1}^{t}\left[q_{ij}+\sum_{s=t+1}^{\ell_i} p_{ijs}\right]\right)!\,\left(1+\sum_{j=1}^{t} p_{ijt}\right)!}{\left(t+1+\sum_{j=1}^{t}\left[q_{ij}+\sum_{s=t}^{\ell_i} p_{ijs}\right]\right)!}. &
\end{aligned} \tag{7.15}$$

For the physical displacements, $z_{v_{i,j}v_{i,j+1}}=0$ for $j<\ell_i$. Taking one arm gives the Taylor coefficients of (5.14), which is exactly the fan kernel result. (A fan is a special type of spider.)

7.3 Kernels of general product forests

As noted in Section 7.1, spiders do not exhaust the graphs of the quantum product: other graphs (general forests) occur from three loops on and are the majority from four loops on, where the trees also begin to branch (Table A.1). The moves in Appendix A.3 reduce all of them.[21]

The reduction of each tree in the forest can be done independently. The details of the star–triangle move are given in Appendix A.

Definition 7.5 (Rooted interaction trees)**.** Root a cubic interaction tree at its unique vertex of full graph degree two. The parent of a non-root vertex v is its neighbour on the path to the root; the other tree neighbours are its children, and vertices with the same parent are siblings. A spoke at a vertex u is an edge uA or uB.

The subtree of a vertex consists of the vertices whose path to the root passes through it.

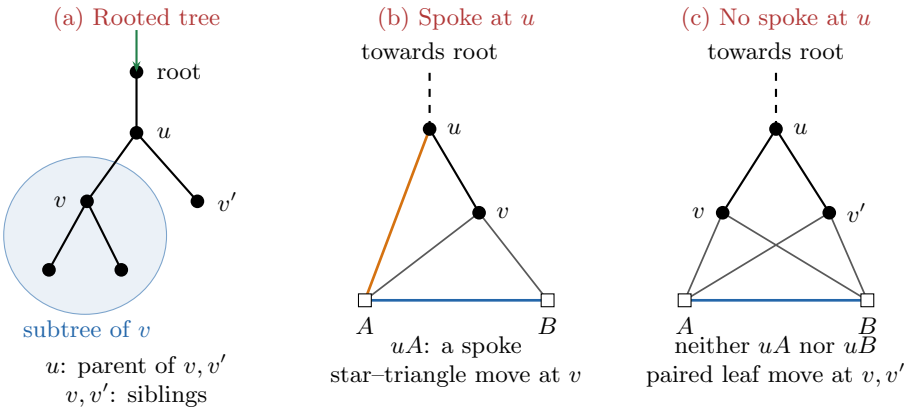


Figure 7.7: Rooted-tree terminology and the two reduction neighborhoods. (a) The shaded subtree contains v and its descendants; u is the parent of v, v', which are siblings. (b) The edge uA is a spoke; after the descendants of v are removed, a star–triangle move removes v. (c) The parent u has no spoke; after both descendant subtrees are removed, v, v' are eliminated together. Dashed edges continue towards the root; the blue edge is AB.

Proposition 7.6 (Kernels of cubic product forests)**.** *Let Γ be a nonzero marked graph of the quantum product with cubic interaction vertices (Proposition 7.3), and root each tree at its vertex of degree two, its only source.*

(i) *If Γ is of type I′, series–parallel moves (Theorem A.1) along each arm reduce it to the marked resistor, which then carries*

$$y^{\mathrm{red}}_{AB,+\dot{a}} = \lambda_{A,+\dot{a}} + \sum_{\text{arms at } B} \left(\prod_{k=1}^{\ell_i} r_{i,2k-1} \right) \lambda_{v_{i,1},+\dot{a}} + \sum_{\text{arms at } A} \left(1 - \prod_{k=1}^{\ell_i} r_{i,2k-1} \right) \lambda_{v_{i,1},+\dot{a}}. \tag{7.16}$$

summed over the arms wired to B and to A. The reduction is similar to that in Figure 7.6, the reduction of spiders.

(ii) *Every Γ is reduced to the marked resistor by removing each vertex after all the vertices of its subtree. A source-free vertex is removed by a star–triangle move (Theorem A.2) when its tree neighbor towards the root has a spoke, and together with its sibling (Theorem A.3) when that neighbor has none; the root is removed last, by a series–parallel move.*

[21] From three loops on, some of these graphs are Laman graphs not of type I′ [13, Definition 4.12]: in the three-loop graph with edges $12, 13, 2A, 2B, 3A, 3B, AB$, the only degree-two vertex, 1, has neighbours that are not joined. These reductions therefore reach beyond the recursion of [13, Theorem 4.13].

(iii) *The kernel of* Γ *is an integral over* $2n$ *coordinates, ranging over intervals and simplices, of a polynomial density times the exponential of polynomial currents (the displacements), and each Taylor coefficient in the displacements is a finite sum of the moments* (A.42). *(The essential point is that no "spanning tree denominators" remain and the calculation is polynomial.)*

Proof. (i) By Proposition A.4, every tree is an arm $v_{i,1} - \cdots - v_{i,\ell_i}$ whose nodes are all wired to one insertion, with the source at $v_{i,1}$ and a resistor from v_{i,ℓ_i} to each insertion. On an arm wired to B, the two resistors of $v_{i,1}$ end at $v_{i,2}$ and B, which are joined, so Theorem A.1 removes $v_{i,1}$ and passes the fraction $r_{i,1}$ of its source to $v_{i,2}$, which then has two resistors in turn. The source that reaches the last node of the arm is $\prod_k r_{i,2k-1}\lambda_{v_{i,1}}$, and the last move passes it to A across the marked resistor, as for the spider (7.10). On an arm wired to A the same moves run with A and B exchanged, and the fraction of the source that reaches A is $1 - \prod_k r_{i,2k-1}$. Conservation at A gives (7.16).

(ii) A source-free vertex v whose subtree is gone has three resistors, to its tree neighbour u towards the root, to A and to B: its own spokes and the sides left by the moves at its children, merged in parallel. If u has a spoke, to $X \in \{A, B\}$, two sides of the triangle that replaces v, namely uX and AB, merge with resistors already there, and Theorem A.2 removes v. See Figure 7.8(a). If u has none, then by (7.5) u has a second child v', and once the subtree of v' is gone, Theorem A.3 removes v and v' together. (The two star–triangle transformations at v and v' are performed together.) In both cases u is then wired to A and to B. The root, once its subtree is gone, has two resistors, to A and to B, which the marked resistor joins, and Theorem A.1 removes it across AB.

(iii) Each move integrates the kernel of the smaller circuit over its ratios with a polynomial (not rational) factor. The currents of the larger circuit are linear in those of the smaller one with polynomial coefficients, and the series–parallel moves pass sources on with the weights r_1 and $1 - r_1$. A move removes one loop with two coordinates, or two loops with four, so the coordinates number $2n$. Expanding the exponential, every Taylor coefficient is a polynomial integral over products of intervals and simplices, a finite sum of (A.42). □

Example 7.7 (Figure 7.8(a))**.** For $\lambda_1 = \lambda_3 = 0$ and $\lambda_B = -\lambda_A - \lambda_2$, apply (A.13) at 3 and 1, then (A.11) at 2. The successive ordered edge lists are

$$\begin{aligned}
\Gamma_1 &= (12, 13, 1A, 2B, 3A, 3B, AB),\\
(\Gamma^{\mathrm{red}})_1 &= (1B, 12, 1A, 2B, AB),\\
(\Gamma^{\mathrm{red\,red}})_1 &= (2A, 2B, AB),\\
(\Gamma^{\mathrm{red\,red\,red}})_1 &= (AB).
\end{aligned}$$

Use r_1, r_2 for the first star–triangle move, r_3, r_4 for the second, and r_5, r_6 for the series–parallel move, all in $(0, 1)$. Their displacement substitutions and source transfer are

$$\begin{aligned}
z_{1B}^{\mathrm{red}} &= z_{13} + z_{3B}, & z_{1A}^{\mathrm{red}} &= r_1 z_{1A} + (1 - r_1)(z_{13} + z_{3A}),\\
z_{AB}^{\mathrm{red}} &= r_2 z_{AB} + (1 - r_2)(z_{3B} - z_{3A}), & z_{12}^{\mathrm{red}} &= z_{12}, \quad z_{2B}^{\mathrm{red}} = z_{2B},\\
z_{2A}^{\mathrm{red\,red}} &= z_{1A}^{\mathrm{red}} - z_{12}, & z_{2B}^{\mathrm{red\,red}} &= r_3 z_{2B} + (1 - r_3)(z_{1B}^{\mathrm{red}} - z_{12}),\\
z_{AB}^{\mathrm{red\,red}} &= r_4 z_{AB}^{\mathrm{red}} + (1 - r_4)(z_{1B}^{\mathrm{red}} - z_{1A}^{\mathrm{red}}),\\
z_{AB}^{\mathrm{red\,red\,red}} &= r_6 z_{AB}^{\mathrm{red\,red}} + (1 - r_6)(z_{2B}^{\mathrm{red\,red}} - z_{2A}^{\mathrm{red\,red}}),\\
\lambda_A^{\mathrm{red\,red\,red}} &= \lambda_A + r_5\lambda_2, & \lambda_B^{\mathrm{red\,red\,red}} &= \lambda_B + (1 - r_5)\lambda_2.
\end{aligned}$$

Derivatives act before these substitutions. With $z^{\mathrm{red}}_{BA} = -z^{\mathrm{red}}_{AB}$, the chain rule gives

$$-\frac{\partial}{\partial z^{\mathrm{red},+\dot a}_{1A}}\frac{\partial}{\partial z^{\mathrm{red},+}_{BA,\dot a}} = -\underbrace{\left(\frac{\partial}{\partial z^{\mathrm{red}\,\mathrm{red},+\dot a}_{2A}} - (1-r_4)\frac{\partial}{\partial z^{\mathrm{red}\,\mathrm{red},+\dot a}_{AB}}\right)}_{\text{displacement Jacobian}}\underbrace{\left(-r_4\frac{\partial}{\partial z^{\mathrm{red}\,\mathrm{red},+}_{AB,\dot a}}\right)}_{\text{displacement Jacobian}}$$
$$= r_4\frac{\partial}{\partial z^{\mathrm{red}\,\mathrm{red},+\dot a}_{2A}}\frac{\partial}{\partial z^{\mathrm{red}\,\mathrm{red},+}_{AB,\dot a}}.$$

$$\mathcal{I}^{\text{raw cut}}_{\Gamma}(\lambda;z) = -\int_{(0,1)^2} dr_1\,dr_2\,r_1r_2\,\frac{\partial}{\partial z^{\mathrm{red},+\dot a}_{1A}}\frac{\partial}{\partial z^{\mathrm{red},+}_{BA,\dot a}}\underbrace{\mathcal{I}^{\text{raw cut}}_{\Gamma^{\mathrm{red}}}(\lambda;z^{\mathrm{red}})}_{\text{two loops}}$$
$$= -\int_{(0,1)^4}\prod_{j=1}^{4} dr_j\,r_1r_2r_3r_4^2$$
$$\times\frac{\partial}{\partial z^{\mathrm{red}\,\mathrm{red},+\dot a}_{2A}}\frac{\partial}{\partial z^{\mathrm{red}\,\mathrm{red},+}_{AB,\dot a}}\frac{\partial}{\partial z^{\mathrm{red}\,\mathrm{red},+\dot c}_{2B}}\frac{\partial}{\partial z^{\mathrm{red}\,\mathrm{red},+}_{AB,\dot c}}$$
$$\times\underbrace{\mathcal{I}^{\text{raw cut}}_{\Gamma^{\mathrm{red}\,\mathrm{red}}}(\lambda;z^{\mathrm{red}\,\mathrm{red}})}_{\text{one loop}}$$
$$= -(\lambda_{2,+\dot a}\lambda_{A,+}{}^{\dot a})^3\int_{(0,1)^6}\prod_{j=1}^{6} dr_j\,r_1r_2r_3r_4^2r_5(1-r_5)r_6^3$$
$$\times e^{-\lambda_{2,+\dot b}\left[r_5 z^{\mathrm{red}\,\mathrm{red}}_{2A}+(1-r_5)z^{\mathrm{red}\,\mathrm{red}}_{2B}\right]^{+\dot b}}\underbrace{e^{-(\lambda_A+r_5\lambda_2)_{+\dot c}z^{\mathrm{red}\,\mathrm{red}\,\mathrm{red},+\dot c}_{AB}}}_{\text{kernel of the final resistor}}.$$

$$\mathcal{I}^{\text{raw cut}}_{\Gamma}(\lambda;0) = -(\lambda_{2,+\dot a}\lambda_{A,+}{}^{\dot a})^3\left(\frac{1}{2}\right)^3\frac{1}{3}\frac{1}{6}\frac{1}{4}$$
$$= -\frac{1}{576}(\lambda_{2,+\dot a}\lambda_{A,+}{}^{\dot a})^3.$$

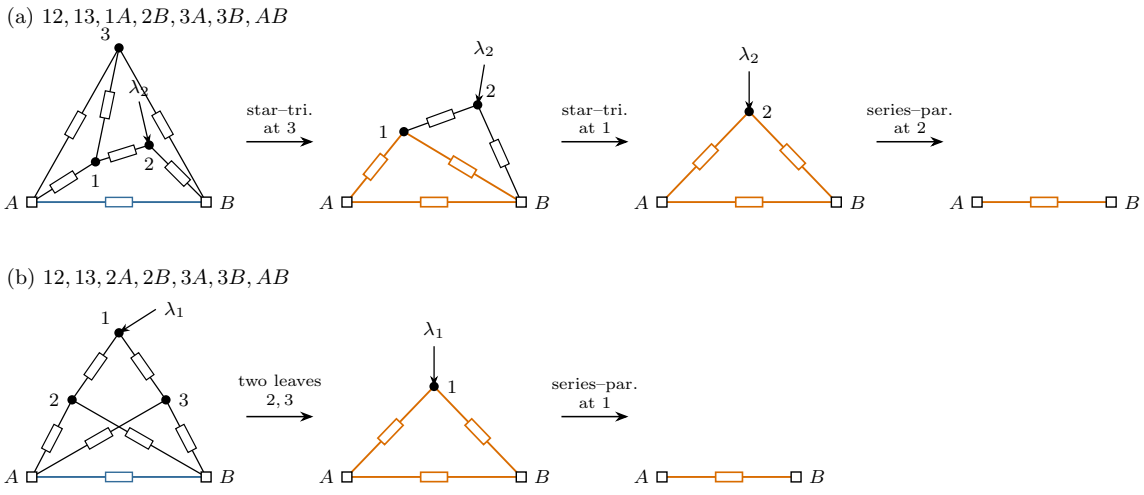


Figure 7.8: The three-loop graphs not of type I′, up to the exchange of A and B, reduced as in Proposition 7.6(ii); orange: the resistors created or merged by the preceding move, blue: the marked resistor. (a) Star–triangle moves (Theorem A.2) at 3 and at 1, then a series–parallel move (Theorem A.1) at the root 2. (b) The leaves $2, 3$ hang from the root 1, which has no spoke, and are removed together (Theorem A.3); the root is removed last.

Proposition 7.8 (Trees with a superpotential vertex)**.** *Let a tree T_j contain one superpotential vertex, with $\deg_\Gamma \bar W$ resistors and $\deg \bar W$ fields, its other vertices being cubic. Then T_j has*

$\deg_\Gamma \bar{W} - 2$ *cubic vertices of degree two, each carrying a source, and the superpotential vertex carries a source when* $\deg \bar{W} > \deg_\Gamma \bar{W}$. *Series–parallel moves pass every source on with the weights* r_1 *and* $1 - r_1$, *so graphs of type I′ keep polynomial currents for any placement of the sources. A star–triangle move at a vertex with a source distributes it with the rational weights* (A.44), *and the currents become rational functions of the coordinates.*

Proof. The counts are (A.43): the terminals of the cubic nodes and of the superpotential node, minus the ends of the resistors, leave $\deg \bar{W} - 2$ free terminals, $\deg \bar{W} - \deg_\Gamma \bar{W}$ of them at the superpotential node and one at each cubic node with two resistors. Theorem A.1 holds for any source at the removed node, with the polynomial weights r_1 and $1 - r_1$. At a star with source λ_v, the source divides among the three outer nodes in proportion to the opposite sides of the triangle, (A.44), and these ratios are rational in the coordinates. □

Example 7.9 (Sourced star–triangle moves)**.** In the graph of Figure 7.8(b), let 2 be a superpotential vertex with $\deg_\Gamma \bar{W} = 3 < \deg \bar{W}$, so that 1, 2, A and B carry sources. The neighbours $2, 3$ of the degree-two vertex 1 are not joined, and the star–triangle move at the only source-free vertex, 3, merges only the side AB. A series step at 1 and this move each leave the complete circuit on four nodes, all of them with sources, so every elimination order needs a star–triangle move at a vertex with a source.

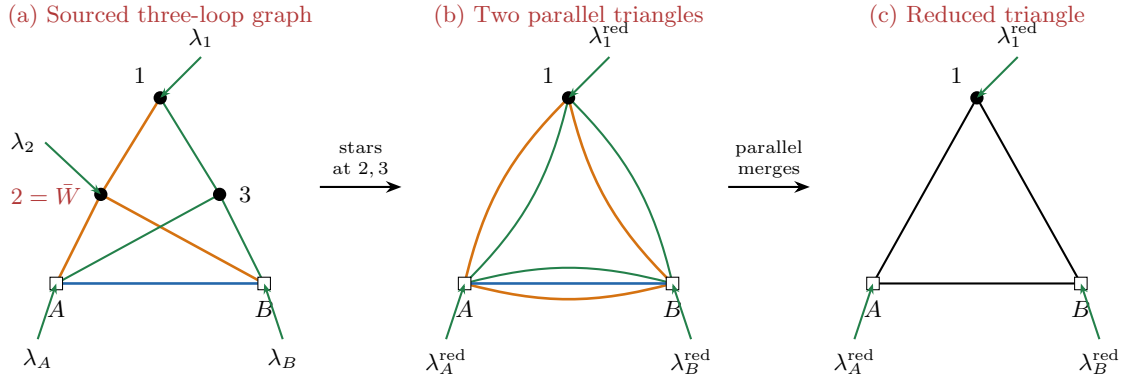


Figure 7.9: The three-loop sourced example, with 2 carrying $\bar{W}$ and $\lambda_2 \neq 0$. Removing the stars at 2 (orange) and 3 (green, $\lambda_3 = 0$) gives two triangles in parallel with the original marked edge AB (blue). Parallel merging leaves the triangle on $1, A, B$ with sources $\lambda_1^{\mathrm{red}}, \lambda_A^{\mathrm{red}}, \lambda_B^{\mathrm{red}}$.

Orient and order the edges as $(21, 2A, 2B, 31, 3A, 3B, AB)$, with $\lambda_3 = 0$. Apply the two star–triangle changes of Theorem A.3, retaining the source terms (A.44) at 2, and merge the parallel sides. The reduced triangle has

$$t_{1A}^{\mathrm{red}} = 1 - r_5 - r_6, \qquad t_{1B}^{\mathrm{red}} = r_5, \qquad t_{AB}^{\mathrm{red}} = r_6,$$
$$0 < r_1, r_2 < 1, \qquad r_3, r_4 > 0, \quad r_3 + r_4 < 1, \qquad r_5, r_6 > 0, \quad r_5 + r_6 < 1,$$
$$\Sigma = \frac{1 - r_5 - r_6}{r_1} + \frac{r_5}{r_2} + \frac{r_6}{r_3}.$$

$$
\begin{aligned}
\lambda_1^{\mathrm{red}} &= \lambda_1 + \frac{r_6/r_3}{\Sigma}\lambda_2,\\
\lambda_A^{\mathrm{red}} &= \lambda_A + \frac{r_5/r_2}{\Sigma}\lambda_2,\\
\lambda_B^{\mathrm{red}} &= \lambda_B + \frac{(1-r_5-r_6)/r_1}{\Sigma}\lambda_2,\\
y_{1A}^{\mathrm{red}} &= r_5\lambda_1^{\mathrm{red}} - r_6\lambda_A^{\mathrm{red}},\\
y_{1B}^{\mathrm{red}} &= \lambda_1^{\mathrm{red}} - y_{1A}^{\mathrm{red}},\\
y_{AB}^{\mathrm{red}} &= \lambda_A^{\mathrm{red}} + y_{1A}^{\mathrm{red}}.
\end{aligned}
$$

$$
\begin{aligned}
y_{21} &= \frac{r_6/r_3}{\Sigma}\lambda_2 - r_1 y_{1A}^{\mathrm{red}} - r_2 y_{1B}^{\mathrm{red}},\\
y_{31} &= -(1-r_1)y_{1A}^{\mathrm{red}} - (1-r_2)y_{1B}^{\mathrm{red}},\\
y_{2A} &= \frac{r_5/r_2}{\Sigma}\lambda_2 + r_1 y_{1A}^{\mathrm{red}} - r_3 y_{AB}^{\mathrm{red}},\\
y_{3A} &= (1-r_1)y_{1A}^{\mathrm{red}} - r_4 y_{AB}^{\mathrm{red}},\\
y_{2B} &= \frac{(1-r_5-r_6)/r_1}{\Sigma}\lambda_2 + r_2 y_{1B}^{\mathrm{red}} + r_3 y_{AB}^{\mathrm{red}},\\
y_{3B} &= (1-r_2)y_{1B}^{\mathrm{red}} + r_4 y_{AB}^{\mathrm{red}},\\
y_{AB} &= (1-r_3-r_4)y_{AB}^{\mathrm{red}}.
\end{aligned}
$$

$$
\begin{aligned}
z_{1A}^{\mathrm{red}} &= r_1(z_{2A}-z_{21}) + (1-r_1)(z_{3A}-z_{31}),\\
z_{1B}^{\mathrm{red}} &= r_2(z_{2B}-z_{21}) + (1-r_2)(z_{3B}-z_{31}),\\
z_{AB}^{\mathrm{red}} &= (1-r_3-r_4)z_{AB} + r_3(z_{2B}-z_{2A}) + r_4(z_{3B}-z_{3A}),
\end{aligned}
$$

$$
\begin{aligned}
\sum_{\substack{e\in\Gamma_1\\ \text{edges}}} y_{e,+\dot{a}} z_e^{+\dot{a}} &= \lambda_{2,+\dot{a}}\left(\frac{r_6 z_{21}/r_3 + r_5 z_{2A}/r_2 + (1-r_5-r_6)z_{2B}/r_1}{\Sigma}\right)^{+\dot{a}}\\
&\quad + y_{1A,+\dot{a}}^{\mathrm{red}} z_{1A}^{\mathrm{red},+\dot{a}} + y_{1B,+\dot{a}}^{\mathrm{red}} z_{1B}^{\mathrm{red},+\dot{a}} + y_{AB,+\dot{a}}^{\mathrm{red}} z_{AB}^{\mathrm{red},+\dot{a}}.
\end{aligned}
$$

Integrate over the displayed domain, at fixed original sources. The derivatives include the

r-dependence of the transferred sources:

$$
\begin{aligned}
\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda;z) &= \int (-dy_{AB,++^{\cdot}} \wedge dy_{AB,+^{\cdot}-}) \wedge (-dy_{2B,++^{\cdot}} \wedge dy_{2B,+^{\cdot}-}) \\
&\qquad \wedge (-dy_{3A,++^{\cdot}} \wedge dy_{3A,+^{\cdot}-}) \exp\left[- \sum_{\substack{e\in\Gamma_1 \\ \text{edges}}} y_{e,+\dot a} z_e^{+\dot a} \right] \\
&= - \int \prod_{j=1}^{6} dr_j \; \underbrace{\det \frac{\partial(y_{AB,++^{\cdot}}, y_{AB,+^{\cdot}-}, y_{2B,++^{\cdot}}, y_{2B,+^{\cdot}-}, y_{3A,++^{\cdot}}, y_{3A,+^{\cdot}-})}{\partial(r_1,r_2,r_3,r_4,r_5,r_6)}}_{\text{Jacobian of the three loop currents}} \\
&\qquad \times \exp\left[- \sum_{\substack{e\in\Gamma_1 \\ \text{edges}}} y_{e,+\dot a}(r) z_e^{+\dot a} \right] \\
&= \int \prod_{j=1}^{6} dr_j \; r_5(1-r_3-r_4)(1-r_5-r_6) \qquad \underbrace{(\lambda^{\text{red}}_{1,+\dot a} \lambda^{\text{red}\;\dot a}_{A,+})}_{\text{reduced triangle current-area factor}} \\
&\qquad \times \left[\lambda^{\text{red}}_{1,+\dot b} \lambda^{\text{red}\;\dot b}_{A,+} + \frac{\lambda_{2,+\dot b} y^{\text{red}\;\dot b}_{AB,+}}{r_1 r_2 \Sigma} \right] \\
&\qquad \times \left[(r_1-r_2)\lambda_{1,+\dot c}\lambda_{A,+}{}^{\dot c} - \frac{(r_1-r_2)r_5(r_2-r_3)+r_2(r_2-1)}{r_2 r_3 \Sigma} \lambda_{1,+\dot c}\lambda_{2,+}{}^{\dot c} \right] \\
&\qquad \times \exp\left[- \sum_{\substack{e\in\Gamma_1 \\ \text{edges}}} y_{e,+\dot d}(r) z_e^{+\dot d} \right] .
\end{aligned}
$$

Normalizing the three sides by their sum makes the source fractions in (A.44) linear, but the subsequent parallel merges reintroduce denominators. In the coordinates above, take $r_2 = r_3 = r_4 = r_5 = r_6 = \frac{1}{4}$ and the source components $\lambda_{1,++^{\cdot}} = \lambda_{2,++^{\cdot}} = \lambda_{A,+^{\cdot}-} = 1$, $\lambda_{1,+^{\cdot}-} = \lambda_{2,+^{\cdot}-} = \lambda_{A,++^{\cdot}} = 0$. Then

$$
\Sigma = \frac{1}{2r_1} + 2, \qquad \frac{r_6/r_3}{\Sigma} = \frac{2r_1}{1+4r_1},
$$

$$
- \det \frac{\partial(y_{AB,++^{\cdot}}, y_{AB,+^{\cdot}-}, y_{2B,++^{\cdot}}, y_{2B,+^{\cdot}-}, y_{3A,++^{\cdot}}, y_{3A,+^{\cdot}-})}{\partial(r_1,r_2,r_3,r_4,r_5,r_6)}\bigg|_{\substack{r_2=\cdots=r_6=1/4 \\ \text{displayed sources}}} = \frac{(4r_1-1)(6r_1+1)(6r_1+7)}{64(4r_1+1)^2}.
$$

The denominator does not cancel at zero displacement in these coordinates. The full integral is nevertheless the six-term polynomial (A.47); Appendix A.6 evaluates it by the graph identity and the reductions of B.3:

$$
\begin{aligned}
&\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda;0)\Big|_{\lambda_3=0} \\
&= - \int dr_1\, dr_2\, dr_3\, dr_4\, dr_5\, dr_6 \; \det \frac{\partial(y_{AB,++^{\cdot}}, y_{AB,+^{\cdot}-}, y_{2B,++^{\cdot}}, y_{2B,+^{\cdot}-}, y_{3A,++^{\cdot}}, y_{3A,+^{\cdot}-})}{\partial(r_1,r_2,r_3,r_4,r_5,r_6)} \\
&= \frac{\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a}}{576} \Big[5(\lambda_{1,+\dot b}\lambda_{A,+}{}^{\dot b})^2 + 12(\lambda_{1,+\dot b}\lambda_{A,+}{}^{\dot b})(\lambda_{2,+\dot c}\lambda_{A,+}{}^{\dot c}) \\
&\qquad - (\lambda_{1,+\dot b}\lambda_{2,+}{}^{\dot b})(\lambda_{1,+\dot c}\lambda_{A,+}{}^{\dot c}) + 3(\lambda_{2,+\dot b}\lambda_{A,+}{}^{\dot b})^2 \\
&\qquad + 3(\lambda_{1,+\dot b}\lambda_{2,+}{}^{\dot b})(\lambda_{2,+\dot c}\lambda_{A,+}{}^{\dot c}) - (\lambda_{1,+\dot b}\lambda_{2,+}{}^{\dot b})^2 \Big].
\end{aligned}
$$

Table 7.1 counts these graphs through seven loops.

Table 7.1: Product graphs whose trees have at most one superpotential node, all other nodes cubic, up to isomorphism fixing A, B. Columns 3 and 6: superpotential node with at least four resistors; last column: those of column 6 reduced without a sourced star–triangle step.

n	graphs	$\bar{W}$ node with	type I′	spiders	$\bar{W}$ node with ≥ 4	reduced without a
		≥ 4 resistors			resistors, not type I′	sourced star–triangle
3	11	3	8	8	0	0
4	46	23	23	19	10	4
5	200	135	59	37	92	29
6	874	680	152	71	557	144
7	3765	3189	370	125	2864	610

8 Quantum supercharges and $\mathcal{N} = 4$ Yang–Mills theory

Combining the graph kernels with the BV pairing gives the quantum supercharge. For a general interaction, let W_Γ denote the ordered fibre contractions, including their Koszul signs. Restoring our scalar normalization (3.19) gives

$$Q_n\mathcal{O} = \frac{(-1)^n}{(n+1)!\,(8\pi^2)^n} \sum_{\Gamma \text{ labelled, rooted, Laman}} \Big[\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda; z) W_\Gamma[\mathcal{I}^{\otimes(n+1)}, \mathcal{O}]\Big]\Big|_{\text{jets, } z_{v,i}=0}. \tag{8.1}$$

At $n = 0$ the single-edge graph gives $Q_0\mathcal{O} = \{\mathcal{I}, \mathcal{O}\}_0$, which is the tree-level (classical) SUSY transformation.

The n-loop Q_n increases the number of derivatives by $2n$ (and by more if $\mathcal{I}$ contains derivatives):

Under $\lambda_{v,+\dot{a}} \to c\,\lambda_{v,+\dot{a}}$ and $z_e^{+\dot{a}} \to z_e^{+\dot{a}}/c$ at fixed $\dot{a}$, the displacement exponential in (3.28) is invariant and the current form, with n factors $dy_{e,+\dot{a}}$, scales as c^n. Since λ_v differentiates output fields and each z_e-derivative takes a derivative from an input letter, Q_n raises the numbers of ∂_{++} and ∂_{+-} derivatives by n each, $2n$ in total. (A similar analysis holds in all dimensions.)

An operator with d_+ and d_- holomorphic derivatives can be Q_n-exact (and exact under the successive differential d_n of Section 9) only if $n \leq \min(d_+, d_-)$, so finitely many orders decide whether it becomes quantum exact.

In this section we present the main result: the loop corrections to the supercharge in $\mathcal{N} = 4$ SYM.

In $\mathcal{N} = 4$ SYM, all the holomorphic fields can be organized into one superfield C [4, Equation (3.52)].

The functional derivatives we defined at displaced points are normalized by

$$\frac{\delta C^B(y, \theta')}{\delta C^A(y', \theta)} = \delta_A{}^B\, \delta^{(4)}(y - y') \prod_{I=1}^{3} (\theta_I - \theta'_I), \tag{8.2}$$

8.1 The all-order anomalous SUSY transformation

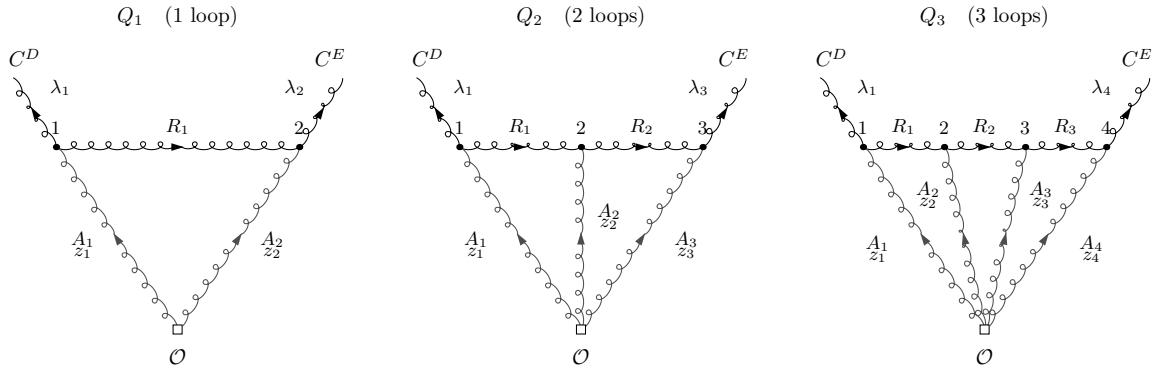


Figure 8.1: One-, two- and three-loop supercharge fans in $\mathcal{N}=4$ Yang–Mills theory. The spoke labelled A_j, z_j contracts an input $C^{A_j}(z_j,\theta)$ at $\mathcal{O}$. The path contracts the colour indices $R_1,\dots,R_n$ and leaves the endpoint fields C^D, C^E, carrying λ_1, λ_{n+1}.

Theorem 8.1. *For the superfield action* (A.3) *and an operator at x, the n-loop supercharge for $n \geq 1$ is*

$$\begin{aligned} Q_n = \frac{(-1)^n i^{n+1}}{2(32\pi^2)^n} \int_{(5.6)} dr_1 \cdots dr_{2n} \left[\prod_{j=1}^{n-1} r_{2j-1}^{n-j} r_{2j}(1-r_{2j})^{j-1}\right] r_{2n}^{n-1} \int d^3\theta \int \prod_{j=1}^{n+1} d^4 z_j \\ \times C_{A_1 D R_1} \left(\prod_{j=2}^{n} C_{A_j R_{j-1} R_j}\right) C_{A_{n+1} R_n E} \\ \times (\partial_{+\dot a_1} \cdots \partial_{+\dot a_n} C^D)(x + x_1(r,z), \theta)(\partial_+{}^{\dot a_1} \cdots \partial_+{}^{\dot a_n} C^E)(x + x_{n+1}(r,z), \theta) \\ \times \frac{\delta}{\delta C^{A_1}(x+z_1,\theta)} \cdots \frac{\delta}{\delta C^{A_{n+1}}(x+z_{n+1},\theta)}, \end{aligned} \tag{8.3}$$

where the output fields are differentiated before they are evaluated at the translated positions.

$$\begin{aligned} x_1^{+\dot a}(r,z) = \sum_{j=1}^{n} (1-r_{2j-1}) \left(\prod_{k=1}^{j-1} r_{2k-1}\right) \\ \times \left[\left(\prod_{k=1}^{j-1}(1-r_{2k})\right) z_1^{+\dot a} + \sum_{i=2}^{j} r_{2i-2} \left(\prod_{k=i}^{j-1}(1-r_{2k})\right) z_i^{+\dot a}\right] \\ + \left(\prod_{k=1}^{n} r_{2k-1}\right) z_{n+1}^{+\dot a}, \\ x_{n+1}^{+\dot a}(r,z) = r_{2n}\left[\left(\prod_{k=1}^{n-1}(1-r_{2k})\right) z_1^{+\dot a} + \sum_{i=2}^{n} r_{2i-2} \left(\prod_{k=i}^{n-1}(1-r_{2k})\right) z_i^{+\dot a}\right] \\ + (1-r_{2n}) z_{n+1}^{+\dot a}. \end{aligned}$$

The (fan) kernel and the complete Taylor expansion in the z_j are derived and listed in Section 5.

Proof.

The cubic term of (A.3) gives the vertex iC_{ABC}, which includes the 3! assignments of its three legs, and the rooted fan has symmetry factor 1/2 for the reversal of its interaction path.

The functional derivatives in (8.3) attach the spokes to the input letters; only the two endpoint fields remain uncontracted, so $\lambda_2 = \cdots = \lambda_n = 0$.

In the endpoint kernel (5.10), the substitution (2.32) gives, with $\partial_{1,+\dot a}$ and $\partial_{n+1,+\dot a}$ acting on the first and second output fields, respectively,

$$
\begin{aligned}
(\lambda_{1,+\dot a}\lambda_{n+1,+}{}^{\dot a})^n &\longmapsto 4^{-n}\partial_{1,+\dot a_1}\cdots\partial_{1,+\dot a_n}\partial_{n+1,+}{}^{\dot a_1}\cdots\partial_{n+1,+}{}^{\dot a_n},\\
e^{\lambda_{1,+\dot a}x_1^{+\dot a}(r,z)+\lambda_{n+1,+\dot a}x_{n+1}^{+\dot a}(r,z)} &\longmapsto e^{-\frac12 x_1^{+\dot a}(r,z)\partial_{1,+\dot a}-\frac12 x_{n+1}^{+\dot a}(r,z)\partial_{n+1,+\dot a}}.
\end{aligned}
$$

$$
\begin{aligned}
&e^{-\frac12 x_1^{+\dot a}(r,z)\partial_{1,+\dot a}-\frac12 x_{n+1}^{+\dot a}(r,z)\partial_{n+1,+\dot a}}C^D(x,\theta)C^E(x,\theta)\\
&\quad= C^D(x+x_1(r,z),\theta)C^E(x+x_{n+1}(r,z),\theta), \qquad \frac{\partial}{\partial x^{+\dot a}} = -\frac12\partial_{+\dot a}.\\
&\underbrace{\frac12}_{\text{path reversal}}\ \underbrace{i^{n+1}}_{\text{cubic vertices}}\ \underbrace{(-8\pi^2)^{-n}}_{\text{scalar-kernel normalization}}\ \underbrace{4^{-n}}_{\text{output derivatives}} = \frac{(-1)^n i^{n+1}}{2(32\pi^2)^n}.
\end{aligned}
$$

The path contracts the colour indices in the order $C_{A_1DR_1}(\prod_{j=2}^n C_{A_jR_{j-1}R_j})C_{A_{n+1}R_nE}$; inserting these factors and the density (5.10) gives (8.3). □

In practice, the colour factor can be simplified further with the trace relations of $SU(N)$ and of the other gauge groups [14].

For the zero-jet sector:

$$
\begin{aligned}
Q_n\big|_{\text{input jets}=0} = \frac{(-1)^n i^{n+1}}{2(32\pi^2)^n (n!)^3 (n+1)} \int d^3\theta\, C_{A_1DR_1}\left(\prod_{j=2}^n C_{A_jR_{j-1}R_j}\right) C_{A_{n+1}R_nE}\\
\times (\partial_{+\dot a_1}\cdots\partial_{+\dot a_n}C^D)(\partial_+{}^{\dot a_1}\cdots\partial_+{}^{\dot a_n}C^E)\prod_{j=1}^{n+1}\frac{\delta}{\delta C^{A_j}(\theta)}.
\end{aligned}
\tag{8.4}
$$

8.2 One- and two-loop examples in $\mathcal{N}=4$ Yang–Mills theory

The general-n operator (8.3) can look abstract. We also display the two-loop result, a problem that has recently attracted attention [3, Section 3.2.3].

$$
\begin{aligned}
Q_2 = &-\frac{i}{2(32\pi^2)^2}\int_0^1 dr_1\int_0^1 dr_2\int_0^1 dr_4\int_0^{1-r_4} dr_3\, r_1r_2r_4\int d^3\theta\int\prod_{j=1}^3 d^4z_j\\
&\times C_{A_1DR}C_{A_2RS}C_{A_3SE}\,(\partial_{+\dot a}\partial_{+\dot b}C^D)(x+x_1(r,z),\theta)(\partial_+{}^{\dot a}\partial_+{}^{\dot b}C^E)(x+x_3(r,z),\theta)\\
&\times\frac{\delta}{\delta C^{A_1}(x+z_1,\theta)}\frac{\delta}{\delta C^{A_2}(x+z_2,\theta)}\frac{\delta}{\delta C^{A_3}(x+z_3,\theta)},
\end{aligned}
\tag{8.5}
$$

where

$$
\begin{aligned}
x_1^{+\dot a}(r,z) &= (1-r_1)z_1^{+\dot a} + r_1(1-r_3)\big[(1-r_2)z_1^{+\dot a}+r_2z_2^{+\dot a}\big]+r_1r_3z_3^{+\dot a},\\
x_3^{+\dot a}(r,z) &= r_4\big[(1-r_2)z_1^{+\dot a}+r_2z_2^{+\dot a}\big]+(1-r_4)z_3^{+\dot a}.
\end{aligned}
$$

This result extends previous results and agrees with them.

For example, the first two components below agree with [4, Equations (3.66) and (3.67)] up to normalization:[22]

[22] On the physical Lagrangian subspace, with $\delta^{AB}(\gamma^I)_B \leftrightarrow \bar\phi_I{}^A$, $(\beta_I)^A \leftrightarrow -\frac{i}{\sqrt2}\psi_+^{IA}$, $\delta^{AB}\mathsf{b}_B \leftrightarrow \frac14 f_{++}^A$, $\partial_{+\dot a}\mathsf{c}^A \leftrightarrow 2\bar\lambda_{\dot a}^A$ and $\partial_{+\dot a}\partial_{+\dot b}\mathsf{c}^A \leftrightarrow 2\nabla_{+(\dot a}\bar\lambda_{\dot b)}^A$, equation (8.6) becomes $Q_1(\bar\phi_1{}^A\psi_+^{1B}) = \frac{i\sqrt2\,g^2}{16\pi^2}C^A{}_{RD}C^{BR}{}_E\,\bar\lambda_{\dot a}^D\bar\lambda^{E\dot a}$, the component form of the Konishi anomaly [7].

In component form:

$$Q_1(\delta^{AA'}(\gamma^1)_{A'}(\beta_1)^B) = \frac{g^2}{64\pi^2} C_{ARD}C_{BRE}(\partial_{+\dot a}\mathsf{c}^D)(\partial_+{}^{\dot a}\mathsf{c}^E), \tag{8.6}$$

$$Q_1((\beta_1)^A(\beta_2)^B) = \frac{ig}{64\pi^2}(C_{ARD}C_{BRE} + C_{BRD}C_{ARE})(\partial_{+\dot a}\mathsf{c}^D)\,\delta^{EF}\partial_+{}^{\dot a}(\gamma^3)_F, \tag{8.7}$$

$$Q_2((\beta_1)^{A_1}(\beta_2)^{A_2}(\beta_3)^{A_3}) = -\frac{ig^2}{48(32\pi^2)^2}\sum_{\sigma\in S_3} C_{A_{\sigma(1)}DR}C_{A_{\sigma(2)}RS}C_{A_{\sigma(3)}SE}(\partial_{+\dot a}\partial_{+\dot b}\mathsf{c}^D)(\partial_+{}^{\dot a}\partial_+{}^{\dot b}\mathsf{c}^E). \tag{8.8}$$

9 Multi-graviton completions and obstruction equations

Quantum multiplication constructs perturbatively closed multi-graviton operators in $\mathcal{N}=4$ Yang–Mills theory. We prove their closure at arbitrary loop order (Theorem 9.1) and, for illustration, evaluate an explicit product (Section 9.2).[23]

We then review the framework that uses the operators Q_n to test which classical classes are lifted: the successive transfers of Section 9.3 compute the perturbative obstructions of classical representatives and decide which of their closed completions are quantum coboundaries. The algorithm and the explicit results are in the supplementary material [14].

Since every graviton class has a closed completion, no differential maps one graviton class to another (Section 9.3); a lift pairs a fortuitous class with a graviton class or with another fortuitous class. This makes the quantum-correction test much lighter than a brute-force sweep. Quantum closure of a classical representative $\mathcal{O}_0$ means that there exist finite corrections $\mathcal{O}_1, \mathcal{O}_2, \dots$ with $\mathbf{Q}(\mathcal{O}_0 + \hbar\mathcal{O}_1 + \hbar^2\mathcal{O}_2 + \cdots) = 0$ at every order in $\hbar$; we remind the reader that $\mathcal{O}_0$ itself is in general not annihilated by $\mathbf{Q}$ (Section 9.3). Nonperturbative corrections can also modify the cohomology.

9.1 Perturbative closure of multi-gravitons

The single-graviton representatives of Chang and Yin [1, Section 4] are components of $\mathrm{Tr}(dC)^p$, where

$$dC = dx^{+\dot a}\frac{\partial C}{\partial x^{+\dot a}} + d\theta_I\frac{\partial_L C}{\partial\theta_I}. \tag{9.1}$$

Theorem 9.1. *Let $G_1, \dots, G_m$ be single-graviton representatives. Every iterated quantum product $\mathcal{O}$ of these operators, with any fixed bracketing, satisfies*

$$\mathbf{Q}\mathcal{O} = 0, \qquad \mathcal{O}|_{\hbar=0} = G_1\cdots G_m. \tag{9.2}$$

For each fixed component, its perturbative completion has finitely many terms.

Proof. The classical differential acts on a single graviton by

$$Q_0(dC) = [C, dC], \qquad Q_0\,\mathrm{Tr}(dC)^p = \mathrm{Tr}[C, (dC)^p] = 0. \tag{9.3}$$

Let us first consider a single graviton G.

[23] The theorem proves that multi-graviton classes stay closed; however, a closed multi-graviton can still become exact, e.g., if there is a nonclosed fortuitous class with the allowed charges. The most direct one-loop example is the lifting of a black-hole class by the Konishi correction: in the $SO(7)$ theory, the one-loop supercharge pairs a multi-graviton with a fortuitous class [2, Section 3], and by the theorem the multi-graviton is the exact member of the pair. Section 10.2 and [14, supplement, Section S3] give the one-loop lifts in $SO(2N+1)$ and their relation to S-duality. There are many more one-loop lifts in $SO(2N+1)$ than in $SU(N)$.

At order $n \geq 1$, the auxiliary θ part of the Q_n kernel reads:

$$\prod_{j=1}^{n} \prod_{I=1}^{3} (\theta_I^{(j)} - \theta_I^{(j+1)}). \tag{9.4}$$

A nonzero coincident value requires at least $3n$ odd derivatives. Each of the $n+1$ contracted factors of dC supplies at most one, leaving degree $3n - (n+1) = 2n - 1 > 0$ in the relative odd coordinates. The kernel therefore vanishes when those coordinates coincide, so all single gravitons are $\mathbf{Q}$-closed at every perturbative order.

$$Q_0 G = 0, \qquad Q_n G = 0 \quad (n \geq 1). \tag{9.5}$$

Applying (6.13) at each binary vertex of a quantum product gives

$$\mathbf{Q}(A \star B) = (\mathbf{Q}A) \star B + (-1)^{|A|} A \star (\mathbf{Q}B) = 0 \qquad (\mathbf{Q}A = \mathbf{Q}B = 0). \tag{9.6}$$

Induction proves the first assertion. The leading term of each binary product is ordinary multiplication, giving the stated classical limit [1, Section 5].

Finiteness follows from the finiteness corollary in Section 6.3. □

For example, for two single gravitons G_1 and G_2, the explicit quantum product shows that $G_1 G_2$ admits a quantum completion (the quantum completion is always nonunique, which is exactly the renormalization freedom), namely $G_1 \star G_2$; equivalently, $G_1 G_2$ is not lifted by a failure of $\mathbf{Q}$-closure.

The graviton part of the cohomology. At finite rank the classical cohomology consists of a graviton part, spanned by multi-graviton classes, and the remaining fortuitous classes [19]. Theorem 9.1 supplies every class of the graviton part with an explicit closed completion: write its classical representative as a polynomial in single-graviton representatives and replace each product by a quantum product. By Theorem 6.4, the resulting class is independent of the bracketing and of the order of the factors. (With more auxiliary pairs one can also define quantum products of three or more factors compatible with $\mathbf{Q}$; they differ from the iterated products $(G_1 \star G_2) \star G_3$ and $G_1 \star (G_2 \star G_3)$ by $\mathbf{Q}$-exact terms.) The same construction applies to a graviton-dressed fortuitous class $G\,\mathcal{O}_0$: once the fortuitous core $\mathcal{O}_0$ has a closed completion $\mathcal{O}[\mathcal{O}_0]$, the product $G \star \mathcal{O}[\mathcal{O}_0]$ is closed.

9.2 An explicit example

Take the two single-graviton representatives

$$A = \delta^{XY} (\gamma^1)_X (\gamma^1)_Y, \qquad B = (\gamma^2)_X (\beta_1)^X, \qquad \mathcal{O}_0 = AB. \tag{9.7}$$

Equation (A.2), with left differentiation, identifies both operators as components of $\mathrm{Tr}(dC)^2$:

$$\begin{aligned} \frac{\partial_L C^A}{\partial \theta_2}\bigg|_{\theta_1=\theta_2=0} &= \delta^{AX} (\gamma^2)_X - \theta_3 (\beta_1)^A, \\ \delta_{AB} \frac{\partial_L C^A}{\partial \theta_2} \frac{\partial_L C^B}{\partial \theta_2}\bigg|_{\theta_1=\theta_2=0} &= \delta^{XY} (\gamma^2)_X (\gamma^2)_Y - 2\theta_3 (\gamma^2)_X (\beta_1)^X. \end{aligned} \tag{9.8}$$

$$\begin{aligned} A &= \delta_{AB} \frac{\partial_L C^A}{\partial \theta_1} \frac{\partial_L C^B}{\partial \theta_1}\bigg|_{\theta=0}, \\ B &= -\frac{1}{2} \frac{\partial_L}{\partial \theta_3} \left(\delta_{AB} \frac{\partial_L C^A}{\partial \theta_2} \frac{\partial_L C^B}{\partial \theta_2} \right)\bigg|_{\theta=0}. \end{aligned} \tag{9.9}$$

Gauge invariance gives $Q_0 A = 0$. For B, the superpotential term gives

$$Q_0 B = -iC^{XYZ}(\gamma^2)_X(\gamma^2)_Y(\gamma^3)_Z = 0. \tag{9.10}$$

Both A and B are automatically Q_1-closed by Theorem 9.1.

Using (8.3), $\mathcal{O}_0$ itself is not Q_1-closed:

$$Q_1\mathcal{O}_0 = \frac{g^2}{32\pi^2} C^X{}_{RD} C^{YR}{}_E (\gamma^1)_X(\gamma^2)_Y(\partial_{+\dot{a}}\mathsf{c}^D)(\partial_+{}^{\dot{a}}\mathsf{c}^E). \tag{9.11}$$

Without the quantum product, whether $\mathcal{O}_0$ is lifted needs to be examined: one has to implement an algorithm to test whether the right-hand side of (9.11) is Q_0-exact. If the answer is yes, RHS $= Q_0\mathcal{O}'$, then $\mathcal{O}_0 - \hbar\mathcal{O}'$ is closed through one loop and the class survives at one loop (it is not **Q**-exact since it has no derivatives); if the answer is no, $\mathcal{O}_0$ is lifted together with another member of the Q_0-cohomology.

However, with the quantum product, we can explicitly answer the question: $\mathcal{O}_0$ is not lifted.

Taking $\mathcal{O}_1 = [\hbar](A \star B)$, we obtain

$$\begin{aligned}
\mathcal{O}_1 &= \{\mathcal{I}, A, B\}^{\gamma,0} \\
&= \frac{i}{8\pi^2} 2(ig)(-1)\frac{1}{2}\frac{1}{4} C_D{}^{XY}(\partial_{+\dot{a}}\mathsf{c}^D)(\partial_+{}^{\dot{a}}(\gamma^1)_X)(\gamma^2)_Y \\
&= \frac{g}{32\pi^2} C_D{}^{XY}(\partial_{+\dot{a}}\mathsf{c}^D)(\partial_+{}^{\dot{a}}(\gamma^1)_X)(\gamma^2)_Y.
\end{aligned} \tag{9.12}$$

Let us verify that $Q_0\mathcal{O}_1 + Q_1\mathcal{O}_0 = 0$.

Expanding $Q_0\mathcal{O}_1$ gives

$$\begin{aligned}
Q_0\mathcal{O}_1 = \frac{g^2}{32\pi^2} C_D{}^{XY} \Big[& -C^D{}_{EF}(\partial_{+\dot{a}}\mathsf{c}^E)\mathsf{c}^F(\partial_+{}^{\dot{a}}(\gamma^1)_X)(\gamma^2)_Y \\
& + C_{XE}{}^F(\partial_{+\dot{a}}\mathsf{c}^D)(\partial_+{}^{\dot{a}}\mathsf{c}^E)(\gamma^1)_F(\gamma^2)_Y \\
& + C_{XE}{}^F(\partial_{+\dot{a}}\mathsf{c}^D)\mathsf{c}^E(\partial_+{}^{\dot{a}}(\gamma^1)_F)(\gamma^2)_Y \\
& + C_{YE}{}^F(\partial_{+\dot{a}}\mathsf{c}^D)(\partial_+{}^{\dot{a}}(\gamma^1)_X)\mathsf{c}^E(\gamma^2)_F \Big].
\end{aligned} \tag{9.13}$$

The first, third and fourth lines cancel by invariance of C_{DXY} (the Jacobi identity):

$$C_{RX}{}^T C_{TYZ} + C_{RY}{}^T C_{XTZ} + C_{RZ}{}^T C_{XYT} = 0. \tag{9.14}$$

The remaining term is

$$\begin{aligned}
Q_0\mathcal{O}_1 &= \frac{g^2}{32\pi^2} C_D{}^{XY} C_{XE}{}^F (\gamma^1)_F(\gamma^2)_Y(\partial_{+\dot{a}}\mathsf{c}^D)(\partial_+{}^{\dot{a}}\mathsf{c}^E) \\
&= -\frac{g^2}{32\pi^2} C^X{}_{RD} C^{YR}{}_E (\gamma^1)_X(\gamma^2)_Y(\partial_{+\dot{a}}\mathsf{c}^D)(\partial_+{}^{\dot{a}}\mathsf{c}^E) \\
&= -Q_1\mathcal{O}_0.
\end{aligned} \tag{9.15}$$

The one-loop term of the quantum product therefore cancels the variation of the ordinary product of the two gravitons. The component degrees $\deg_\theta \mathcal{O}_0 = 5$ and $\deg_\theta \mathcal{O}_1 = 2$ make the quantum product terminate after its one-loop term:

$$\begin{gathered}
\mathcal{O} = A \star B = \mathcal{O}_0 + \hbar\mathcal{O}_1, \qquad Q_{n\geq 2}\mathcal{O}_0 = Q_{n\geq 1}\mathcal{O}_1 = 0, \\
\mathbf{Q}\mathcal{O} = \hbar(Q_1\mathcal{O}_0 + Q_0\mathcal{O}_1) = 0.
\end{gathered} \tag{9.16}$$

The displayed one-loop completion is the full perturbative product.

For other, more complicated classes, the two-loop term $Q_2\mathcal{O}_0$ might not vanish automatically, but the quantum product provides the completion at every order.

9.3 Homotopy transfer and successive obstructions

Having derived the full Q_2 and Q_n results, we now review the foundations of homotopy transfer.

$$\mathbf{Q}(\mathcal{O}_0+\hbar\mathcal{O}_1+\hbar^2\mathcal{O}_2+\cdots)=0 \quad\Longleftrightarrow\quad \sum_{\substack{j=0\\ \text{loop orders}}}^{s} Q_j\mathcal{O}_{s-j}=0 \qquad (s\geq 0). \tag{9.17}$$

Successive contractions. Let E_0 be the complex of holomorphic local operators and set $d_0=Q_0$. The pages are iterated cohomologies,

$$\begin{gathered} E_1=H(Q_0), \qquad E_2=H\big(H(Q_0),d_1\big), \qquad E_3=H\big(H\big(H(Q_0),d_1\big),d_2\big),\\ \ldots, \qquad E_{r+1}=H(E_r,d_r), \end{gathered} \tag{9.18}$$

The first differential is $d_1[\mathcal{O}_0]_1=[Q_1\mathcal{O}_0]_1$, and $Q_1^2=-(Q_0Q_2+Q_2Q_0)$ gives $d_1^2=0$ on $H(Q_0)$. For $r\geq 2$, the page obstruction includes the corrections to the representative:

$$d_r[\mathcal{O}_0]_r=\left[\sum_{\substack{j=1\\ \text{loop orders}}}^{r} Q_j\mathcal{O}_{r-j}\right]_r, \qquad \sum_{\substack{j=0\\ \text{loop orders}}}^{s} Q_j\mathcal{O}_{s-j}=0 \quad (0\leq s<r). \tag{9.19}$$

Successive transfer constructs these maps from $\mathbf{Q}$; the calculation below gives $d_r^2=0$ on E_r. Choose an inclusion $i_r: E_{r+1}\to E_r$ of cohomology representatives, a projection $\pi_r: E_r\to E_{r+1}$, and a homotopy $h_r: E_r\to E_r$ of degree -1, satisfying

$$\begin{gathered} \pi_r i_r=1, \qquad d_r i_r=\pi_r d_r=0, \qquad d_r h_r+h_r d_r=1-i_r\pi_r,\\ h_r i_r=\pi_r h_r=h_r^2=0. \end{gathered} \tag{9.20}$$

For a d_r-closed representative $a\in E_r$, the contraction separates its chosen cohomology representative from its exact part:

$$d_r a=0 \quad\Longrightarrow\quad a=i_r\pi_r a+\underbrace{d_r h_r a}_{\text{exact part}}, \qquad Q_0\mathcal{O}_0=0 \quad\Longrightarrow\quad \mathcal{O}_0=i_0\pi_0\mathcal{O}_0+Q_0h_0\mathcal{O}_0. \tag{9.21}$$

Write $\mathbf{Q}^{(0)}=\mathbf{Q}$. The superscript (r) counts transfers, while Q_n is the original n-loop coefficient; $[\hbar^j]$ extracts the coefficient of $\hbar^j$.

$$\begin{gathered} \mathbf{Q}^{(0)}=Q_0+\hbar Q_1+\hbar^2Q_2+\cdots,\\ \mathbf{Q}^{(r)}=d_r+\hbar([\hbar]\mathbf{Q}^{(r)})+\hbar^2([\hbar^2]\mathbf{Q}^{(r)})+\cdots, \qquad d_r=[\hbar^0]\mathbf{Q}^{(r)},\\ Q_n: E_0^k\to E_0^{k+1}, \qquad \mathbf{Q}^{(r)}: E_r^k[[\hbar]]\to E_r^{k+1}[[\hbar]], \qquad d_r: E_r^k\to E_r^{k+1}. \end{gathered} \tag{9.22}$$

Here k is the cohomological degree and $E_r[[\hbar]]$ consists of formal series with coefficients in E_r; the coefficient $[\hbar^0]$ is the part independent of $\hbar$.

The transferred differential and corrected maps are

$$\begin{aligned} i_r(\hbar)&=\big[1+h_r(\mathbf{Q}^{(r)}-d_r)\big]^{-1}i_r,\\ \pi_r(\hbar)&=\pi_r\big[1+(\mathbf{Q}^{(r)}-d_r)h_r\big]^{-1},\\ h_r(\hbar)&=\big[1+h_r(\mathbf{Q}^{(r)}-d_r)\big]^{-1}h_r,\\ \mathbf{Q}^{(r+1)}&=\frac{1}{\hbar}\pi_r(\mathbf{Q}^{(r)}-d_r)i_r(\hbar), \qquad d_{r+1}=\mathbf{Q}^{(r+1)}\Big|_{\hbar=0}. \end{aligned} \tag{9.23}$$

$$
\begin{aligned}
E^k_{r+1} &= \frac{\ker(d_r : E^k_r \to E^{k+1}_r)}{\operatorname{im}(d_r : E^{k-1}_r \to E^k_r)},\\
E_1 \xrightarrow{i_0} E_0 \xrightarrow{Q_1} E_0 \xrightarrow{\pi_0} E_1, &\qquad d_1 = \pi_0 Q_1 i_0,\\
i_r(\hbar) &= i_r - \hbar h_r([\hbar]\mathbf{Q}^{(r)}) i_r + \cdots .
\end{aligned}
\tag{9.24}
$$

$$
\begin{aligned}
\mathbf{Q}^{(r)} i_r(\hbar) &= \hbar i_r(\hbar)\mathbf{Q}^{(r+1)},\\
\pi_r(\hbar)\mathbf{Q}^{(r)} &= \hbar \mathbf{Q}^{(r+1)}\pi_r(\hbar),\\
\mathbf{Q}^{(r)} h_r(\hbar) + h_r(\hbar)\mathbf{Q}^{(r)} &= 1 - i_r(\hbar)\pi_r(\hbar),\\
\pi_r(\hbar) i_r(\hbar) &= 1, \qquad\qquad (\mathbf{Q}^{(r+1)})^2 = 0.
\end{aligned}
\tag{9.25}
$$

Starting from $\mathbf{Q}^2 = 0$, these identities prove nilpotence successively:

$$
\begin{aligned}
0 &= (\mathbf{Q}^{(r)})^2 i_r(\hbar) = \hbar^2 i_r(\hbar)(\mathbf{Q}^{(r+1)})^2,\\
0 &= \hbar^2 \pi_r(\hbar) i_r(\hbar)(\mathbf{Q}^{(r+1)})^2 = \hbar^2 (\mathbf{Q}^{(r+1)})^2,\\
(\mathbf{Q}^{(r+1)})^2 &= 0, \qquad d^2_{r+1} = [\hbar^0](\mathbf{Q}^{(r+1)})^2 = 0,\\
&\operatorname{im} d_{r+1} \subseteq \ker d_{r+1}.
\end{aligned}
\tag{9.26}
$$

The third line follows by equating coefficients of formal series.

The first four obstruction maps make the successive transfers explicit.

The first obstruction maps. Writing $[\hbar^k]$ for the coefficient of $\hbar^k$, the first transfer has coefficients

$$
\begin{aligned}
[\hbar^0]\mathbf{Q}^{(1)} &= \pi_0 Q_1 i_0,\\
[\hbar]\mathbf{Q}^{(1)} &= \pi_0 (Q_2 - Q_1 h_0 Q_1) i_0,\\
[\hbar^2]\mathbf{Q}^{(1)} &= \pi_0 (Q_3 - Q_2 h_0 Q_1 - Q_1 h_0 Q_2 + Q_1 h_0 Q_1 h_0 Q_1) i_0.
\end{aligned}
\tag{9.27}
$$

The maps π_0, i_0, h_0 form the original contraction of the full physical operator complex onto its classical Q_0-cohomology.

In particular,

$$
\begin{aligned}
d_{r+1} &= \pi_r([\hbar]\mathbf{Q}^{(r)}) i_r,\\
d_1 &= \pi_0([\hbar]\mathbf{Q}^{(0)}) i_0,\\
d_2 &= \pi_1([\hbar]\mathbf{Q}^{(1)}) i_1,\\
d_3 &= \pi_2([\hbar]\mathbf{Q}^{(2)}) i_2,\\
d_4 &= \pi_3([\hbar]\mathbf{Q}^{(3)}) i_3.
\end{aligned}
\tag{9.28}
$$

$$
\begin{aligned}
d_1 &= \pi_0 Q_1 i_0,\\
d_2 &= \pi_1 \pi_0 (Q_2 - Q_1 h_0 Q_1) i_0 i_1,\\
d_3 &= \pi_2 \pi_1 \Big([\hbar^2]\mathbf{Q}^{(1)} - ([\hbar]\mathbf{Q}^{(1)}) h_1([\hbar]\mathbf{Q}^{(1)}) \Big) i_1 i_2,\\
d_4 &= \pi_3 \pi_2 \Big([\hbar^2]\mathbf{Q}^{(2)} - ([\hbar]\mathbf{Q}^{(2)}) h_2([\hbar]\mathbf{Q}^{(2)}) \Big) i_2 i_3.
\end{aligned}
\tag{9.29}
$$

The coefficients on each page are obtained from the same expansion:

$$
\begin{aligned}
[\hbar]\mathbf{Q}^{(r+1)} &= \pi_r \Big([\hbar^2]\mathbf{Q}^{(r)} - ([\hbar]\mathbf{Q}^{(r)}) h_r([\hbar]\mathbf{Q}^{(r)}) \Big) i_r,\\
[\hbar^2]\mathbf{Q}^{(r+1)} &= \pi_r \Big([\hbar^3]\mathbf{Q}^{(r)} - ([\hbar^2]\mathbf{Q}^{(r)}) h_r([\hbar]\mathbf{Q}^{(r)})\\
&\qquad - ([\hbar]\mathbf{Q}^{(r)}) h_r([\hbar^2]\mathbf{Q}^{(r)}) + ([\hbar]\mathbf{Q}^{(r)}) h_r([\hbar]\mathbf{Q}^{(r)}) h_r([\hbar]\mathbf{Q}^{(r)}) \Big) i_r.
\end{aligned}
\tag{9.30}
$$

$$
\begin{aligned}
[\hbar^3]\mathbf{Q}^{(1)} = \pi_0\Big(&Q_4 - Q_3h_0Q_1 - Q_2h_0Q_2 - Q_1h_0Q_3 \\
&+ Q_2h_0Q_1h_0Q_1 + Q_1h_0Q_2h_0Q_1 + Q_1h_0Q_1h_0Q_2 \\
&- Q_1h_0Q_1h_0Q_1h_0Q_1\Big)i_0.
\end{aligned}
\tag{9.31}
$$

The second term in d_2 is the contribution of the first correction to the representative. To see this directly, take $\mathcal{O}_0 = i_0 i_1 a$ with $a \in E_2$, and choose $\mathcal{O}_1 = -h_0 Q_1 \mathcal{O}_0$. Then

$$
\begin{aligned}
Q_0\mathcal{O}_1 &= -(1 - i_0\pi_0)Q_1\mathcal{O}_0 = -Q_1\mathcal{O}_0, \\
Q_0(Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1) &= -Q_1^2\mathcal{O}_0 - Q_1Q_0\mathcal{O}_1 = 0, \\
\pi_1\pi_0(Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1) &= d_2 a.
\end{aligned}
\tag{9.32}
$$

Equivalently, with $[\cdot]_r$ denoting a class in E_r,

$$
d_2[\mathcal{O}_0]_2 = [Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1]_2. \tag{9.33}
$$

Changing $\mathcal{O}_1$ by a classical cocycle changes the projected obstruction by a d_1-boundary, so it does not affect the d_2 obstruction equation.

Corrected representatives. For a class $a \in E_3$, the condition $d_2 i_2 a = 0$ gives the following choice of corrections through two loops:

$$
\begin{aligned}
\mathcal{O}_0 &= i_0 i_1 i_2 a, \\
\mathcal{O}_1 &= -h_0 Q_1 \mathcal{O}_0 - i_0 h_1 \pi_0 (Q_2 - Q_1 h_0 Q_1)\mathcal{O}_0, \\
\mathcal{O}_2 &= -h_0(Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1).
\end{aligned}
\tag{9.34}
$$

$$
\begin{gathered}
\{\mathcal{O}_1 : Q_0\mathcal{O}_1 = -Q_1\mathcal{O}_0\} = -h_0Q_1\mathcal{O}_0 + \ker Q_0, \\
Q_0\mathcal{O}_2 = -(Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1), \qquad \pi_0(Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1) = 0, \\
Q_0\Big[-i_0h_1\pi_0(Q_2 - Q_1h_0Q_1)\mathcal{O}_0\Big] = 0.
\end{gathered}
\tag{9.35}
$$

The second-order equation selects a Q_0-closed addition to the first-order solution:

$$
\begin{aligned}
\pi_0(Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1) &= (1 - d_1h_1)\pi_0(Q_2 - Q_1h_0Q_1)\mathcal{O}_0 \\
&= i_1\pi_1\pi_0(Q_2 - Q_1h_0Q_1)\mathcal{O}_0 \\
&= i_1 d_2 i_2 a = 0, \qquad a \in E_3.
\end{aligned}
\tag{9.36}
$$

The choice of $\mathcal{O}_2$ above solves the second-order equation; a further Q_0-closed term gives another such choice.

For a homogeneous component in $\mathcal{N} = 4$ theory, the grading of Appendix B gives

$$
\begin{aligned}
&\deg_\theta \mathcal{O}_s = \deg_\theta \mathcal{O}_0 - 3s, \qquad \mathcal{O}_s = 0 \quad \text{if } 3s > \deg_\theta \mathcal{O}_0, \\
&\deg_\theta(d_r[\mathcal{O}_0]_r) = \deg_\theta \mathcal{O}_0 - 3r \quad \text{whenever } d_r[\mathcal{O}_0]_r \neq 0.
\end{aligned}
\tag{9.37}
$$

On every page the same construction gives

$$
\mathbf{Q}\, i_0(\hbar)\cdots i_{r-1}(\hbar) = \hbar^r i_0(\hbar)\cdots i_{r-1}(\hbar)\mathbf{Q}^{(r)}. \tag{9.38}
$$

For $a \in E_{r+1}$, choose $\mathcal{O}_s$ as the coefficient of $\hbar^s$ in $i_0(\hbar)\cdots i_{r-1}(\hbar) i_r a$. Then

$$
\sum_{\substack{k=0 \\ \text{loop orders}}}^{s} Q_k\mathcal{O}_{s-k} = 0 \quad (0 \le s \le r), \qquad d_{r+1}[\mathcal{O}_0]_{r+1} = \left[\sum_{\substack{k=1 \\ \text{loop orders}}}^{r+1} Q_k\mathcal{O}_{r+1-k}\right]_{r+1}. \tag{9.39}
$$

For $a \in E_4$, expand $i_0(\hbar)i_1(\hbar)i_2(\hbar)i_3 a$ to first order:

$$\begin{aligned}
[\hbar] i_r(\hbar) &= -h_r([\hbar]\mathbf{Q}^{(r)}) i_r, \\
\mathcal{O}_0 &= i_0 i_1 i_2 i_3 a, \\
\mathcal{O}_1 &= -h_0 Q_1 \mathcal{O}_0 - i_0 h_1([\hbar]\mathbf{Q}^{(1)}) i_1 i_2 i_3 a \\
&\quad - i_0 i_1 h_2([\hbar]\mathbf{Q}^{(2)}) i_2 i_3 a.
\end{aligned} \tag{9.40}$$

$$\begin{aligned}
Q_0 i_0 i_1 h_2 = 0, \qquad d_1 \pi_0 i_0 i_1 h_2 = d_1 i_1 h_2 = 0, \\
\pi_2([\hbar]\mathbf{Q}^{(2)}) i_2 i_3 a = d_3 i_3 a = 0, \\
d_2\Big[-h_2([\hbar]\mathbf{Q}^{(2)}) i_2 i_3 a\Big] = -([\hbar]\mathbf{Q}^{(2)}) i_2 i_3 a.
\end{aligned} \tag{9.41}$$

The last term in $\mathcal{O}_1$ preserves the preceding closure equations and supplies the next compatibility condition.

Products and classical representatives. Suppose the operators $A = \sum_{s<r} \hbar^s A_s$ and $B = \sum_{s<r} \hbar^s B_s$ already satisfy the closure equations through order $r-1$. Write $\mathcal{O}_s = [\hbar^s](A \star B)$. The leading term is $\mathcal{O}_0 = A_0 B_0$. Taking coefficients in (6.13) gives

$$\begin{aligned}
&\sum_{\substack{k=0 \\ \text{loop orders}}}^{s} Q_k \mathcal{O}_{s-k} = 0 \qquad (0 \le s < r), \\
Q_0 \mathcal{O}_r + &\sum_{\substack{k=1 \\ \text{loop orders}}}^{r} Q_k \mathcal{O}_{r-k} = \Big(\sum_{\substack{k=1 \\ \text{loop orders}}}^{r} Q_k A_{r-k} \Big) B_0 \\
&\qquad + (-1)^{|A_0|} A_0 \Big(\sum_{\substack{k=1 \\ \text{loop orders}}}^{r} Q_k B_{r-k} \Big).
\end{aligned} \tag{9.42}$$

Passing to E_r gives

$$\begin{aligned}
[A_0]_r [B_0]_r &= [A_0 B_0]_r, \\
d_r([A_0]_r [B_0]_r) &= (d_r[A_0]_r)[B_0]_r + (-1)^{|A_0|}[A_0]_r (d_r[B_0]_r).
\end{aligned} \tag{9.43}$$

Starting from multiplication on $H(Q_0)$, this identity gives an associative graded-commutative product on each successive page.

$$E_{r+1}^k = \frac{\ker d_r^k}{\operatorname{im} d_r^{k-1}}, \qquad d_r^k : E_r^k \to E_r^{k+1}. \tag{9.44}$$

$$\begin{aligned}
d_r[A_0]_r = d_r[B_0]_r = 0 \quad &\Longrightarrow \quad d_r[A_0 B_0]_r = 0, \\
[A_0]_r = d_r a, \quad d_r[B_0]_r = 0 \quad &\Longrightarrow \quad [A_0 B_0]_r = d_r(a[B_0]_r), \\
[A_0]_{r+1}[B_0]_{r+1} &= [A_0 B_0]_{r+1}, \\
[A_0]_{r+1}[B_0]_{r+1} = 0 \quad &\Longleftrightarrow \quad [A_0 B_0]_r \in \operatorname{im} d_r.
\end{aligned} \tag{9.45}$$

Higher differentials on the graviton part. Applying the projection identity in (9.25) on successive pages gives

$$\pi_{r-1}(\hbar) \cdots \pi_0(\hbar)\, \mathbf{Q} = \hbar^r\, \mathbf{Q}^{(r)}\, \pi_{r-1}(\hbar) \cdots \pi_0(\hbar). \tag{9.46}$$

If $\mathbf{Q}\mathcal{O} = 0$, dividing (9.46) applied to $\mathcal{O}$ by $\hbar^r$ gives $\mathbf{Q}^{(r)}\pi_{r-1}(\hbar)\cdots\pi_0(\hbar)\mathcal{O} = 0$. Its constant term is $d_r\pi_{r-1}\cdots\pi_0\mathcal{O}_0$, and by induction on r the projection $\pi_{r-1}\cdots\pi_0\mathcal{O}_0$ is the class $[\mathcal{O}_0]_r$. Hence, a class with a closed completion satisfies

$$d_r[\mathcal{O}_0]_r = 0 \qquad (r \geq 1). \tag{9.47}$$

By Theorem 9.1, every d_r with $r \geq 1$ vanishes on the graviton part. For $d_r^k : E_r^k \to E_r^{k+1}$, the dimensions satisfy

$$\begin{aligned} \operatorname{rank} d_r^k &= \dim E_r^k - \dim\ker d_r^k, \\ \dim E_{r+1}^k &= \dim E_r^k - \operatorname{rank} d_r^k - \operatorname{rank} d_r^{k-1}. \end{aligned} \tag{9.48}$$

$$\begin{aligned} \operatorname{rank} d_r^k = \dim \left\{ [\mathcal{O}_0]_1 \in E_1^k : \begin{array}{c} \exists\, \mathcal{O}_1, \dots, \mathcal{O}_{r-1}, \\ \sum\limits_{\substack{j=0 \\ \text{loop orders}}}^{s} Q_j \mathcal{O}_{s-j} = 0 \quad (0 \leq s < r) \end{array} \right\} \\ - \dim \left\{ [\mathcal{O}_0]_1 \in E_1^k : \begin{array}{c} \exists\, \mathcal{O}_1, \dots, \mathcal{O}_{r}, \\ \sum\limits_{\substack{j=0 \\ \text{loop orders}}}^{s} Q_j \mathcal{O}_{s-j} = 0 \quad (0 \leq s \leq r) \end{array} \right\}. \end{aligned} \tag{9.49}$$

$$d_r[\mathcal{O}_0]_r = 0 \quad (\mathcal{O}_0 \text{ a multi-graviton}), \qquad [\mathcal{O}_0]_{r+1} = 0 \iff [\mathcal{O}_0]_r \in \operatorname{im} d_r^{k-1}. \tag{9.50}$$

The extension spaces are nested and contain the graviton part, so for any finite number R of pages,

$$\sum_{\substack{r=1 \\ \text{pages}}}^{R} \operatorname{rank} d_r^k \leq \dim E_1^k - \dim \operatorname{span} \left\{ [\mathcal{O}_0]_1 \in E_1^k : \mathcal{O}_0 \text{ is a multi-graviton} \right\}. \tag{9.51}$$

Fortuitous classes are sparse among the classical classes known so far [19, 20], so the successive differentials are extremely sparse on the known cohomology.

For completed operators satisfying

$$\mathbf{Q}X = \hbar^r \mathcal{O}, \qquad \mathbf{Q}\mathcal{O} = 0, \qquad \mathcal{O} = \mathbf{Q}(\hbar^{-r}X), \tag{9.52}$$

the last equality holds after inverting $\hbar$. Over formal power series, the first equality records a class annihilated by $\hbar^r$. The successive pages record these d_r-exact classes at their respective orders in $\hbar$.

$$\mathbf{Q}X = \hbar^r\mathcal{O} \implies d_r[X_0]_r = [\mathcal{O}_0]_r, \qquad [\mathcal{O}_0]_{r+1} = 0, \qquad X_0 = [\hbar^0]X. \tag{9.53}$$

Proposition 9.2 (Validity of successive homotopy transfer)**.**

Let $\mathbf{Q}^{(r)}$ be a formal differential on $E_r[[\hbar]]$, with $(\mathbf{Q}^{(r)})^2 = 0$ and constant coefficient d_r, and choose a contraction as in (9.20).

The maps defined by (9.23) are well defined coefficientwise and satisfy all identities in (9.25); consequently the successive transfers and the representative chain (9.38) hold.

For the homogeneous $\mathcal{N} = 4$ contractions of (B.8), both inverse series terminate on every fixed homogeneous polynomial input.

Proof.

The starting contraction can be chosen degree by degree: select representatives of $\ker d_r / \operatorname{im} d_r$ and a complement of $\ker d_r$ in E_r.

Take i_r to be the representative inclusion, π_r the projection onto these representatives followed by their cohomology classes, and h_r the inverse of d_r from its image to the chosen

complement, extended by zero on the representatives and the complement.

On the representatives, boundaries and complement, $d_r h_r + h_r d_r$ acts respectively as $0, 1, 1$, while $1 - i_r \pi_r$ has the same action; the remaining identities in (9.20) follow from these definitions.

Since $\mathbf{Q}^{(r)} - d_r$ has zero constant coefficient, the inverses are the following formal series, with k counting powers of the displayed operator:

$$\begin{aligned} \big[1 + h_r(\mathbf{Q}^{(r)} - d_r)\big]^{-1} &= \sum_{\substack{k \geq 0 \\ \text{operator powers}}} (-1)^k \big[h_r(\mathbf{Q}^{(r)} - d_r)\big]^k, \\ \big[1 + (\mathbf{Q}^{(r)} - d_r)h_r\big]^{-1} &= \sum_{\substack{k \geq 0 \\ \text{operator powers}}} (-1)^k \big[(\mathbf{Q}^{(r)} - d_r)h_r\big]^k. \end{aligned} \tag{9.54}$$

At coefficient order $\hbar^n$, terms with $k > n$ vanish, so each coefficient is a finite sum; multiplying the series by its denominator cancels adjacent powers and gives the identity.

$$\begin{aligned} i_r(\hbar) &= i_r - h_r(\hbar)(\mathbf{Q}^{(r)} - d_r)i_r, \\ \pi_r(\hbar) &= \pi_r - \pi_r(\mathbf{Q}^{(r)} - d_r)h_r(\hbar), \\ h_r(\hbar) &= \big[1 + h_r(\mathbf{Q}^{(r)} - d_r)\big]^{-1} h_r \\ &= h_r \big[1 + (\mathbf{Q}^{(r)} - d_r)h_r\big]^{-1}. \end{aligned} \tag{9.55}$$

The last equality follows term by term from associativity of composition; using $h_r^2 = h_r i_r = \pi_r h_r = 0$ gives

$$\begin{aligned} h_r\big[1 + h_r(\mathbf{Q}^{(r)} - d_r)\big]^{-1} &= h_r, \\ h_r(\hbar)^2 = \big[1 + h_r(\mathbf{Q}^{(r)} - d_r)\big]^{-1} h_r^2 &= 0, \\ h_r(\hbar) i_r = 0, \qquad \pi_r h_r(\hbar) &= 0. \end{aligned} \tag{9.56}$$

$$\begin{aligned} \pi_r(\hbar) i_r(\hbar) = \pi_r i_r &- \pi_r h_r(\hbar)(\mathbf{Q}^{(r)} - d_r) i_r \\ &- \pi_r(\mathbf{Q}^{(r)} - d_r) h_r(\hbar) i_r \\ &+ \pi_r(\mathbf{Q}^{(r)} - d_r) h_r(\hbar)^2 (\mathbf{Q}^{(r)} - d_r) i_r = 1, \\ h_r(\hbar) i_r(\hbar) = 0, \qquad \pi_r(\hbar) h_r(\hbar) &= 0. \end{aligned} \tag{9.57}$$

From $(\mathbf{Q}^{(r)})^2 = d_r^2 = 0$,

$$\begin{aligned} &(\mathbf{Q}^{(r)} - d_r)\mathbf{Q}^{(r)} + \mathbf{Q}^{(r)}(\mathbf{Q}^{(r)} - d_r) \\ &= -d_r \mathbf{Q}^{(r)} - \mathbf{Q}^{(r)} d_r = (\mathbf{Q}^{(r)} - d_r)^2. \end{aligned} \tag{9.58}$$

Multiplication by the two inverse denominators gives the explicit homotopy calculation:

$$\begin{aligned} &\big[1 + h_r(\mathbf{Q}^{(r)} - d_r)\big](\mathbf{Q}^{(r)} h_r(\hbar) + h_r(\hbar)\mathbf{Q}^{(r)}) \\ &\qquad \times \big[1 + (\mathbf{Q}^{(r)} - d_r)h_r\big] \\ &= \mathbf{Q}^{(r)} h_r + h_r \mathbf{Q}^{(r)} \\ &\quad + h_r\big[(\mathbf{Q}^{(r)} - d_r)\mathbf{Q}^{(r)} + \mathbf{Q}^{(r)}(\mathbf{Q}^{(r)} - d_r)\big] h_r \\ &= 1 - i_r \pi_r + (\mathbf{Q}^{(r)} - d_r)h_r + h_r(\mathbf{Q}^{(r)} - d_r) \\ &\quad + h_r(\mathbf{Q}^{(r)} - d_r)^2 h_r \\ &= \big[1 + h_r(\mathbf{Q}^{(r)} - d_r)\big]\big[1 + (\mathbf{Q}^{(r)} - d_r)h_r\big] - i_r \pi_r. \end{aligned} \tag{9.59}$$

Applying the inverses already established yields

$$\mathbf{Q}^{(r)}h_r(\hbar) + h_r(\hbar)\mathbf{Q}^{(r)} = 1 - i_r(\hbar)\pi_r(\hbar). \tag{9.60}$$

$$\begin{aligned}\mathbf{Q}^{(r)}i_r(\hbar)\pi_r(\hbar) &= \mathbf{Q}^{(r)} - \mathbf{Q}^{(r)}h_r(\hbar)\mathbf{Q}^{(r)}\\ &= i_r(\hbar)\pi_r(\hbar)\mathbf{Q}^{(r)},\\ h_r(\hbar)\mathbf{Q}^{(r)}i_r(\hbar) &= \big[1 - i_r(\hbar)\pi_r(\hbar) - \mathbf{Q}^{(r)}h_r(\hbar)\big]i_r(\hbar) = 0.\end{aligned} \tag{9.61}$$

The differential induced on $E_{r+1}[[\hbar]]$ is therefore

$$\begin{aligned}\pi_r(\hbar)\mathbf{Q}^{(r)}i_r(\hbar) &= \pi_r\mathbf{Q}^{(r)}i_r(\hbar)\\ &\quad - \pi_r(\mathbf{Q}^{(r)} - d_r)h_r(\hbar)\mathbf{Q}^{(r)}i_r(\hbar)\\ &= \pi_r(\mathbf{Q}^{(r)} - d_r)i_r(\hbar) = \hbar\mathbf{Q}^{(r+1)}.\end{aligned} \tag{9.62}$$

The retraction and invariance identities now give both chain maps:

$$\begin{aligned}\mathbf{Q}^{(r)}i_r(\hbar) &= i_r(\hbar)\pi_r(\hbar)\mathbf{Q}^{(r)}i_r(\hbar)\\ &= \hbar i_r(\hbar)\mathbf{Q}^{(r+1)},\\ \pi_r(\hbar)\mathbf{Q}^{(r)} &= \pi_r(\hbar)\mathbf{Q}^{(r)}i_r(\hbar)\pi_r(\hbar)\\ &= \hbar\mathbf{Q}^{(r+1)}\pi_r(\hbar).\end{aligned} \tag{9.63}$$

$$\begin{aligned}0 = \pi_r(\hbar)(\mathbf{Q}^{(r)})^2 i_r(\hbar) &= \hbar^2(\mathbf{Q}^{(r+1)})^2,\\ (\mathbf{Q}^{(r+1)})^2 = 0, \qquad d_{r+1}^2 &= [\hbar^0](\mathbf{Q}^{(r+1)})^2 = 0,\\ d_{r+1} &= \pi_r\big([\hbar]\mathbf{Q}^{(r)}\big)i_r.\end{aligned} \tag{9.64}$$

Multiplication by $\hbar^2$ shifts formal-series coefficients injectively, giving the second line; iterating the proved inclusion identity gives (9.38).

For a homogeneous polynomial $a \in E_r$, $\deg_\theta a$ is the component degree (B.2); the homogeneous $\mathcal{N} = 4$ contractions (B.8) give, for $m \geq 1$ and nonzero outputs,

$$\begin{aligned}\deg_\theta\Big(h_r\big([\hbar^m]\mathbf{Q}^{(r)}\big)a\Big) &= \deg_\theta a - 3m,\\ \deg_\theta\Big(\big([\hbar^m]\mathbf{Q}^{(r)}\big)h_r a\Big) &= \deg_\theta a - 3m.\end{aligned} \tag{9.65}$$

Nonnegative component degree therefore implies

$$\begin{aligned}\big[h_r(\mathbf{Q}^{(r)} - d_r)\big]^k a = \big[(\mathbf{Q}^{(r)} - d_r)h_r\big]^k a &= 0 \qquad (3k > \deg_\theta a),\\ \big[1 + h_r(\mathbf{Q}^{(r)} - d_r)\big]^{-1} a &= \sum_{\substack{0\le k\le \lfloor \deg_\theta a/3\rfloor\\ \text{operator powers}}} (-1)^k\big[h_r(\mathbf{Q}^{(r)} - d_r)\big]^k a,\\ \big[1 + (\mathbf{Q}^{(r)} - d_r)h_r\big]^{-1} a &= \sum_{\substack{0\le k\le \lfloor \deg_\theta a/3\rfloor\\ \text{operator powers}}} (-1)^k\big[(\mathbf{Q}^{(r)} - d_r)h_r\big]^k a.\end{aligned} \tag{9.66}$$

Each surviving coefficient of order $\hbar^n$ lowers the component degree by $3n$, so both expressions are polynomials in $\hbar$ of degree at most $\lfloor \deg_\theta a/3\rfloor$. □

10 Fortuitous classes and quantum lifting

The tables below list the finite-rank classes and their successive obstruction maps. The algorithm, the computations and the certificates are described in the supplementary material [14], which also contains the results report cited below.

Several of the classes are new. For $SU(2)$, beyond the threshold class $\mathcal{O}_0$ [20, 21] and the class $\mathcal{O}_1$ [19], we find the classes P39 and X40. For $SU(3)$, beyond the towers F_1 and B_1 of [22, 23], we find the charge-conjugation-odd class X_7 and four primary sectors with ten and eleven letters. X40 and X_7 are visible to Cartan restriction, $h = \rho$, although they are not multi-gravitons. For $SU(4)$ we find the first fortuitous class.[24] For $SO(2N+1)$ and $Sp(N)$, the $SO(7)$ class is that of [27], and the first fortuitous classes of $SO(5) = Sp(2)$, $Sp(3)$, $SO(9)$ and $SO(11)$ are new. For G_2, the class (10.5) is a monotone class that is invisible to Cartan restriction. The one-loop lifts of $\mathcal{O}_1$ and of the $SO(7)$ class are those of [20] and [2]; the one-loop lifts in $SO(9)$ and $SO(11)$ and all the higher-loop lifts below are new.

The census builds on the following work. The classical cohomology and its fortuitous classes are organized as in [1, 19, 21, 28], and the enumerations extend those of [22, 23, 27, 29]; they reproduce the published $SU(2)$ census [30], the one-loop lifts of [2, 20] and the mass-deformed classes of [31]. The classes invisible to Cartan restriction are analysed through the commuting scheme of the Lie algebra, with the higher-dimensional Chevalley restriction results of [32–41], the proofs of the conjecture of Cachazo, Douglas, Seiberg and Witten [8, 42–46], the super-commuting schemes of [47] and standard invariant theory [48–51]. The Leigh–Strassler family is that of [15, 52]. The one- and two-loop operators used in the tests were compared with [4, 20], with the two-loop kernel of [13, Section 4.5] and with the two-loop coefficient of the fortuity report [25]. Exact linear algebra over the integers and the rationals uses FLINT [53].

The general component action is (3.3), with the gauge group, the matter representation and the potential specified for each theory below:

$$S_{\mathrm{tw}} = \int_{\mathbb{C}^2} \Omega \wedge \Big[\mathsf{b}_A \left(\bar\partial \mathsf{c}^A - \frac{g}{2} C^A{}_{BC} \mathsf{c}^B \mathsf{c}^C \right) + \left(\bar\partial \gamma_i - i g \mathsf{c}^A (t_A)^j{}_i \gamma_j \right) \beta^i + \bar W(\gamma, \mathsf{w}) \Big]_{(0,2)} . \tag{10.1}$$

For the $\mathcal{N} = 4$ theories, the sectors are $(\lambda)_Y = (\theta_1, \theta_2, \theta_3; n_+, n_-)_Y$, which record the three component weights, the two holomorphic derivative totals and the number Y of letters, as in (B.1); the level $E + J_L$ is given by (B.5).

The fortuitous dimension f is the dimension of E_1 modulo the image of the stable cohomology, the monotone classes [19, 28]; it is computed as in [14, supplement, Section S1.5]. The Cartan-kernel dimension is $h - \rho$, with $h = \dim E_1$ and ρ the rank of Cartan restriction. Cartan restriction is useful in the computation, since most monotone classes are visible to it. One might expect that every monotone class is visible to it and that every fortuitous class lies in its kernel. Both statements have counterexamples: the classes X40 and X_7 below are visible but are not multi-gravitons, and the G_2 class (10.5) is monotone and lies in the kernel.

The labels $[p]$, $[p, p']$ and $[\mathbb{Q}]$ denote finite-field rank computations, independent checks at a second prime, and rational certificates, as specified in [14, supplement, Section S1.7]. Unmarked census ranks are computed modulo $p = 2^{31} - 1$. These labels describe how the linear independence behind each rank is established: a result marked $[\mathbb{Q}]$ is proved, and one marked $[p]$ or $[p, p']$ is a lower bound on the rational rank that is exact unless p divides the relevant minors.

[24] Earlier $SU(4)$ searches [22, 24, 25] cover the BMN sector, whose operators carry no derivatives or gaugini; the class found here has $n_+ = n_- = 2$. A fortuitous $SU(4)$ class built from two $SU(2)$ cores at much higher charges is constructed in [26].

10.1 $\mathcal{N} = 4$ Yang–Mills: $SU(N)$

The three chiral multiplets are adjoint, and the superfield C is (A.2); the action is in (A.3) and [14, results report, §1.1].

$$S_{\text{tw}} = \int_{\mathbb{C}^2} \Omega \wedge \left[\int d^3\theta \left(\frac{1}{2} \delta_{AB} C^A \bar{\partial} C^B + \frac{i}{3!} C_{ABC} C^A C^B C^C \right) \right]_{(0,2)}, \qquad \mathfrak{g} = \mathfrak{su}(N). \tag{10.2}$$

The classes and their lifting chains are listed in [14, supplement, Sections S2.2 and S2.3]; $R_r = \sum^{r}_{\substack{k=1 \\ \text{loop orders}}} Q_k \mathcal{O}_{r-k}$ is the order-r residual of Section 9.3.

We list only the fortuitous generators: classes that are not derivatives or other descendants of lower classes, nor products of lower fortuitous classes and gravitons. With the quantum product, the remaining classes need not be tested separately for outgoing differentials.

In the tables, an incoming test asks whether a class is d_r-exact, and an outgoing test whether d_r of the class vanishes.

We find no lift beyond one loop in any of the $\mathcal{N} = 4$ theories.

Table 10.1: Fortuitous generators of $SU(N)$ $\mathcal{N} = 4$ Yang–Mills theory and their lifting.

G	Class / sector	Classical result	Lifting result
$SU(2)$	$\mathcal{O}_0 : (4,4,4;0,0)_7$	Known threshold class [20, 21]	Closed completion through $\hbar^4$; all-order survival [$\mathbb{Q}$]
	$\mathcal{O}_1 : (6,6,6;0,0)_9$	Known class [19]	$d_1 \neq 0$ [20], lifted at one loop
	P39: $(5,5,5;3,0)_9$	One new class beyond 12 products and descendants at this level	All-order test: $d_1 = 0$; $R_2, R_3, R_4 \in \operatorname{im} Q_0$; fifth target empty; no incoming map $[p, p']$. So all-order survival [p,p']
	X40: $(7,5,5;1,1)_{10}$	$h = \rho = 8$, graviton dimension 7	All admissible outgoing obstructions vanish; incoming sources empty $[p, p']$ through $r = 2$, $[p]$ above; survival [p,p']
$SU(3)$	$X_7 : (2,2,2;2,2)_7$	$h = \rho = 26$, graviton dimension 25; charge-conjugation odd	All admissible incoming and outgoing maps vanish $[p]$. Survival [p]
	X_7 member: $(0,0,0;5,5)_7$	Fortuitous relative Lie-algebra class [$\mathbb{Q}$]	All-order survival in the X_7 multiplet $[p]$
	F_1, $n = 3$: $(7,4,4;0,0)_{10}$	Known tower [22, 23]	Closed completion through $\hbar^4$; no higher or incoming obstruction $[p]$; survival [p]
	$(5,4,4;1,1)_{10}$	New primary sector; $h = 2$, $\rho = 0$	$d_1 = d_2 = d_3 = 0$ on both classes; $d_4 = 0$ if R_4 has zero Cartan restriction; this condition remains unresolved; no incoming map $[p]$
	$(7,4,4;1,0)_{11}$	$h = 2$, $\rho = 0$; one class beyond the F_1 descendant	$d_1 = d_3 = 0$; $d_2 = d_4 = 0$ conditional; no incoming map $[p]$
	$(6,4,4;1,1)_{11}$	$h = 3$, $\rho = 0$	$d_3 = 0$; $d_4 = 0$ conditional; d_1, d_2 open; no incoming map $[p]$
	$(7,5,4;1,0)_{11}$	$h = 4$, $\rho = 0$	$d_4 = 0$ conditional; d_1, d_2, d_3 open; no incoming map $[p]$

Table 10.1: Fortuitous generators of $SU(N)$ $\mathcal{N}=4$ Yang–Mills theory and their lifting.

G	Class / sector	Classical result	Lifting result
	B_1, $n=3$: $(8,5,4;0,0)_{11}$	Known tower [22, 23]	d_1 open
$SU(4)$	$(4,2,2;2,2)_9$	$f=1$ [$\mathbb{Q}$]; charge-conjugation odd	R_1, R_2 exact; higher outgoing targets empty; incoming ranks zero [p]

Here "conditional" assumes zero Cartan restriction of R_r; because $h=\rho$, this gives $R_r \in \operatorname{im} Q_0$ and hence $d_r = 0$ [14, results report, §1.6].

The frontiers of the complete census, the partial scans at higher levels and the null results at higher rank are tabulated in [14, supplement, Section S2.1].

Explicit representatives of the threshold class and of its graviton hair, and the multi-graviton checks at low letter number, are in [14, supplement, Sections S8.2 and S8.3].

10.2 $\mathcal{N}=4$ Yang–Mills: $SO(2N+1)$ and $Sp(N)$

These gauge groups are related by S-duality, while their Lie algebras are not isomorphic for $N \geq 3$.

$$S_{\mathrm{tw}} = \int_{\mathbb{C}^2} \Omega \wedge \left[\int d^3\theta \left(\frac{1}{2} \delta_{AB} C^A \bar{\partial} C^B + \frac{i}{3!} C_{ABC} C^A C^B C^C \right) \right]_{(0,2)}, \qquad \mathfrak{g} = \mathfrak{so}_{2N+1} \text{ or } \mathfrak{sp}_{2N}. \tag{10.3}$$

The first fortuitous sectors and the rank-one lifts are given in [14, supplement, Sections S3.1 and S3.2].

Table 10.2: First fortuitous sectors of $SO(2N+1)$ and $Sp(N)$ $\mathcal{N}=4$ Yang–Mills theory and their lifting.

G	Source sector	Classical result	Lifting result
$SO(5)=Sp(2)$	$(3,2,2;1,1)_7$	$f=1$, $\rho=0$	All admissible incoming and outgoing maps vanish [p]
$SO(7)$	$O_f : (3,3,3;0,0)_8$	Known fortuitous source [27]	$\operatorname{rank} d_1 = 1$ into $(2,2,2;1,1)_8$ [2]
$Sp(3)$	$(5,2,2;1,1)_9$	$h=f=1$	All admissible incoming and outgoing maps vanish [p]
$Sp(3)$	$(5,0,0;3,3)_9$	$h=1$, $\rho=0$	Outgoing maps excluded; incoming d_2 unresolved
$SO(9)$	$(5,3,3;0,0)_{10}$	$h=6$; one-class excess over $Sp(4)$	$\operatorname{rank} d_1 = 1$ into $(4,2,2;1,1)_{10}$ [p]
$SO(11)$	$(7,3,3;0,0)_{12}$	$h=23$, $f=1$	$\operatorname{rank} d_1 = 1$ into $(6,2,2;1,1)_{12}$ [p]

Through $E+J_L=11$, the computed $SO(7)/Sp(3)$ and $SO(9)/Sp(4)$ spectra agree charge by charge after the rank-one d_1 pairings, with no d_2 candidate pair and zero rank on every d_3 candidate [14, supplement, Section S3.3].

10.3 $\mathcal{N}=4$ Yang–Mills: G_2

$$S_{\mathrm{tw}} = \int_{\mathbb{C}^2} \Omega \wedge \left[\int d^3\theta \left(\frac{1}{2} \delta_{AB} C^A \bar{\partial} C^B + \frac{i}{3!} C_{ABC} C^A C^B C^C \right) \right]_{(0,2)}, \qquad \mathfrak{g} = \mathfrak{g}_2. \tag{10.4}$$

The new scalar result is a monotone class in the kernel of Cartan restriction [14, supplement, Section S5.2], with traces in the seven-dimensional representation:

$$\left[\det_{I,J} \operatorname{tr}_7(\gamma^I\gamma^J)\right] = \left[\frac{1}{6}\varepsilon_{IJK}\varepsilon_{LMN} \operatorname{tr}_7(\gamma^I\gamma^L) \operatorname{tr}_7(\gamma^J\gamma^M) \operatorname{tr}_7(\gamma^K\gamma^N)\right] \neq 0. \tag{10.5}$$

Table 10.3: Results for G_2 $\mathcal{N} = 4$ Yang–Mills theory.

Sector	Classical result	Lifting result
$(2,2,2;0,0)_6$	$\dim C^6 = 4$, $\operatorname{rank} Q_0 = 3$, $h = 1$, $\rho = 0$ $[\mathbb{Q}]$	All-order survival: closed multi-graviton completion; incoming maps excluded by $n_+ = n_- = 0$
All sectors through $E + J_L = \frac{23}{2}$	At $\frac{23}{2}$: $\sum_{\text{sectors}} h = 2075$, $\sum_{\text{sectors}}(h-\rho) = 170$	$d_1 = 0$; no d_2 pairs; all d_3 candidate ranks zero $[p]$

The level-by-level census of the classes in the kernel of Cartan restriction is in [14, supplement, Section S5.3].

10.4 $SU(3)$ Leigh–Strassler deformation

This theory, a deformation of $\mathcal{N} = 4$ Yang–Mills theory, gave the first two-loop lifts that we found.

The matter consists of three adjoint chiral multiplets [15]. In the trace convention $\operatorname{tr}(t_A t_B) = \frac{1}{2}\delta_{AB}$ of [14, results report, §6.1], the potential in the common twisted action is

$$\begin{aligned}\bar{W}(\gamma, \mathsf{w}) = {} & \operatorname{tr}(\gamma^1\gamma^2\gamma^3) - q \operatorname{tr}(\gamma^1\gamma^3\gamma^2) \\ & + \frac{h}{3}\left[\operatorname{tr}((\gamma^1)^3) + \operatorname{tr}((\gamma^2)^3) + \operatorname{tr}((\gamma^3)^3)\right] - \frac{1}{4}\mathsf{w}_{\dot{a}}^{A}\mathsf{w}_A{}^{\dot{a}}.\end{aligned} \tag{10.6}$$

The sectors $(\Theta, c; n_+, n_-)_Y$ use $\Theta = \theta_1 + \theta_2 + \theta_3$ and $c = \theta_2 + 2\theta_3 \pmod 3$.

The census is carried out at $\rho = (1-q)/(1+q) = 12345/678$ and $h_e = 2h/(1+q) = 4321/97$, as a model of generic q and h; the checks at independent parameter values and primes are listed in [14, supplement, Section S6.1].

Table 10.4: Sources of the two-loop lifts in the $SU(3)$ Leigh–Strassler theory: finite-rank cohomology, comparison with the stable range, and fortuitous dimension.

Source sector	Finite-rank classes	Stable comparison	Fortuitous dimension	Two-loop rank
$P_{14} : (7,0;0,0)_5$	$h = 5$	$h_{SU(6)} = 0$	$f = 5$	1
$P_{16} : (8,0;0,0)_6$	$h = 4$	$h_{SU(7)} = 2$	$f \geq 2$	1
$\partial_{++}P_{14} : (7,0;1,0)_5$	$h = 5$	$h_{SU(6)} = 0$	$f = 5$	1
$\partial^2_{++}P_{14} : (7,0;2,0)_5$	$h = 5$	$h_{SU(6)} = 0$	$f = 5$	1
$(8,0;1,0)_6$	$h = 6$	$h_{SU(7)} = 4$	$f \geq 2$	1
$P_{19} : (8,0;1,0)_7$	$h = 19$	$h_{SU(8)} = 6$	$f \geq 13$	1
$P_{21} : (9,0;1,0)_7$	$h = 40$	$h_{SU(8)} = 5$	$f \geq 35$	6
$P_{21} : (9,1;1,0)_7$	$h = 36$	$h_{SU(8)} = 0$	$f = 36$	6

These comparisons with the stable ranks are in [14, supplement, Section S6.6]; where only a lower bound on f is known for a sector, the classification of the individual lifted vectors awaits the stable-image test.

Table 10.5: Two-loop lifts in the $SU(3)$ Leigh–Strassler theory through $E + J_L = 11$.

Source	Target	rank d_2	Origin
$(7,0;0,0)_5$	$(1,0;2,2)_4$	1	P_{14}
$(8,0;0,0)_6$	$(2,0;2,2)_5$	1	P_{16}
$(7,0;1,0)_5$	$(1,0;3,2)_4$	1	$\partial_{++\dot{+}}P_{14}$
$(8,0;1,0)_6$	$(2,0;3,2)_5$	1	$\partial_{++\dot{+}}P_{16}$
$(8,0;1,0)_7$	$(2,0;3,2)_6$	1	P_{19}
$(7,0;2,0)_5$	$(1,0;4,2)_4$	1	$\partial^2_{++\dot{+}}P_{14}$
$(9,c;1,0)_7$, $c=0,1$	$(3,c;3,2)_6$	6 each	P_{21}
$(8,0;1,1)_7$	$(2,0;3,3)_6$	5	$E + J_L = 11$
$(8,0;2,0)_6$	$(2,0;4,2)_5$	1	$\partial^2_{++\dot{+}}P_{16}$
$(8,0;2,0)_7$	$(2,0;4,2)_6$	5	Includes $\partial_{++\dot{+}}P_{19}$
$(8,0;2,0)_8$	$(2,0;4,2)_7$	1	$E + J_L = 11$

The ranks and symmetry multiplicities are those of [14, supplement, Section S6.2]; for $\Theta \not\equiv 0 \pmod 3$, the $c = 0$ sector represents three cyclic images.

For the lowest source,

$$\begin{gathered} P_{14}: \quad \dim E_1 = 5 \xrightarrow{\text{rank } d_1 = 3} \dim E_2 = 2 \xrightarrow{\text{rank } d_2 = 1} \dim E_3 = 1, \\ Q_2\mathcal{O}_0 = 0, \qquad Q_0\mathcal{O}_1 = -Q_1\mathcal{O}_0, \qquad d_2[\mathcal{O}_0]_2 = [Q_1\mathcal{O}_1]_2 \neq 0. \end{gathered} \tag{10.7}$$

The independence of these results from the two-loop scheme, for the lowest lifts, and the parameter loci on which they vanish are in [14, supplement, Sections S6.4 and S6.5].

10.5 $SU(3)$ β-deformation

$$\bar{W}(\gamma, \mathsf{w}) = \operatorname{tr}(\gamma^1\gamma^2\gamma^3) - q\operatorname{tr}(\gamma^1\gamma^3\gamma^2) - \frac{1}{4}\mathsf{w}^A_{\dot{a}}\mathsf{w}_A{}^{\dot{a}}, \qquad h = 0. \tag{10.8}$$

The three flavour weights are separately conserved, and the census [14, supplement, Section S6.7] uses $\rho = (1-q)/(1+q) = 12345/678$.

Table 10.6: Two-loop tests in the $SU(3)$ β-deformation.

Scope	Source	Target	Lifting result
All sectors through $E + J_L = 10$	All d_2 candidates	-	$\operatorname{rank} d_2 = 0$
$E + J_L = \frac{21}{2}$	$(5,2,2;1,0)_8$, $h = 1$	$(3,0,0;3,2)_7$, $\dim E_2 = 4$	$\operatorname{rank} d_2 = 1$ $[p, p']$

$$Q_2\mathcal{O}_0 = 0, \qquad d_2[\mathcal{O}_0]_2 = [Q_1\mathcal{O}_1]_2 \neq 0. \tag{10.9}$$

10.6 $\mathcal{N}=2$ orbifold quiver: $SU(N)\times SU(N)$

At $g_1=g_2=g$, the twisted action is the $SU(2N)$ superfield action restricted by the orbifold projection [14, results report, §3.1]:

$$S_{\rm tw}=\int_{\mathbb{C}^2}\Omega\wedge\left[\int d^3\theta\left(\frac{1}{2}\delta_{AB}C^A\bar{\partial}C^B+\frac{i}{3!}C_{ABC}C^AC^BC^C\right)\right]_{(0,2)},$$
$$\mathrm{diag}(\mathbf{1}_N,-\mathbf{1}_N)\,C(\theta_1,\theta_2,\theta_3)\,\mathrm{diag}(\mathbf{1}_N,-\mathbf{1}_N)=C(-\theta_1,-\theta_2,\theta_3). \tag{10.10}$$

The fields $\mathsf{c},\mathsf{b},\gamma^3,\beta_3$ are block diagonal, with each diagonal block traceless, while $\gamma^{1,2},\beta_{1,2}$ are off diagonal; this gives two adjoint chirals and two bifundamental hypermultiplets.

$$\kappa=\frac{g_2}{g_1}\neq 0\quad\Longrightarrow\quad E_r(\kappa)\cong E_r(1)\qquad(r\geq 1). \tag{10.11}$$

Table 10.7: Lifting tests in the $\mathcal{N}=2$ orbifold quiver.

Gauge group	Complete census through	d_1	Higher-loop test
$SU(2)\times SU(2)$	$E+J_L=\frac{21}{2}$	Nonzero in the listed sectors	Every d_2 candidate has rank 0 at $\kappa^2=1,9$
$SU(3)\times SU(3)$	$E+J_L=9$	Nonzero in the listed sectors	Every d_2 candidate has rank 0; d_3 candidates at levels $9,10$ have rank 0

The sector dimensions and the tested pairs are in [14, supplement, Section S7.1]; the fortuitous or monotone character of these classes awaits the stable-image computation.

10.7 $\mathcal{N}=2$ SQCD: $SU(N)$ with $N_f=2N$

The first gauge node remains dynamical: $\mathsf{c},\mathsf{b},\gamma^3,\beta_3$ occupy its traceless diagonal block, and $\gamma^{1,2},\beta_{1,2}$ occupy both off-diagonal blocks, furnishing $2N$ fundamental hypermultiplets [14, results report, §4.1].

Here C_{ABC} and δ^{AB} are inherited from the ambient $\mathfrak{su}(2N)$ algebra. In the cubic term, D,E run over the off-diagonal matter blocks and F over the first-node adjoint; the gauge kinetic indices run over that same first-node adjoint.

$$\bar{W}(\gamma,\mathsf{w})=iC_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^1)_D(\gamma^2)_E(\gamma^3)_F-\frac{1}{4}\mathsf{w}^A_{\dot{a}}\mathsf{w}_A{}^{\dot{a}},$$
$$G=SU(N),\qquad N_f=2N. \tag{10.12}$$

Table 10.8: Lifting tests in $\mathcal{N}=2$ SQCD.

Gauge group	Complete census through	d_1	Higher-loop test
$SU(2)$, $N_f=4$	$E+J_L=12$	$d_1=0$ throughout the census	Every d_2 candidate has rank 0
$SU(3)$, $N_f=6$	$E+J_L=9$	Nonzero in the sectors listed in [14]	Every d_2 candidate has rank 0

The flavour-singlet spectra and the candidate counts are in [14, supplement, Section S7.1].

10.8 Adjoint-mass deformations

For the one-node theories, the mass term added to the displayed twisted action is [14, results report, §5.1]

$$\Delta S_{\mathrm{tw}} = \int_{\mathbb{C}^2} \Omega \wedge \left[\frac{M}{2} \delta^{AB} (\gamma^3)_A (\gamma^3)_B \right]_{(0,2)}, \qquad M \neq 0. \tag{10.13}$$

For the two-node quiver, the same term has mass M on the first adjoint block and $-M$ on the second.

Eliminating the massive adjoint gives the quartic $\mathcal{N} = 1$ theories of [14, supplement, Section S7.2]; for the one-mass $\mathcal{N} = 4$ theory,

$$\begin{aligned} 0 &= \frac{\partial \bar{W}}{\partial (\gamma^3)_F} = M \delta^{FG} (\gamma^3)_G + i C_{ABC} \delta^{AD} \delta^{BE} \delta^{CF} (\gamma^1)_D (\gamma^2)_E, \\ \bar{W}(\gamma, 0) &= \frac{1}{2M} C_{ABC} C_{DEF} \delta^{CF} \delta^{AG} \delta^{BH} \delta^{DI} \delta^{EJ} (\gamma^1)_G (\gamma^2)_H (\gamma^1)_I (\gamma^2)_J. \end{aligned} \tag{10.14}$$

The mass-deformed sectors are labelled $(A, B; n_+, n_-; \mathrm{gh})$, with infrared weight $w = A + B + 2(n_+ + n_-)$.

Table 10.9: Classes of the mass-deformed $SU(2)$ theory.

Theory / class	Sector	Classification	Lifting result
$SU(2)$, w^2	$(6, 6; 0, 0; -4)$	Product of the graviton w	Closed completion through $\hbar^3$; all-order survival
$SU(2)$, $\mathcal{O}_7$	$(5, 2; 0, 0; -2)$	Class of [31]	$d_1 = 0$; higher targets empty; no incoming map $[p]$
$SU(2)$, $\mathcal{O}_{16}$	$(8, 8; 0, 0; -5)$	Non-graviton class of [31]	$d_1 = d_2 = d_3 = d_4 = 0$; higher targets empty; no incoming map $[p]$

Table 10.10: Census of the mass-deformed theories.

Ultraviolet theory and mass	Complete weight bound	Nonzero one-loop maps	Higher-loop results
$\mathcal{N} = 4$ $SU(2)$, M	$w \leq 16$	$(6, 6; 1, 1; -3)$	All tested d_2, d_3 candidate ranks zero
$\mathcal{N} = 4$ $SU(3)$, M	$w \leq 13$	None in the census	All tested d_2, d_3 candidate ranks zero
SQCD $SU(2)$, $N_f = 4$, M	$w \leq 15$	None in the census	All tested d_2, d_3 candidate ranks zero
SQCD $SU(3)$, $N_f = 6$, M	$w \leq 13$	None in the census	All tested d_2 candidate ranks zero
$SU(2) \times SU(2)$ quiver, $(M, -M)$	$w \leq 13$	$(2, 2; k, 1; 1)$, $k = 1, 2, 3$	All tested d_2, d_3 candidate ranks zero

The representatives of w^2 and $\mathcal{O}_{16}$ and the full mass-deformed census are in [14, supplement, Sections S7.2 and S8.4].

10.9 Massive SQED with a cubic superpotential

The gauge group is $U(1)$, with chiral multiplets of charges $+1, -1, 0$ and $gmy \neq 0$. The charge matrix is $t = \mathrm{diag}(-1, 1, 0)$ in $(+, -, s)$ order [14, results report, §5.4]:

$$S_{\mathrm{tw}} = \int_{\mathbb{C}^2} \Omega \wedge \Big[\mathsf{b}\bar\partial \mathsf{c} + (\bar\partial\gamma_+)\beta^+ + (\bar\partial\gamma_-)\beta^- + (\bar\partial\gamma_s)\beta^s + ig\mathsf{c}(\gamma_+\beta^+ - \gamma_-\beta^-) + \frac{m}{2}\gamma_s^2 + y\gamma_s\gamma_+\gamma_- - \frac{1}{4}\mathsf{w}_{\dot a}\mathsf{w}^{\dot a} \Big]_{(0,2)}. \tag{10.15}$$

Table 10.11: The two-loop lift in massive SQED.

Source	First page	Second obstruction	Certificate
The weight-14 class, derivatives $(1,0)$	$\dim E_1 = 18 - 17 = 1$; $d_1 = 0$; no incoming d_1	$d_2[\mathcal{O}_0]_2 \neq 0$	Quotient rank $394 \to 395$ $[\mathbb{Q}]$

$$d_2[\mathcal{O}_0]_2 = -\frac{2(ig)^2 y^4}{m^2(64\pi^2)^2} \Big[\partial_{+\dot a}(\gamma_+\gamma_-)\, \partial_{+}{}^{\dot b}\partial_{+\dot c}(\gamma_+\gamma_-) \times \partial_{+}{}^{\dot b}\partial_{+}{}^{\dot c}(\gamma_+\gamma_-) \Big]_2 \neq 0. \tag{10.16}$$

The exact elimination of the massive pair, the first correction and the rational quotient certificate are in [14, supplement, Section S8.1]. Further abelian theories with a gauge-neutral chiral multiplet have two-loop lifts, and three-loop lifts $d_3 \neq 0$ occur in $\mathcal{N} = 2$ SQED with one flavour and in its deformation by a cubic superpotential of the neutral multiplet, in each case first at weight 11, where they are certified over $\mathbb{Q}$; two-loop lifts $d_2 \neq 0$ also occur in the $\mathcal{N} = 2$ $U(1)^2$ linear quiver, and a four-loop lift $d_4 \neq 0$ occurs in the cubic-superpotential deformation at weight 16, certified over $\mathbb{Q}$ [14, results report, §8]. In non-abelian component tests, two-loop lifts $d_2 \neq 0$ occur in pure $\mathcal{N} = 2$ $SU(2)$ and $SU(3)$ Yang–Mills theory, certified over $\mathbb{Q}$ first at weight 10 for $SU(2)$, and both two-loop ($d_2 \neq 0$, weight 9) and three-loop ($d_3 \neq 0$, weight 14) lifts occur in $\mathcal{N} = 2$ $SU(2)$ Yang–Mills theory coupled to a flipping singlet, certified over $\mathbb{Q}$ [14, results report, §9].

11 Discussion

We close with three directions.

The physical theory. All our results are obtained in the holomorphically twisted theory. It remains to understand the same structures directly in the physical theory: the supersymmetry transformation of composite operators and, more generally, the full BV transformation that includes every symmetry together with its ghosts and auxiliary fields. This requires complete control of the counterterms and of the ultraviolet divergences of composite operators, and would give a complete account of quantum anomalies in this generalized sense and of the exact structure of local operators in supersymmetric gauge theories. It is considerably harder than the twisted computation suggests.

Lifting beyond one loop. The most immediate question is whether $\mathcal{N} = 4$ Yang–Mills theory has a two-loop lift at all. In the ranges that we have scanned, the $\mathcal{N} = 4$ theories have a nonzero d_1 in nineteen sectors, one for $SU(2)$ [20], sixteen for $SO(7)$ through $E + J_L = 11$, the first of them found in [2], and one each for $SO(9)$ and $SO(11)$, and no nonzero d_r with $r \geq 2$ [14, supplement, Section S3.3]. By contrast, the complete census of the $SU(3)$ Leigh–Strassler theory

through $E + J_L = 11$ has 53 pairs of sectors with $d_1 \neq 0$ and 12 with $d_2 \neq 0$ [14, supplement, Section S6.2]. The scans should be pushed to higher levels and ranks, and the algorithm improved. Most of our ranks are computed modulo a large prime; the results marked [$\mathbb{Q}$] are certified over the rational numbers, and the others remain to be certified in the same way. Nonperturbative lifts of perturbative cohomology classes by instantons [3, 4], and their interplay with the higher brackets and the quantum product, are also worth studying.

Other dimensions and the gravity dual. The circuit reduction is not tied to four dimensions: the graph integrals and their current equations exist on $\mathbb{C}^d$ and in holomorphic–topological theories [5, 54], and the iterated network reductions should evaluate the corresponding Feynman graphs. It would also be interesting to identify the counterparts of the loop corrections, of the quantum product and of lifting in the gravity dual.

Acknowledgments

AI tools were used to explore simplification routes for the Feynman graphs and to improve the algorithms and the code of the lifting tests.

A Graph integrals and network reductions

We reduce each graph integral to an integro-differential operator acting on the kernel of a smaller graph.

A.1 The $\mathcal{N} = 4$ Yang–Mills superfield

The $\mathcal{N} = 4$ SYM theory has the superpotential

$$\bar{W}(\gamma, 0) = \frac{i}{3!}\varepsilon_{IJK} C_{ABC} \delta^{AD}\delta^{BE}\delta^{CF}(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F. \tag{A.1}$$

For the twist by Q_-^4, the three adjoint chiral multiplets can be further combined with the gauge fields into one superfield $C^A(\theta)$ [1, 4].

The odd coordinates obey $\theta_I\bar{\partial} = -\bar{\partial}\theta_I$, with $\int d^3\theta\, \theta_1\theta_2\theta_3 = 1$ and $\varepsilon^{123} = \varepsilon_{123} = 1$.

$$C^A(\theta) = ig\,\mathsf{c}^A + \theta_I\delta^{AB}(\gamma^I)_B - \frac{1}{2}\varepsilon^{IJK}\theta_I\theta_J(\beta_K)^A + \frac{\theta_1\theta_2\theta_3}{ig}\delta^{AB}\mathsf{b}_B, \tag{A.2}$$

The physical meaning of each θ_I is $Q_+^I C^A = \partial_L C^A/\partial\theta_I$.

With $(t_A^{\mathrm{adj}})^B{}_C = iC^B{}_{AC}$, the action and differential are

$$\begin{aligned} S_{\mathrm{tw}} &= \int_{\mathbb{C}^2} \Omega \wedge \left[\int d^3\theta \left(\frac{1}{2}\delta_{AB}C^A\bar{\partial}C^B + \frac{i}{3!}C_{ABC}C^AC^BC^C\right)\right]_{(0,2)}, \\ Q_0C^A &= \frac{i}{2}C^A{}_{BC}C^BC^C. \end{aligned} \tag{A.3}$$

The left fibre derivative and odd integration reproduce the component pairing,

$$\begin{aligned} \frac{\delta}{\delta C^A(\theta)} &= \frac{\theta_1\theta_2\theta_3}{ig}\frac{\partial_L}{\partial \mathsf{c}^A} - \frac{1}{2}\varepsilon^{IJK}\theta_J\theta_K\frac{\partial}{\partial(\gamma^I)_A} - \theta_I\frac{\partial_L}{\partial(\beta_I)^A} - ig\frac{\partial}{\partial \mathsf{b}_A}, \\ \frac{\delta C^B(\theta^{(2)})}{\delta C^A(\theta^{(1)})} &= \delta_A{}^B\prod_{I=1}^{3}(\theta_I^{(1)} - \theta_I^{(2)}). \end{aligned} \tag{A.4}$$

$$\Delta_{vw} = -\delta^{DE}\int d^3\theta\, \frac{\delta}{\delta C_v^D(\theta)}\frac{\delta}{\delta C_w^E(\theta)}. \tag{A.5}$$

A.2 Spanning trees

In principle, a formalism already exists for computing all integrals of the form (3.28) [5, Theorem 4 and Corollary 2], which solves the discrete Laplace equation on the circuit exactly and expresses all edge currents in terms of the external currents λ and the resistances t_e. However, this formula is cumbersome and lengthy, with exponential complexity. We review it briefly.

To evaluate the current integral at fixed resistances, first solve the conservation law on each spanning tree. Then we sum over the spanning trees with the weights in (A.6); the result automatically satisfies the loop voltage equation.

Deleting $e \in T$ gives its current as the source sum on the component containing $e(0)$.

The weighted average over spanning trees solves the full network [5],

$$y^T_{e,+\dot{a}} = \begin{cases} \sum\limits_{\substack{v \text{ connected to } e(0) \\ \text{in } T\setminus\{e\}}} \lambda_{v,+\dot{a}}, & e \in T, \\ 0, & e \notin T, \end{cases}$$

$$y_{e,+\dot{a}}(t) = \frac{\sum\limits_{\substack{T \\ \text{spanning trees}}} \left(\prod\limits_{h\in\Gamma_1\setminus T} t_h \right) y^T_{e,+\dot{a}}}{\sum\limits_{\substack{T \\ \text{spanning trees}}} \prod\limits_{h\in\Gamma_1\setminus T} t_h}. \tag{A.6}$$

Proof.

Every y^T satisfies the conservation law.

The normalized tree weights in the second line of (A.6) sum to one, so the average still obeys the conservation law.

We now check the voltage law, orienting the edges of a loop L along its traversal.

Group the numerator of $\sum_{e\in L} t_e y_{e,+\dot{a}}(t)$ by the two-component spanning forest $T \setminus \{e\}$.

$$t_e \prod_{h\in\Gamma_1\setminus T} t_h = \prod_{h\in\Gamma_1\setminus(T\setminus\{e\})} t_h, \qquad e \in T.$$

For each fixed cut forest, pair each exit from one component with the next entry into it; these crossing edges e, f give spanning trees T, T' with $T \setminus \{e\} = T' \setminus \{f\}$.

$$t_e \left(\prod_{h\in\Gamma_1\setminus T} t_h \right) y^T_{e,+\dot{a}} + t_f \left(\prod_{h\in\Gamma_1\setminus T'} t_h \right) y^{T'}_{f,+\dot{a}}$$
$$= \left(\prod_{h\in\Gamma_1\setminus(T\setminus\{e\})} t_h \right) \sum_{v\in\Gamma_0} \lambda_{v,+\dot{a}} = 0,$$

$$\sum_{e\in L} t_e y_{e,+\dot{a}}(t)$$
$$= \frac{\sum\limits_{\substack{T \\ \text{spanning trees}}} \sum\limits_{e\in L\cap T} t_e \left(\prod\limits_{h\in\Gamma_1\setminus T} t_h \right) y^T_{e,+\dot{a}}}{\sum\limits_{\substack{T \\ \text{spanning trees}}} \prod\limits_{h\in\Gamma_1\setminus T} t_h} = 0. \qquad \square$$

A common rescaling of the resistances does not affect the formula, so we may impose $\sum_{e\in\Gamma_1} t_e = 1$.

Then the integral is just the Jacobian of the change of variables from the y's to the independent t's.

A.3 Local reductions as kernel operators

This section gives the details of (5.7) and of the reduction procedure of Proposition 7.6. Each move replaces a circuit by a reduced one, whose quantities carry the superscript red; the kernel operator of the move follows from the corresponding change of variables.

Series elimination

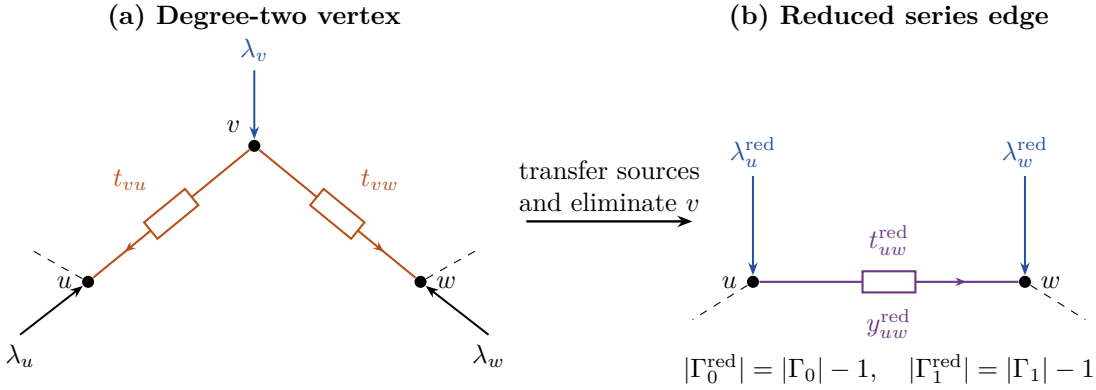


Figure A.1: Series elimination. The source at v is transferred to u, w as in (A.8); the loop number is unchanged.

If v has only two neighbours, u, w, we can eliminate v by first transferring the external current injected at v to u and w.

Conservation and the voltage between them give

$$\begin{aligned} y_{vu,+\dot{a}} + y_{vw,+\dot{a}} &= \lambda_{v,+\dot{a}}, \\ \frac{(x_u - x_w)^-_{\dot{a}}}{2} &= -t_{vu} y_{vu,+\dot{a}} + t_{vw} y_{vw,+\dot{a}} \\ &= t_{vw}\lambda_{v,+\dot{a}} - (t_{vu} + t_{vw}) y_{vu,+\dot{a}}. \end{aligned} \tag{A.7}$$

For the elimination labelled j, take $r_j = t_{vu}/(t_{vu} + t_{vw})$.

The reduced edge $u \to w$ and transferred sources are

$$\begin{aligned} r_j &= \frac{t_{vu}}{t_{vu} + t_{vw}}, & t_{uw}^{\mathrm{red}} &= t_{vu} + t_{vw}, \\ \lambda_{u,+\dot{a}}^{\mathrm{red}} &= \lambda_{u,+\dot{a}} + (1 - r_j)\lambda_{v,+\dot{a}}, & \lambda_{w,+\dot{a}}^{\mathrm{red}} &= \lambda_{w,+\dot{a}} + r_j \lambda_{v,+\dot{a}}, \\ y_{vu,+\dot{a}} &= (1 - r_j)\lambda_{v,+\dot{a}} - y_{uw,+\dot{a}}^{\mathrm{red}}, & y_{vw,+\dot{a}} &= r_j \lambda_{v,+\dot{a}} + y_{uw,+\dot{a}}^{\mathrm{red}}. \end{aligned} \tag{A.8}$$

The intermediate current form.

For an n-loop Laman graph, series elimination preserves the loop number h_1:

$$|\Gamma_0^{\mathrm{red}}| = n+1, \qquad |\Gamma_1^{\mathrm{red}}| = 2n = 2|\Gamma_0^{\mathrm{red}}| - 2,$$
$$h_1(\Gamma^{\mathrm{red}}) = 2n - (n+1) + 1 = n, \qquad \dim\left\{ t_e^{\mathrm{red}} > 0,\ \sum_{e\in\Gamma_1^{\mathrm{red}}} t_e^{\mathrm{red}} = 1 \right\} = 2n-1.$$

At fixed reduced sources, the $2n$-current form vanishes on this $(2n-1)$-dimensional resistance-ratio space; retain the r_j-dependence of the transferred sources in (A.8).

Order vu, vw first and choose a spanning tree T containing both; put the new series edge first in the reduced order, followed by the retained edges in their original order.

Evaluate their currents y_e on the reduced circuit, keeping y_{uw}^{red} for the new series edge, distinct from any pre-existing uw edge.

$$\sum_{e\in\Gamma_1} y_{e,+\dot a} z_e^{+\dot a} = \lambda_{v,+\dot a}[(1-r_j)z_{vu} + r_j z_{vw}]^{+\dot a} + y_{uw,+\dot a}^{\mathrm{red}}(z_{vw}-z_{vu})^{+\dot a} + \sum_{e\in\Gamma_1\setminus\{vu,vw\}} y_{e,+\dot a} z_e^{+\dot a}.$$

Here d varies r_j and the positive reduced resistance ratios $\mathbb{RP}^{2n-1}_{>0}$; $\iota_{\partial/\partial r_j}$ extracts the coefficient of dr_j.

$$\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda; z) = -\int_0^1 dr_j\, e^{-\lambda_{v,+\dot a}[(1-r_j)z_{vu}+r_j z_{vw}]^{+\dot a}} \times \int_{\mathbb{RP}^{2n-1}_{>0}} e^{-y_{uw,+\dot a}^{\mathrm{red}}(z_{vw}-z_{vu})^{+\dot a} - \sum_{e\in\Gamma_1\setminus\{vu,vw\}} y_{e,+\dot a} z_e^{+\dot a}} \times \iota_{\partial/\partial r_j} \bigwedge_{e\notin T} \left(-dy_{e,++} \wedge dy_{e,+-} \right).$$

The inner integral uses the positive reduced resistance ratios, with the orientation $dt_{e_2} \wedge \cdots \wedge dt_{e_{|\Gamma_1|}}$ on the simplex $t_{e_1} = 1 - \sum_{j\geq 2} t_{e_j}$; its overall minus sign follows from

$$t_{vu} = r_j t_{uw}^{\mathrm{red}}, \qquad t_{vw} = (1-r_j) t_{uw}^{\mathrm{red}}, \qquad t_{uw}^{\mathrm{red}} = 1 - \sum_{e\in\Gamma_1\setminus\{vu,vw\}} t_e,$$
$$dt_{vw} \wedge \bigwedge_{e\in\Gamma_1\setminus\{vu,vw\}} dt_e = -t_{uw}^{\mathrm{red}} dr_j \wedge \bigwedge_{e\in\Gamma_1\setminus\{vu,vw\}} dt_e.$$

The intermediate object is the displayed $(2n-1)$-form on the reduced resistance-ratio space.

Parallel merge

Orient the q parallel edges from u to w, labelled by $j = 1, \ldots, q$.

Their common voltage gives

$$t_{uw}^{(j)} y_{uw,+\dot a}^{(j)} = \frac{(x_u - x_w)_{\dot a}^-}{2},$$
$$y_{uw,+\dot a}^{\mathrm{red}} = \sum_{j=1}^q y_{uw,+\dot a}^{(j)} = \left(\sum_{j=1}^q \frac{1}{t_{uw}^{(j)}} \right) \frac{(x_u - x_w)_{\dot a}^-}{2}. \tag{A.9}$$

$$\frac{1}{t_{uw}^{\mathrm{red}}} = \sum_{j=1}^{q} \frac{1}{t_{uw}^{(j)}}, \qquad r_j = \frac{t_{uw}^{\mathrm{red}}}{t_{uw}^{(j)}}, \qquad \sum_{j=1}^{q} r_j = 1, \tag{A.10}$$
$$y_{uw,+\dot{a}}^{(j)} = r_j y_{uw,+\dot{a}}^{\mathrm{red}}.$$

One-step kernel reductions

We now integrate out the eliminated vertices.

Each move becomes an operator on the reduced kernel.

In successive moves, derivatives act on the entire expression to their right.

Theorem A.1 (Series–parallel)**.** Let a Laman circuit have a degree-two vertex v, edges vu, vw, and an edge uw. Order vu, vw first. After elimination, retain the other edge positions, including the merged uw. The current fractions satisfy $0 < r_1, r_2 < 1$. The new circuit is still a Laman circuit.

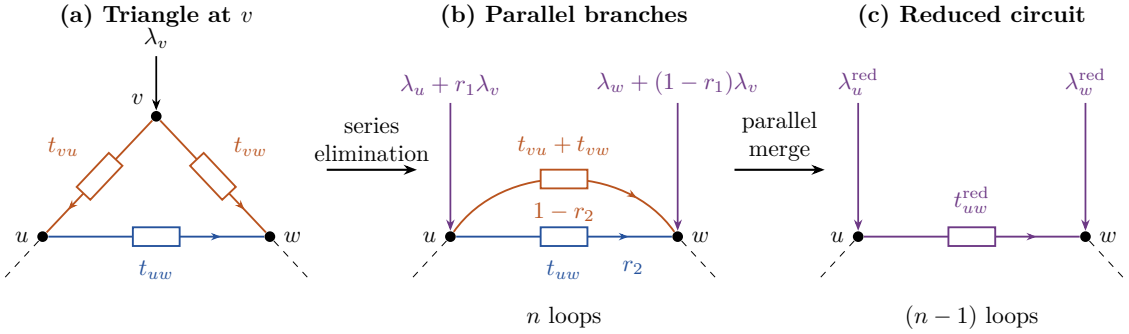


Figure A.2: Series–parallel reduction. The series branch (orange) merges with the original uw edge (blue), removing one loop.

$$r_1 = \frac{t_{vw}}{t_{vu} + t_{vw}}, \qquad r_2 = \frac{t_{vu} + t_{vw}}{t_{vu} + t_{vw} + t_{uw}},$$
$$\lambda_u^{\mathrm{red}} = \lambda_u + r_1\lambda_v, \qquad \lambda_w^{\mathrm{red}} = \lambda_w + (1 - r_1)\lambda_v.$$

$$\mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda; z) = \int_0^1 dr_1 \int_0^1 dr_2 \times e^{-\lambda_{v,+\dot{a}}(r_1 z_{vu} + (1-r_1) z_{vw})^{+\dot{a}}} \underbrace{r_2 \lambda_{v,+\dot{a}} \frac{\partial}{\partial z_{uw,\dot{a}}^{\mathrm{red},+}}}_{\text{local current Jacobian}} \mathcal{I}_{\Gamma^{\mathrm{red}}}^{\text{raw cut}}(\lambda^{\mathrm{red}}; z^{\mathrm{red}}). \tag{A.11}$$

$$z_{uw}^{\mathrm{red}} = r_2 z_{uw} + (1 - r_2)(z_{vw} - z_{vu}), \qquad z_e^{\mathrm{red}} = z_e \quad (e \neq uw).$$

Proof. The resistance change fixes the orientation.

$$t_{vu} = \frac{(1 - r_1) t_{uw}^{\mathrm{red}}}{1 - r_2}, \quad t_{vw} = \frac{r_1 t_{uw}^{\mathrm{red}}}{1 - r_2}, \quad t_{uw} = \frac{t_{uw}^{\mathrm{red}}}{r_2},$$
$$d\left(\frac{t_{vu}}{t_{uw}}\right) \wedge d\left(\frac{t_{vw}}{t_{uw}}\right) = -\frac{r_2}{(1 - r_2)^3}\, dr_1 \wedge dr_2.$$

$$y_{uw} = r_2 y_{uw}^{\mathrm{red}}, \qquad y_{vu} = r_1\lambda_v - (1 - r_2) y_{uw}^{\mathrm{red}}, \qquad y_{vw} = \lambda_v - y_{vu}. \tag{A.12}$$

$$\sum_{\substack{e\in\Gamma_1\\ \text{edges}}} y_{e,+\dot a} z_e^{+\dot a} = \lambda_{v,+\dot a}(r_1 z_{vu} + (1-r_1) z_{vw})^{+\dot a} + \sum_{\substack{e\in\Gamma_1^{\mathrm{red}}\\ \text{edges}}} y^{\mathrm{red}}_{e,+\dot a} z_e^{\mathrm{red},+\dot a}.$$

Choose a reduced spanning tree containing uw; conservation at fixed λ makes $d(y^{\mathrm{red}}_{uw,+\dot a} - r_1\lambda_{v,+\dot a})$ a linear combination of the retained loop-current differentials; $\equiv$ denotes equality modulo those differentials.

$$\begin{aligned}
dy^{\mathrm{red}}_{uw,+\dot a} &= \lambda_{v,+\dot a} dr_1 + d\Big(y^{\mathrm{red}}_{uw,+\dot a} - r_1\lambda_{v,+\dot a}\Big),\\
dy_{vu,+\dot a} &= r_2\lambda_{v,+\dot a} dr_1 + y^{\mathrm{red}}_{uw,+\dot a} dr_2 - (1-r_2) d\Big(y^{\mathrm{red}}_{uw,+\dot a} - r_1\lambda_{v,+\dot a}\Big),\\
-dy_{vu,+\dot+} \wedge dy_{vu,+\dot-} &= -r_2\left(\lambda_{v,+\dot a} y^{\mathrm{red}}_{uw,+}{}^{\dot a}\right) dr_1\wedge dr_2\\
&\quad + (1-r_2)\left(r_2\lambda_{v,+\dot a} dr_1 + y^{\mathrm{red}}_{uw,+\dot a} dr_2\right) \wedge d\Big(y^{\mathrm{red}}_{uw,+}{}^{\dot a} - r_1\lambda_{v,+}{}^{\dot a}\Big)\\
&\quad - \frac{(1-r_2)^2}{2} d\Big(y^{\mathrm{red}}_{uw,+\dot a} - r_1\lambda_{v,+\dot a}\Big) \wedge d\Big(y^{\mathrm{red}}_{uw,+}{}^{\dot a} - r_1\lambda_{v,+}{}^{\dot a}\Big)\\
&\equiv -r_2\left(\lambda_{v,+\dot a} y^{\mathrm{red}}_{uw,+}{}^{\dot a}\right) dr_1\wedge dr_2.
\end{aligned}$$

$$\begin{aligned}
&\lambda_{v,+\dot a}\frac{\partial}{\partial z^{\mathrm{red},+}_{uw,\dot a}} e^{-\sum_{\substack{e\in\Gamma_1^{\mathrm{red}}\\ \text{edges}}} y^{\mathrm{red}}_{e,+\dot b} z_e^{\mathrm{red},+\dot b}}\\
&= (\lambda_{v,+\dot a} y^{\mathrm{red}}_{uw,+}{}^{\dot a}) e^{-\sum_{\substack{e\in\Gamma_1^{\mathrm{red}}\\ \text{edges}}} y^{\mathrm{red}}_{e,+\dot b} z_e^{\mathrm{red},+\dot b}}.
\end{aligned}$$

The two orientation signs cancel, giving (A.11). □

Theorem A.2 (Star–triangle, the first case: two edges present)**.** In a Laman circuit, let v have degree three, neighbours u, w, x, and source λ_v. Assume ux, wx are present and uw is absent. Let r_1, r_2 be the old ux, wx current fractions. The reduction removes one vertex, two edges, and one loop, preserving the Laman condition.

For $\lambda_v = 0$, the operator is (A.13); the complete sourced version (A.17) follows below.

$$\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda; z) = \int_0^1 dr_1 \int_0^1 dr_2 \underbrace{-r_1 r_2 \frac{\partial}{\partial z^{\mathrm{red},+\dot a}_{ux}} \frac{\partial}{\partial z^{\mathrm{red},+}_{wx,\dot a}}}_{\text{local current Jacobian}} \mathcal{I}_{\Gamma^{\mathrm{red}}}^{\text{raw cut}}(\lambda; z^{\mathrm{red}}). \tag{A.13}$$

$$\begin{aligned}
z^{\mathrm{red}}_{uw} &= z_{vw} - z_{vu},\\
z^{\mathrm{red}}_{ux} &= r_1 z_{ux} + (1-r_1)(z_{vx} - z_{vu}),\\
z^{\mathrm{red}}_{wx} &= r_2 z_{wx} + (1-r_2)(z_{vx} - z_{vw}),\\
z^{\mathrm{red}}_e &= z_e \quad (e \neq uw, ux, wx).
\end{aligned}$$

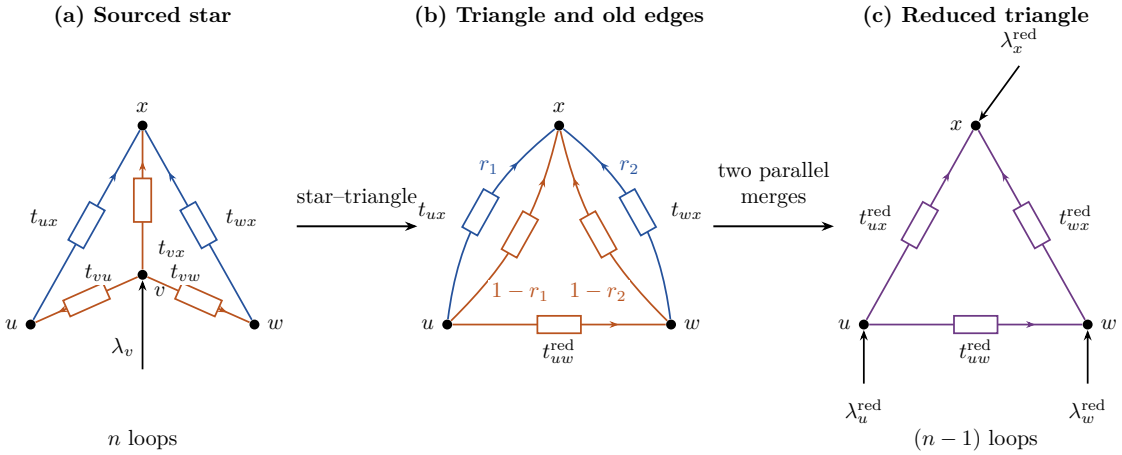


Figure A.3: Sourced star–triangle reduction. The source at v is transferred to u, w, x; the blue and orange labels are fractions of the final merged currents.

Proof for $\lambda_v = 0$. The star–triangle change and the two merges give

$$
\begin{aligned}
t^{\mathrm{red}}_{uw} &= t_{vu} + t_{vw} + \frac{t_{vu}t_{vw}}{t_{vx}}, \\
\frac{1}{t^{\mathrm{red}}_{ux}} &= \frac{1}{t_{ux}} + \frac{t_{vw}}{t_{vu}t_{vw} + t_{vw}t_{vx} + t_{vx}t_{vu}}, \\
\frac{1}{t^{\mathrm{red}}_{wx}} &= \frac{1}{t_{wx}} + \frac{t_{vu}}{t_{vu}t_{vw} + t_{vw}t_{vx} + t_{vx}t_{vu}}, \\
r_1 &= \frac{t^{\mathrm{red}}_{ux}}{t_{ux}}, \qquad r_2 = \frac{t^{\mathrm{red}}_{wx}}{t_{wx}}.
\end{aligned}
$$

The sign of this resistance change fixes the current-form orientation:

$$
\operatorname{sgn}\det \frac{\partial(t_{vu}, t_{vw}, t_{vx}, t_{ux}, t_{wx})}{\partial(r_1, r_2, t^{\mathrm{red}}_{uw}, t^{\mathrm{red}}_{ux}, t^{\mathrm{red}}_{wx})} = -1.
$$

$$
\begin{aligned}
y_{vu} &= -y^{\mathrm{red}}_{uw} - (1-r_1)y^{\mathrm{red}}_{ux}, \\
y_{vw} &= y^{\mathrm{red}}_{uw} - (1-r_2)y^{\mathrm{red}}_{wx}, \\
y_{vx} &= (1-r_1)y^{\mathrm{red}}_{ux} + (1-r_2)y^{\mathrm{red}}_{wx}, \\
y_{ux} &= r_1 y^{\mathrm{red}}_{ux}, \qquad y_{wx} = r_2 y^{\mathrm{red}}_{wx}.
\end{aligned}
\tag{A.14}
$$

These currents give the stated displacement substitutions.

Choose a reduced spanning tree containing ux, wx; conservation at fixed λ makes $d(y^{\mathrm{red}}_{ux,+\dot{a}} + y^{\mathrm{red}}_{uw,+\dot{a}})$ and $d(y^{\mathrm{red}}_{wx,+\dot{a}} - y^{\mathrm{red}}_{uw,+\dot{a}})$ linear combinations of the other retained loop-current differentials; $\equiv$ denotes equality modulo those differentials.

$$
\begin{aligned}
dy_{ux,+\dot{a}} &= y^{\mathrm{red}}_{ux,+\dot{a}} dr_1 - r_1 dy^{\mathrm{red}}_{uw,+\dot{a}} + r_1 d\left(y^{\mathrm{red}}_{ux,+\dot{a}} + y^{\mathrm{red}}_{uw,+\dot{a}}\right), \\
dy_{wx,+\dot{a}} &= y^{\mathrm{red}}_{wx,+\dot{a}} dr_2 + r_2 dy^{\mathrm{red}}_{uw,+\dot{a}} + r_2 d\left(y^{\mathrm{red}}_{wx,+\dot{a}} - y^{\mathrm{red}}_{uw,+\dot{a}}\right).
\end{aligned}
$$

$$
\begin{aligned}
&(-dy_{ux,++}\wedge dy_{ux,+-})\wedge(-dy_{wx,++}\wedge dy_{wx,+-})\\
&=\frac{1}{4}\Big[y^{\mathrm{red}}_{ux,+\dot a}dr_1-r_1dy^{\mathrm{red}}_{uw,+\dot a}+r_1d\Big(y^{\mathrm{red}}_{ux,+\dot a}+y^{\mathrm{red}}_{uw,+\dot a}\Big)\Big]\\
&\quad\wedge\Big[y^{\mathrm{red}}_{ux,+}{}^{\dot a}dr_1-r_1dy^{\mathrm{red}}_{uw,+}{}^{\dot a}+r_1d\Big(y^{\mathrm{red}}_{ux,+}{}^{\dot a}+y^{\mathrm{red}}_{uw,+}{}^{\dot a}\Big)\Big]\\
&\quad\wedge\Big[y^{\mathrm{red}}_{wx,+\dot b}dr_2+r_2dy^{\mathrm{red}}_{uw,+\dot b}+r_2d\Big(y^{\mathrm{red}}_{wx,+\dot b}-y^{\mathrm{red}}_{uw,+\dot b}\Big)\Big]\\
&\quad\wedge\Big[y^{\mathrm{red}}_{wx,+}{}^{\dot b}dr_2+r_2dy^{\mathrm{red}}_{uw,+}{}^{\dot b}+r_2d\Big(y^{\mathrm{red}}_{wx,+}{}^{\dot b}-y^{\mathrm{red}}_{uw,+}{}^{\dot b}\Big)\Big]\\
&\equiv -r_1r_2(y^{\mathrm{red}}_{ux,+\dot a}y^{\mathrm{red}}_{wx,+}{}^{\dot a})dr_1\wedge dr_2\wedge(-dy^{\mathrm{red}}_{uw,++}\wedge dy^{\mathrm{red}}_{uw,+-}).
\end{aligned}
$$

Replacing the paired currents by the two displacement derivatives proves (A.13). □

Theorem A.2, with a source. The transferred source depends on the reduced resistances. The sum of the three triangle resistances before the parallel merges is

$$
\Sigma=t^{\mathrm{red}}_{uw}+\frac{t^{\mathrm{red}}_{ux}}{1-r_1}+\frac{t^{\mathrm{red}}_{wx}}{1-r_2}. \tag{A.15}
$$

$$
\begin{aligned}
&\lambda^{\mathrm{red}}_u=\lambda_u+\frac{t^{\mathrm{red}}_{wx}}{(1-r_2)\Sigma}\lambda_v, && \lambda^{\mathrm{red}}_w=\lambda_w+\frac{t^{\mathrm{red}}_{ux}}{(1-r_1)\Sigma}\lambda_v,\\
&\lambda^{\mathrm{red}}_x=\lambda_x+\frac{t^{\mathrm{red}}_{uw}}{\Sigma}\lambda_v,\\
&y_{vu}=\frac{t^{\mathrm{red}}_{wx}}{(1-r_2)\Sigma}\lambda_v-y^{\mathrm{red}}_{uw}-(1-r_1)y^{\mathrm{red}}_{ux},\\
&y_{vw}=\frac{t^{\mathrm{red}}_{ux}}{(1-r_1)\Sigma}\lambda_v+y^{\mathrm{red}}_{uw}-(1-r_2)y^{\mathrm{red}}_{wx},\\
&y_{vx}=\frac{t^{\mathrm{red}}_{uw}}{\Sigma}\lambda_v+(1-r_1)y^{\mathrm{red}}_{ux}+(1-r_2)y^{\mathrm{red}}_{wx},\\
&y_{ux}=r_1y^{\mathrm{red}}_{ux}, && y_{wx}=r_2y^{\mathrm{red}}_{wx}.
\end{aligned}
\tag{A.16}
$$

$$
\begin{aligned}
\sum_{\substack{e\in\Gamma_1\\ \text{edges}}}y_{e,+\dot a}z_e^{+\dot a}&=\frac{\lambda_{v,+\dot a}}{\Sigma}\left(\frac{t^{\mathrm{red}}_{wx}z_{vu}}{1-r_2}+\frac{t^{\mathrm{red}}_{ux}z_{vw}}{1-r_1}+t^{\mathrm{red}}_{uw}z_{vx}\right)^{+\dot a}\\
&\quad+\sum_{\substack{e\in\Gamma^{\mathrm{red}}_1\\ \text{edges}}}y^{\mathrm{red}}_{e,+\dot a}z_e^{\mathrm{red},+\dot a}.
\end{aligned}
$$

Use the same z^{red} as above, and leave all other vertex sources unchanged. Choose a reduced spanning tree T^{red} containing ux, wx; evaluate its currents at the displayed λ^{red}. Here d varies r_1, r_2 and the positive reduced resistance ratios $\mathbb{RP}^{2n-2}_{>0}$, at fixed original λ; $\iota_{\partial/\partial r_2}\iota_{\partial/\partial r_1}$ extracts

the coefficient of $dr_1 \wedge dr_2$.

$$\begin{aligned}
\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda; z) = & -\int_0^1 dr_1 \int_0^1 dr_2 \int_{\mathbb{RP}^{2n-2}_{>0}} \\
& \times \exp\left[-\frac{\lambda_{v,+\dot{a}}}{\Sigma} \left(\frac{t^{\text{red}}_{wx} z_{vu}}{1-r_2} + \frac{t^{\text{red}}_{ux} z_{vw}}{1-r_1} + t^{\text{red}}_{uw} z_{vx} \right)^{+\dot{a}} \right] \\
& \times \exp\left[- \sum_{\substack{e \in \Gamma_1^{\text{red}} \\ \text{edges}}} y^{\text{red}}_{e,+\dot{a}} z_e^{\text{red},+\dot{a}} \right] \iota_{\partial/\partial r_2} \iota_{\partial/\partial r_1} \\
& \left[(-d(r_1 y^{\text{red}}_{ux,++}) \wedge d(r_1 y^{\text{red}}_{ux,+-})) \right. \\
& \quad \wedge (-d(r_2 y^{\text{red}}_{wx,++}) \wedge d(r_2 y^{\text{red}}_{wx,+-})) \\
& \quad \left. \wedge \bigwedge_{\substack{e \in \Gamma_1^{\text{red}} \setminus T^{\text{red}} \\ e \neq uw}} (-dy^{\text{red}}_{e,++} \wedge dy^{\text{red}}_{e,+-}) \right].
\end{aligned} \tag{A.17}$$

The source and exponential stay inside this integral. In particular, the source differentials are

$$\begin{aligned}
d\Sigma &= dt^{\text{red}}_{uw} + \frac{dt^{\text{red}}_{ux}}{1-r_1} + \frac{t^{\text{red}}_{ux}\, dr_1}{(1-r_1)^2} + \frac{dt^{\text{red}}_{wx}}{1-r_2} + \frac{t^{\text{red}}_{wx}\, dr_2}{(1-r_2)^2}, \\
d\lambda^{\text{red}}_{u,+\dot{a}} &= \frac{\lambda_{v,+\dot{a}}}{\Sigma} \left[\frac{dt^{\text{red}}_{wx}}{1-r_2} + \frac{t^{\text{red}}_{wx}\, dr_2}{(1-r_2)^2} - \frac{t^{\text{red}}_{wx}\, d\Sigma}{(1-r_2)\Sigma} \right], \\
d\lambda^{\text{red}}_{w,+\dot{a}} &= \frac{\lambda_{v,+\dot{a}}}{\Sigma} \left[\frac{dt^{\text{red}}_{ux}}{1-r_1} + \frac{t^{\text{red}}_{ux}\, dr_1}{(1-r_1)^2} - \frac{t^{\text{red}}_{ux}\, d\Sigma}{(1-r_1)\Sigma} \right], \\
d\lambda^{\text{red}}_{x,+\dot{a}} &= \frac{\lambda_{v,+\dot{a}}}{\Sigma} \left[dt^{\text{red}}_{uw} - \frac{t^{\text{red}}_{uw}\, d\Sigma}{\Sigma} \right].
\end{aligned}$$

Proof of the sourced version.

Replace ux, wx in T^{red} by vu, vw, vx. The original chords are ux, wx and the reduced chords other than uw, giving exactly the displayed wedge. The resistance change reverses orientation, supplying its overall minus sign. Conservation gives the following exact one-form identities:

$$\begin{aligned}
dy_{ux,+\dot{a}} &= y^{\text{red}}_{ux,+\dot{a}} dr_1 - r_1 dy^{\text{red}}_{uw,+\dot{a}} + r_1 d\lambda^{\text{red}}_{u,+\dot{a}} \\
&\quad + r_1 d\Big(y^{\text{red}}_{ux,+\dot{a}} + y^{\text{red}}_{uw,+\dot{a}} - \lambda^{\text{red}}_{u,+\dot{a}} \Big), \\
dy_{wx,+\dot{a}} &= y^{\text{red}}_{wx,+\dot{a}} dr_2 + r_2 dy^{\text{red}}_{uw,+\dot{a}} + r_2 d\lambda^{\text{red}}_{w,+\dot{a}} \\
&\quad + r_2 d\Big(y^{\text{red}}_{wx,+\dot{a}} - y^{\text{red}}_{uw,+\dot{a}} - \lambda^{\text{red}}_{w,+\dot{a}} \Big).
\end{aligned}$$

The final differential in each line is a linear combination of the other retained chord-current differentials. For $\lambda_v = 0$, also $d\lambda^{\text{red}} = 0$. Modulo those retained differentials,

$$\begin{aligned}
& (-dy_{ux,++} \wedge dy_{ux,+-}) \wedge (-dy_{wx,++} \wedge dy_{wx,+-})\Big|_{\lambda_v = 0} \\
& \equiv -r_1 r_2 \left(y^{\text{red}}_{ux,+\dot{a}} y^{\text{red}}_{wx,+}{}^{\dot{a}} \right) dr_1 \wedge dr_2 \wedge \left(-dy^{\text{red}}_{uw,++} \wedge dy^{\text{red}}_{uw,+-} \right).
\end{aligned}$$

The two orientation signs cancel. Replacing the paired currents by the two displacement derivatives recovers (A.13). □

Theorem A.3 (Star–triangle, the second case: two star–triangle reductions at a time)**.** In a Laman circuit, let v_1, v_2 be source-free degree-three vertices joined to u, A, B, with AB present and uA, uB absent. The conductance fractions contributed by the v_1 star to uA, uB are r_1, r_2; those contributed by the v_2 star are $1-r_1, 1-r_2$. The fractions on AB contributed by v_1, v_2 and the old edge are $r_3, r_4, 1-r_3-r_4$. The reduction removes two vertices, four edges and two loops, preserving the Laman condition.

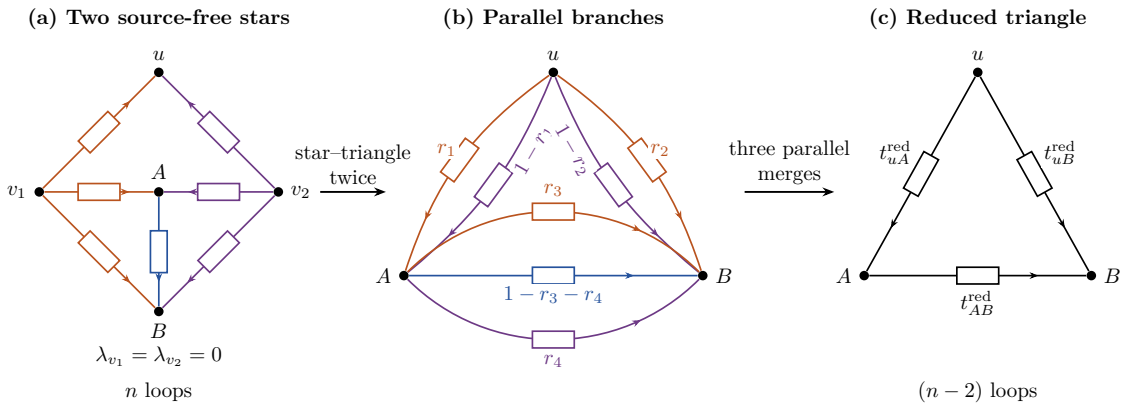


Figure A.4: Two source-free stars reduce to a triangle by two star–triangle moves and three parallel merges. Orange and purple branches come from v_1 and v_2; blue is the old AB edge. The labels in the middle panel are fractions of the final merged currents.

$$\begin{aligned}\mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda; z) = &\int_0^1 dr_1 \int_0^1 dr_2 \int_{\substack{r_3>0,\ r_4>0\\ r_3+r_4<1}} dr_3\, dr_4 \\ &\times (r_1 - r_2)(1 - r_3 - r_4) \\ &\times \frac{\partial}{\partial z_{uA}^{\mathrm{red},+\dot{a}}} \frac{\partial}{\partial z_{AB,\dot{a}}^{\mathrm{red},+}} \frac{\partial}{\partial z_{uB}^{\mathrm{red},+\dot{c}}} \frac{\partial}{\partial z_{AB,\dot{c}}^{\mathrm{red},+}} \mathcal{I}_{\Gamma^{\mathrm{red}}}^{\text{raw cut}}(\lambda; z^{\mathrm{red}}).\end{aligned} \tag{A.18}$$

$$\begin{aligned} z_{uA}^{\mathrm{red}} &= r_1(z_{v_1A} - z_{v_1u}) + (1 - r_1)(z_{v_2A} - z_{v_2u}), \\ z_{uB}^{\mathrm{red}} &= r_2(z_{v_1B} - z_{v_1u}) + (1 - r_2)(z_{v_2B} - z_{v_2u}), \\ z_{AB}^{\mathrm{red}} &= (1 - r_3 - r_4) z_{AB} + r_3(z_{v_1B} - z_{v_1A}) + r_4(z_{v_2B} - z_{v_2A}), \\ z_e^{\mathrm{red}} &= z_e \quad (e \neq uA, uB, AB). \end{aligned}$$

Proof. The two star–triangle changes and three merges preserve orientation.

The branch resistances are

$$\begin{aligned}
\frac{t^{\mathrm{red}}_{uA}}{r_1} &= t_{v_1u}+t_{v_1A}+\frac{t_{v_1u}t_{v_1A}}{t_{v_1B}},\\
\frac{t^{\mathrm{red}}_{uA}}{1-r_1} &= t_{v_2u}+t_{v_2A}+\frac{t_{v_2u}t_{v_2A}}{t_{v_2B}},\\
\frac{t^{\mathrm{red}}_{uB}}{r_2} &= t_{v_1u}+t_{v_1B}+\frac{t_{v_1u}t_{v_1B}}{t_{v_1A}},\\
\frac{t^{\mathrm{red}}_{uB}}{1-r_2} &= t_{v_2u}+t_{v_2B}+\frac{t_{v_2u}t_{v_2B}}{t_{v_2A}},\\
\frac{t^{\mathrm{red}}_{AB}}{r_3} &= t_{v_1A}+t_{v_1B}+\frac{t_{v_1A}t_{v_1B}}{t_{v_1u}},\\
\frac{t^{\mathrm{red}}_{AB}}{r_4} &= t_{v_2A}+t_{v_2B}+\frac{t_{v_2A}t_{v_2B}}{t_{v_2u}},\\
\frac{t^{\mathrm{red}}_{AB}}{1-r_3-r_4} &= t_{AB}.
\end{aligned}$$

$$\begin{aligned}
y_{v_1u} &= -r_1y^{\mathrm{red}}_{uA}-r_2y^{\mathrm{red}}_{uB}, & y_{v_2u} &= -(1-r_1)y^{\mathrm{red}}_{uA}-(1-r_2)y^{\mathrm{red}}_{uB},\\
y_{v_1A} &= r_1y^{\mathrm{red}}_{uA}-r_3y^{\mathrm{red}}_{AB}, & y_{v_2A} &= (1-r_1)y^{\mathrm{red}}_{uA}-r_4y^{\mathrm{red}}_{AB},\\
y_{v_1B} &= r_2y^{\mathrm{red}}_{uB}+r_3y^{\mathrm{red}}_{AB}, & y_{v_2B} &= (1-r_2)y^{\mathrm{red}}_{uB}+r_4y^{\mathrm{red}}_{AB},\\
y_{AB} &= (1-r_3-r_4)y^{\mathrm{red}}_{AB}.
\end{aligned} \tag{A.19}$$

Choose a reduced spanning tree containing uA, uB. At fixed λ, conservation makes $d(y^{\mathrm{red}}_{uA,+\dot a}-y^{\mathrm{red}}_{AB,+\dot a})$ and $d(y^{\mathrm{red}}_{uB,+\dot a}+y^{\mathrm{red}}_{AB,+\dot a})$ linear combinations of the other reduced chord-current differentials. The determinant is

$$\begin{aligned}
&\det\frac{\partial(y_{AB,++},y_{AB,+-},y_{v_1B,++},y_{v_1B,+-},y_{v_2A,++},y_{v_2A,+-})}{\partial(r_1,r_2,r_3,r_4,y^{\mathrm{red}}_{AB,++},y^{\mathrm{red}}_{AB,+-})}\\
&=(r_1-r_2)(1-r_3-r_4)(y^{\mathrm{red}}_{uA,+\dot a}y^{\mathrm{red}}_{AB,+}{}^{\dot a})(y^{\mathrm{red}}_{uB,+\dot b}y^{\mathrm{red}}_{AB,+}{}^{\dot b}).
\end{aligned}$$

$$\begin{aligned}
dy_{AB,+\dot a} &= (1-r_3-r_4)dy^{\mathrm{red}}_{AB,+\dot a}-y^{\mathrm{red}}_{AB,+\dot a}(dr_3+dr_4),\\
dy_{v_1B,+\dot a} &= y^{\mathrm{red}}_{uB,+\dot a}dr_2+y^{\mathrm{red}}_{AB,+\dot a}dr_3+(r_3-r_2)dy^{\mathrm{red}}_{AB,+\dot a}\\
&\quad+r_2d\Big(y^{\mathrm{red}}_{uB,+\dot a}+y^{\mathrm{red}}_{AB,+\dot a}\Big),\\
dy_{v_2A,+\dot a} &= -y^{\mathrm{red}}_{uA,+\dot a}dr_1-y^{\mathrm{red}}_{AB,+\dot a}dr_4+(1-r_1-r_4)dy^{\mathrm{red}}_{AB,+\dot a}\\
&\quad+(1-r_1)d\Big(y^{\mathrm{red}}_{uA,+\dot a}-y^{\mathrm{red}}_{AB,+\dot a}\Big).
\end{aligned}$$

In the following wedge products, $\equiv$ omits terms containing $dy^{\mathrm{red}}_{e,++}$ or $dy^{\mathrm{red}}_{e,+-}$, where $e\neq AB$ lies outside the chosen reduced spanning tree. They vanish after wedging with the area forms of all those remaining chord currents.

$$\begin{aligned}
&dy_{AB,++}\wedge dy_{AB,+-}\\
&=(1-r_3-r_4)^2dy^{\mathrm{red}}_{AB,++}\wedge dy^{\mathrm{red}}_{AB,+-}\\
&\quad-(1-r_3-r_4)(dr_3+dr_4)\wedge(y^{\mathrm{red}}_{AB,+\dot a}dy^{\mathrm{red}}_{AB,+}{}^{\dot a}).
\end{aligned}$$

$$
\begin{aligned}
&dy_{v_1B,+\dot+}\wedge dy_{v_1B,+\dot-}\\
&\equiv (y^{\mathrm{red}}_{uB,+\dot a}y^{\mathrm{red}}_{AB,+}{}^{\dot a})dr_2\wedge dr_3\\
&\quad+(r_3-r_2)dr_2\wedge (y^{\mathrm{red}}_{uB,+\dot a}dy^{\mathrm{red}}_{AB,+}{}^{\dot a})\\
&\quad+(r_3-r_2)dr_3\wedge (y^{\mathrm{red}}_{AB,+\dot a}dy^{\mathrm{red}}_{AB,+}{}^{\dot a})\\
&\quad+(r_3-r_2)^2dy^{\mathrm{red}}_{AB,+\dot+}\wedge dy^{\mathrm{red}}_{AB,+\dot-}.
\end{aligned}
$$

$$
\begin{aligned}
&dy_{v_2A,+\dot+}\wedge dy_{v_2A,+\dot-}\\
&\equiv (y^{\mathrm{red}}_{uA,+\dot a}y^{\mathrm{red}}_{AB,+}{}^{\dot a})dr_1\wedge dr_4\\
&\quad-(1-r_1-r_4)dr_1\wedge (y^{\mathrm{red}}_{uA,+\dot a}dy^{\mathrm{red}}_{AB,+}{}^{\dot a})\\
&\quad-(1-r_1-r_4)dr_4\wedge (y^{\mathrm{red}}_{AB,+\dot a}dy^{\mathrm{red}}_{AB,+}{}^{\dot a})\\
&\quad+(1-r_1-r_4)^2dy^{\mathrm{red}}_{AB,+\dot+}\wedge dy^{\mathrm{red}}_{AB,+\dot-}.
\end{aligned}
$$

$$
\begin{aligned}
&dy_{AB,+\dot+}\wedge dy_{AB,+\dot-}\wedge dy_{v_1B,+\dot+}\wedge dy_{v_1B,+\dot-}\wedge dy_{v_2A,+\dot+}\wedge dy_{v_2A,+\dot-}\\
&\equiv\Big[(1-r_3-r_4)^2-(1-r_3-r_4)(1-r_1-r_4)\\
&\qquad\qquad+(1-r_3-r_4)(r_3-r_2)\Big]\\
&\quad\times (y^{\mathrm{red}}_{uA,+\dot a}y^{\mathrm{red}}_{AB,+}{}^{\dot a})(y^{\mathrm{red}}_{uB,+\dot b}y^{\mathrm{red}}_{AB,+}{}^{\dot b})\\
&\quad\times dr_1\wedge dr_2\wedge dr_3\wedge dr_4\wedge dy^{\mathrm{red}}_{AB,+\dot+}\wedge dy^{\mathrm{red}}_{AB,+\dot-}\\
&=(r_1-r_2)(1-r_3-r_4)(y^{\mathrm{red}}_{uA,+\dot a}y^{\mathrm{red}}_{AB,+}{}^{\dot a})(y^{\mathrm{red}}_{uB,+\dot b}y^{\mathrm{red}}_{AB,+}{}^{\dot b})\\
&\quad\times dr_1\wedge dr_2\wedge dr_3\wedge dr_4\wedge dy^{\mathrm{red}}_{AB,+\dot+}\wedge dy^{\mathrm{red}}_{AB,+\dot-},\\
&\quad\text{modulo } dy^{\mathrm{red}}_{e,+\dot+},\ dy^{\mathrm{red}}_{e,+\dot-}\quad (e\neq AB \text{ a reduced chord}).
\end{aligned}
$$

The three loop-area minus signs reduce to the one on the retained loop. The four displacement derivatives insert the displayed current product. □

A.4 Fan kernels from the one-loop kernel

Repeated application of (A.11) gives integral–derivative operators acting on the known triangle kernel (5.3):

$$
\mathcal{I}^{\text{raw cut}}_{\Gamma}(\lambda;z)=\left[\prod_{i=1}^{n-1}\left(\int_0^1 dr_{2i-1}\int_0^1 dr_{2i}\,e^{-\lambda^{\mathrm{red}}_{i,+\dot a}[r_{2i-1}z_{i,i+1}+(1-r_{2i-1})z^{\mathrm{red}}_{i,n+2}]^{+\dot a}}\right.\right.
\left.\left.\times r_{2i}\lambda^{\mathrm{red}\,\dot b}_{i,+}\frac{\partial}{\partial z^{\mathrm{red},+\dot b}_{i+1,n+2}}\right)\right]\underbrace{\mathcal{I}^{\text{raw cut}}_{\triangle}(\lambda^{\mathrm{red}};z^{\mathrm{red}})}_{\text{vertices } n,n+1,n+2}.
$$

The factor $i=1$ acts outermost; red refers to the indicated stage, and each derivative acts on everything to its right before that stage's displacement substitution.

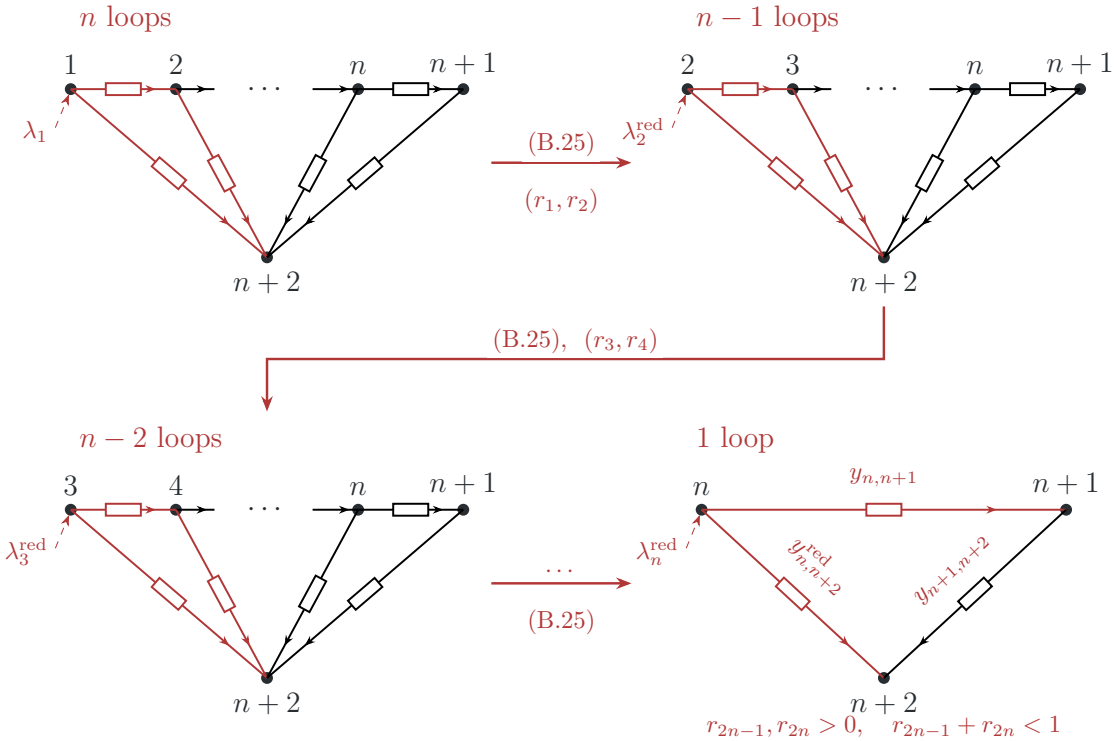


Figure A.5: The operators in (A.11) remove vertices $1, 2, \ldots, n-1$, leaving the one-loop kernel. Source transfer follows the arrows; evaluation and current reconstruction start from the final triangle and run backwards.

The first two operators

For $n \geq 2$, the first factor is (A.11) with $(v, u, w) = (1, 2, n+2)$:

$$\mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda; z) = \int_0^1 dr_1 \int_0^1 dr_2\, e^{-\lambda_{1,+\dot{a}}[r_1 z_{12} + (1-r_1) z_{1,n+2}]^{+\dot{a}}} \times r_2 \lambda_{1,+}{}^{\dot{b}} \frac{\partial}{\partial z_{2,n+2}^{\text{red},+\dot{b}}} \underbrace{\mathcal{I}_{\Gamma^{\text{red}}}^{\text{raw cut}}(\lambda^{\text{red}}; z^{\text{red}})}_{(n-1)\text{-loop fan}},$$

$$\lambda_2^{\text{red}} = \lambda_2 + r_1 \lambda_1, \qquad \lambda_{n+2}^{\text{red}} = \lambda_{n+2} + (1 - r_1)\lambda_1,$$

$$z_{2,n+2}^{\text{red}} = r_2 z_{2,n+2} + (1 - r_2)(z_{1,n+2} - z_{12}).$$

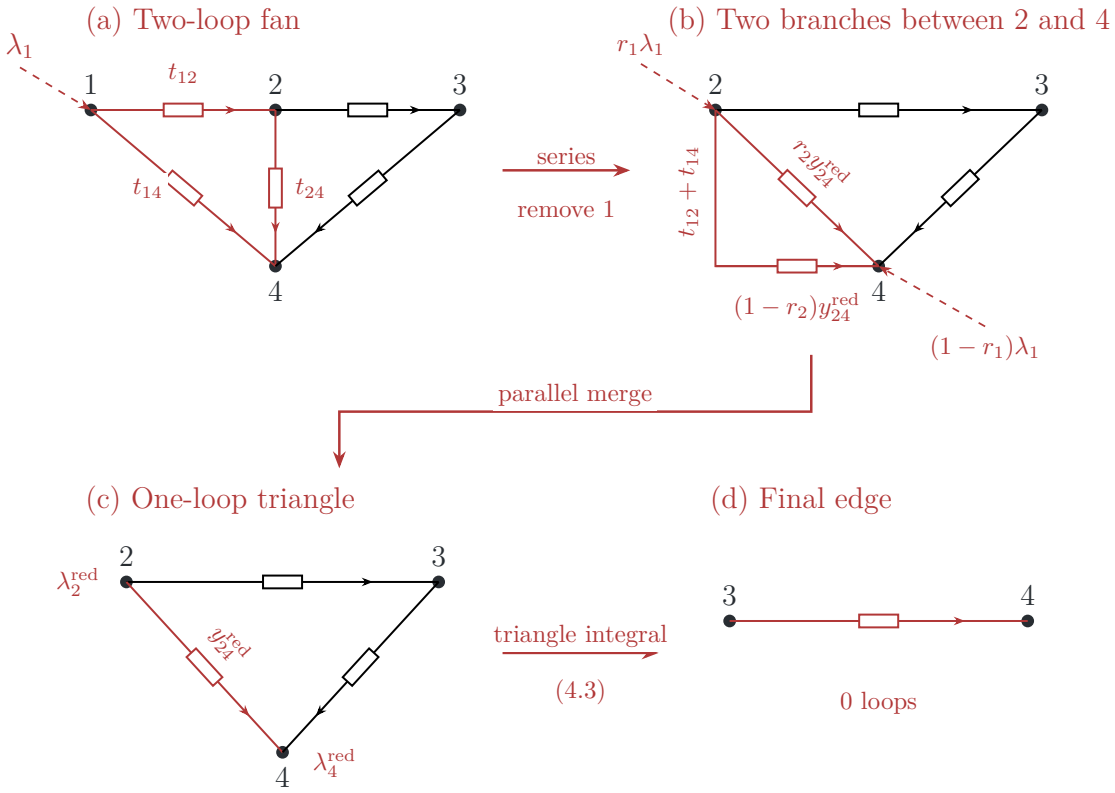


Figure A.6: The $n = 2$ step: series elimination, parallel merge, and the remaining triangle. Dashed arrows show the additional source transfers; the existing sources at $2, 4$ remain. The old and new branches carry $r_2 y_{24}^{\mathrm{red}}$ and $(1 - r_2) y_{24}^{\mathrm{red}}$.

For $n \geq 3$, apply the same formula to the smaller fan at $(v, u, w) = (2, 3, n + 2)$:

$$\begin{aligned}
\mathcal{I}^{\text{raw cut}}_{\Gamma^{\mathrm{red}}}(\lambda^{\mathrm{red}}; z^{\mathrm{red}}) = \int_0^1 dr_3 \int_0^1 dr_4 \, e^{-\lambda^{\mathrm{red}}_{2,+\dot{a}}[r_3 z_{23} + (1-r_3) z^{\mathrm{red}}_{2,n+2}]^{+\dot{a}}} \\
\times r_4 \lambda^{\mathrm{red}\,\dot{b}}_{2,+} \frac{\partial}{\partial z^{\mathrm{red},+\dot{b}}_{3,n+2}} \underbrace{\mathcal{I}^{\text{raw cut}}_{(\Gamma^{\mathrm{red}})^{\mathrm{red}}}(\lambda^{\mathrm{red}}; z^{\mathrm{red}})}_{(n-2)\text{-loop fan}}, \\
\lambda^{\mathrm{red}}_3 = \lambda_3 + r_3 \lambda^{\mathrm{red}}_2, \\
\lambda^{\mathrm{red}}_{n+2} = \lambda_{n+2} + (1 - r_1)\lambda_1 + (1 - r_3)\lambda^{\mathrm{red}}_2, \\
z^{\mathrm{red}}_{3,n+2} = r_4 z_{3,n+2} + (1 - r_4)(z^{\mathrm{red}}_{2,n+2} - z_{23}).
\end{aligned}$$

The sources passed inward obey the following recurrence, with empty products equal to one:

$$\begin{aligned}
\lambda^{\mathrm{red}}_{1,+\dot{a}} &= \lambda_{1,+\dot{a}}, \\
\lambda^{\mathrm{red}}_{i+1,+\dot{a}} &= \lambda_{i+1,+\dot{a}} + r_{2i-1} \lambda^{\mathrm{red}}_{i,+\dot{a}} \\
&= \lambda_{i+1,+\dot{a}} + \sum_{\substack{1 \leq j \leq i \\ \text{vertices}}} \left(\prod_{k=j}^{i} r_{2k-1} \right) \lambda_{j,+\dot{a}} \\
&= \sum_{\substack{1 \leq j \leq i+1 \\ \text{vertices}}} \left(\prod_{k=j}^{i} r_{2k-1} \right) \lambda_{j,+\dot{a}}, \qquad 1 \leq i < n.
\end{aligned} \tag{A.20}$$

For $n \geq 3$, the first derivative acts on both the second phase and its displacement substitution;

the two terms reconstruct the reduced current $y^{\mathrm{red}}_{2,n+2}$ inserted in (A.23):

$$\begin{aligned}
&r_2\lambda_{1,+}{}^{\dot a}\frac{\partial}{\partial z^{\mathrm{red},+\dot a}_{2,n+2}}e^{-\lambda^{\mathrm{red}}_{2,+\dot b}(r_3z_{23}+(1-r_3)z^{\mathrm{red}}_{2,n+2})^{+\dot b}}\\
&\quad= r_2(1-r_3)\lambda_{1,+\dot a}\lambda^{\mathrm{red}\,\dot a}_{2,+}e^{-\lambda^{\mathrm{red}}_{2,+\dot b}(r_3z_{23}+(1-r_3)z^{\mathrm{red}}_{2,n+2})^{+\dot b}},\\
&z^{\mathrm{red}}_{3,n+2}=r_4z_{3,n+2}+(1-r_4)(z^{\mathrm{red}}_{2,n+2}-z_{23}),\\
&\frac{\partial}{\partial z^{\mathrm{red},+\dot a}_{2,n+2}}\mathcal{I}^{\text{raw cut}}_{(\Gamma^{\mathrm{red}})^{\mathrm{red}}}=(1-r_4)\frac{\partial}{\partial z^{\mathrm{red},+\dot a}_{3,n+2}}\mathcal{I}^{\text{raw cut}}_{(\Gamma^{\mathrm{red}})^{\mathrm{red}}}.
\end{aligned}\tag{A.21}$$

The Leibniz rule gives the bracket $r_2[(1-r_3)\lambda_{1,+\dot a}\lambda^{\mathrm{red}\,\dot a}_{2,+}+(1-r_4)\lambda_{1,+}{}^{\dot a}\partial/\partial z^{\mathrm{red},+\dot a}_{3,n+2}]$ in the second equality of (5.7).

Evaluate from the triangle outward

Start with the one-loop kernel (5.3), relabelled to $n, n+1, n+2$:

$$\begin{aligned}
\mathcal{I}^{\text{raw cut}}_{\triangle}(\lambda^{\mathrm{red}};z^{\mathrm{red}})&=(\lambda^{\mathrm{red}}_{n,+\dot a}\lambda_{n+1,+}{}^{\dot a})\int_0^1 dr_{2n-1}\int_0^{1-r_{2n-1}}dr_{2n}\,e^{-\sum_{e\in\triangle_1}^{\text{edges}}y^{\mathrm{red}}_{e,+\dot b}z^{\mathrm{red},+\dot b}_e},\\
y_{n,n+1,+\dot a}&=r_{2n-1}\lambda^{\mathrm{red}}_{n,+\dot a}-r_{2n}\lambda_{n+1,+\dot a}.
\end{aligned}\tag{A.22}$$

Evaluation of one operator.

Fix the inner integration ratios and sources; the factors already obtained are independent of the displacements, so the next derivative acts on their exponential:

$$\begin{aligned}
&r_{2i}\lambda^{\mathrm{red}\,\dot a}_{i,+}\frac{\partial}{\partial z^{\mathrm{red},+\dot a}_{i+1,n+2}}e^{-\sum_{e\in\Gamma_1^{\mathrm{red}}}^{\text{edges}}y^{\mathrm{red}}_{e,+\dot b}z^{\mathrm{red},+\dot b}_e}\\
&\quad=r_{2i}(\lambda^{\mathrm{red}}_{i,+\dot a}y^{\mathrm{red}}_{i+1,n+2,+}{}^{\dot a})e^{-\sum_{e\in\Gamma_1^{\mathrm{red}}}^{\text{edges}}y^{\mathrm{red}}_{e,+\dot b}z^{\mathrm{red},+\dot b}_e}.
\end{aligned}\tag{A.23}$$

Multiplying by the phase of the vertex being restored and substituting the reduced displacement gives

$$\begin{aligned}
&\lambda^{\mathrm{red}}_{i,+\dot a}[r_{2i-1}z_{i,i+1}+(1-r_{2i-1})z^{\mathrm{red}}_{i,n+2}]^{+\dot a}\\
&\quad+y^{\mathrm{red}}_{i+1,n+2,+\dot a}[r_{2i}z_{i+1,n+2}+(1-r_{2i})(z^{\mathrm{red}}_{i,n+2}-z_{i,i+1})]^{+\dot a}\\
&=[r_{2i-1}\lambda^{\mathrm{red}}_{i,+\dot a}-(1-r_{2i})y^{\mathrm{red}}_{i+1,n+2,+\dot a}]z^{+\dot a}_{i,i+1}\\
&\quad+[(1-r_{2i-1})\lambda^{\mathrm{red}}_{i,+\dot a}+(1-r_{2i})y^{\mathrm{red}}_{i+1,n+2,+\dot a}]z^{\mathrm{red},+\dot a}_{i,n+2}+r_{2i}y^{\mathrm{red}}_{i+1,n+2,+\dot a}z^{+\dot a}_{i+1,n+2}.
\end{aligned}$$

The coefficients of the three displacements are the restored currents: the new series branch carries $(1-r_{2i})y^{\mathrm{red}}_{i+1,n+2}$, and the old spoke carries $r_{2i}y^{\mathrm{red}}_{i+1,n+2}$.

Conservation at $i+1$ identifies the factor inserted by the derivative:

$$\begin{aligned}
y^{\mathrm{red}}_{i+1,n+2,+\dot a}&=\lambda_{i+1,+\dot a}+r_{2i-1}\lambda^{\mathrm{red}}_{i,+\dot a}-y_{i+1,i+2,+\dot a},\\
r_{2i}\lambda^{\mathrm{red}}_{i,+\dot a}y^{\mathrm{red}}_{i+1,n+2,+}{}^{\dot a}&=r_{2i}\lambda^{\mathrm{red}}_{i,+\dot a}(\lambda_{i+1,+}{}^{\dot a}-y_{i+1,i+2,+}{}^{\dot a}),\qquad 1\le i<n.
\end{aligned}\tag{A.24}$$

Repeat for $i=n-1,n-2,\dots,1$; each operator multiplies the displacement-independent factors by this contraction and restores one more circuit. □

The resulting nested integral is (5.7):

$$\mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda;z)=\int_{(5.6)} d^{2n}r\prod_{i=1}^{n-1}\Big[r_{2i}\lambda^{\text{red}}_{i,+\dot{a}_i}\big(\lambda_{i+1,+}{}^{\dot{a}_i}-y_{i+1,i+2,+}{}^{\dot{a}_i}(r)\big)\Big]\\ \times\big(\lambda^{\text{red}}_{n,+\dot{b}}\lambda_{n+1,+}{}^{\dot{b}}\big)e^{-\sum_{e\in\Gamma_1\atop\text{edges}}y_{e,+\dot{a}}(r)z_e^{+\dot{a}}}.$$

Insert the source sum (A.20); the interval pairs come from (A.11), and the final simplex comes from (A.22).

Recover the currents in (5.7)

Compute the final triangle current first, then read the coefficient of $z_{i,i+1}$ above backwards through the circuit:

$$\begin{aligned}y_{n,n+1,+\dot{a}}&=r_{2n-1}\lambda^{\text{red}}_{n,+\dot{a}}-r_{2n}\lambda_{n+1,+\dot{a}},\\ y_{i,i+1,+\dot{a}}&=r_{2i-1}\lambda^{\text{red}}_{i,+\dot{a}}-(1-r_{2i})y^{\text{red}}_{i+1,n+2,+\dot{a}}\\ &=r_{2i-1}r_{2i}\lambda^{\text{red}}_{i,+\dot{a}}-(1-r_{2i})\lambda_{i+1,+\dot{a}}+(1-r_{2i})y_{i+1,i+2,+\dot{a}},\qquad i=n-1,\dots,1.\end{aligned}\tag{A.25}$$

The spoke currents follow from conservation in the restored fan:

$$\begin{aligned}y_{1,n+2,+\dot{a}}&=\lambda_{1,+\dot{a}}-y_{12,+\dot{a}},\\ y_{i,n+2,+\dot{a}}&=\lambda_{i,+\dot{a}}+y_{i-1,i,+\dot{a}}-y_{i,i+1,+\dot{a}},\qquad 2\le i\le n,\\ y_{n+1,n+2,+\dot{a}}&=\lambda_{n+1,+\dot{a}}+y_{n,n+1,+\dot{a}}.\end{aligned}\tag{A.26}$$

For $n=1$, the operator product is empty and gives (5.3); for $n=2$, the single operator restores the two-loop circuit in Figure 24(b).

A.5 Iterating the spider operators

For the spider (7.8), apply (A.11) along each arm, then remove the book pages until one triangle remains.

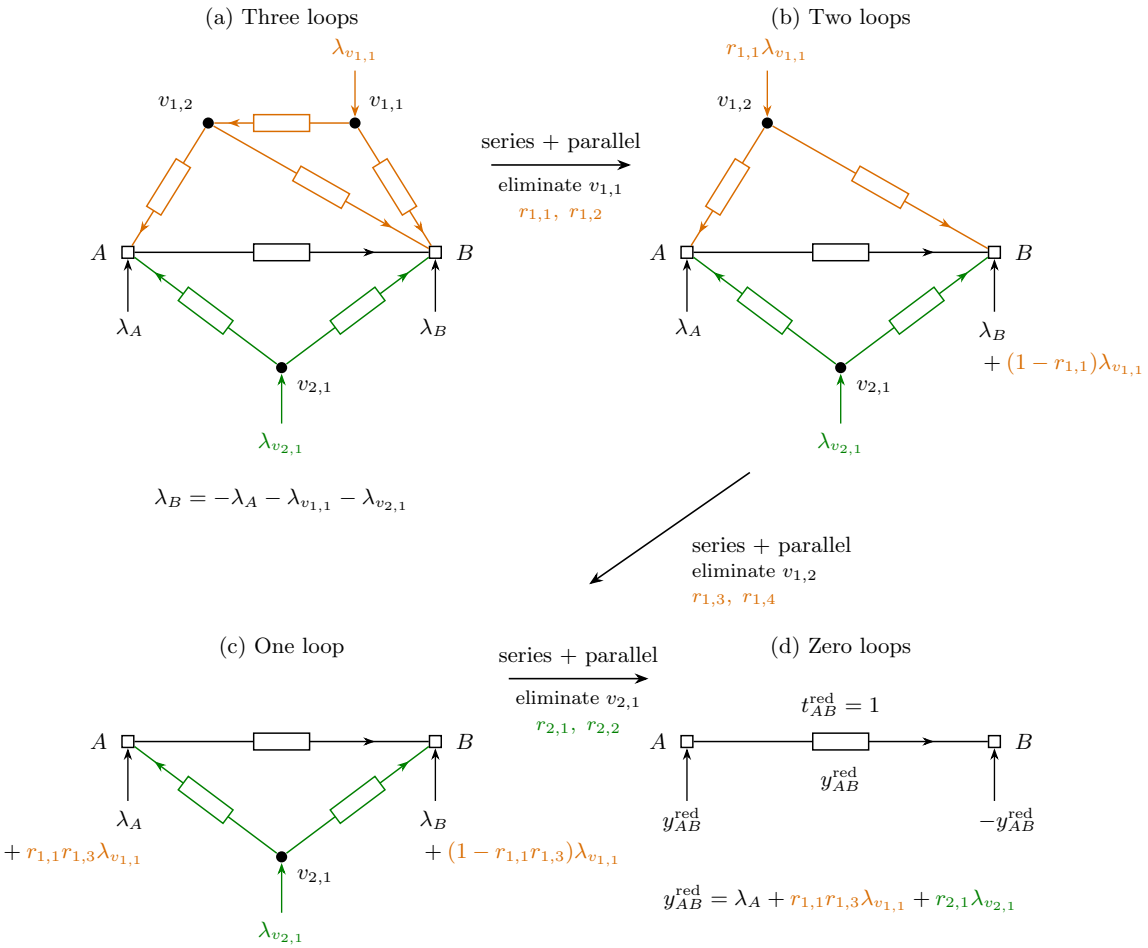


Figure A.7: Successive applications of Theorem A.1: a three-loop spider, a two-page book, a triangle, and the final edge. Each arrow removes one vertex and one loop. The terminal current shares are related to the local parameters by (A.29).

Eliminate the interior vertices $v_{i,1}, \dots, v_{i,\ell_i-1}$, arm by arm, followed by the terminal vertices $v_{1,\ell_1}, \dots, v_{a-1,\ell_{a-1}}$; put $v_{i,\ell_i+1} = A$ and $u = v_{i,j+1}$ when $v = v_{i,j}$.

Each factor below has its own local parameters r_1, r_2 of (A.11), acts on everything to its right, and substitutes the source and displacement updates after differentiating:

$$\mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda; z) = \prod_{v \text{ in the displayed elimination order}} \left[\int_0^1 dr_1 \int_0^1 dr_2 \, e^{-\lambda_{v,+\dot{a}}^{\mathrm{red}} [r_1 z_{vu} + (1-r_1) z_{vB}^{\mathrm{red}}]^{+\dot{a}}} \times r_2 \lambda_{v,+}^{\mathrm{red}\,\dot{b}} \frac{\partial}{\partial z_{uB}^{\mathrm{red},+\dot{b}}} \right] \underbrace{\mathcal{I}_{\triangle}^{\text{raw cut}}(\lambda^{\mathrm{red}}; z^{\mathrm{red}})}_{\substack{\text{vertices } v_{a,\ell_a}, A, B \\ \text{known one-loop kernel (5.3)}}} .$$

We evaluate these derivatives before expanding the final exponential.

Derivatives along an arm

The internal arm sources vanish as in (7.8), and $v_{i,\ell_i+1} = A$.

$$\begin{aligned} \lambda^{\mathrm{red}}_{v_{i,j+1},+\dot{a}} &= r_{i,2j-1}\lambda^{\mathrm{red}}_{v_{i,j},+\dot{a}} \qquad (j < \ell_i), \\ \lambda^{\mathrm{red}}_{v_{i,j},+\dot{a}} &= \left(\prod_{k=1}^{j-1} r_{i,2k-1}\right)\lambda_{v_{i,1},+\dot{a}}, \qquad 1 \le j \le \ell_i. \end{aligned} \tag{A.27}$$

For $j < \ell_i$, use (A.11) with $(v,u,w) = (v_{i,j}, v_{i,j+1}, B)$ and local parameters $(r_1, r_2) = (r_{i,2j-1}, 1 - r_{i,2j})$.

Each superscript red refers to the circuit at the indicated step; the incoming source and spoke displacement already include the preceding reductions.

$$\begin{aligned} \mathcal{I}^{\text{raw cut}}_{\Gamma}(\lambda; z) = \int_0^1 dr_{i,2j-1} \int_0^1 dr_{i,2j} \\ \times e^{-\lambda^{\mathrm{red}}_{v_{i,j},+\dot{a}}[r_{i,2j-1} z_{v_{i,j}v_{i,j+1}} + (1-r_{i,2j-1}) z^{\mathrm{red}}_{v_{i,j}B}]^{+\dot{a}}} \\ \times \underbrace{(1 - r_{i,2j})\lambda^{\mathrm{red}}_{v_{i,j},+}{}^{\dot{b}} \frac{\partial}{\partial z^{\mathrm{red},+\dot{b}}_{v_{i,j+1}B}}}_{\text{operator supplied by (A.11)}} \underbrace{\mathcal{I}^{\text{raw cut}}_{\Gamma^{\mathrm{red}}}(\lambda^{\mathrm{red}}; z^{\mathrm{red}})}_{\text{one fewer arm vertex and loop}} . \end{aligned}$$

$$z^{\mathrm{red}}_{v_{i,j+1}B} = (1 - r_{i,2j}) z_{v_{i,j+1}B} + r_{i,2j}(z^{\mathrm{red}}_{v_{i,j}B} - z_{v_{i,j}v_{i,j+1}}).$$

The later source on the same arm is a multiple of $\lambda_{v_{i,1}}$, so an earlier directional derivative annihilates its source phase:

$$\lambda_{v_{i,1},+\dot{a}} \lambda^{\mathrm{red}}_{v_{i,k},+}{}^{\dot{a}} = 0.$$

$$\lambda_{v_{i,1},+}{}^{\dot{a}} \frac{\partial}{\partial z^{\mathrm{red},+\dot{a}}_{v_{i,k}B}} e^{-\lambda^{\mathrm{red}}_{v_{i,k},+\dot{b}}[r_{i,2k-1} z_{v_{i,k}v_{i,k+1}} + (1-r_{i,2k-1}) z^{\mathrm{red}}_{v_{i,k}B}]^{+\dot{b}}} = 0.$$

Its action on the next kernel passes through the affine displacement substitution:

$$\lambda_{v_{i,1},+}{}^{\dot{a}} \frac{\partial}{\partial z^{\mathrm{red},+\dot{a}}_{v_{i,k}B}} = r_{i,2k} \lambda_{v_{i,1},+}{}^{\dot{a}} \frac{\partial}{\partial z^{\mathrm{red},+\dot{a}}_{v_{i,k+1}B}} \qquad (k < \ell_i).$$

Repeat for $j = 1, \ldots, \ell_i - 1$ on each arm; the remaining graph is the book on $A, B, v_{1,\ell_1}, \ldots, v_{a,\ell_a}$.

Evaluate the book from one triangle

Use the book labels $1, \ldots, n$, $A = n+1$, $B = n+2$, with $n = a$ pages and $\lambda_i = \lambda^{\mathrm{red}}_{v_{i,\ell_i}}$.

Remove pages $1, \ldots, n-1$, leaving the triangle (n, A, B); the sources and the AB displacement update at each step by

$$\begin{aligned} \lambda^{\mathrm{red}}_{A}\Big|_{\text{after } i} &= \lambda^{\mathrm{red}}_{A}\Big|_{\text{before } i} + r_1 \lambda_i, \\ \lambda^{\mathrm{red}}_{B}\Big|_{\text{after } i} &= \lambda^{\mathrm{red}}_{B}\Big|_{\text{before } i} + (1 - r_1)\lambda_i, \\ z^{\mathrm{red}}_{AB}\Big|_{\text{after } i} &= r_2 \, z^{\mathrm{red}}_{AB}\Big|_{\text{before } i} + (1 - r_2)(z_{iB} - z_{iA}). \end{aligned}$$

The later page phases are independent of z^{red}_{AB}; therefore the composed derivatives act directly on the known triangle kernel:

$$\begin{aligned}\mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda;z) = \int_{(0,1)^{2n-2}} \prod_{i=1}^{n-1} \left[d(r_1|_{v=i})\, d(r_2|_{v=i})\right] \\ \times e^{-\sum\limits_{\substack{1\le i<n \\ \text{removed pages}}} \lambda_{i,+\dot a}[(r_1|_{v=i})z_{iA}+(1-r_1|_{v=i})z_{iB}]^{+\dot a}} \\ \times \prod_{i=1}^{n-1}\left[\left(\prod_{k=i}^{n-1} r_2|_{v=k}\right)\lambda_{i,+}{}^{\dot a_i}\frac{\partial}{\partial z^{\mathrm{red},+\dot a_i}_{AB}}\right]\underbrace{\mathcal{I}_{\triangle}^{\text{raw cut}}(\lambda^{\mathrm{red}};z^{\mathrm{red}})}_{(n,A,B)}.\end{aligned} \tag{A.28}$$

The product of local r_2 factors follows at every substitution from

$$\left.\frac{\partial}{\partial z^{\mathrm{red},+\dot a}_{AB}}\right|_{\text{before page } k} = r_2|_{v=k}\left.\frac{\partial}{\partial z^{\mathrm{red},+\dot a}_{AB}}\right|_{\text{after page } k}.$$

In the last triangle, use the local coordinates r_1, r_2 of (5.3); its currents are

$$\begin{aligned} y^{\mathrm{red}}_{nA,+\dot a} &= r_1\lambda_{n,+\dot a} - r_2\lambda^{\mathrm{red}}_{A,+\dot a}, \\ y^{\mathrm{red}}_{nB,+\dot a} &= \lambda_{n,+\dot a} - y^{\mathrm{red}}_{nA,+\dot a}, \\ y^{\mathrm{red}}_{AB,+\dot a} &= (1-r_2)\lambda^{\mathrm{red}}_{A,+\dot a} + r_1\lambda_{n,+\dot a}. \end{aligned}$$

Every derivative in (A.28) inserts this AB current:

$$\begin{aligned}&\prod_{i=1}^{n-1}\left[\lambda_{i,+}{}^{\dot a_i}\frac{\partial}{\partial z^{\mathrm{red},+\dot a_i}_{AB}}\right]\mathcal{I}_{\triangle}^{\text{raw cut}}(\lambda^{\mathrm{red}};z^{\mathrm{red}}) \\ &= (\lambda_{n,+\dot b}\lambda^{\mathrm{red}\ \dot b}_{A,+})\int_0^1 dr_1\int_0^{1-r_1} dr_2 \prod_{i=1}^{n-1}(\lambda_{i,+\dot a_i}y^{\mathrm{red}}_{AB,+}{}^{\dot a_i})e^{-\sum\limits_{e\in\{nA,nB,AB\}\ \text{triangle edges}} y^{\mathrm{red}}_{e,+\dot c}z_e^{\mathrm{red},+\dot c}}.\end{aligned}$$

To write the result in the current shares of (7.9), include the final triangle's local parameters and use

$$\begin{aligned} r_1|_{v=n} &= \frac{r_1}{1-r_2}, \qquad r_2|_{v=n} = 1-r_2. \\ y^{\mathrm{red}}_{AB,+\dot a}\Big|_{\text{triangle}} &= r_2|_{v=n}\, y^{\mathrm{red}}_{AB,+\dot a}\Big|_{\text{final edge}}, \\ \lambda_{n,+\dot a}\,\lambda^{\mathrm{red}\ \dot a}_{A,+}\Big|_{\text{triangle}} &= \lambda_{n,+\dot a}\, y^{\mathrm{red}}_{AB,+}{}^{\dot a}\Big|_{\text{final edge}}. \\ r_1|_{v=i} &= r_{2i-1}, \qquad r_2|_{v=i} = \frac{1-\sum_{i\le k\le n\ \text{pages}} r_{2k}}{1-\sum_{i<k\le n\ \text{pages}} r_{2k}}, \\ r_{2i} &= (1-r_2|_{v=i})\prod_{k=i+1}^{n} r_2|_{v=k}. \end{aligned} \tag{A.29}$$

The resulting currents are

$$\begin{aligned} y^{\mathrm{red}}_{n+1,n+2,+\dot a} &= \lambda_{n+1,+\dot a} + \sum_{j=1}^{n} r_{2j-1}\lambda_{j,+\dot a}, \\ y_{i,n+1,+\dot a} &= r_{2i-1}\lambda_{i,+\dot a} - r_{2i}y^{\mathrm{red}}_{n+1,n+2,+\dot a}, \\ y_{i,n+2,+\dot a} &= (1-r_{2i-1})\lambda_{i,+\dot a} + r_{2i}y^{\mathrm{red}}_{n+1,n+2,+\dot a}, \\ y_{n+1,n+2,+\dot a} &= \left(1-\sum_i r_{2i}\right)y^{\mathrm{red}}_{n+1,n+2,+\dot a}. \end{aligned} \tag{A.30}$$

The successive derivatives and the last triangle supply $\prod_{i=1}^{n}(r_2|_{v=i})^i$; substituting (A.29) gives the positive integration densities

$$\begin{aligned}
&\prod_{i=1}^{n}\left[(r_2|_{v=i})^i\,|d(r_2|_{v=i})|\right]\\
&=\prod_{i=1}^{n}\left(1-\sum_{\substack{i\le k\le n\\ \text{pages}}} r_{2k}\right)\Bigg/\prod_{i=1}^{n}\left(1-\sum_{\substack{i<k\le n\\ \text{pages}}} r_{2k}\right)\prod_{i=1}^{n} dr_{2i}\\
&=\left(1-\sum_{\substack{1\le i\le n\\ \text{pages}}} r_{2i}\right)\prod_{i=1}^{n} dr_{2i}.
\end{aligned}$$

The phase substitutions restore the original edge exponential, and the accumulated operators give

$$\begin{aligned}
\mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda;z)=\int_{\substack{r_{2i}>0,\ \sum_{\substack{1\le i\le n\\ \text{pages}}} r_{2i}<1}}\ \prod_{i=1}^{n}(dr_{2i-1}\,dr_{2i})\left(1-\sum_{\substack{1\le i\le n\\ \text{pages}}} r_{2i}\right)\\
\times\prod_{i=1}^{n}(\lambda_{i,+\dot a_i}y_{AB,+}^{\text{red}\ \dot a_i})e^{-\sum_{\substack{e\in\Gamma_1\\ \text{edges}}} y_{e,+\dot b}z_e^{+\dot b}}.
\end{aligned}\tag{A.31}$$

For the marked product, set $z_{AB}=0$ after differentiating.

Multiply the arm factors

Restore the spider labels, $n=\sum_{\substack{1\le i\le a\\ \text{arms}}}\ell_i$; at the book, the final AB displacement contains the terminal spoke with coefficient $r_{i,2\ell_i}$.

Combining this coefficient with the successive arm derivatives gives the contribution from vertex $v_{i,j}$:

$$(1-r_{i,2j})\left(\prod_{k=1}^{j-1} r_{i,2k-1}\right)\left(\prod_{k=j+1}^{\ell_i-1} r_{i,2k}\right)r_{i,2\ell_i}(\lambda_{v_{i,1},+\dot a}y_{AB,+}^{\text{red}\ \dot a}),\qquad j<\ell_i.\tag{A.32}$$

The terminal page contributes $\left(\prod_{k=1}^{\ell_i-1} r_{i,2k-1}\right)(\lambda_{v_{i,1},+\dot a}y_{AB,+}^{\text{red}\ \dot a})$; the remaining coefficients multiply as

$$\begin{aligned}
&\left[\prod_{j=1}^{\ell_i-1}(1-r_{i,2j})\left(\prod_{k=1}^{j-1} r_{i,2k-1}\right)\left(\prod_{k=j+1}^{\ell_i-1} r_{i,2k}\right)r_{i,2\ell_i}\right]\left(\prod_{k=1}^{\ell_i-1} r_{i,2k-1}\right)\\
&\qquad=r_{i,2\ell_i}^{\ell_i-1}\prod_{j=1}^{\ell_i-1}\left[r_{i,2j-1}^{\ell_i-j}r_{i,2j}^{j-1}(1-r_{i,2j})\right].
\end{aligned}\tag{A.33}$$

The source phases and reduced exponents combine as in Appendix (3.7); the evaluated integral

is

$$\begin{aligned}\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda;z) = \int_{(7.9)} d^{2n}r \left(1 - \sum_{\substack{1\le i\le a\\ \text{arms}}} r_{i,2\ell_i}\right)\\ \times \prod_{i=1}^{a}\left\{ r_{i,2\ell_i}^{\ell_i-1} \prod_{j=1}^{\ell_i-1}\left[r_{i,2j-1}^{\ell_i-j} r_{i,2j}^{j-1}(1-r_{i,2j})\right]\left(\lambda_{v_{i,1},+\dot a_i} y^{\text{red}}_{AB,+}{}^{\dot a_i}\right)^{\ell_i}\right\}\\ \times e^{-\sum_{e\in\Gamma_1}\ \sum_{\text{edges}} y_{e,+\dot b} z_e^{+\dot b}}.\end{aligned} \tag{A.34}$$

This is (7.11); every factor comes from a displayed derivative acting on the remaining kernel.

Recover the currents by the same iteration

The final transferred source determines the final edge current:

$$y^{\text{red}}_{AB,+\dot a} = \lambda_{A,+\dot a} + \sum_{\substack{1\le i\le a\\ \text{arms}}} \left(\prod_{k=1}^{\ell_i} r_{i,2k-1}\right)\lambda_{v_{i,1},+\dot a}. \tag{A.35}$$

Reverse the current update in (A.11), starting at each terminal page:

$$\begin{aligned} y^{\text{red}}_{v_{i,\ell_i}B,+\dot a} &= (1-r_{i,2\ell_i-1})\left(\prod_{k=1}^{\ell_i-1} r_{i,2k-1}\right)\lambda_{v_{i,1},+\dot a} + r_{i,2\ell_i} y^{\text{red}}_{AB,+\dot a},\\ y^{\text{red}}_{v_{i,j}B,+\dot a} &= (1-r_{i,2j-1})\lambda^{\text{red}}_{v_{i,j},+\dot a} + r_{i,2j} y^{\text{red}}_{v_{i,j+1}B,+\dot a} \quad (j<\ell_i).\end{aligned} \tag{A.36}$$

$$\begin{aligned} y^{\text{red}}_{v_{i,j}B,+\dot a} = \sum_{r=j}^{\ell_i}\left(\prod_{k=j}^{r-1} r_{i,2k}\right)(1-r_{i,2r-1})\left(\prod_{k=1}^{r-1} r_{i,2k-1}\right)\lambda_{v_{i,1},+\dot a}\\ + \left(\prod_{k=j}^{\ell_i-1} r_{i,2k}\right) r_{i,2\ell_i}\, y^{\text{red}}_{AB,+\dot a}.\end{aligned} \tag{A.37}$$

$$\begin{aligned} y_{v_{i,j}v_{i,j+1},+\dot a} &= \lambda^{\text{red}}_{v_{i,j},+\dot a} - y^{\text{red}}_{v_{i,j}B,+\dot a},\\ y_{v_{i,1}B,+\dot a} &= y^{\text{red}}_{v_{i,1}B,+\dot a},\\ y_{v_{i,j}B,+\dot a} &= (1-r_{i,2j-2}) y^{\text{red}}_{v_{i,j}B,+\dot a} \quad (j\ge 2),\\ y_{AB,+\dot a} &= \left(1-\sum_i r_{i,2\ell_i}\right) y^{\text{red}}_{AB,+\dot a}.\end{aligned} \tag{A.38}$$

$$\begin{aligned} y_{v_{i,j}v_{i,j+1},+\dot a} = \sum_{s=j}^{\ell_i-1}\left(\prod_{k=1}^{s} r_{i,2k-1}\right)\left(\prod_{k=j}^{s-1} r_{i,2k}\right)(1-r_{i,2s})\,\lambda_{v_{i,1},+\dot a}\\ + \left(\prod_{k=1}^{\ell_i} r_{i,2k-1}\right)\left(\prod_{k=j}^{\ell_i-1} r_{i,2k}\right)\lambda_{v_{i,1},+\dot a} - \left(\prod_{k=j}^{\ell_i} r_{i,2k}\right) y^{\text{red}}_{AB,+\dot a}.\end{aligned} \tag{A.39}$$

$$\begin{aligned}\sum_e y_{e,+\dot a} z_e^{+\dot a} = \lambda_{A,+\dot a} z_{AB}^{+\dot a} + \sum_i \lambda_{v_{i,1},+\dot a} z_{v_{i,1}B}^{+\dot a}\\ + \sum_{i=1}^{a}\sum_{j=1}^{\ell_i} y_{v_{i,j}v_{i,j+1},+\dot a}(z_{v_{i,j}v_{i,j+1}} + z_{v_{i,j+1}B} - z_{v_{i,j}B})^{+\dot a}, \qquad v_{i,\ell_i+1} = A.\end{aligned} \tag{A.40}$$

With the multiplicities $p_{ijs}, q_{ijA}, q_{ijk}, m_{iA}, m_{ik}, q_{ij}$ and M_i defined before (7.15), the coordinate exponents are

$$\begin{aligned} \kappa_{i,2t-1} &= M_i + \sum_{s=t}^{\ell_i} \sum_{j=1}^{s} p_{ijs}, \qquad \kappa_{i,2t} = \sum_{j=1}^{t} \left[q_{ij} + \sum_{s=t+1}^{\ell_i} p_{ijs} \right] \quad (t < \ell_i), \\ \bar{\kappa}_{i,2t} &= \sum_{j=1}^{t} p_{ijt}, \qquad \kappa_{i,2\ell_i} = \sum_{j=1}^{\ell_i} q_{ij}. \end{aligned} \tag{A.41}$$

A.6 Forest reductions

Root each cubic interaction tree at its unique source and process its descendants first.

A source-free leaf uses (A.13) when its parent has an insertion spoke; two sibling leaves use (A.18); a remaining degree-two source uses (A.11).

Keep applying the operator to the smaller kernel inside the integral, stop at the final triangle, and insert its known one-loop kernel.

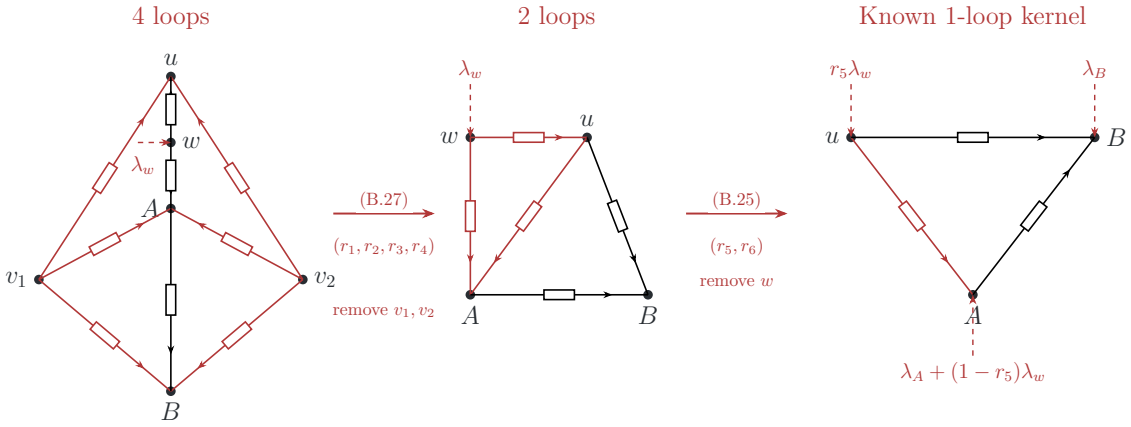


Figure A.8: The four-loop forest is reduced by (A.18) and (A.11) to the known one-loop kernel. The sources at u, v_1, v_2 vanish; the sources at w, A, B are retained and transferred as shown.

For the four-loop forest with edges $(v_1u, v_1A, v_1B, v_2u, v_2A, v_2B, AB, wu, wA)$ and $\lambda_u = \lambda_{v_1} = \lambda_{v_2} = 0$, remove v_1, v_2 by (A.18), then w by (A.11):

$$\begin{aligned} \mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda; z) = \int_0^1 dr_1 \int_0^1 dr_2 \int_{\substack{r_3,r_4>0 \\ r_3+r_4<1}} dr_3\, dr_4\, (r_1 - r_2)(1 - r_3 - r_4) \\ \times \frac{\partial}{\partial z_{uA}^{\text{red},+\dot{a}}} \frac{\partial}{\partial z_{AB,\dot{a}}^{\text{red},+}} \frac{\partial}{\partial z_{uB}^{\text{red},+\dot{c}}} \frac{\partial}{\partial z_{AB,\dot{c}}^{\text{red},+}} \\ \times \left[\int_0^1 dr_5 \int_0^1 dr_6\, e^{-\lambda_{w,+\dot{d}}[r_5 z_{wu} + (1-r_5) z_{wA}]^{+\dot{d}}} \right. \\ \left. \times\, r_6 \lambda_{w,+\dot{b}} \frac{\partial}{\partial (z_{uA,\dot{b}}^{\text{red}})^{\text{red},+}} \underbrace{\mathcal{I}_{\triangle}^{\text{raw cut}}((\lambda^{\text{red}})^{\text{red}}; (z^{\text{red}})^{\text{red}})}_{\text{known one-loop kernel on } u,A,B} \right]. \end{aligned}$$

The first red is the two-loop graph and the second is the triangle; make these substitutions after the inner derivative and before the four outer derivatives, then use the first-stage substitutions

of (A.18):

$$
\begin{aligned}
&(\lambda_u^{\mathrm{red}})^{\mathrm{red}} = r_5\lambda_w, && (\lambda_A^{\mathrm{red}})^{\mathrm{red}} = \lambda_A + (1-r_5)\lambda_w, \quad (\lambda_B^{\mathrm{red}})^{\mathrm{red}} = \lambda_B,\\
&(z_{uA}^{\mathrm{red}})^{\mathrm{red}} = r_6 z_{uA}^{\mathrm{red}} + (1-r_6)(z_{wA} - z_{wu}),\\
&(z_{uB}^{\mathrm{red}})^{\mathrm{red}} = z_{uB}^{\mathrm{red}}, && (z_{AB}^{\mathrm{red}})^{\mathrm{red}} = z_{AB}^{\mathrm{red}}.
\end{aligned}
$$

Evaluate the nested derivatives

Insert (5.3) for the final triangle, using its coordinates $r_7, r_8 > 0$, $r_7 + r_8 < 1$:

$$
\begin{aligned}
&\mathcal{I}_{\triangle}^{\text{raw cut}}((\lambda^{\mathrm{red}})^{\mathrm{red}}; (z^{\mathrm{red}})^{\mathrm{red}})\\
&\quad = r_5(\lambda_{w,+\dot a}\lambda_{A,+}{}^{\dot a}) \int_{\substack{r_7,r_8>0\\ r_7+r_8<1}} dr_7\, dr_8\, e^{-\sum_{\substack{e\in\{uA,uB,AB\}\\ \text{triangle edges}}} (y_{e,+\dot b}^{\mathrm{red}})^{\mathrm{red}} (z_e^{\mathrm{red}})^{\mathrm{red},+\dot b}},\\
(y_{uA}^{\mathrm{red}})^{\mathrm{red}} &= r_5 r_7 \lambda_w - r_8[\lambda_A + (1-r_5)\lambda_w],\\
(y_{uB}^{\mathrm{red}})^{\mathrm{red}} &= r_5\lambda_w - (y_{uA}^{\mathrm{red}})^{\mathrm{red}},\\
(y_{AB}^{\mathrm{red}})^{\mathrm{red}} &= \lambda_A + (1-r_5)\lambda_w + (y_{uA}^{\mathrm{red}})^{\mathrm{red}}.
\end{aligned}
$$

The inner derivative first gives

$$
\begin{aligned}
\lambda_{w,+\dot a}(y_{uA,+}^{\mathrm{red}})^{\mathrm{red}\dot a} &= -r_8\lambda_{w,+\dot a}\lambda_{A,+}{}^{\dot a},\\
&\quad r_6\lambda_{w,+\dot a}\frac{\partial}{\partial(z_{uA,\dot a}^{\mathrm{red}})^{\mathrm{red},+}}\mathcal{I}_{\triangle}^{\text{raw cut}}((\lambda^{\mathrm{red}})^{\mathrm{red}}; (z^{\mathrm{red}})^{\mathrm{red}})\\
&\quad = -r_5 r_6(\lambda_{w,+\dot a}\lambda_{A,+}{}^{\dot a})^2 \int_{\substack{r_7,r_8>0\\ r_7+r_8<1}} dr_7\, dr_8\, r_8\, e^{-\sum_{\substack{e\in\{uA,uB,AB\}\\ \text{triangle edges}}} (y_{e,+\dot b}^{\mathrm{red}})^{\mathrm{red}} (z_e^{\mathrm{red}})^{\mathrm{red},+\dot b}}.
\end{aligned}
$$

After the inner displacement substitution, the three currents differentiated by the outer operator are

$$
y_{uA}^{\mathrm{red}} = r_6(y_{uA}^{\mathrm{red}})^{\mathrm{red}}, \qquad y_{uB}^{\mathrm{red}} = (y_{uB}^{\mathrm{red}})^{\mathrm{red}}, \qquad y_{AB}^{\mathrm{red}} = (y_{AB}^{\mathrm{red}})^{\mathrm{red}}.
$$

The phase at w is independent of these three displacements; the four outer derivatives therefore insert

$$
\begin{aligned}
y_{uA,+\dot a}^{\mathrm{red}} y_{AB,+}^{\mathrm{red}}{}^{\dot a} &= r_5 r_6 r_7 \lambda_{w,+\dot a}\lambda_{A,+}{}^{\dot a},\\
y_{uB,+\dot a}^{\mathrm{red}} y_{AB,+}^{\mathrm{red}}{}^{\dot a} &= r_5(1-r_7-r_8)\lambda_{w,+\dot a}\lambda_{A,+}{}^{\dot a}.
\end{aligned}
$$

Multiplying these factors and combining the phases as in Appendix (3.7) gives

$$
\begin{aligned}
\mathcal{I}_{\Gamma}^{\text{raw cut}}(\lambda; z) = &-(\lambda_{w,+\dot a}\lambda_{A,+}{}^{\dot a})^4 \int_0^1 dr_1 \int_0^1 dr_2 \int_{\substack{r_3,r_4>0\\ r_3+r_4<1}} dr_3\, dr_4\\
&\times \int_0^1 dr_5 \int_0^1 dr_6 \int_{\substack{r_7,r_8>0\\ r_7+r_8<1}} dr_7\, dr_8\\
&\times (r_1-r_2)(1-r_3-r_4) r_5^3 r_6^2 r_7 r_8 (1-r_7-r_8)\, e^{-\sum_{e\in\Gamma_1\ \text{edges}} y_{e,+\dot b}(r) z_e^{+\dot b}}.
\end{aligned}
$$

Reverse the same two reductions to recover the original currents, starting with the root w and

then its source-free descendants:

$$\begin{aligned}
y_{wu} &= r_5\lambda_w - (1-r_6)(y_{uA}^{\mathrm{red}})^{\mathrm{red}},\\
y_{wA} &= (1-r_5)\lambda_w + (1-r_6)(y_{uA}^{\mathrm{red}})^{\mathrm{red}},\\
y_{v_1u} &= -r_1 y_{uA}^{\mathrm{red}} - r_2 y_{uB}^{\mathrm{red}},\\
y_{v_1A} &= r_1 y_{uA}^{\mathrm{red}} - r_3 y_{AB}^{\mathrm{red}},\\
y_{v_1B} &= r_2 y_{uB}^{\mathrm{red}} + r_3 y_{AB}^{\mathrm{red}},\\
y_{AB} &= (1-r_3-r_4) y_{AB}^{\mathrm{red}}.
\end{aligned}$$

The v_2 currents follow from the three v_1 expressions by $r_1 \mapsto 1-r_1$, $r_2 \mapsto 1-r_2$, and $r_3 \mapsto r_4$, as in Appendix (3.7).

For a larger forest, substitute the next integral of Appendix A.3 for the reduced kernel and continue to a triangle; evaluate the derivatives from the innermost brackets outwards, including their action on later phases and displacement substitutions.

The edge currents are then recovered in reverse reduction order by the series–parallel (A.12), star–triangle (A.14), and paired-leaf (A.19) current maps of Appendix A.3.

Finite factorial sums

For a cubic product forest reduced by (A.11)–(A.18), the n-loop kernel uses $2n$ coordinates in unit intervals or standard simplices Δ; each monomial is evaluated by

$$\int_{\substack{r_j>0\\ \sum_{1\le j\le \dim\Delta \text{ coordinates}} r_j<1}} \prod_{j=1}^{\dim\Delta} (r_j^{k_j}\, dr_j) = \frac{\prod_{j=1}^{\dim\Delta} k_j!}{\Big(\dim\Delta + \sum_{\substack{1\le j\le \dim\Delta\\ \text{coordinates}}} k_j\Big)!}. \tag{A.42}$$

Apply the derivatives of Appendix A.3, with zero source at every vertex removed by (A.13) or (A.18), to the final one-loop kernel. At displacement degree $m \ge 0$, write the resulting polynomial as

$$\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda;z)\Big|_{\deg z=m} = \int_{\prod\Delta} d^{2n}r \sum_{\substack{0\le k_j\le m+2n-1\ (1\le j\le 2n)\\ \text{reduction monomials}}} c_k(\lambda;z)\, r^k.$$

Here $k=(k_1,\dots,k_{2n})$, $r^k = \prod_{j=1}^{2n} r_j^{k_j}$, and each c_k is homogeneous of degree m in z and $2n+m$ in λ.

$$\boxed{\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda;z)\Big|_{\deg z=m} = \sum_{\substack{0\le k_j\le m+2n-1\ (1\le j\le 2n)\\ \text{reduction monomials}}} c_k(\lambda;z) \prod_{\Delta \text{ in the reduction}} \frac{\prod_{j:\, r_j\in\Delta} k_j!}{\Big(\dim\Delta + \sum_{\substack{j:\, r_j\in\Delta\\ \text{coordinates}}} k_j\Big)!}.}$$

Finite bound from the operators.

The triangle coefficient of displacement degree m has source degree $2+m$ and coordinate degree at most m; the three operators of Appendix A.3 give the induction

formula	loops removed	derivatives	largest reduced degree needed
(A.11)	1	1	$m+1$
(A.13)	1	2	$m+2$
(A.18)	2	4	$m+4$

For a coordinate of the smaller graph, the induction bound is

$$\begin{aligned}(\text{A.11}):&\quad (m+1)+2(n-1)-1=m+2n-2,\\ (\text{A.13}):&\quad (m+2)+2(n-1)-1=m+2n-1,\\ (\text{A.18}):&\quad (m+4)+2(n-2)-1=m+2n-1.\end{aligned}$$

For the new coordinates of (A.11), take its phase term of degree ℓ, $0 \le \ell \le m$; the reduced term has displacement degree $m-\ell+1$, source degree $2(n-1)+m-\ell+1$, and sources affine in r_1.

$$\begin{aligned}\deg_{r_1} &\le \ell + 2(n-1) + m - \ell + 1 = m + 2n - 1,\\ \deg_{r_2} &\le m - \ell + 1 \le m + 2n - 1.\end{aligned}$$

In (A.13) and (A.18), the displacement substitutions are affine and each new-coordinate coefficient has degree at most one, giving degree at most $m+1$; hence

$$\deg_{r_j}(\text{degree-}m\text{ integrand}) \le m+2n-1, \qquad \#\{\text{candidate monomials}\} \le (m+2n)^{2n}.$$

Integrating the monomials by (A.42) proves the finite sum; the same operator induction applies to arbitrary-source type I′ graphs reduced entirely by (A.11). □

Graphs of type I′

Definition. A graph of type I′ is built from one edge by adding vertices joined to both endpoints of an existing edge [13, Definition 4.12].

Proposition A.4. Let Γ contain AB, interaction trees T_j, and at most one spoke per vertex to each insertion, with $|E_j| = |T_j| + 1$ spokes. Then Γ is of type I′ exactly when the endpoints of every internal edge share an insertion neighbour. For cubic vertices, each tree is then an arm with a common insertion neighbour, its source at one endpoint, and both insertion spokes at the other. For $n \ge 3$, it is a spider exactly when all arms of length at least two share the same insertion neighbour.

Proof.

Every edge of a type I′ graph lies in a triangle. A triangle containing an internal edge must meet an insertion, since the internal graph is a forest. Conversely, the common-neighbour condition gives each interaction vertex a spoke. If every leaf had two, a tree with $L \ge 2$ leaves would have

$$|E_j| \ge 2L + (|T_j| - L) \ge |T_j| + 2,$$

contrary to the hypothesis. A leaf with one spoke is removed by a series–parallel step. Induction proves the first assertion. For cubic vertices, a vertex of tree degree three has zero spokes by (7.5). Thus every tree is a path, and its common insertion propagates along the path. The spider condition says precisely that this insertion is shared by all nontrivial arms. □

Table A.1 gives the marked-graph census through seven loops, up to isomorphism fixing A, B. The fan and spider columns exclude the book.

Table A.1: Census of the marked graphs through seven loops, up to isomorphism fixing A, B. The fan and spider columns exclude the book.

n	graphs	books	fans	spiders	type I′,	not type I′,	branching	$A \leftrightarrow B$
				(≥ 3 arms)	not spiders	paths	trees	symmetric
1	1	1	0	0	0	0	0	1
2	3	1	2	0	0	0	0	1
3	8	1	4	0	0	3	0	2
4	23	1	6	2	1	11	2	3
5	65	1	6	6	3	38	11	5
6	194	1	8	12	8	112	53	8
7	576	1	8	20	16	313	218	14

Higher-degree vertices

For one superpotential vertex, $\deg_\Gamma \bar{W}$ counts contracted fields and $\deg \bar{W}$ counts its monomial factors. The field count is

$$\begin{aligned}
\#\{\text{degree-two cubic vertices}\} &= \underbrace{3(|T_j|-1)}_{\text{three per cubic vertex}} - \underbrace{(3|T_j|-1-\deg_\Gamma \bar{W})}_{\substack{\text{their degrees,}\\ \text{by (7.6)}}} = \deg_\Gamma \bar{W} - 2,\\
\#\{\text{fields surviving at } \bar{W}\} &= \deg \bar{W} - \deg_\Gamma \bar{W},\\
\#\{\text{fields surviving in } T_j\} &= \underbrace{3(|T_j|-1)+\deg \bar{W}}_{\text{fields}} - \underbrace{(3|T_j|-1)}_{\substack{\text{contracted ends,}\\ \text{(7.6)}}} = \deg \bar{W} - 2.
\end{aligned} \tag{A.43}$$

A.6.1 sourced trivalent step

Before parallel merges, the triangle distributes the source by its opposite-side resistances,

$$\begin{aligned}
\Sigma &= t_{uw}^{\mathrm{red}} + t_{ux}^{\mathrm{red}} + t_{wx}^{\mathrm{red}},\\
\lambda_u^{\mathrm{red}} &= \lambda_u + \frac{t_{wx}^{\mathrm{red}}}{\Sigma}\lambda_v, \qquad \lambda_w^{\mathrm{red}} = \lambda_w + \frac{t_{ux}^{\mathrm{red}}}{\Sigma}\lambda_v,\\
\lambda_x^{\mathrm{red}} &= \lambda_x + \frac{t_{uw}^{\mathrm{red}}}{\Sigma}\lambda_v.
\end{aligned} \tag{A.44}$$

In (A.44), red labels the triangle before parallel merges; in Appendix A.3 it labels the merged graph. By (A.10),

$$\begin{aligned}
t_{uw}^{\mathrm{red}}\Big|_{\text{before merges}} &= t_{uw}^{\mathrm{red}}\Big|_{\text{after merges}},\\
t_{ux}^{\mathrm{red}}\Big|_{\text{before merges}} &= \frac{t_{ux}^{\mathrm{red}}}{1-r_1}\Big|_{\text{after merges}}, \qquad t_{wx}^{\mathrm{red}}\Big|_{\text{before merges}} = \frac{t_{wx}^{\mathrm{red}}}{1-r_2}\Big|_{\text{after merges}}.
\end{aligned}$$

These substitutions give the same Σ (A.15) and source transfer (A.16) as in Appendix A.3. Use the sourced operator (A.17), including the differentials of this resistance-dependent transfer.

The sourced transfer (A.44) preserves the linear dependence of every current on λ; the kernel of Appendix A.3 has source degree $2n+m$ at displacement degree m.

For the sourced three-loop coefficient below, we combine the reductions with the quadratic identity of the graph kernels [9, Section 3.2], [5, Theorem 7] to evaluate the rational density of Example 7.9.

The sourced three-loop graph

Keep the ordered edges $(21, 2A, 2B, 31, 3A, 3B, AB)$ of Example 7.9, set $\lambda_3 = 0$, and extract the degree-zero displacement term.

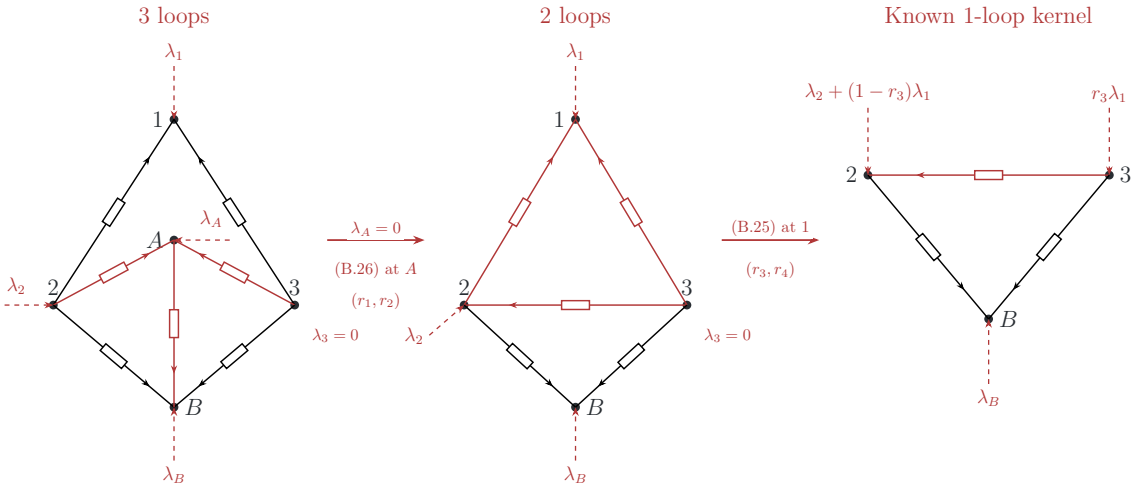


Figure A.9: The sourced three-loop graph and its boundary calculation at $\lambda_A = 0$: (A.13) at A, then (A.11) at 1, leave the known triangle kernel. The arrows retain the edge orientations and show the transferred sources.

Insert a vertex v on AB, with ordered edges $(21, 2A, 2B, 31, 3A, 3B, Av, vB)$; its proper induced Laman subgraphs are the eight edges, and we choose the sources below on this six-vertex graph.

$$0 = \sum_{\substack{1\le k\le 8 \\ e_k \text{ in } (21,2A,2B,31,3A,3B,Av,vB)}} (-1)^{k-1}\, \mathcal{I}^{\text{raw cut}}_{\Gamma_{AB\mapsto Av,vB}/e_k}(\lambda; 0), \qquad \lambda_3 = \lambda_B = 0, \qquad \lambda_v = -\lambda_1 - \lambda_2 - \lambda_A. \tag{A.45}$$

Each contraction inherits the remaining edge order and adds its endpoint sources; contracting vB restores the target source $\lambda_B^{\text{red}} = \lambda_v = -\lambda_1 - \lambda_2 - \lambda_A$.

Contractions of $2A, 2B, 3A, 3B$ are fans, evaluated by (5.7), and the other four are copies of the target graph; with the sources of (A.45), the signed fan sum is

$$\begin{aligned}
&\sum_{\substack{k\in\{2,3,5,6\} \\ \text{fan contractions}}} (-1)^{k-1} \mathcal{I}^{\text{raw cut}}_{\Gamma_{AB\mapsto Av,vB}/e_k}(\lambda; 0) \\
&= \frac{\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a} + \lambda_{1,+\dot a}\lambda_{A,+}{}^{\dot a}}{144}\Big[(\lambda_{1,+\dot b}\lambda_{2,+}{}^{\dot b})(\lambda_{1,+\dot c}\lambda_{A,+}{}^{\dot c}) \\
&\quad + 2(\lambda_{1,+\dot b}\lambda_{2,+}{}^{\dot b})(\lambda_{2,+\dot c}\lambda_{A,+}{}^{\dot c}) + (\lambda_{1,+\dot b}\lambda_{A,+}{}^{\dot b})(\lambda_{2,+\dot c}\lambda_{A,+}{}^{\dot c}) + (\lambda_{2,+\dot b}\lambda_{A,+}{}^{\dot b})^2\Big].
\end{aligned}$$

Coefficient calculation.

The zero-displacement kernel is a cubic polynomial in source spinor contractions; the even edge permutation $A \leftrightarrow B$ sends both remaining contractions to their negatives when $\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a} = 0$.

$$\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a} = 0 \quad\Longrightarrow\quad \lambda_{1,+\dot a}\lambda_{B,+}{}^{\dot a} = -\lambda_{1,+\dot a}\lambda_{A,+}{}^{\dot a}, \qquad \lambda_{2,+\dot a}\lambda_{B,+}{}^{\dot a} = -\lambda_{2,+\dot a}\lambda_{A,+}{}^{\dot a},$$
$$\mathcal{I}^{\text{raw cut}}_{\Gamma}(\lambda; 0)\Big|_{\lambda_3=0,\,\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a}=0} = -\,\mathcal{I}^{\text{raw cut}}_{\Gamma}(\lambda; 0)\Big|_{\lambda_3=0,\,\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a}=0} = 0.$$

Thus the kernel contains the factor $\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a}$; denote the six numerical coefficients inside the brackets of (A.47), in displayed order, by $c_0,\dots,c_5$.

To fix the last coefficient, set $\lambda_A=0$. Apply (A.13) with $(v,u,w,x)=(A,3,2,B)$ and coordinates r_1,r_2, then (A.11) with $(v,u,w)=(1,3,2)$ and coordinates r_3,r_4. On the final triangle $3,2,B$, use $r_5,r_6>0$, $r_5+r_6<1$:

$$
\begin{aligned}
(\lambda_3^{\mathrm{red}})^{\mathrm{red}} &= r_3\lambda_1, & (\lambda_2^{\mathrm{red}})^{\mathrm{red}} &= \lambda_2+(1-r_3)\lambda_1,\\
(y_{32}^{\mathrm{red}})^{\mathrm{red}} &= r_5r_3\lambda_1 - r_6[\lambda_2+(1-r_3)\lambda_1],\\
\lambda_{1,+\dot a}(y_{32,+}^{\mathrm{red}})^{\mathrm{red}\dot a} &= -r_6\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a},\\
(y_{3B,+\dot a}^{\mathrm{red}})^{\mathrm{red}}(y_{2B,+}^{\mathrm{red}})^{\mathrm{red}\dot a} &= r_3(1-r_5-r_6)\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a}.
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda;0)\Big|_{\lambda_A=\lambda_3=0} &= (\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a})^3\int_{[0,1]^4\times\{r_5,r_6>0,\ r_5+r_6<1\}} d^6r\\
&\quad\times \underbrace{r_1r_2r_3(1-r_5-r_6)}_{\text{(A.13)}}\underbrace{(-r_4r_6)}_{\text{(A.11)}}\underbrace{r_3}_{\text{triangle}}\\
&= (\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a})^3\left(\frac12\right)^3\frac13\left(\frac{1}{24}+\frac{1}{12}-\frac16\right) = -\frac{(\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a})^3}{576}.
\end{aligned}
$$

Multiply (A.45) by 576 and compare, in order, the coefficients of

$$
\begin{gathered}
(\lambda_{1,+\dot b}\lambda_{2,+}{}^{\dot b})^2(\lambda_{1,+\dot c}\lambda_{A,+}{}^{\dot c}),\qquad (\lambda_{1,+\dot b}\lambda_{2,+}{}^{\dot b})^2(\lambda_{2,+\dot c}\lambda_{A,+}{}^{\dot c}),\\
(\lambda_{1,+\dot b}\lambda_{2,+}{}^{\dot b})(\lambda_{1,+\dot c}\lambda_{A,+}{}^{\dot c})(\lambda_{2,+\dot d}\lambda_{A,+}{}^{\dot d}),\qquad (\lambda_{1,+\dot b}\lambda_{2,+}{}^{\dot b})(\lambda_{2,+\dot c}\lambda_{A,+}{}^{\dot c})^2.
\end{gathered}
$$

Dividing the third equality by 2 gives the first four rows below; $A\leftrightarrow B$ gives the fifth, and the boundary integral gives the sixth:

$$
\begin{pmatrix}
1 & 0 & 1 & 0 & 0 & 0\\
1 & 0 & 0 & 0 & 1 & 0\\
1 & -1 & -1 & 0 & 0 & 0\\
2 & -1 & -1 & -1 & 0 & 0\\
-2 & 1 & 2 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}
\begin{pmatrix} c_0\\ c_1\\ c_2\\ c_3\\ c_4\\ c_5 \end{pmatrix}
=
\begin{pmatrix} 4\\ 8\\ -6\\ -4\\ 0\\ -1 \end{pmatrix},
\qquad \det = 2,
\tag{A.46}
$$

$$
(c_0,c_1,c_2,c_3,c_4,c_5) = (5,12,-1,3,3,-1).
$$

The finite system determines all six coefficients. □

$$
\begin{aligned}
&\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda;0)\Big|_{\lambda_3=0}\\
&= \frac{\lambda_{1,+\dot a}\lambda_{2,+}{}^{\dot a}}{576}\Big[5(\lambda_{1,+\dot b}\lambda_{A,+}{}^{\dot b})^2+12(\lambda_{1,+\dot b}\lambda_{A,+}{}^{\dot b})(\lambda_{2,+\dot c}\lambda_{A,+}{}^{\dot c})\\
&\quad-(\lambda_{1,+\dot b}\lambda_{2,+}{}^{\dot b})(\lambda_{1,+\dot c}\lambda_{A,+}{}^{\dot c})+3(\lambda_{2,+\dot b}\lambda_{A,+}{}^{\dot b})^2\\
&\quad+3(\lambda_{1,+\dot b}\lambda_{2,+}{}^{\dot b})(\lambda_{2,+\dot c}\lambda_{A,+}{}^{\dot c})-(\lambda_{1,+\dot b}\lambda_{2,+}{}^{\dot b})^2\Big].
\end{aligned}
\tag{A.47}
$$

B Charge sectors and the grading of the loop corrections

For $\mathcal{N}=4$ Yang–Mills theory, we organize the cochains by their letter weights and derivative counts.

Counting the vertices and contractions gives the grading of each loop correction, which

determines its target sector and bounds the possible obstruction orders.

For the $\mathcal{N}=4$ letters, θ_I counts the Ith odd-coordinate weight in C, and $n_\pm$ counts $\partial_{+\pm}$ derivatives.

For a monomial with Y letters, let e_I be the Ith unit vector and write

$$\begin{aligned} &\lambda=(\theta_1,\theta_2,\theta_3;n_+,n_-), \qquad \Theta=\theta_1+\theta_2+\theta_3, \qquad \mathsf{d}=(1,1,1;-1,-1),\\ &\theta(\gamma^I)=e_I, \qquad \theta(\beta_I)=(1,1,1)-e_I, \qquad \theta(\mathsf{b})=(1,1,1), \qquad \theta(\mathsf{c})=0, \end{aligned} \tag{B.1}$$

$C^Y(\lambda)$, also written $(\lambda)_Y$, is the cochain sector with weight λ and Y letters. Its component degree is $\Theta=\deg_\theta$.

$$\deg_\theta \mathsf{c}=0, \quad \deg_\theta \gamma=1, \quad \deg_\theta \beta=2, \quad \deg_\theta \mathsf{b}=3, \qquad \deg_\theta(\partial_{+\dot a}X)=\deg_\theta X, \tag{B.2}$$

At order n, a product graph has n cubic vertices and $2n$ physical edges, each pairing component degrees that sum to three.

$$\deg_\theta \mathcal{O}_n=\deg_\theta \mathcal{O}_0+3n-6n=\deg_\theta \mathcal{O}_0-3n, \qquad \mathcal{O}_n=0 \quad \text{if } 3n>\deg_\theta \mathcal{O}_0. \tag{B.3}$$

The degree losses add under iterated products.

For a nonzero Q_nX, the $n+1$ cubic vertices and $2n+1$ edges give

$$\deg_\theta(Q_nX)=\deg_\theta X+3(n+1)-3(2n+1)=\deg_\theta X-3n \tag{B.4}$$

The physical charges are the dimension E, the three $SU(4)_R$ Cartan charges R_I, and the spins J_L, J_R.

$$\begin{aligned} E&=\tfrac12(\Theta+Y)+n_++n_-, \qquad R_I=\theta_I-\tfrac12(\Theta-Y),\\ J_L&=\tfrac12(\Theta-Y+n_++n_-), \qquad J_R=\tfrac12(n_+-n_-),\\ E&=R_1+R_2+R_3+2J_L, \qquad 2(E+J_L)=2\Theta+3(n_++n_-). \end{aligned} \tag{B.5}$$

Table B.1: Letters of $\mathcal{N}=4$ Yang–Mills theory and their charges.

letter	physical field	θ	(n_+,n_-)	E	(R_1,R_2,R_3)	J_L	J_R
γ^I	scalar $\bar\phi_I$	e_I	$(0,0)$	1	e_I	0	0
β_I	fermion ψ_+^I	$(1,1,1)-e_I$	$(0,0)$	$\frac32$	$(\frac12,\frac12,\frac12)-e_I$	$\frac12$	0
b	field strength f_{++}	$(1,1,1)$	$(0,0)$	2	$(0,0,0)$	1	0
$\partial_{+\pm}\mathsf{c}$	gaugino $\bar\lambda_\pm$	0	$(1,0),(0,1)$	$\frac32$	$(\frac12,\frac12,\frac12)$	0	$\pm\frac12$
$\partial_{+\pm}$	covariant derivative $\nabla_{+\pm}$	0	$(1,0),(0,1)$	1	$(0,0,0)$	$\frac12$	$\pm\frac12$

The level is $E+J_L$. Equation (B.5) makes its set of sectors finite.

Example Threshold charges

The $SU(2)$ threshold class $\mathcal{O}_0$ [20, 21] has two γ, five β and zero derivatives.

$$\lambda=(4,4,4;0,0), \quad Y=7,$$
$$E=\frac{19}{2}, \qquad R_I=\frac32, \qquad J_L=\frac52, \qquad E+J_L=12.$$

The current form (3.28) has degree n in each dotted component, giving

$$\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda_{v,+\dot a}\mapsto c\lambda_{v,+\dot a};\ z_e^{+\dot a}\mapsto c^{-1}z_e^{+\dot a})=c^n\,\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda;z), \qquad c>0. \tag{B.6}$$

The λ powers supply output derivatives and the z powers remove input derivatives. Hence Q_n adds n derivatives in each direction [3, Equations (3.44)–(3.45)].

Q_0 preserves both totals. Choose homogeneous contractions (i_r, π_r, h_r) in (9.23).

Proposition B.1.

For $n \geq 0$ and $r \geq 1$,

$$Q_n : C^Y(\lambda) \to C^{Y+1-n}(\lambda - n\mathsf{d}), \qquad d_r : E_r^Y(\lambda) \to E_r^{Y+1-r}(\lambda - r\mathsf{d}). \tag{B.7}$$

Both maps have $(\Delta E; \Delta R_I; \Delta J_L, \Delta J_R) = (\frac{1}{2}; \frac{1}{2}, \frac{1}{2}, \frac{1}{2}; -\frac{1}{2}, 0)$.

Proof

$$\Delta Y = 3(n+1) - 2(2n+1) = 1 - n, \qquad \theta_I(Q_n X) - \theta_I(X) = (n+1) - (2n+1) = -n.$$

The vertex and edge counts give the letter and flavour shifts. Equation (B.6) supplies n derivatives in each direction.

The homogeneous contractions and transferred coefficients obey

$$\begin{aligned} h_r &: E_r^Y(\lambda) \longrightarrow E_r^{Y+r-1}(\lambda + r\mathsf{d}), \\ [\hbar^m]\mathbf{Q}^{(r)} &: E_r^Y(\lambda) \longrightarrow E_r^{Y+1-r-m}(\lambda - (r+m)\mathsf{d}). \end{aligned} \tag{B.8}$$

For $[\hbar^m]\mathbf{Q}^{(r+1)}$, let ℓ count perturbation factors. Their orders sum to $m+1$, and the term contains $\ell - 1$ copies of h_r.

$$\begin{aligned} \Delta Y &= \ell(1-r) - (m+1) + (\ell-1)(r-1) = -r - m, \\ \Delta\lambda &= -(\ell r + m + 1)\mathsf{d} + (\ell - 1)r\mathsf{d} = -(r+m+1)\mathsf{d}. \end{aligned} \tag{B.9}$$

This proves the induction from $\mathbf{Q}^{(0)} = \mathbf{Q}$. Set $m = 0$ for d_{r+1} and insert the shifts into (B.5).

$$\Delta\Theta = -3n, \qquad \Delta(n_+ + n_-) = 2n, \qquad (\Delta E; \Delta R_I; \Delta J_L, \Delta J_R) = \left(\frac{1}{2}; \frac{1}{2}, \frac{1}{2}, \frac{1}{2}; -\frac{1}{2}, 0\right).$$

□

The sectors $(\lambda + k\mathsf{d})_{Y+k}$, $k \in \mathbb{Z}$, have the same physical charges. The cochain grading distinguishes the loop orders.

Corollary B.2.

An outgoing d_r requires $r \leq \min_I \theta_I$, and an incoming d_r requires $r \leq \min(n_+, n_-)$.

A class with zero input derivatives has no incoming d_r for $r \geq 1$.

Proof

$$\theta_I - r \geq 0 \iff r \leq \min_I \theta_I, \qquad n_\pm - r \geq 0 \iff r \leq \min(n_+, n_-).$$

□

References

[1] C.-M. Chang and X. Yin, "1/16 BPS States in $\mathcal{N} = 4$ Super-Yang–Mills Theory," *Phys. Rev. D* **88** (2013), 106005, arXiv:1305.6314.

[2] J. Choi and E. Lee, "Konishi Lifts a Black Hole," arXiv:2511.09519.

[3] K. Budzik, D. Gaiotto, J. Kulp, B. R. Williams, J. Wu and M. Yu, "Semi-Chiral Operators in 4d $\mathcal{N} = 1$ Gauge Theories," *JHEP* **05** (2024), 245, arXiv:2306.01039.

[4] K. Budzik and J. Kulp, "Loop Corrected Supercharges from Holomorphic Anomalies," *JHEP* **08** (2026), 089, arXiv:2512.07771.

[5] M. Wang, "Feynman Graph Integrals on $\mathbb{C}^d$," *Commun. Math. Phys.* **406** (2025), 116, arXiv:2401.08113.

[6] D. Gaiotto, J. Kulp and J. Wu, "Higher Operations in Perturbation Theory," *JHEP* **05** (2025), 230, arXiv:2403.13049.

[7] K. Konishi, "Anomalous Supersymmetry Transformation of Some Composite Operators in SQCD," *Phys. Lett. B* **135** (1984), 439–444.

[8] F. Cachazo, M. R. Douglas, N. Seiberg and E. Witten, "Chiral Rings and Anomalies in Supersymmetric Gauge Theory," *JHEP* **12** (2002), 071, arXiv:hep-th/0211170.

[9] K. Budzik, D. Gaiotto, J. Kulp, J. Wu and M. Yu, "Feynman Diagrams in Four-Dimensional Holomorphic Theories and the Operatope," *JHEP* **07** (2023), 127, arXiv:2207.14321.

[10] J. Mathews, "Application of Linear Network Analysis to Feynman Diagrams," *Phys. Rev.* **113** (1959), 381–385.

[11] I. Prlina, M. Spradlin and S. Stanojevic, "All-Loop Singularities of Scattering Amplitudes in Massless Planar Theories," *Phys. Rev. Lett.* **121** (2018), 081601, arXiv:1805.11617.

[12] E. Panzer, *Feynman Integrals and Hyperlogarithms*, PhD thesis, Humboldt-Universität zu Berlin (2015), arXiv:1506.07243.

[13] Z. Gui, M. Wang and B. R. Williams, "Higher-dimensional Chiral Algebras in the Jouanolou Model and Free-Field Realization," arXiv:2510.26608.

[14] B. Li, K. Liu and C.-M. Chang, "Quantum Lifts: Results Report and Supplementary Material on the Algorithm of the Lifting Tests," `https://github.com/lbotao399-stack/quantum-lifts-results`.

[15] R. G. Leigh and M. J. Strassler, "Exactly Marginal Operators and Duality in Four Dimensional $\mathcal{N} = 1$ Supersymmetric Gauge Theory," *Nucl. Phys. B* **447** (1995), 95–136, arXiv:hep-th/9503121.

[16] S. Weinberg, *The Quantum Theory of Fields*, Vol. III: *Supersymmetry*, Cambridge University Press (2000).

[17] K. Costello, *Renormalization and Effective Field Theory*, Mathematical Surveys and Monographs **170**, American Mathematical Society (2011).

[18] D. Gaiotto, J. Kulp and M. Yu, *Laman Loopstrap*, version 1.0.0 (2022), `https://github.com/TwistedQFTs/laman-loopstrap`.

[19] S. Choi, S. Kim, E. Lee, S. Lee and J. Park, "Towards Quantum Black Hole Microstates," *JHEP* **11** (2023), 175, arXiv:2304.10155.

[20] S. Kim, S. Kim, S. Lee and J. Park, "Quantum Black Hole Cohomologies," arXiv:2606.27955.

[21] C.-M. Chang and Y.-H. Lin, "Words to Describe a Black Hole," *JHEP* **02** (2023), 109, arXiv:2209.06728.

[22] S. Kim, J. Lee, S. Lee and J. Song, "Black Hole Cohomologies in $\mathcal{N} = 4$ SYM," arXiv:2610.20340.

[23] R. de Mello Koch, M. Kim, S. Kim, J. Lee and S. Lee, "Brane-Fused Black Hole Operators," *JHEP* **07** (2025), 216, arXiv:2412.08695.

[24] J. Choi, S. Choi, S. Kim, J. Lee and S. Lee, "Finite N Black Hole Cohomologies," *JHEP* **12** (2024), 029, arXiv:2312.16443.

[25] "Fortuity: Fortuitous BPS States of $\mathcal{N} = 4$ SYM at Weak Coupling," report and code, `https://github.com/davidsd/fortuity-ai` (2026).

[26] R. de Mello Koch, S. Duary, M. Kim and S. Liu, "Fortuitous BMN Operators from Multiple Cores," arXiv:2611.04722.

[27] A. Gadde, E. Lee, R. Raj and S. Tomar, "Probing Non-Graviton Spectra in $\mathcal{N} = 4$ SYM via BMN Truncation and S-Duality," *JHEP* **02** (2026), 026, arXiv:2506.13887.

[28] C.-M. Chang, "Super-Chevalley Restriction and Relative Lie Algebra Cohomology over the 2|3 Algebra," arXiv:2604.23549.

[29] C.-M. Chang and Y.-H. Lin, "Violation of S-duality in Classical Q-Cohomology," arXiv:2510.24008.

[30] "bps-counting: Counting of 1/16 BPS States and One-Loop Anomalous Dimensions in 3 + 1d Maximally Supersymmetric Yang–Mills with Gauge Group $SU(N)$," data accompanying [21], `https://github.com/yinhslin/bps-counting`.

[31] J. Choi and S. Kim, "Fortuity and Relevant Deformation," arXiv:2512.12674.

[32] T.-H. Chen and B. C. Ngô, "On the Hitchin Morphism for Higher Dimensional Varieties," *Duke Math. J.* **169** (2020), 1971–2004, arXiv:1905.04741.

[33] T.-H. Chen and B. C. Ngô, "Invariant Theory for the Commuting Scheme of Symplectic Lie Algebras," arXiv:2102.01849.

[34] L. Song, X. Xia and J. Xu, "A Higher-Dimensional Chevalley Restriction Theorem for Orthogonal Groups," *Adv. Math.* **426** (2023), 109104, arXiv:2207.03147.

[35] F. Vaccarino, "Linear Representations, Symmetric Products and the Commuting Scheme," *J. Algebra* **317** (2007), 634–641.

[36] M. Domokos, "Vector Invariants of a Class of Pseudoreflection Groups and Multisymmetric Syzygies," *J. Lie Theory* **19** (2009), 507–525.

[37] M. Hunziker, "Classical Invariant Theory for Finite Reflection Groups," *Transform. Groups* **2** (1997), 147–163.

[38] R. W. Richardson, "Conjugacy Classes of n-Tuples in Lie Algebras and Algebraic Groups," *Duke Math. J.* **57** (1988), 1–35.

[39] P. Li, D. Nadler and Z. Yun, "Functions on the Commuting Stack via Langlands Duality," *Ann. of Math.* **200** (2024), 609–748, arXiv:2301.02618.

[40] J. Sempliner and X. Yang, "On the Normality of the Commuting Scheme," arXiv:2609.27360.

[41] P. Goyal, "The Commuting Scheme of $\mathfrak{g}_2$," arXiv:2610.05018.

[42] P. Etingof and V. Kac, "On the Cachazo–Douglas–Seiberg–Witten Conjecture for Simple Lie Algebras," arXiv:math/0305175.

[43] M. Cederwall and G. Ferretti, "The Chiral Ring of $D = 4$, $\mathcal{N} = 1$ SYM with Exceptional Gauge Groups," *Fortsch. Phys.* **72** (2024), 2400027, arXiv:2311.12119.

[44] S. Kumar, "On Cachazo–Douglas–Seiberg–Witten Conjecture for Simple Lie Algebras," *J. Amer. Math. Soc.* **21** (2008), 797–808, arXiv:math/0412201.

[45] E. Witten, "Chiral Ring of $Sp(N)$ and $SO(N)$ Supersymmetric Gauge Theory in Four Dimensions," arXiv:hep-th/0302194.

[46] P. Etingof, "On the Cachazo–Douglas–Seiberg–Witten Conjecture for Simple Lie Algebras, II," arXiv:math/0312452.

[47] T. Kibe, S. van Leuven and K. Mathieson, "Non-Coulomb Q-Cohomology and Quantum Lifting," arXiv:2610.22332.

[48] A. N. Sergeev, "The Tensor Algebra of the Identity Representation as a Module over the Lie Superalgebras $\mathfrak{gl}(n,m)$ and $Q(n)$," *Math. USSR Sb.* **51** (1985), 419–427.

[49] A. Berele and A. Regev, "Hook Young Diagrams with Applications to Combinatorics and to Representations of Lie Superalgebras," *Adv. Math.* **64** (1987), 118–175.

[50] R. Goodman and N. R. Wallach, *Symmetry, Representations, and Invariants*, Graduate Texts in Mathematics **255**, Springer (2009).

[51] N. Bourbaki, *Groupes et algèbres de Lie*, Chapitres IV–VI, Hermann (1968).

[52] D. Bundzik and T. Månsson, "The General Leigh–Strassler Deformation and Integrability," *JHEP* **01** (2006), 116, arXiv:hep-th/0512093.

[53] The FLINT team, *FLINT: Fast Library for Number Theory*, `https://flintlib.org`, used through `https://github.com/flintlib/python-flint`.

[54] M. Wang and B. R. Williams, "On the Renormalization and Quantization of Topological–Holomorphic Field Theories," *Adv. Math.* **496** (2026), 110970, arXiv:2407.08667.